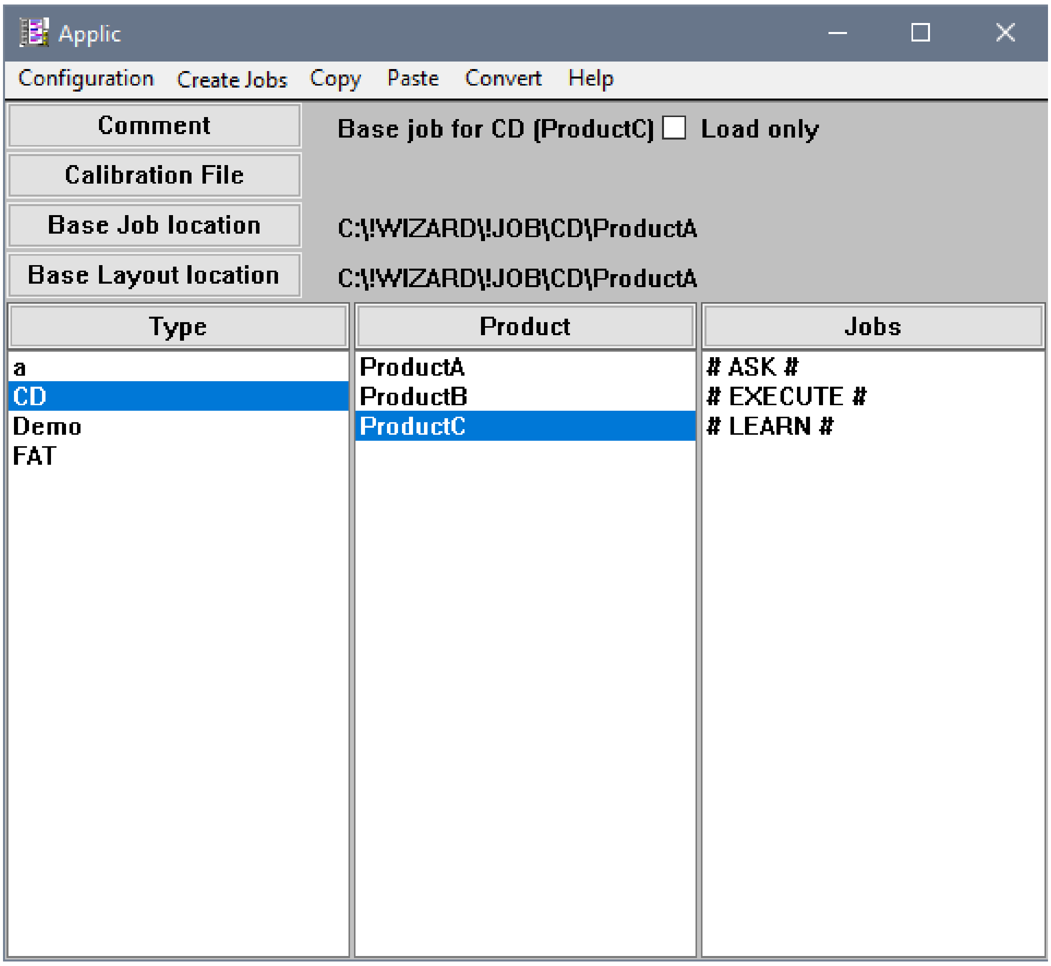
MueTec
Metrology and Inspection



The Wizard

for MT3000, MT2010, MT270UV, IRIS 2100, Spector M/A

User Manual

Legal Notices

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Publication ID

Publication ID: MTWizard-UMWEP-210728

Contents

Introduction

The wizard is a macro for executing or learning a job with MueTec systems. Depending on the job setup, the Wizard executes measurement jobs of critical dimension (CD), overlay or registration (OVL, RG), film thickness (FT), and defect inspection (SDI, Sealing Inspection).

With different system configurations, ranging from light path (reflected or transmitted) and illumination sources (UV, visible or infrared) to wafer or mask handling (manual or robotic), the wizard handles all commands for system hardware, software, job specific options, parameters and results. The wizard works in local mode as well as remotely with SECS/GEM.

This document explains how the Wizard is used for job execution and learning. The Wizard is designed to guide the user step by step through the processes.

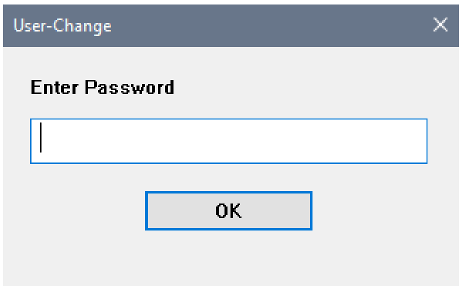
About MueTec

MueTec develops high precision optical measurement and inspection solutions that help customers in the semiconductor and microelectronics industries to improve their manufacturing yield and overall process control. Our team of technical experts knows, fulfills and anticipates the needs of the global producers of wafers, masks and MEMS and offers custom-tailored and standard solutions that distinguish themselves through their extremely low maintenance requirements. Scalable systems allow for a combination of many different applications in a single system, including critical dimension, overlay and film thickness measurement as well as defect inspection and review.

In the last 25 years, MueTec has installed more than 300 measurement and inspection systems worldwide. There is hardly a leading semiconductor manufacturer that does not use our technology in production. Due to our ability to adapt systems precisely to customers' needs, our customers include not just large semiconductor companies, but also research institutions all around the world.

1 User Access and Function Keys

The Wizard functionality and appearance depends on the user access level. Users may login during software start or click **User Change** at the menu of the **Nanostar** window to change the access level.



For **operator level access**, user can start and stop a measurement or inspection job only. Do not have access for job learning or modification.

For **user access higher than operator level**, user may execute and learn a job. If there is an existing job, user can modify the job parameter as well. In the learn mode, user is guided step by step to define all the essential job parameters.

Besides learning or modifying job, higher access user also can use the function keys of keyboard to execute different functions (see 2.8.4). Following is the function keys table of the wizard (applicable to user access above operator level only):

Function Key	Function
F1	To show the Main Menu **Docu** Window
F2	To start the job from certain job sequence and select the job mode
F3	For CD/OVL/FT: To show complete protocol For Defect Inspection: Offline cluster evaluation
F4	For CD/OVL/FT: To show summary protocol For Defect Inspection: Offline review
F5	To unload sample or clear the wafer from the chuck
F9	To check or change all numerical parameters of the job. It is possible to edit some of those parameters that cannot be edited directly during job learning.
F10	To select a recipe
ESC	To stop a running job and the wizard stays at the command line where it is stopped. Click **Yes** – The job stops and **Messages** window is displayed as below.  Click **No** – The job continues from the exact step where it was stopped.
ALT + F8	To resume executing the wizard command where it stopped before, repeating the last command that was executed during stop
ALT + F9	To resume executing the wizard command where it stopped before, skipping the last command that was executed during stop

Typically, the Wizard is named as **Wizard.mac** and stored at the directory of *C:/!Wizard/!Job*, together with other jobs.

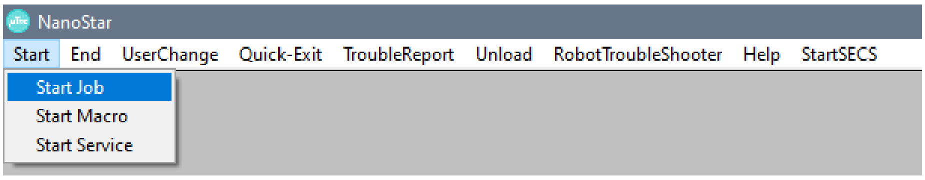
2 Job Execution

A job execution can be started with any user level.

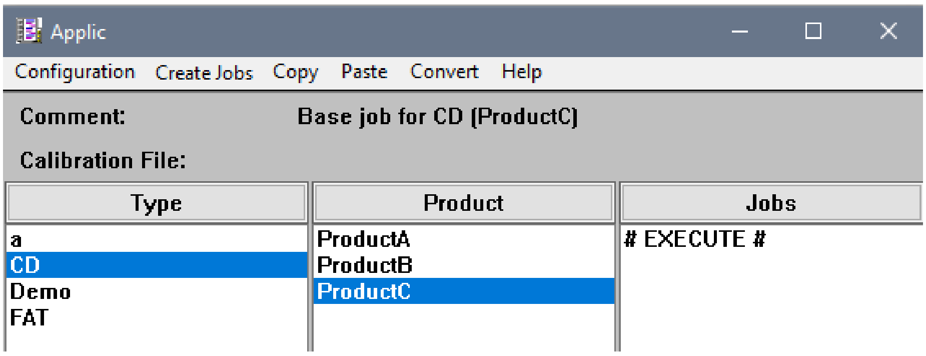
2.1 Start Job

2.1.1 Select Job and Job Mode

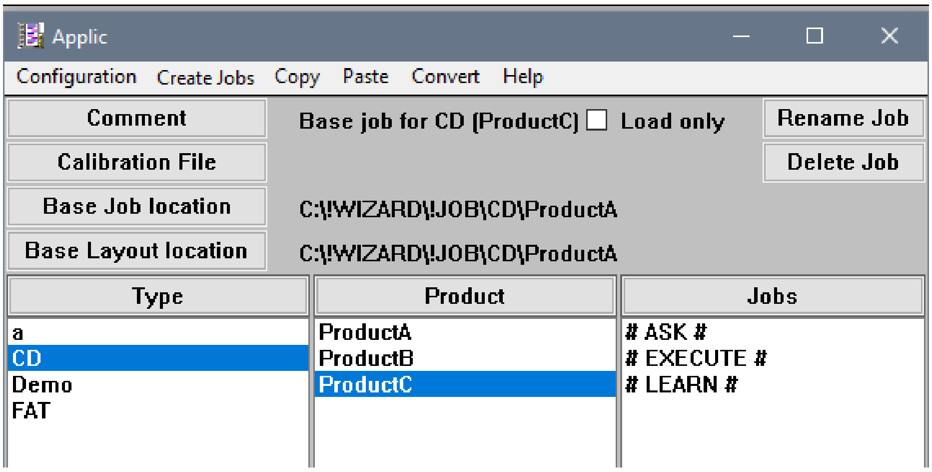
In the menu of the **Nanostar** window, select **Start – Start Job.** The **Applic** Window is shown with the list of the existing jobs. All subdirectories corresponding to the customer specific job names are defined and displayed in the **Applic** window. The **Applic** window is used to start an existing job, to learn new job or to modify job parameters. Each of the job folder contains its job parameter files that are stored at the *C:\!Wizard\!Job* directory. In the **Applic** window, the level columns are the same as the folder directories where the jobs are stored.



The **Applic** window with its specific layout is corresponding to the user access level. The **Applic** window for operator access is shown as below. There is only the **# EXECUTE #** function, offered in the rightmost column of the **Applic** window.



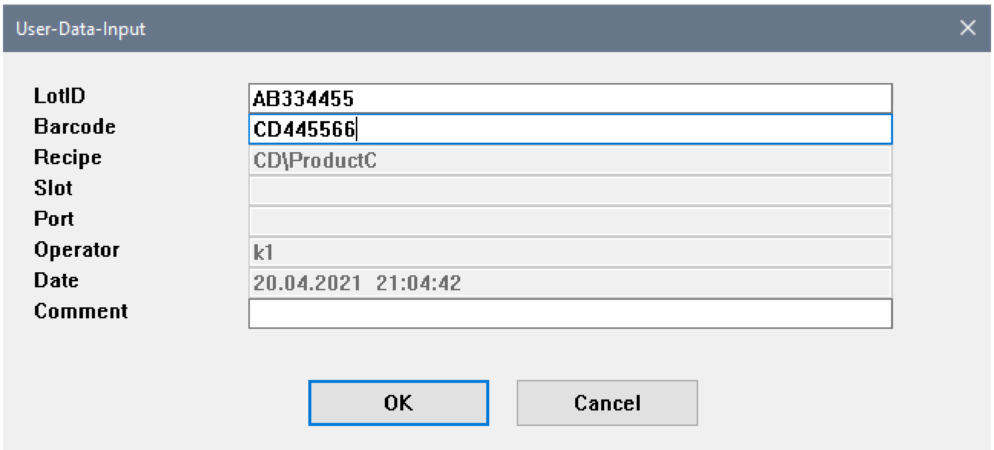
The **Applic** window for user access higher than operator level is shown as below. The **# LEARN #** function is offered in the rightmost column of the **Applic** window.



Select the job by columns from left to right (job folder and its subfolders) and click **# EXECUTE #** to start the job.

2.1.2 User Data Input

After # **Execute #** is selected, the **User Data Input** window is shown. It is mandatory to fill in the user data information, especially the first two entry fields (LotID and Barcode in the example below). These user data parameters may vary and are customized with customer preference, which can be predefined in the wizard macro. The grayed-out fields are filled automatically and locked by default. Users have no access to edit the contents of the grayed-out fields. The user data is stored and used for result files and directory naming. After measurement, the results can be identified by the user data. It is optional for users to fill in a **Comment**. Click **OK** to continue.



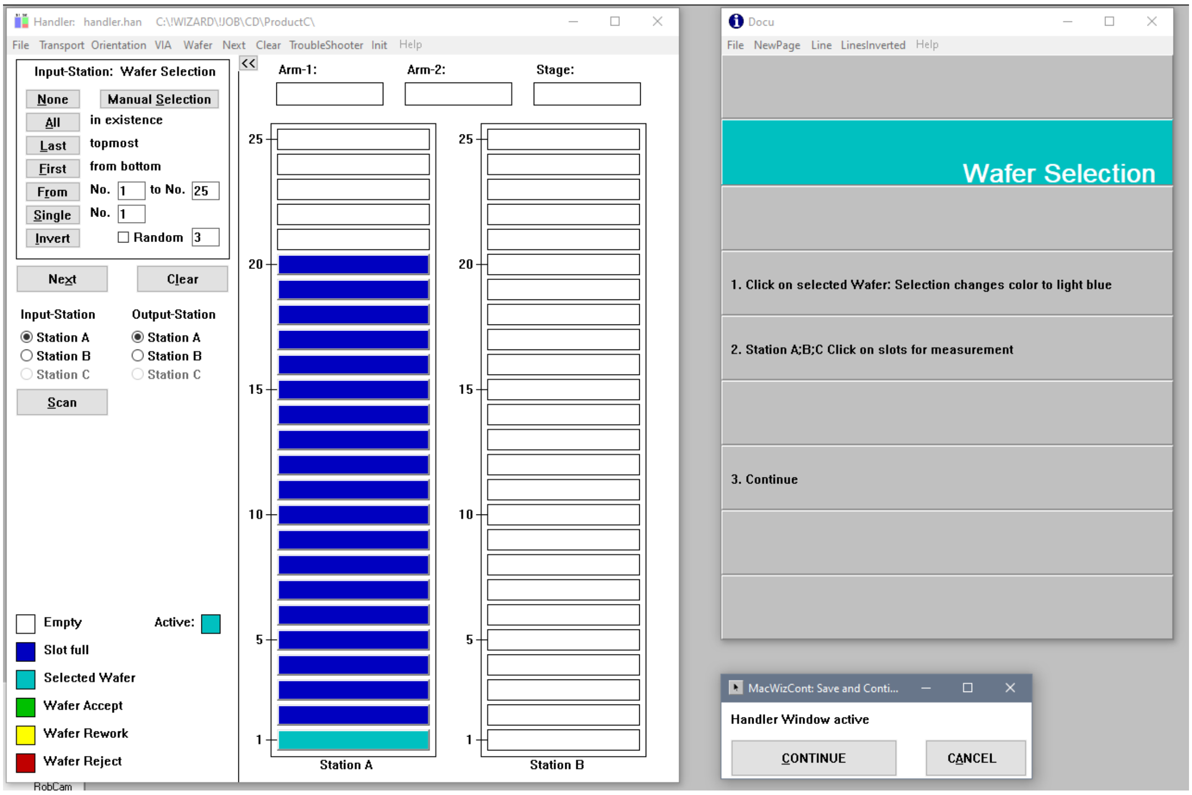
2.1.3 Wafer Handling

For systems with robotic handling, the wafer is handled by a robot with handler arms. The cassette with wafers has to be placed at one of the cassette stations. In the **Handler** window, click the **Scan** button with a selected station for cassette mapping. The cassette is scanned with the mapper to check which slot is full or empty. When mapping is done, the mapping result is displayed in the **Handler** window. A filled slot is displayed in dark blue colour whereas an empty slot is in white colour.

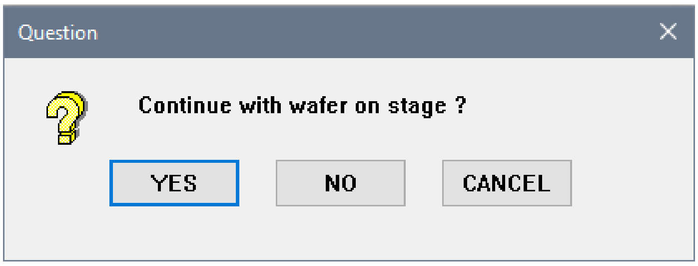
To activate the wafer selection, click the light blue square box named **Selected Wafer** at the lower left of the **Handler** window. The **Active box** is then shown in light blue colour. Select the wafer to be loaded and the selection changes the colour from dark blue to light blue in the station map. To deactivate the slot selection, click the dark blue square box named **Slot Full** at the lower left of the **Handler** window and select the slot to be deactivated. The deactivated slots are changed back to dark blue colour. It is also possible to click a selected wafer a second time to deselect it.

The sequence of wafer loading is configurable, it either starts from slot01 (bottom) or slot25 (top). In the menu of the **Handler** window, select **Transport-Input order** to define the sequence of wafer loading. It is possible to save the wafer selection by clicking **File-Save** in the **Handler** window. In the **MacWizCont** window, click **Continue** to start the wafer handling from cassette to chuck. Clicking the **Continue** button will also store the handler settings.

First, the robot handler gets the first selected wafer from the cassette and puts it onto prealigner for the flat or notch detection, followed by wafer orientation adjustment, either for wafer orientation with the barcode reader (if present) or orientation on chuck. Depending on the setting of wafer orientation on chuck, the wafer may once again transfer back to the prealigner to adjust the wafer orientation on the chuck. Lastly, the wafer will be placed on the chuck and fixed by vacuum. The stage is aligned automatically. At the menu of the **Handler** window, it is possible to turn off the barcode reader for specific recipes by selecting **VIA-via Barcode reader**. This avoids errors on wafers which do not have barcodes at all.



Through vacuum state checking, the software can detect the presence of wafer on chuck. If there is a wafer on stage, the **Question** window is shown and askes to proceed with the wafer on the stage. If declined, the user can load any other wafer.



Selection	Description
YES	To continue with the wafer on the stage for measurement or inspection
NO	The wafer on stage is unloaded back to cassette and **Handler** window will be ready for user to select another wafer
CANCEL	The wizard stops and the wafer stays on stage

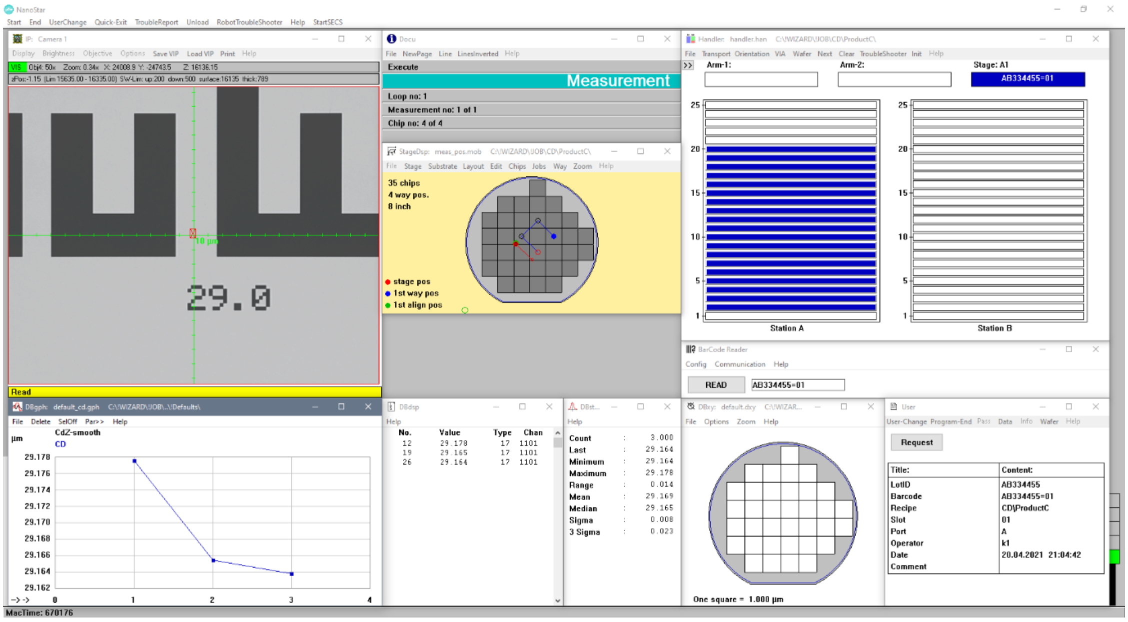
2.2 CD/OVL/FT/Image Measurement

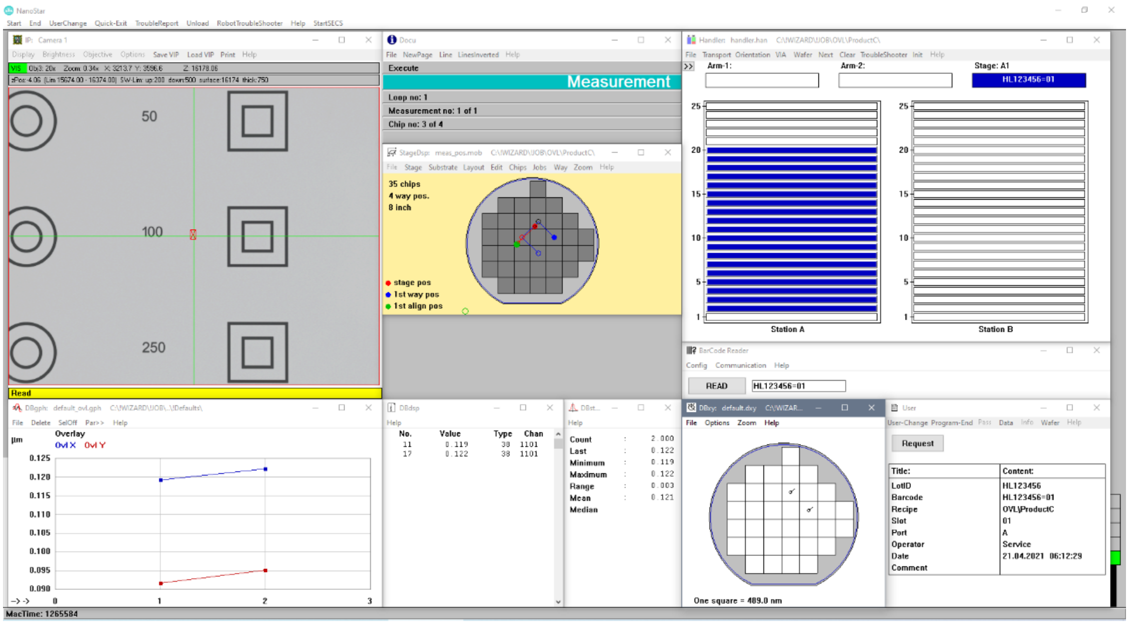
At first, the stage moves to the learned alignment positions, typically at the left, center or right of the wafer to perform stage alignment. **CFinder** searches for the alignment pattern automatically by referring to the learned reference image. The zero position (0,0) of the wafer coordinate system is referring to the first alignment position. The **Alfa** camera is also turned and adjusted to be parallel with the substrate coordinate system, either horizontally or vertically.

After the stage alignment is successful, the stage drives to the measurement positions and the job performs all the programmed tasks for measuring critical dimension, overlay, film thickness or performing a defect inspection automatically until the job is completed. The stage movement is plotted as a red line on the wafer layout in the **StageDsp** window. The **IP** window displays the current camera image.

2.2.1 CD/OVL Measurement

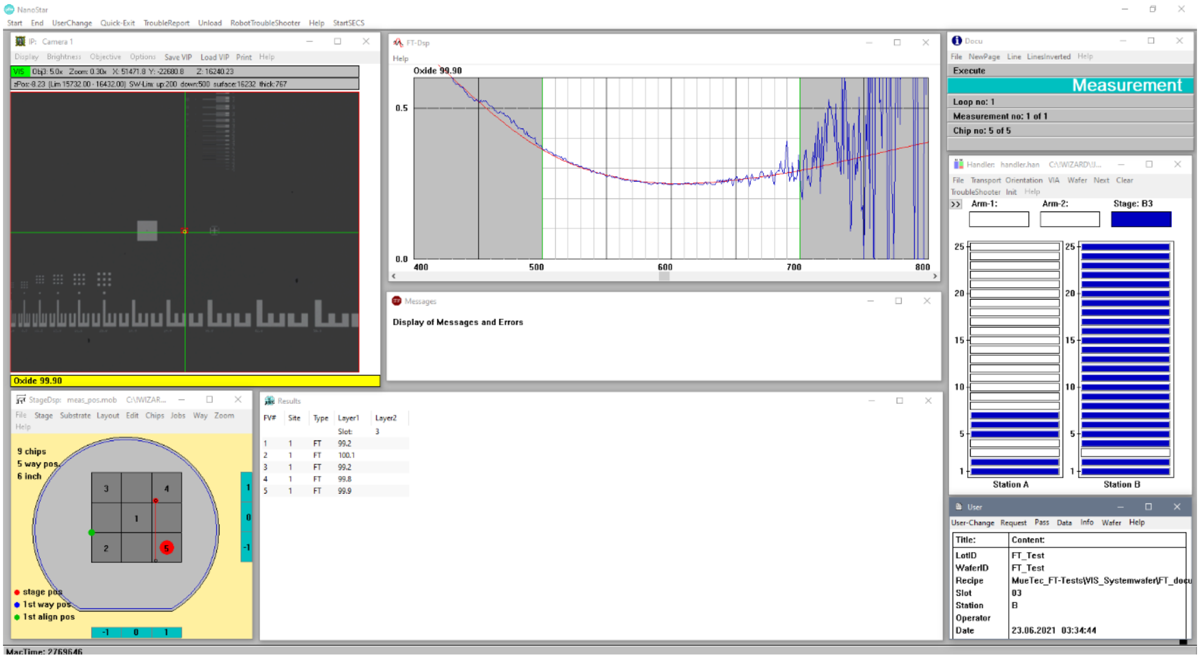
After the scanning stage moves to the learned measurement positions, performing the brightness adjustment of the lamp and the focussing automatically. The **CFinder** searches for the alignment pattern by referring to the learned reference image, followed by the CD/OVL or FT measurement. The measurements can be performed on multiple structures within the same chip and are usually repeated on other chips of the same type, as defined by the **way** and shown in the **StageDsp** window. During job measurement, all the results are continually updated in the **DBdsp & DBgph** window. In the **User** window, the **User Data Input** entered at job start is shown. This information is later used to assign the measurement results output to the measured substrate and job.





2.2.2 Film Thickness Measurement

The measurement job flow of film thickness measurement is similar to the CD/OVL measurement (see 2.2.1). At the measurement position, film thickness measurement is performed instead of CD or OVL measurement. Before starting the first film thickness measurement, the stage drives to the pre-defined calibration positions to perform background calibration (usually above a dark holder surface near to the mini chuck) and substrate calibration (usually at the substrate samples on the mini chuck). The calibration is renewed after a specified time interval (e.g. 10 minutes, if configured) has elapsed since the last calibration. The FT measurement is performed by recording the reflectance spectrum with the spectrometer and comparing a range of calculated model spectra to the measured data. During measurement, the results and spectra are continually updated in the **FT-Dsp** window.



2.2.3 Taking Images Only

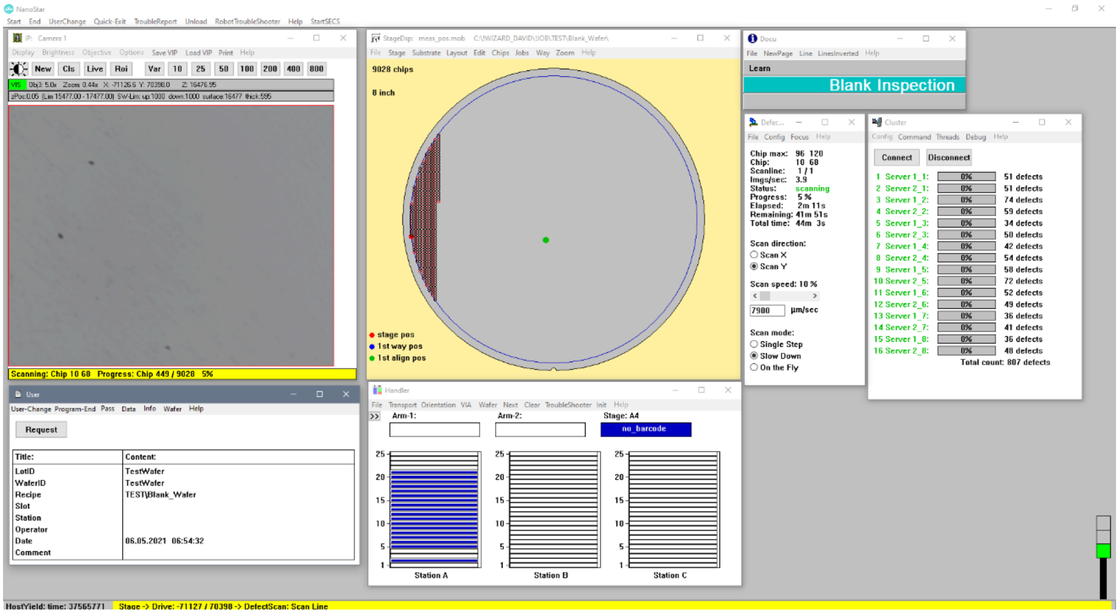
The **Image** option is a special function. The feature alignment is similar to CD/OVL but it takes only images and store them to disk without any kind of measurement.

2.3 Defect Inspection

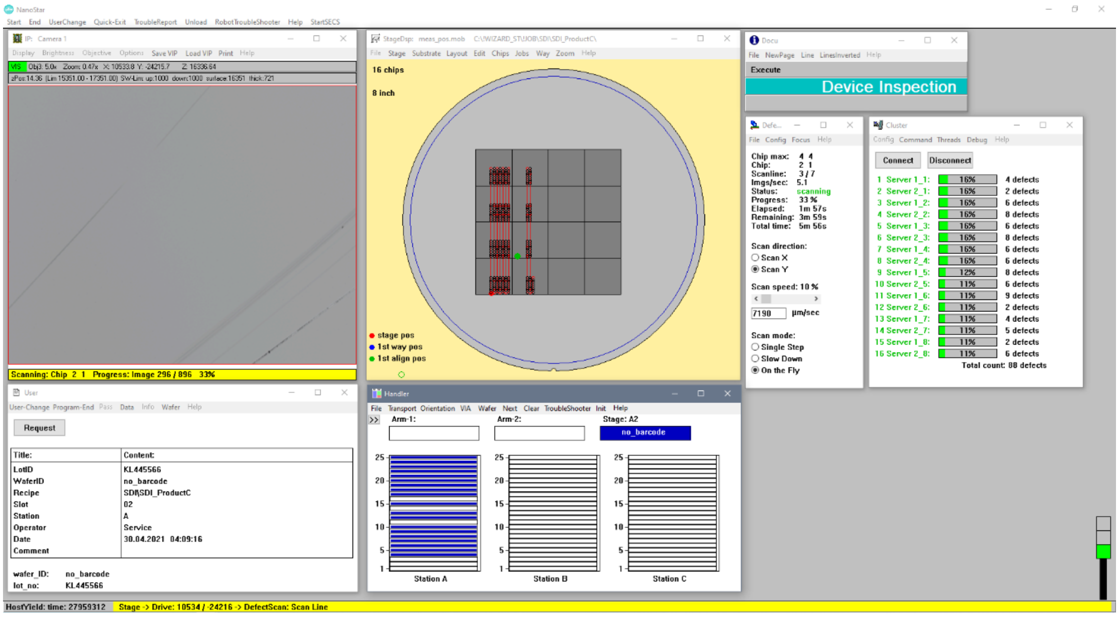
There are three types of defect inspection, **Smart Device Inspection (SDI), Blank Inspection** and **Sealing Inspection**. The availability of inspection types is a matter of purchased system options and can differ. The types of the inspection job can be defined during the job learning procedure. Sealing inspection requires a system with infrared (IR) camera and illumination. The connected cluster servers handle all data and image files for defect evaluation.

2.3.1 Smart Defect Inspection (SDI)

The scan starts immediately after the **User Data Input** is entered.



For structured wafer scan (**Smart Device Inspection**, SDI), the wafer is first aligned with the alignment pattern as CD/OVL before scanning starts.



During scanning, the overall progress of the wafer scan is graphically indicated in the **StageDsp** window with the red scan lines. The **IP** window continuously shows the images recorded by the camera. The detected defects in the individual cluster instances and the total sum of defects are displayed in the **Cluster** window. The images are evaluated using separate cluster servers connected by TCP/IP. In the **User** window, the **User Data Input** entered before scans is shown for identifying the inspection result. The paths of result data are also generated using a predefined combination of these user data.

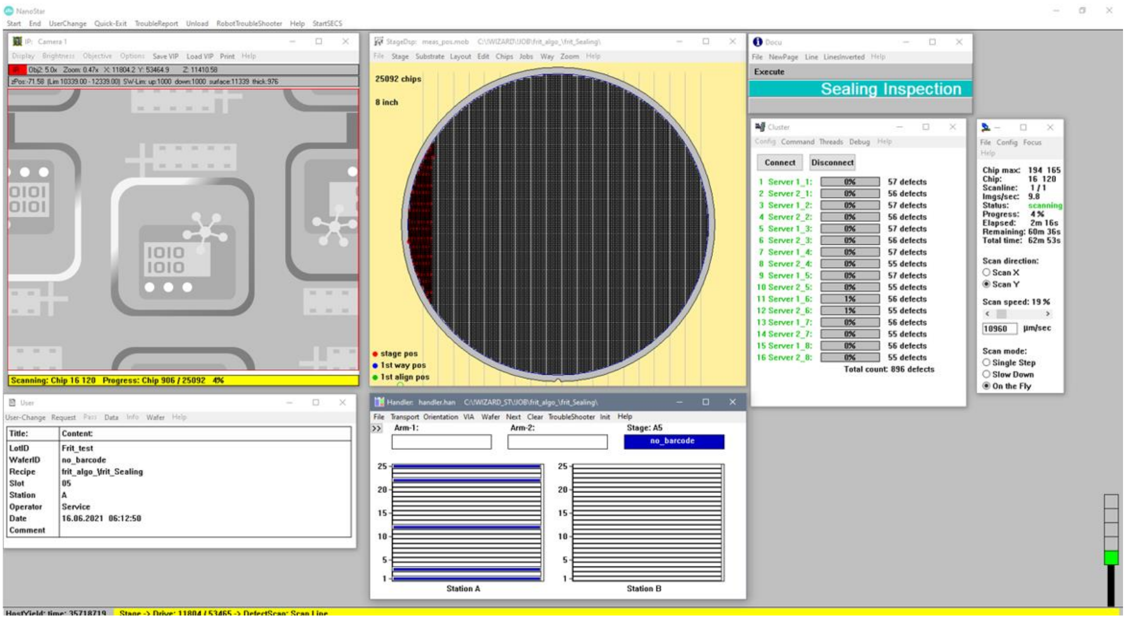
For defect detection, an average image (golden image) is first determined from a predefined number of scanned images. The golden image is compared to the scanned images. Areas with different gray values are identified as defect. During job learning, a defect is defined by the gray value threshold, the number of contributing pixels and defect size.

2.3.2 Blank Inspection

The blank inspection skips wafer alignment and chip layout definition because of the lack of structures. The scanned area refers to the wafer center definition.

2.3.3 Sealing Inspection

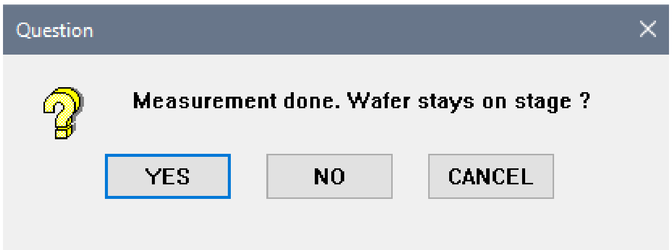
The defect scan is executed immediately after the stage alignment. In the **StageDsp** window all scanned chips (areas) are marked with red circle and lines.



2.4 Job Finished

After measurement, scan and inspection completion, the wafer will be automatically unloaded back to cassette by robot and the next wafer will be loaded onto the stage. If all wafers are evaluated, the **Applic** window is shown and the results are displayed in the **Draw** window as a protocol for users to check the measured value immediately (except for inspection jobs).

If the job is started to execute with the wafer on stage, the **Question** window is shown and askes for the wafer to stay on stage after the measurement is done. The user decides to unload the wafer back to cassette or to leave the wafer on stage. For defect inspection, click **YES** if you wish to perform an online review of the wafer, the wafer will stay on stage.

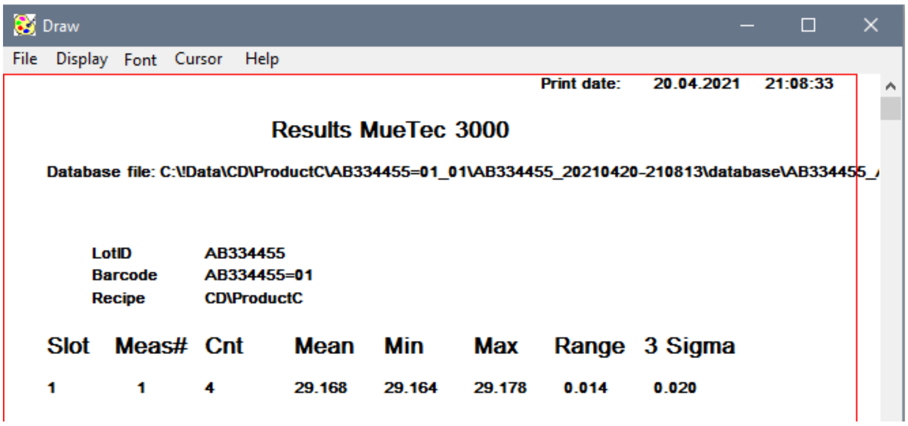


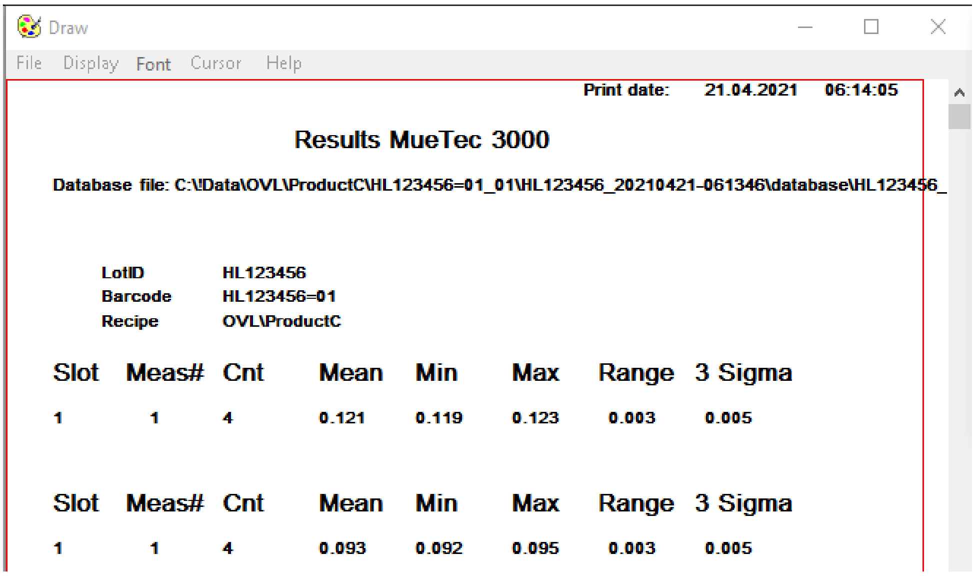
Selection	Description
YES	To leave the wafer on the stage
NO	The wafer on stage is unloaded back to cassette
CANCEL	The wizard stops and the wafer stays on stage

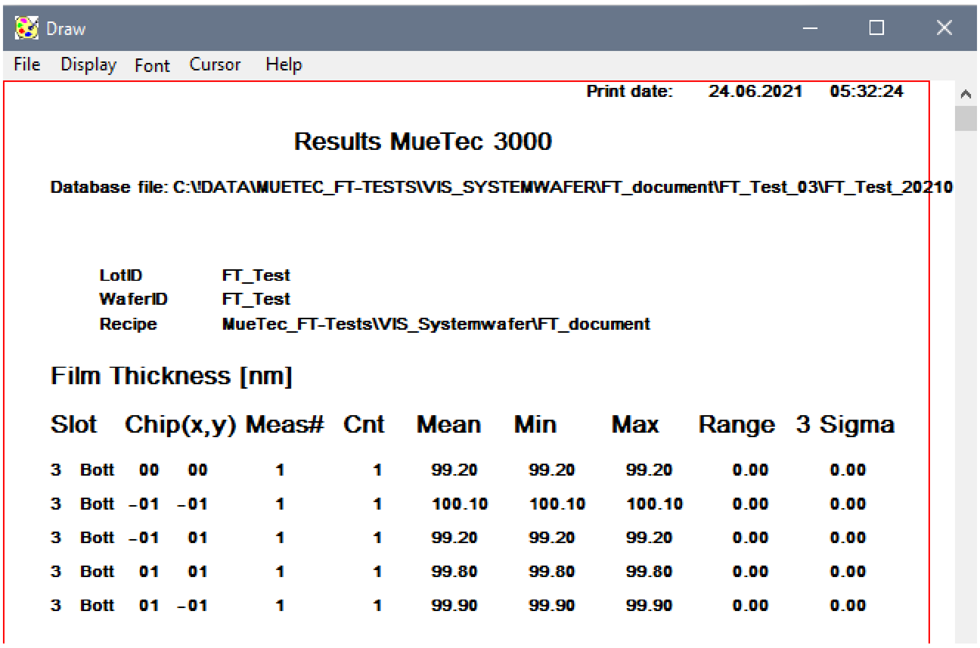
2.5 Measurement Result Review

2.5.1 Displaying and Saving Measurement Results

After measurement completion, the results are displayed as in the examples below. The result protocol looks different for different measurement types.







Different formats and file types of the results are saved named according to **User Data input** (LotID, Wafer ID and Job name) in the following directories:

Result Path	Description
C:\!Data\......\all_values	The results are stored as a text file as the database format. The individual results are stored as text file by measurement type and chip.
C:\!Data\......\database	The individual results are stored as database file.
C:\!Data\......\production_values	The individual results are stored as text file by measurement type and chip.

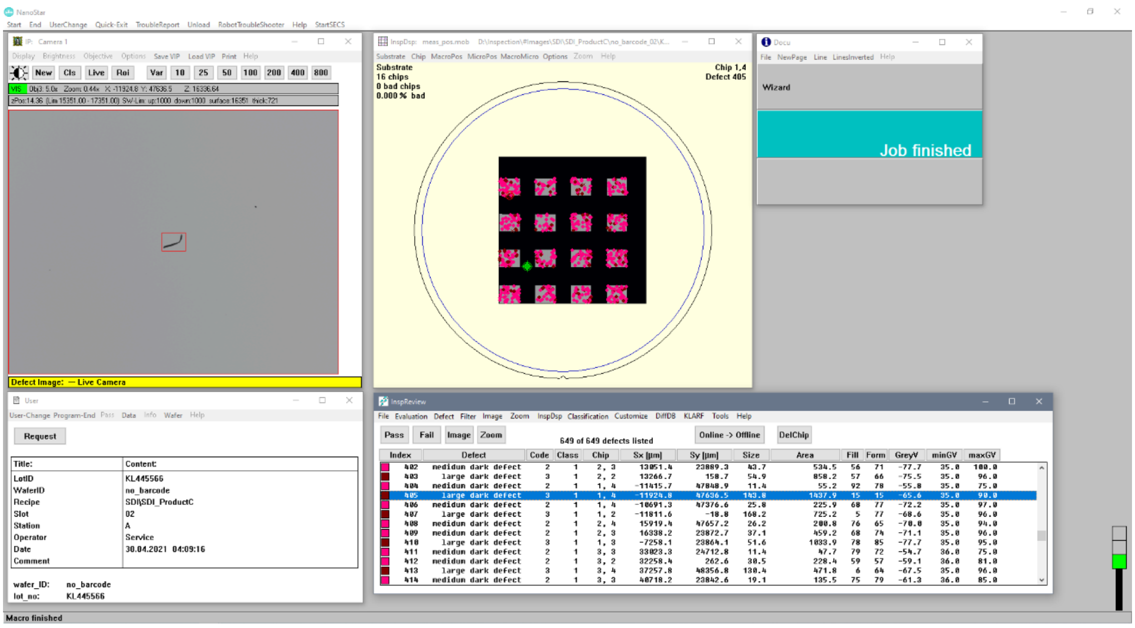
C:\!Data\......\protocols	The summary results (Mean, Min, Max, Range and 3Sigma) are stored as bitmap image file.
C:\!Data\......\Stepper	The stepper correction results are stored as text file for overlay measure-ment.
D:\!Data	The individual results are stored as text file (same content as production value, but different file naming).

Special configurations may apply. It is also possible to store all data into one directory per recipe or into one subdirectory of the recipe named by the LotID. A second path can be configured to store all data twice (firstly on the local computer and secondly on a network drive for fab-automated analysis).

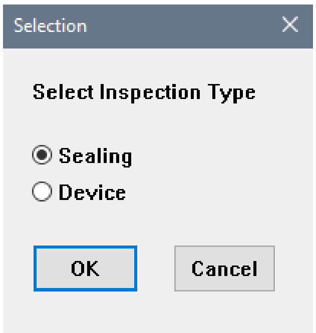
2.6 Inspection Result Review

2.6.1 Displaying and Saving Defect Inspection Results

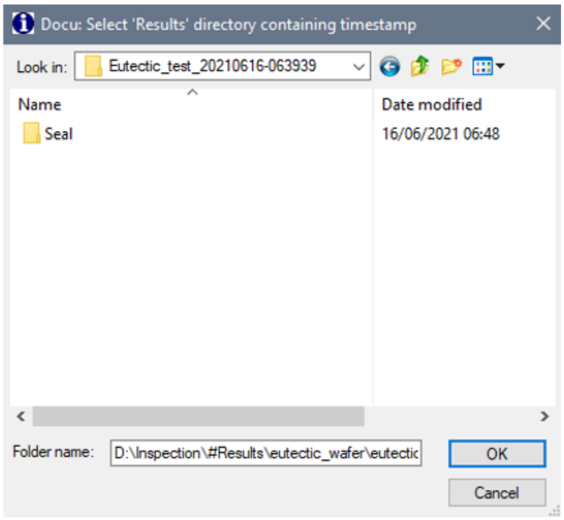
After defect inspection completion, the results are displayed in the **InspReview** Window as shown be-low. If the wafer is staying on the chuck, **Online** review can be performed. By clicking on the defect in the **InspDsp** Window or the defect list in the **InspReview** window, the scanning table moves to the corresponding position and the defect is displayed in the red frame in the **IP** window. By switching to **Offline** review the stage does not move and the defects are shown using saved images, <u>provided that image saving was activated in the used job</u>. In the **InspReview** Window, the defect can be sorted by size, area, or gray value by clicking its respective button in the defect list headline as user preference. The sort order is inverted by clicking again. Depending on the number of defects, magnification, size and image saving setting, storage space of multiple GigaBytes is needed.



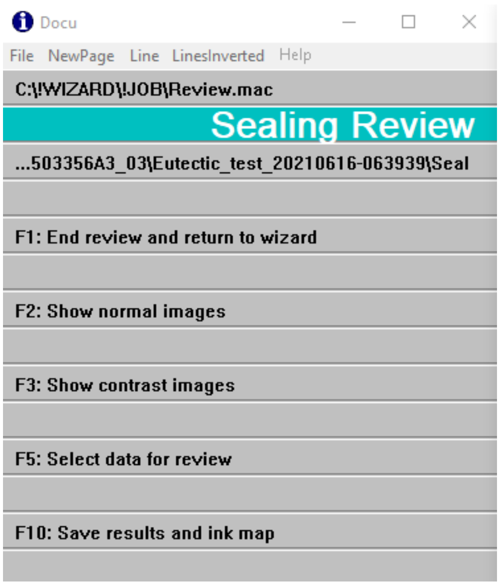
For higher access level users, by pressing the **F1** function key on keyboard, the **Docu** Window is shown (see 2.8.4). Higher access level user can perform **Offline Review** by pressing the **F4** function key. The **Selection** window is shown to select the inspection type, which is either **Sealing** inspection or **Device** inspection (SDI).



Subsequently, select the **Results** folder to review to load data and images. There are a number of folders and sub-folders. Please select the folder that contains the time stamp.

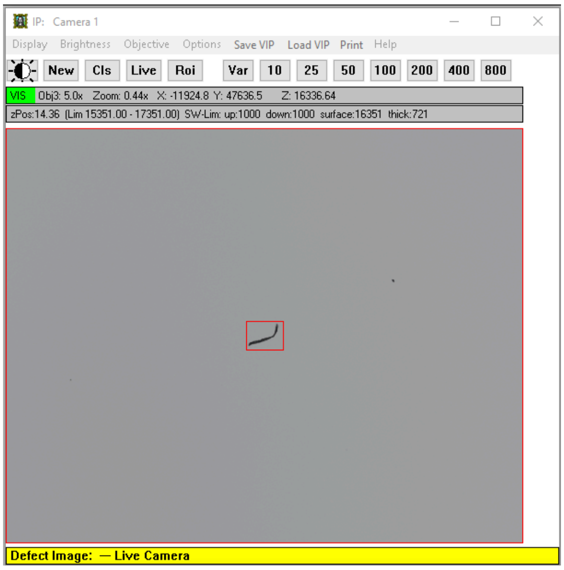


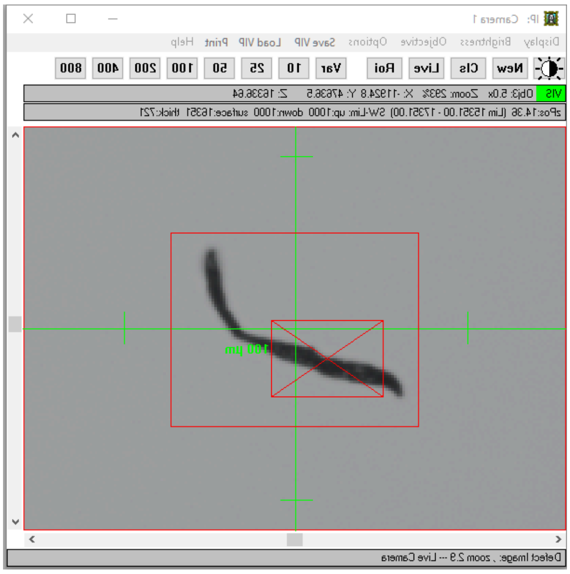
The **Docu** window of Review is shown. It has more functions for selection during review.

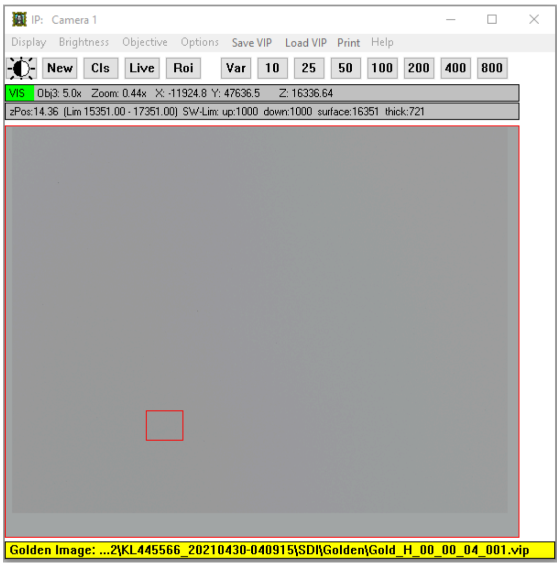


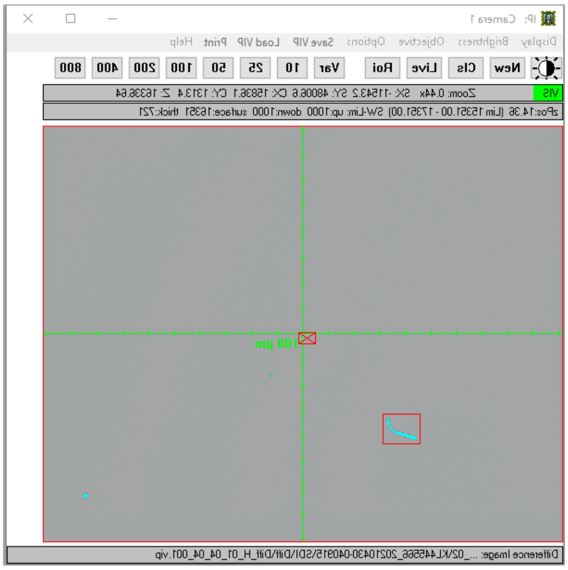
Function Key	Function
F1	To end the review and return to the Wizard. Same function as clicking **Continue** button in **MacWizCont** window.
F2	To display normal images in the IP window
F3	To display contrast enhanced images in the IP window
F5	To select data for review
F10	To save results and ink map

Under **Offline Review** mode, you can view defect images in the **IP** window by selecting the defect list in the **InspReview** window. You may identify the type of defect by clicking the **Zoom** button with higher magnification. The availability of offline images depends on the image saving setting of the executed job. By clicking the **Image** button, the defect image, golden image, and difference image can be switched around for comparison purpose. See the SDI examples below.

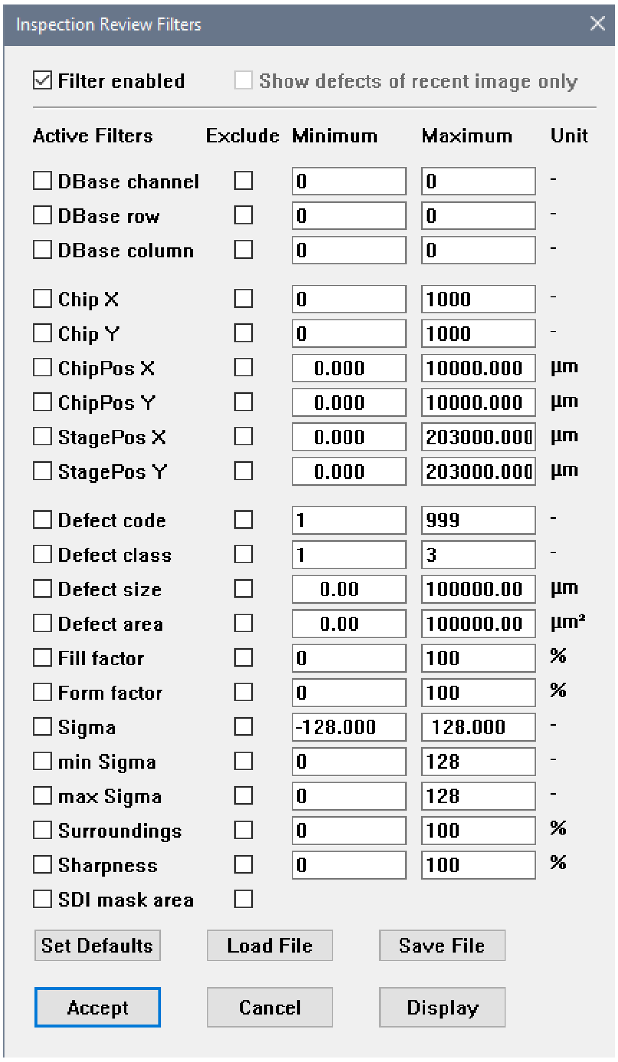




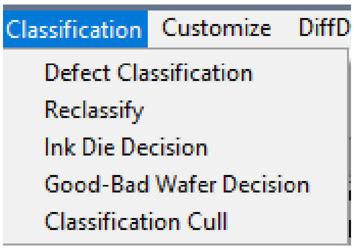


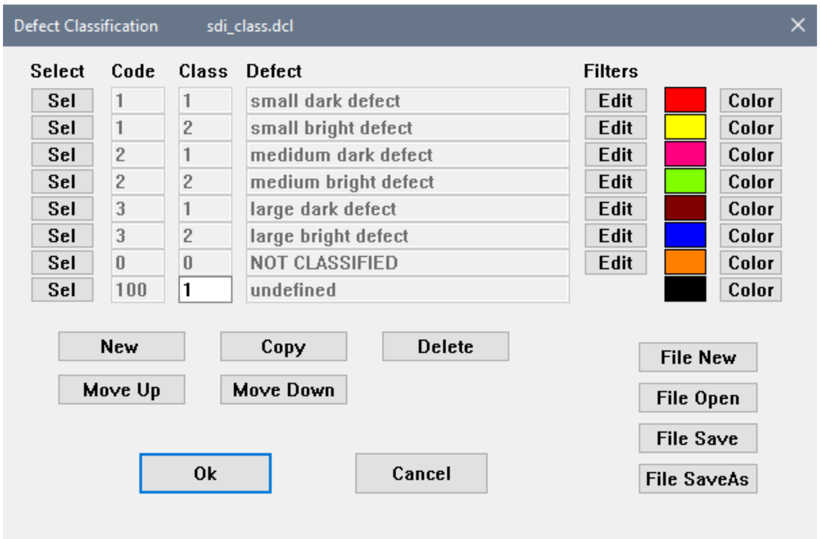


In the menu of the **InspReview** window, the **Inspection Review Filter** window will be shown by clicking the **Filter** button. This is another useful feature for defect analysis, where the filtered defects can be hidden. Different types of filters can be applied according to user preferences. With the filter function, different classes of defects can be examined separately.

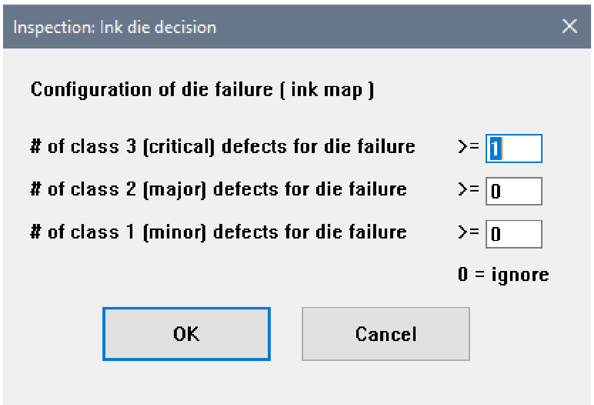


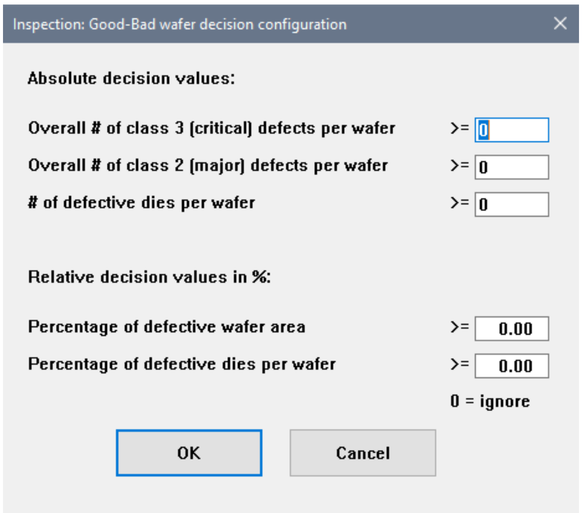
In the menu of the **InspReview** window you can select **Classification - Defect Classification** to perform classification of defects with different colors. The defect classification is done during evaluation on the clusters. A manual subsequent reclassification is possible by selecting **Classification - Reclassify** in the menu of the **InspReview** window.





Besides **Defect Classification**, the decision making of reject the die or wafer can be configured with the **Ink Die Decision** and **Good-Bad Wafer Decision**. In the **Ink Die Decision** window, you can define the minimum number of defects to reject a die by defect classes. Similarly, user can define the minimum number of defects to reject a wafer by defect classes, by defective dies and by defective area percentage in the **Good-Bad Wafer Decision** window. The **Good-Bad Wafer Decision** requires the usage of fixed defect classes (class 1: minor defects, class 2: major defect, class 3: critical defect).





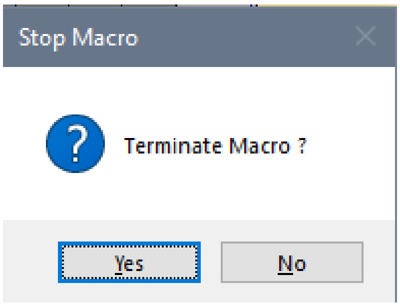
In the **MacWizCont** window, click **Continue** button to end the review.

In the *D:\Inspection* directory, there are two main subdirectories, *#Images* and *#Results*. Different formats and file types of the results are saved named according to **User Data input** (LotID, Wafer ID and Job name) in the following directories:

Result Path	Description
D:\Inspection\#Images\......	The defect images are stored as *vip* file (Nanostar specific data format, containing image and meta data).
D:\Inspection\#Results\......\CompositeMap	The composite map is stored as *bmp* file and *vip* file.
D:\Inspection\#Results\......\DataBase	The inspection results are stored as *db* file (Nanostar specific data format).
D:\Inspection\#Results\......\DefectMap	The defect map is stored as *bmp* file, same as the **InspDsp** window.
D:\Inspection\#Results\......\InkMap	The ink map with the number of defective dies result is stored as *txt* file.
D:\Inspection\#Results\......\Results	The results of defect scan are stored as text file, same format as in the **InspReview** window.

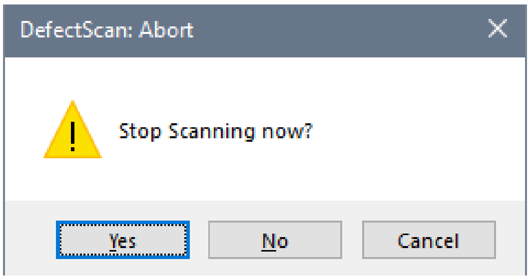
2.7 Job Termination

If you wish to stop a running job before it is completed, press the **ESC** button on the keyboard. The job stops and the **Stop Macro** window is shown.



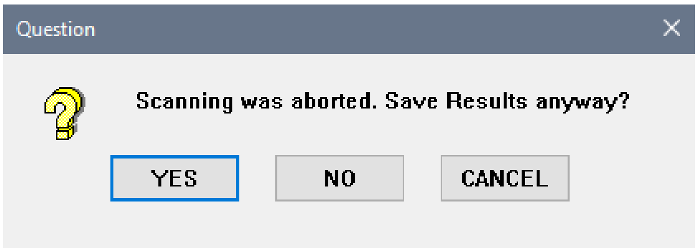
Selection	Description
Yes	To terminate the job.The job is stopped
No	The job continues from the exact step where it was stopped

For defect inspection, the scanning stops and the **DefectScan: Abort** window is shown.



Selection	Description
Yes	To stop the scanning. The Wizard continues as described below.
No	The job continues from the exact step where it was stopped.
Cancel	The wizard stops and the wafer stays on stage.

After scanning is aborted, the **Question** window is shown and askes for saving the results.



Selection	Description
YES	To save the existing scan results.
NO	Do not save the existing scan results.
CANCEL	The Wizard stops and the wafer stays on stage.

2.8 Actions After Job Interruption

There are several possibilities to continue after an interrupted job.

2.8.1 Resuming the Existing Job Run With Wafer on Stage

By pressing **Alt + F8**, the wizard will continue executing the recent command line and the job will continue from where it was stopped. This works in most cases because the last instruction is repeated. If the last instruction which was aborted, should be skipped, then use **Alt + F9** to continue. This is better if you stopped wafer-handling or oscillating lamp alignments.

2.8.2 Starting a New Job With the Wafer on Stage or Another Wafer

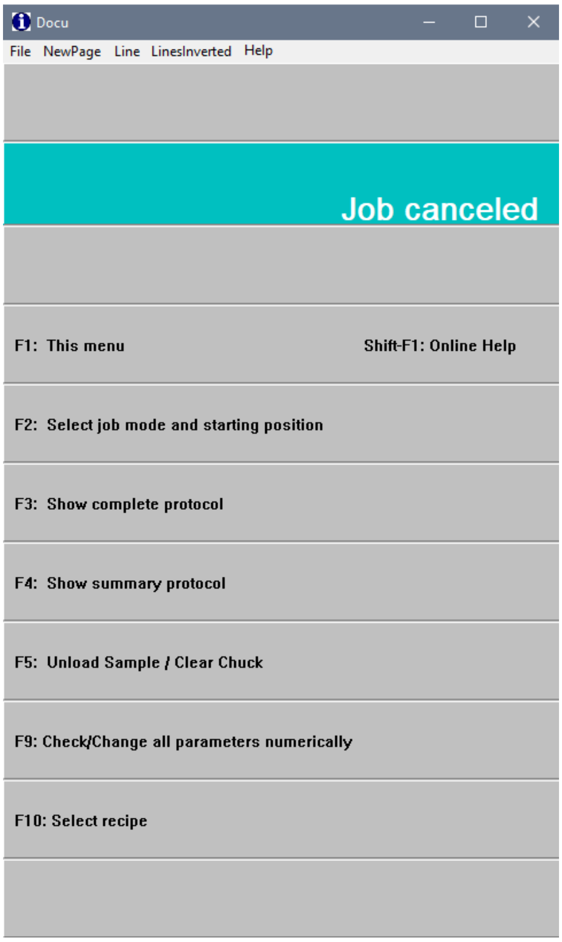
Typically, you can start a new job as usual (see 2.1). When the **Question** window is shown and askes to continue with the wafer on stage, click **YES** to continue with the wafer on chuck. Click **NO** if you want to select another wafer. In that case, the wafer on stage will be unloaded back to cassette and the **Handler** window is shown for wafer selection (see 2.1.3).

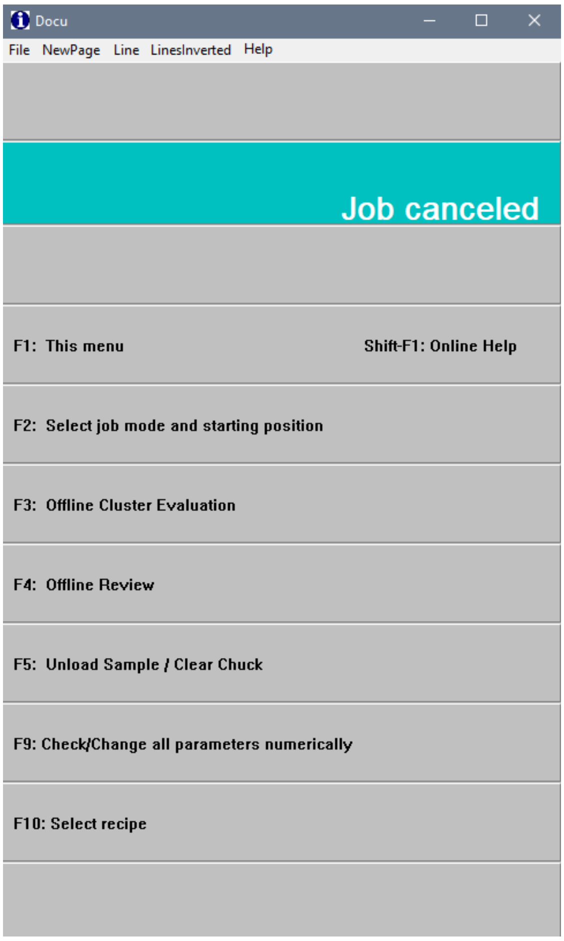
2.8.3 Clearing the Wafer from Stage

To remove the wafer from stage, click the **Clear** button in the **Handler** window and the robot handler will unload the wafer back to cassette. **Alt + F5** does also unload with robot or manual unload with the holder routines for masks or manual tools.

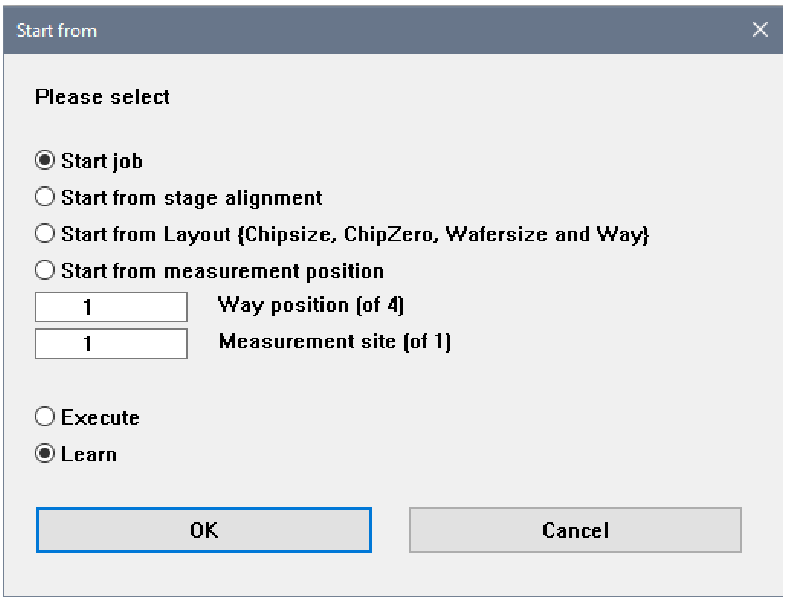
2.8.4 Continuing with Other Options with Higher Level Access

For higher access than operator level, a user has more options to continue by pressing the **F1** function key on keyboard to show the **Docu** Window. You may press the function keys as displayed in the **Docu** window or click on the function line in the **Docu** Window, according to the function to proceed. Then the Wizard will start the corresponding function. Depending on the last job type (CD/OVL/FT or inspection), the functions of the **F3** and **F4** key differ.

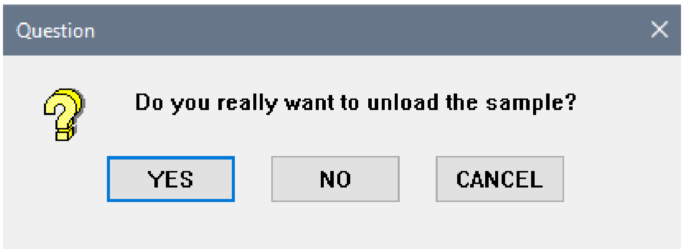




For example, by pressing **F2**, the **Start From** window is shown and user can continue from here with the job start selection.



By pressing the **F5** function key, the **Question** window is shown and you decide to unload the wafer from stage.



Selection	Description
YES	The wafer on stage will be unloaded back to cassette.
NO	Leaves the wafer on the stage for later use.
CANCEL	The Wizard stops and the wafer stays on stage.

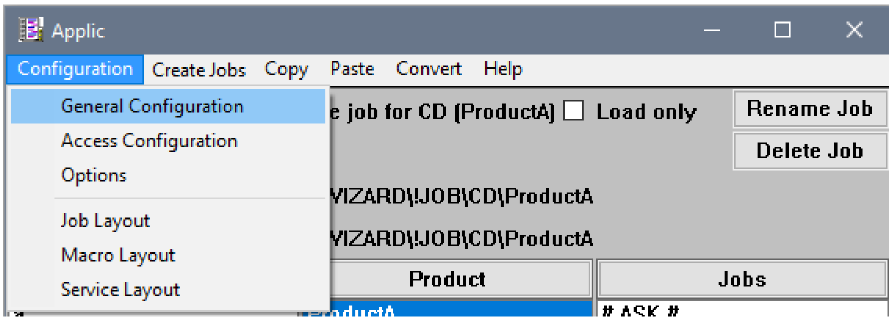
F9 is very useful when some numerical parameters need to be checked or modified after test run (see 3.6).

3 Job Learning

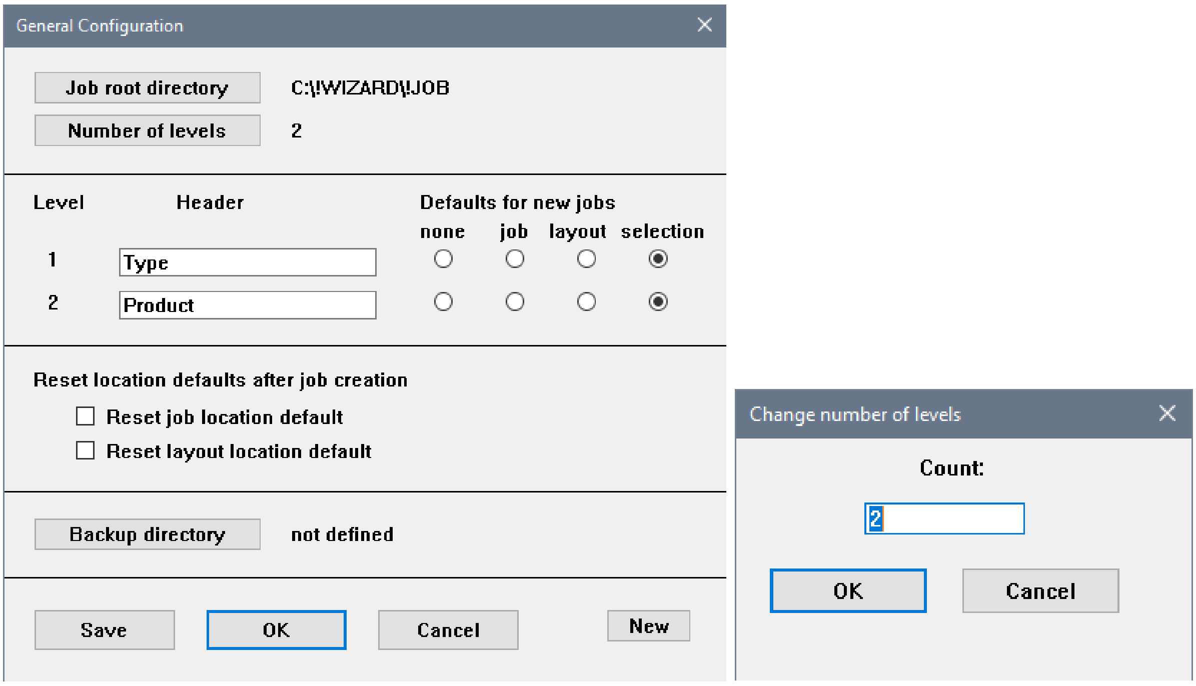
A job can be learned with any user level above operator access. The learn mode of the wizard helps user to setup a new job or modify all parameters to optimize an existing job which is specific to a product.

Before starting job learning, it is advisable to have some experiences or guidance on the following windows: Applic, Lamp, Alfa, CFinder, Stage, StageDsp, CD-Meas, CD-Cal, OVL, FT and all windows related to defect inspection (DefectScan, SDI, Sealing Inspection, Object, Cluster). Please refer to the Nanostar User's Manual or use the Online Help.

At the beginning, you may need to define the **Job root directory** where all the jobs going to be stored and the number of levels for job folder categorization. However, these settings are usually already done by the system manufacturer. In the menu of the **Nanostar** window, select **Start – Start Job** (see 2.1) to show the **Applic** window. Select **Configuration – General Configuration** in the menu of the **Applic** window.



The **General Configuration** window is shown to set the root directory by clicking the **Job root directory** button. By clicking the **Number of Levels** button, you will be able to key in the number of levels (usually 2 or 3 levels are used). The job folder categorization depends on user preference, whether the jobs should be categorized by measurement type, product or user specific naming.



In the **General Configuration** window, you can rename the header of the columns (e.g. Type or Product) displayed in the **Applic** window, depending on the number of levels set. After setup, press **Save** in **General Configuration** window.

3.1 Starting a Learning Job

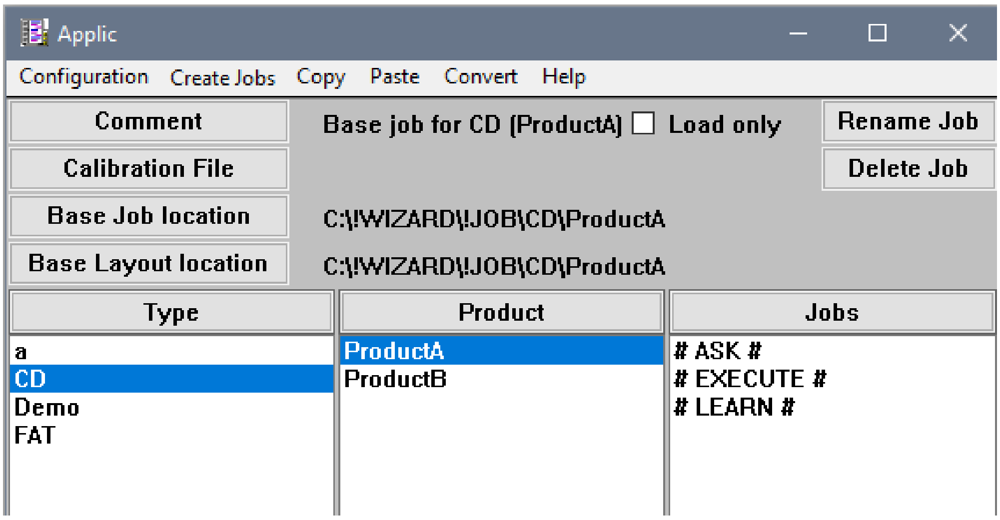
In the **Applic** Window, you can setup a fresh new job (without any known parameters and definition) or to copy from an existing job or to optimize a specific job. Before modifying an existing job, it is advisable to create a backup copy of the original job folder. During modifying the job, the existing parameters will be overwritten. The job that was running well previously, might no longer work anymore if modifying went wrong.

The main principle in the learning flow is that pressing the **Enter** key**,** the **OK** or **Continue** button leaves the current parameters unchanged in order to go quickly through already learned parts of the job.

3.1.1 Creating a New Job

It is recommended to set up a new job for new parameters or to copy from an existing job. It is often easier and time-saving to copy from existing job when the new job is for a similar product family or has the same layout.

To create a new job based on an existing job, select the existing job (so-called **Base Job**) in the corresponding columns. Select each of the column from left to right and do not select **# ASK #**, **# EXECUTE #** or **# LEARN #** in the last column. Then, click the **Base Job Location** and **Base Layout Location** button, the directories of selected job are shown next to the buttons. It is possible to copy **Base Job** and **Base Layout** from different existing jobs respectively. Click **Create Jobs** in the menu of the **Applic** window to create a new job.



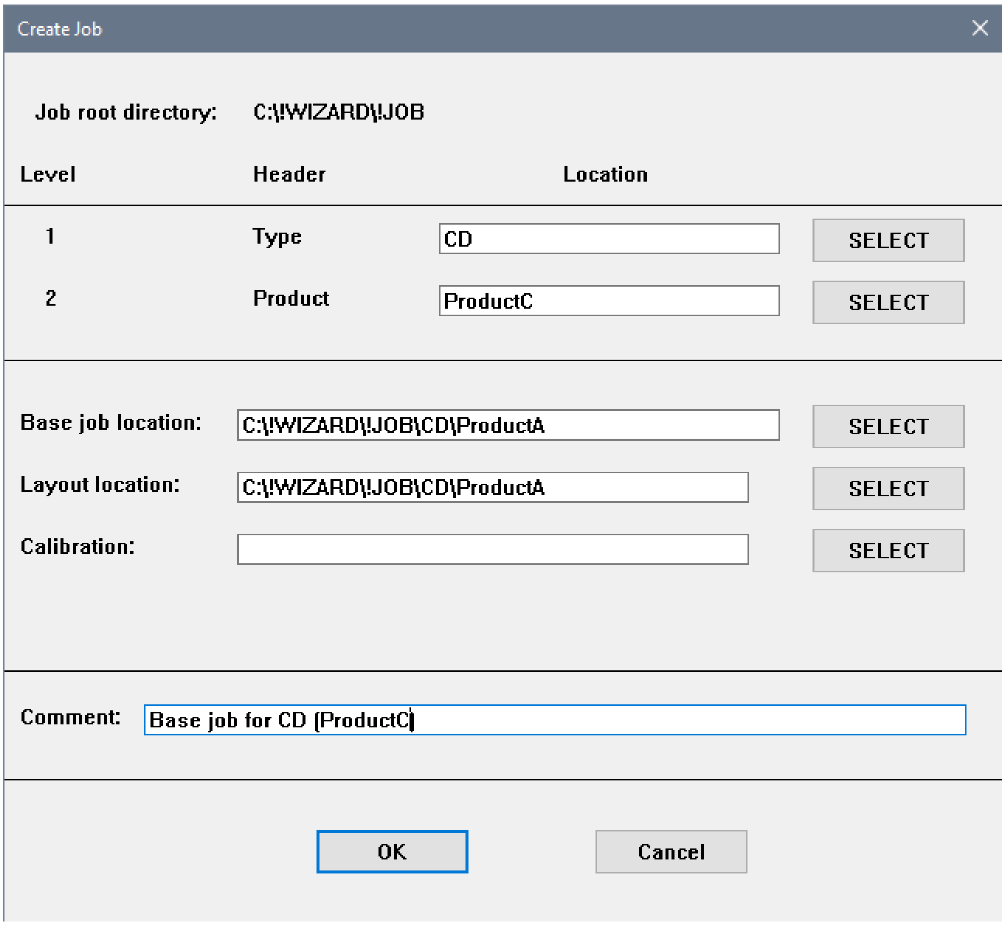
If you prefer to setup a fresh new job from an empty job, select **Demo/EmptyJob** as base for the job and layout. It contains no data files in the beginning.

In the **Create Job** window, you define new names in the entry field (**Location**) of Level 1 (Type) and Level 2 (Product) for the new job. As an example (see figure below) in the **Create Job** window, the **Level1** entry field is named as **CD** and **Level2** entry field is named as **ProductC**. The new **ProductC** folder is stored in the **CD** folder in the same location as **ProductA** folder.

Alternatively, you can set the **Base Job Location** and **Base Layout Location** in the **Create Job** window if you wish to copy from an existing job. For example, the new job named as **ProductC** is created based on the base job and base layout of **ProductA**. All the parameter files of base job and base layout are duplicated from the directory of *C:\!Wizard\!Job\CD\ProductA* (base job) to *C:\!Wizard\!Job\CD\ProductC* (new job).

In the **Create Job** window, a calibration file can be assigned to this job, for example the Multipoint calibration for structure width measurement. The calibration file must be available or created before being used here. It is recommended to fill in the comment.

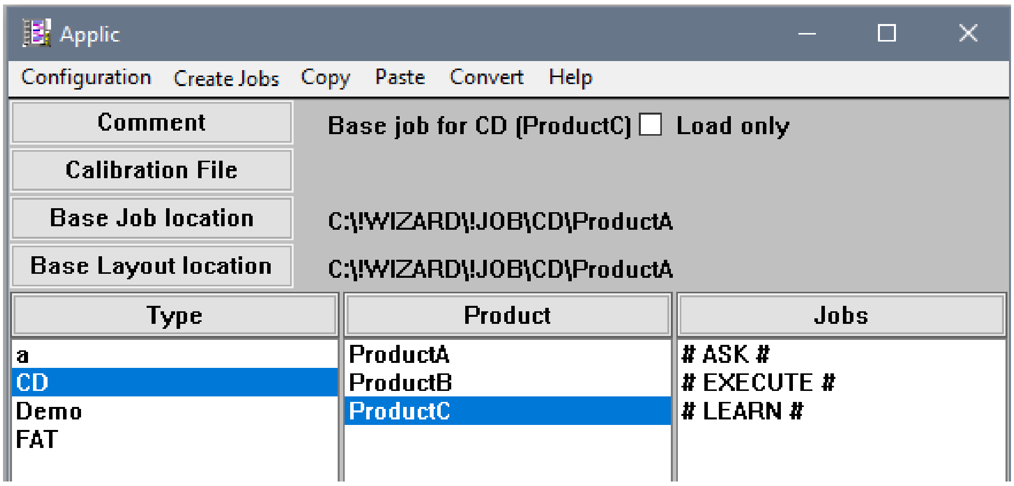
After everything is defined in the **Create Job** window, click **OK** to create the new job.



Parameter	Description
Job Level	The categorization of the job folders in the job root directory. You may select the folder if already exist, otherwise a new folder will be created based on the entered folder name.
Base Job Location	The location of base job to be copied from. Duplication of the files to the new job.
Base Layout Location	The location of base layout to be copied from. Duplication of the files to the new job.
Calibration File	You may assign a calibration to this job if you have one. The file extension of the calibration files is *cde,* e.g. *CD_CAL.cde*.
Comment	Comment for this job should be filled in as a brief job description for other users.

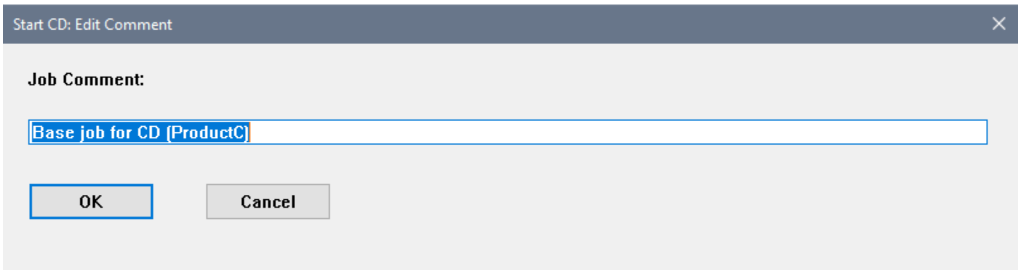
The new entry (**ProductC**) is shown in the second column of the **Applic** window as defined. The job level displayed in the **Applic** window is the same as the folder stored in *C:\!Wizard\!Job*. The created job can be copied or renamed by renaming its folder name in Window Explorer. The currently selected job is locked and cannot be renamed, moved or deleted.

Do not move, rename or delete the *C:\!Wizard\!Job* folder nor all the files that have the extensions *mac* and *ini* directly in the **!Job** folder.

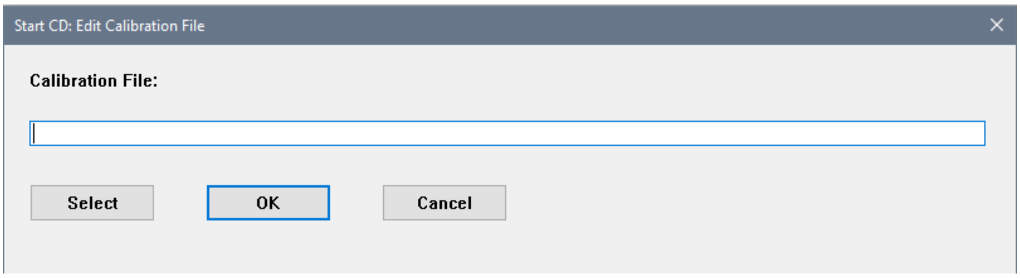


Alternatively, the **Calibration File** and **Comment** can be amended the at **Applic** window after job is created.

By clicking the **Comment** button, you can add a **Job Comment** for this job.



By clicking the **Calibration File** button, user may change (add or remove) the calibration file for this job.



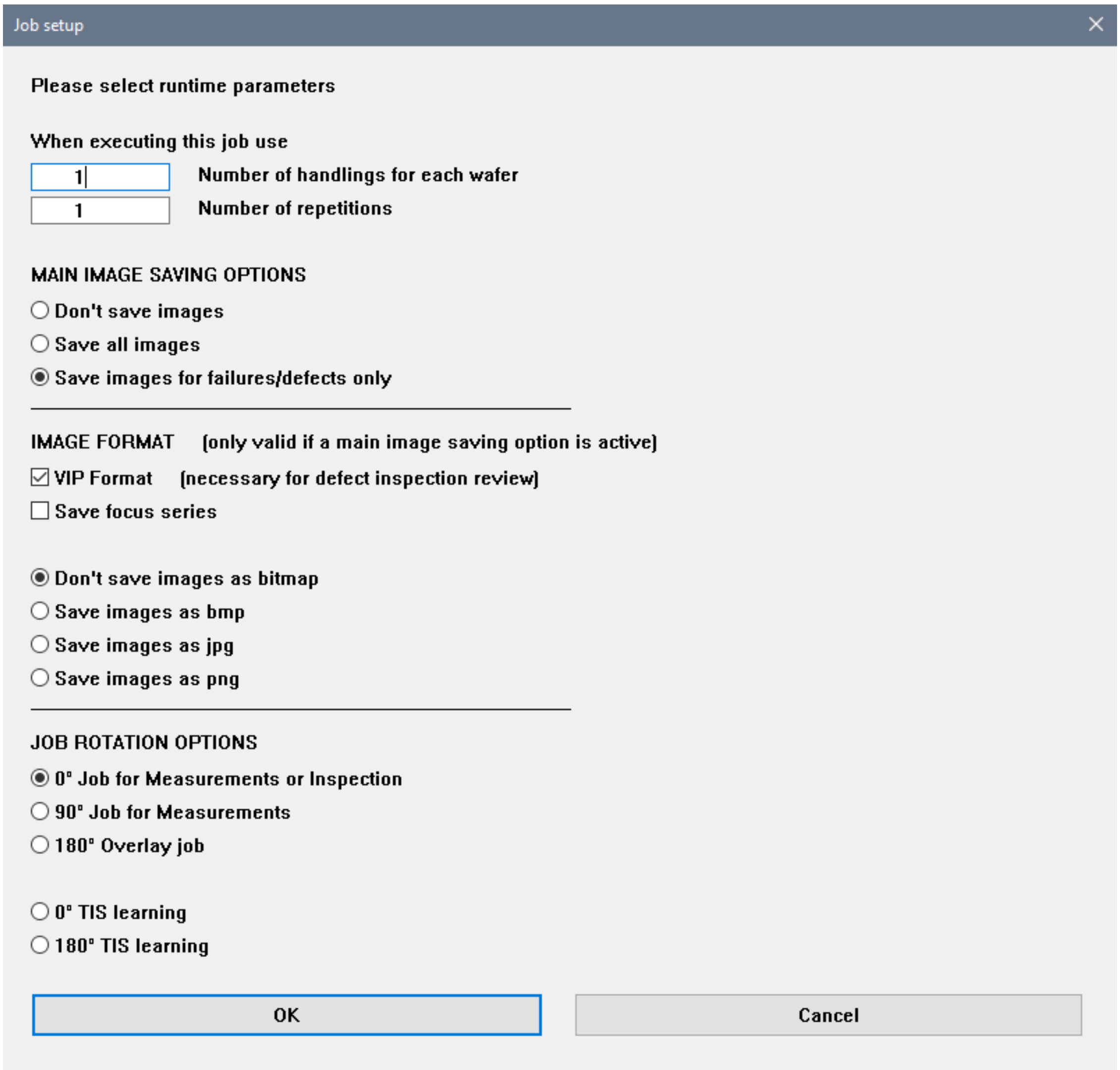
To start learning the newly created job, select the job in the column of **Applic** window and click **# LEARN #** at the last column.

3.1.2 Job Parameters

3.1.2.1 Job Setup

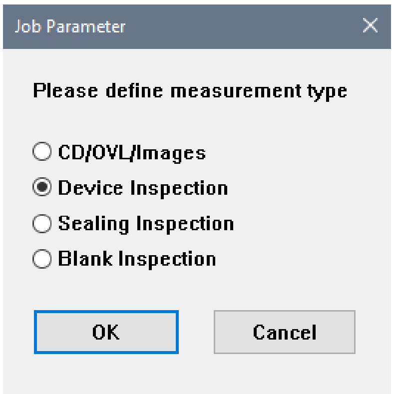
For job setup, enter the number of handlings and the number of repetitions in the entry field. Typically, only 1 handling and 1 repetition will be entered, except for special jobs for repeatability test purpose. Select the radio button and activate the checkbox as user preference. Click **OK** to continue the job learning.

For TIS calibration learning a correctly learned OVL job is needed. Goto **# ASK #** in **Applic** window, select the **Job execute** mode and 0° TIS learning for the first run. After this run is successfully finished, you will be guided to start the next run with the wafer rotated by 180°. The TIS calibration file is generated after both, the 0° and the 180° job, have successfully run.



Selection	Description
Number of handlings for each wafer	The whole job including wafer handling from/to cassette is repeated as defined to test repeatability of the measurement. For systems without automated handling the entire job is started several times without removing the substrate in between (job repeatability).
Number of repetitions	The number of repetitions of each measurement at every measurement position before moving to the next measurement type or position (instant repeatability).
Main Image Saving options	Either do not save images, save all images, or save defect images only. The defect images are useful for tests or documentation purpose, so usually only the defect images are saved.
Image Format VIP Format	To save the images of device inspection in VIP format, necessary to be activated for defect inspection review
Save focus series	To save the focus series images for CD/OVL, if necessary for support or offline evaluation
Don't save images as bitmap	Use this radio button if you do not want to save images in other formats. Images are stored in VIP format then. In such images a later analysis is possible.
Save images as bmp, as jpg, as png	Use this radio button if you wish to save images as *bmp* or *jpeg* or *png* format. Such images are only for documentation and tracing. No measurements are possible.
0° Job for measurements or Inspection	To execute the measurement with zero-degree orientation. Select this for normal measurement job (critical dimension, film thickness & inspection).
90° Job rotation for measurements	To execute the measurement with 90 degrees orientation
180° Overlay Job	To execute the overlay measurement with 180 degrees orientation. This mode is used only to verify whether a job can run in 180° (for Overlay+TIS).
0° TIS learning	To learn the TIS calibration with zero-degree orientation
180° TIS learning	To learn the TIS calibration with 180 degrees orientation
OK	To continue the job learning
Cancel	To interrupt the job learning. The wizard stops

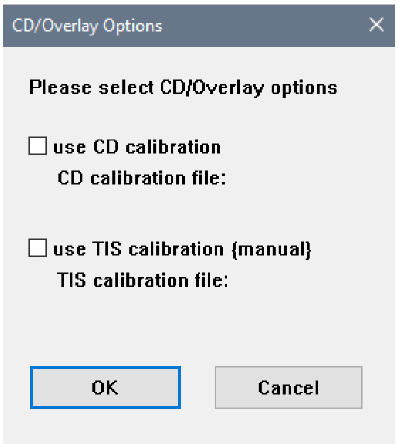
With different configuration setting, the **Job Parameter** window is shown for users to define the measurement type after **Job setup** window. The shown measurement types depend on purchased system options. Click **OK** to continue.



Selection	Description
CD/OVL/Images	For CD/OVL/Image saving
Device Inspection	For defect inspection of structured wafer
Sealing Inspection	For sealing inspection of structured wafer
Blank Inspection	For defect inspection of blank wafer
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

3.1.2.2 CD/Overlay Options

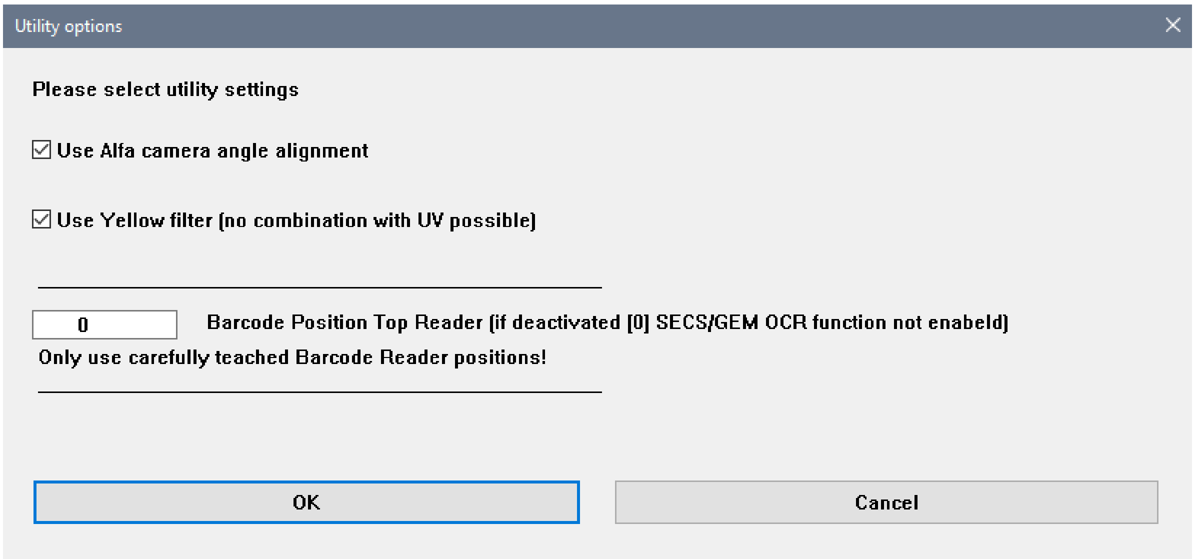
These options are used to activate a calibration file for CD/OVL measurement. After activating the checkbox and clicking **OK**, select the intended calibration file for this job in the **Select Cal file** window.



Selection	Description
Use CD calibration	To use CD calibration file for this CD measurement job. If a multipoint calibration file is selected for this job, the CDE file name is displayed
Use TIS calibration	To use TIS calibration file for this OVL measurement job. If a TIS calibration file is selected for this job, the TIS file name is displayed.
OK	To continue the job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.1.2.3 *Utility Options*

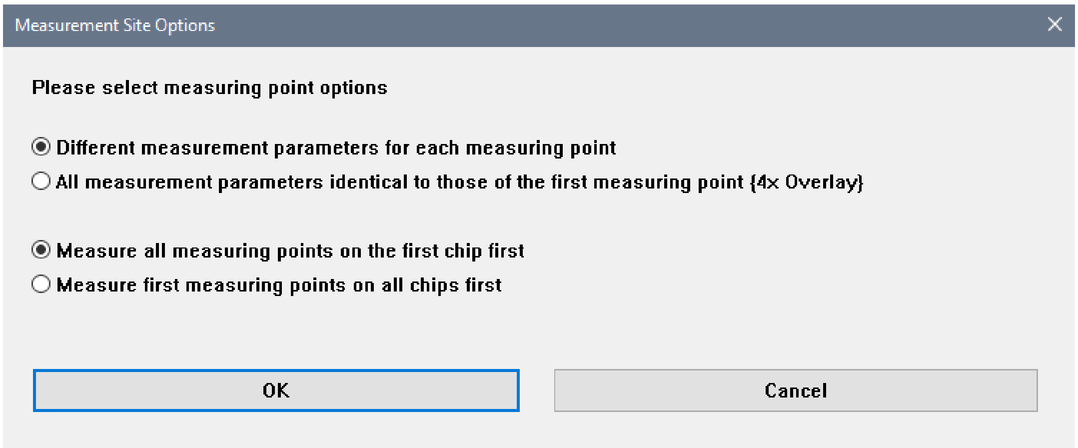
The utility options are used to activate **Alfa camera angle alignment** and **Yellow filter.** For wafers without photoresist layer, the yellow filter can be deactivated. The yellow filter must be off for infrared and UV-light measurements. Click **OK** to continue.



Selection	Description
Use alfa camera angle alignment	To use the alfa camera to align to structure edge on the wafer, either horizontally or vertically. This is done for nearly all jobs.
Use Yellow filter (no combination with UV/IR possible)	To use the yellow filter. This option is recommended for the wafer coated with light sensitive layers, e.g. for photoresist on wafer.
Barcode Position Top Reader	This is the barcode position for the top reader. If the entry is zero, the barcode reading is deactivated for SECS/GEM.
OK	To continue the job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.1.2.4 Measurement Site Options

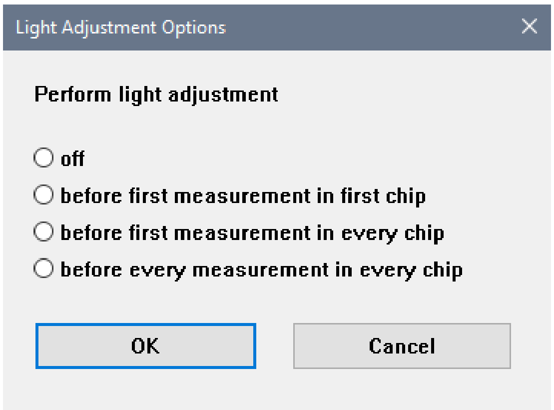
These options are used for CD/OVL measurement site setup.



Selection	Description
Different measurement parameters for each measuring point	To use different measurement parameters at different measuring position
All measurement parameter identical to those of the first measuring point	To use the same measurement parameters at all measuring positions. This is selected for overlay measurement with 2/4/8 identical overlay structures at 4 corners of a shot.
Measure all measuring points on the first chip first	To finish all the measurements in the first chip before continuing to the next chip at another way position.
Measure first measuring point on all chips first	To finish all the same types of measurement in the first round of all way positions, then continue with another type of measurement in a second round and so on, until all measurements complete. This mode is recommended if different microscope-objectives are used. Rotating the revolver is pretty slow and costs lifetime.
OK	To continue the job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.1.2.5 Light Adjustment Options

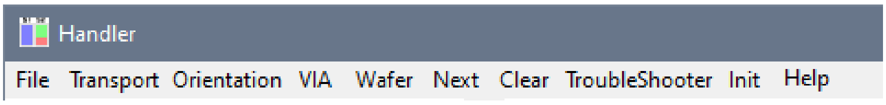
These options are used to define where to perform light adjustment. Option 4 is the slowest.



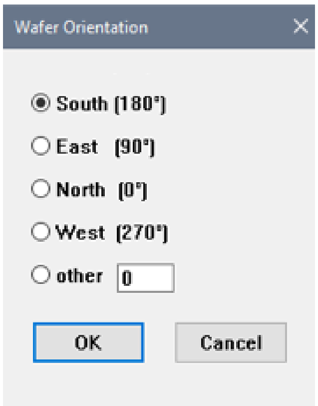
Selection	Description
Off	To switch off the light adjustment
Before first measurement in first chip	To perform the light adjustment before first measurement in first chip
Before first measurement in every chip	To perform the light adjustment before first measurement in every chip
Before every measure-ment in every chip	To perform the light adjustment before every measurement in every chip
OK	To continue the job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.1.3 Wafer Handling

The wafer handling procedure and routine here are the same as described in wafer handling for job executing (see 2.1.3). The wafer orientation can be changed by clicking **Orientation** in the menu of the **Handler** window.



The **Wafer Orientation** window is shown and you can select the intended wafer flat or notch orientation on the chuck. It is possible to save the wafer selection and wafer orientation by clicking **File-Save** in the **Handler** window. In the **MacWizCont** window, click **Continue** to start the wafer handling from cassette to chuck.



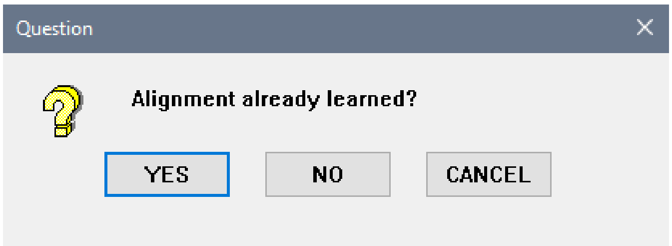
3.2 Stage Alignment

To automatically execute the alignment during job run, the alignment reference structure and its positions must be defined during job learning. The alignment of the wafer structure to the stage axis is based on two or three alignment positions on the wafer.

It is recommended to choose the alignment structure and its position as follows:

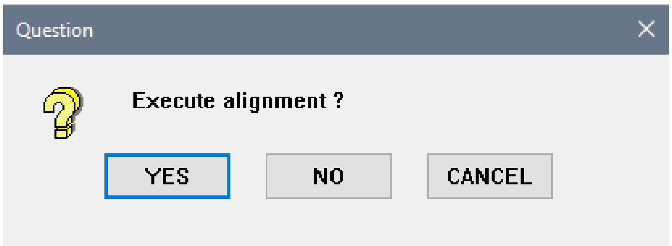
- The reference structure for alignment is simple but unique.
- Do not have a similar structure nearby.
- Preferable is a structure with rectangle shape and must contain both horizontal and vertical edges (for better stage alignment and for angle alignment use later). Circle structure is not advisable.
- The structure does not change its appearance with process variations between wafer to wafer.
- At least two identical reference structures are required for alignment.
- The alignment structures must lay on identical coordinates either in x or y direction nominally.
- For practical application, the first alignment structure can be defined at the center of the wafer and the second alignment structure near to the wafer edge. The distance between the two structures is about half of the wafer diameter. This will help in reference structure searching and reducing the handling effect resulting from placement and angle deviation.
- The alignment structures are better far apart for highest alignment accuracy but not too far away (e.g. close to wafer diameter). Depending on the placement and angle deviation, the second structure may not be found instantly which leads to the scan searching slowly. The distance apart also must not be less than 30 mm.
- In some cases, three alignment positions are required (left, nearby the first, right). It is useful as an interim alignment when the angle between wafer and stage is relatively large, for example for manual loading. The second alignment position that is far away and may not be found in the image field after driving a long distance between first and second alignment positions.

If the job is copied based on an existing job or to modify the existing job, the **Question** window is shown and asking is the stage alignment is already learned.



Selection	Description
YES	To skip alignment learning and perform alignment based on existing parameter
NO	To continue with stage alignment learning
CANCEL	To interrupt the job learning. The Wizard stops.

If the alignment is already learned and by answering **YES**, the next **Question** window is shown and askes for executing the alignment.

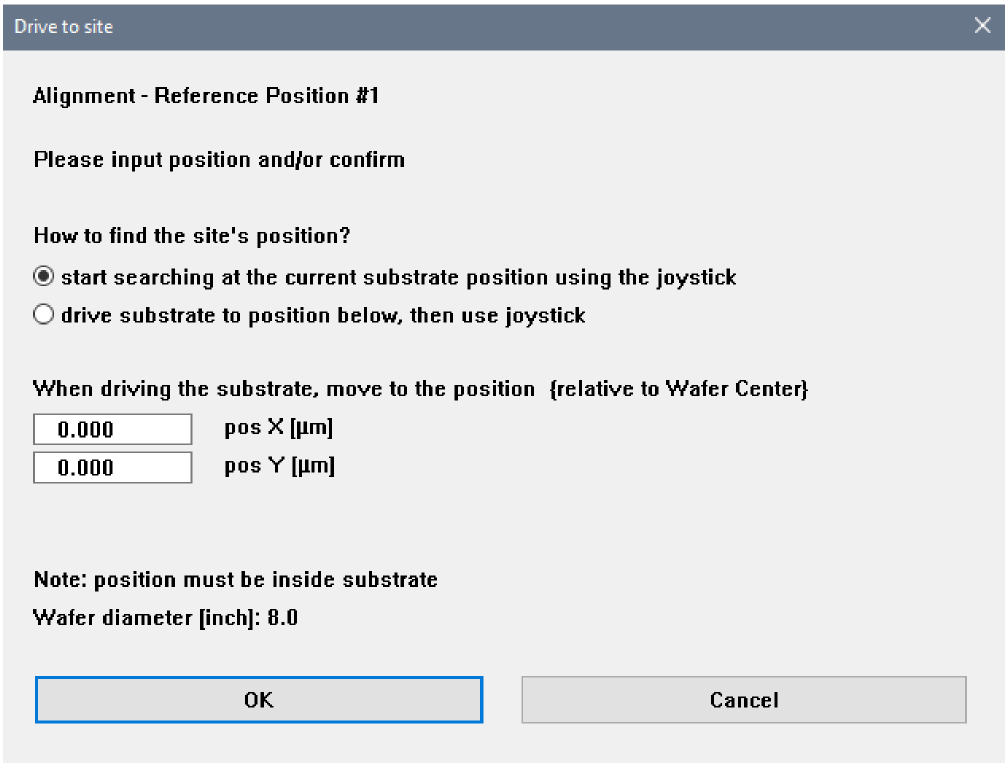


Selection	Description
YES	To proceed with executing alignment
NO	To continue with alignment learning
CANCEL	To interrupt the job learning. The Wizard stops.

3.2.1 First Alignment

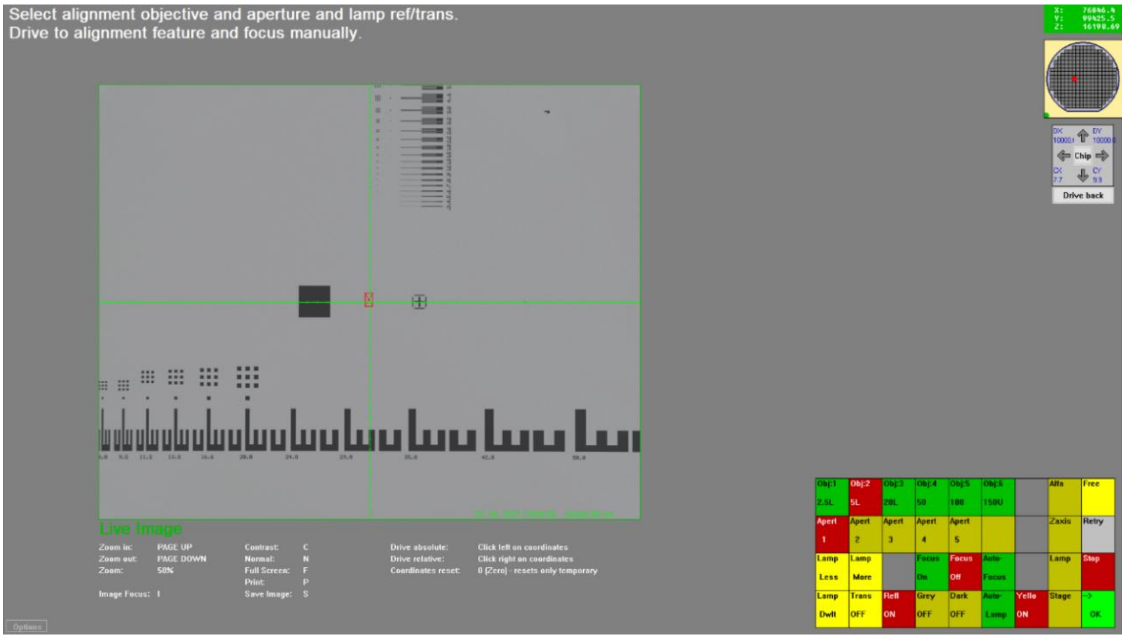
To learn a new stage alignment, the first alignment position is needed to be defined with guidance in the **Drive to site** window. By clicking **OK**, you will be guided to **Free mode**.

For teaching jobs from remote without a joystick there is a special configuration flag (use_remote_control).



Selection	Description
Start searching at the current substrate position using the joystick	The stage does not drive to any predefined position. The user starts looking for a new position of alignment structure with joystick from current position in **Free mode**.
Drive substrate to the position below, then use joystick	If there is any predefined position, the stage will move to the position as stated in Pos X and Pos Y (typically the existing position is from base job). You can also use a joystick to make necessary correction of the position in **Free mode**.
OK	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

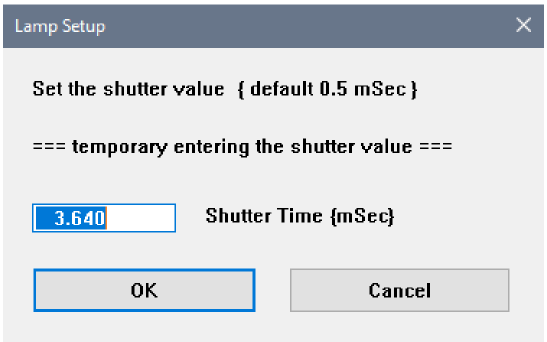
In **Free mode**, the alignment objective, aperture and lamp path (reflected or transmitted) need to be defined. Then drive to the new alignment structure and adjust the focus level with z-drive manually. Do not place the focus spot on or close to any edges. Maybe it is necessary to adjust the light with the **DWIT** button on the keypad if the brightness is not in good condition. Click **OK** on the keypad to continue with brightness adjustment.



If the message **Alignment error – First Reference Structure not found** is shown in the **Message** window frequently during test run or job executing, the stage alignment position and the focus spot are needed to be re-checked. Do not place the focus spot near to edges or hilly topography area. Another option is to change from 5x to 10x objective for better alignment and focusing. The 5x objective is more sensitive to horizontal edges.

3.2.1.1 Brightness Adjustment

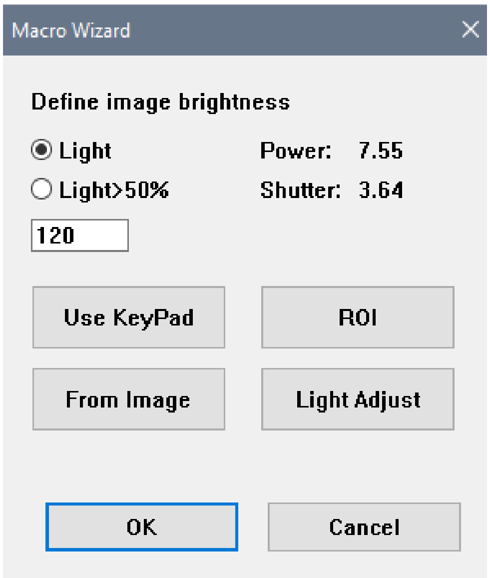
Well-defined brightness is an important basic parameter and should be defined for the first time at the first alignment position. The **Lamp Setup** window is shown and set the shutter value (for the camera shutter) with the suggested shutter time by clicking **OK**.



Selection	Description
OK	Set the shutter time and continue with job learning.
Cancel	To interrupt the job learning. The Wizard stops.

Next, the **Macro Wizard** window is shown for defining the image brightness. The average intensity should be set around 80-150 gv without red pixels in the **IP** window. The range is 0-255 grey values. Red pixels indicate image saturation (overflow of the camera, more than 200 gv) and the quality of measurement and pattern finder is reduced. If red pixels are observed, lower down the intensity value.

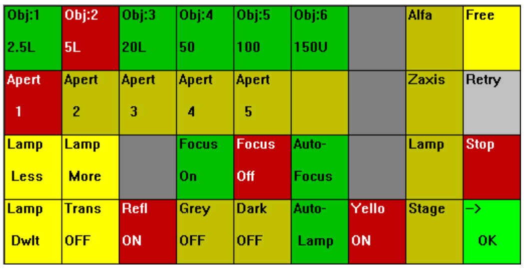
While defining the ROI, it is advisable to choose the entire screen in the **IP** window and set the gray value to around 100 gv. The ROI should contain both bright and dark areas, to avoid extremely bright or dark image. If the ROI rectangle is too small, it may be located entirely within a complete bright or dark area. For the best result of brightness adjustment, the region should be uniform and bright.

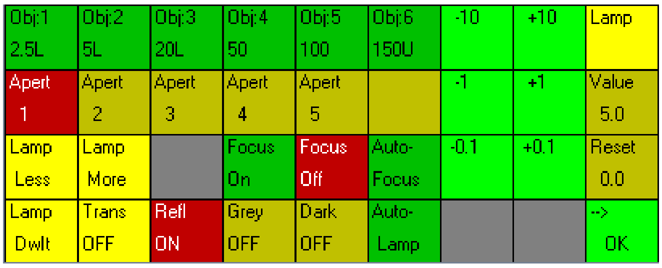


Selection	Description
Light	This function is recommended to be used for homogeneous areas, especially with reflected light. Use this for wafers with reflected light.
Light> 50%	This function is recommended to be used for inhomogeneous areas with stable contrast condition (2 gray levels, black & white), especially with transmitted light. Use this for photomasks in transmitted light.
Use KeyPad	To display and use the keypad on screen.
ROI	To define the region of interest. Brightness is adjusted based on this region.
From Image	To read out the brightness from the current image inside the ROI in the IP window.
Light Adjust	The brightness is adjusted to the defined value in the entry field. Do this to test user settings before clicking OK
OK	Sets the brightness and continues with job learning.
Cancel	To interrupt the job learning. The Wizard stops.

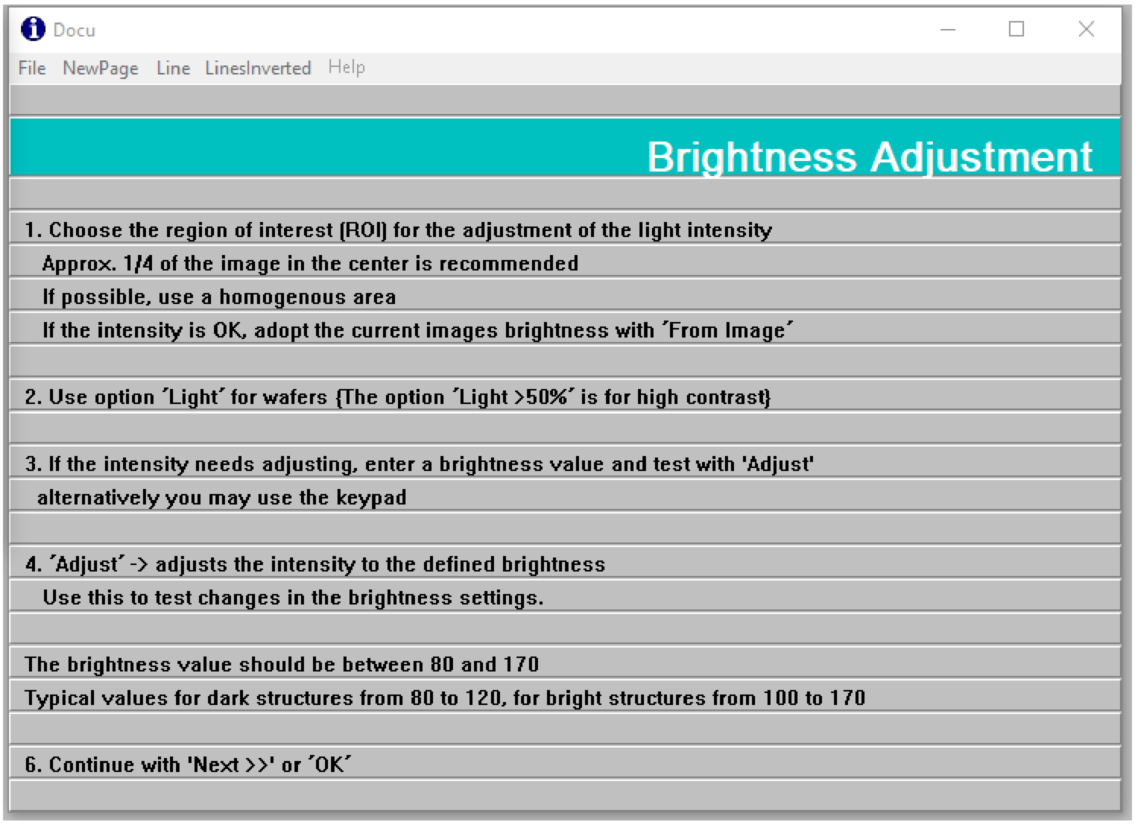
A few options to adjust the image brightness:

- Change the value in the entry field and click **Light Adjust**. Click **OK** to set the brightness.
- Use the keypad with **Lamp Less**, **Lamp More** or **DWIT**. Click **OK** to set the brightness.
- Use the keypad with **Lamp**. The last three column of the keypad layout changes and offer additional keys. Press the **+/- 0.1 to +/-10** button to adjust the brightness and click **OK** on the keypad to accept the current brightness. Click **OK** to set the brightness.





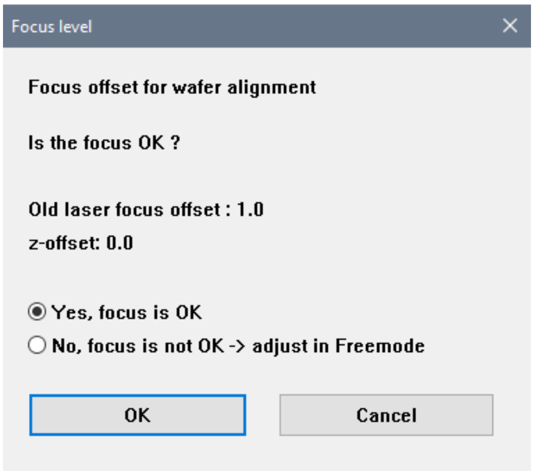
The **Docu** window has some information and guideline for brightness adjustment.



If the message **Warning: Lamp is oscillating i**s shown in the **Message** window frequently during test run or job executing, the image brightness setting is needed to be re-checked. Do not use **Light>50%** for wafers with many different gray levels.

3.2.1.2 Focus Adjustment

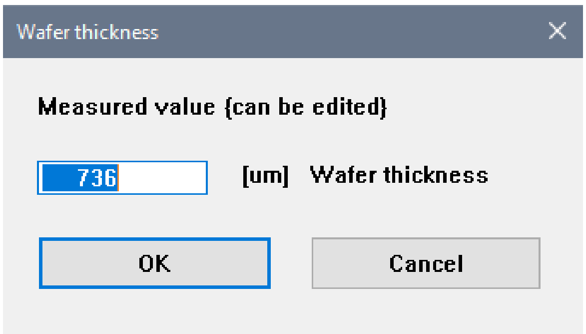
The next parameter to learn is the focus. Be reminded do not place the focus spot on an edge or close to any edge. The live image will be displayed in the **IP** window and the **Focus level** window is shown. Evaluate the sharpness of the live image.



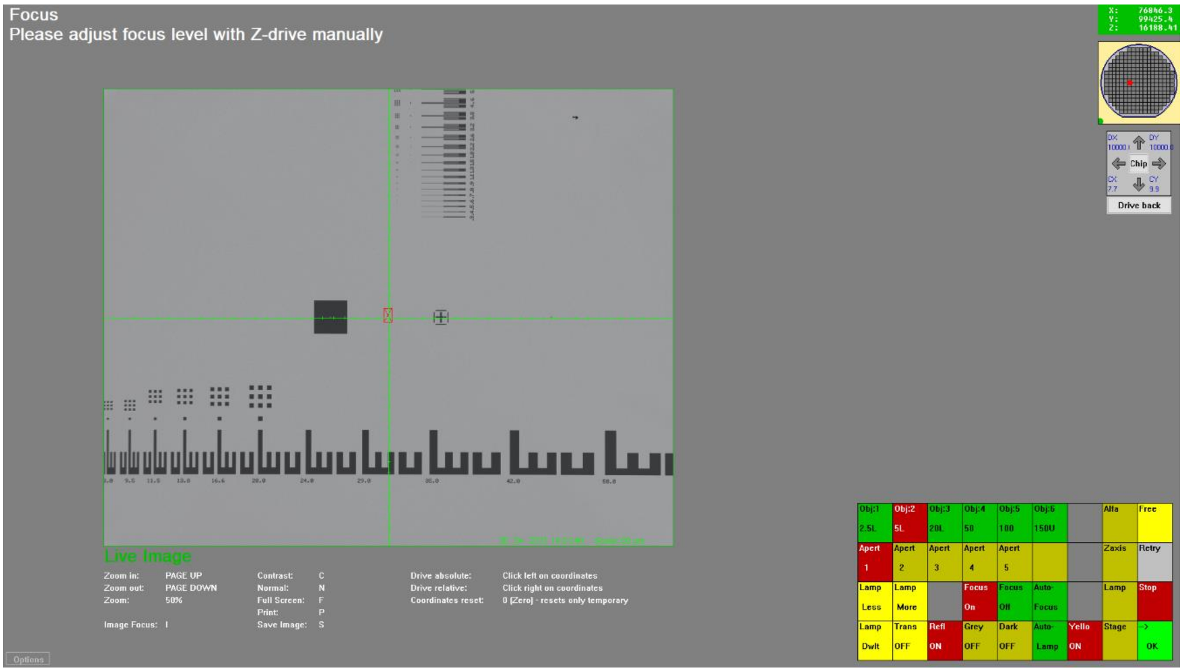
Selection	Description
Yes, the focus is OK	The current focus level is acceptable.
No, the focus is not OK	The current focus level is not acceptable.
OK	If yes, the job will continue with displaying wafer thickness. If no, the job will go to **Free mode** for manual focus readjustment.
CANCEL	To interrupt the job learning. The Wizard stops.

If the live image is in focus, select **Yes, focus is OK** and click **OK** in the **Focus level** window. Based on the z-height corresponding to the focused image, the wafer thickness is calculated and displayed in the **Wafer thickness** window. Click **OK** to accept the measured wafer thickness and continue with job learning.

If the shown value is far away from the expected value, then enter the correct value here. Call service. Wrong thickness values prevent jobs from running due to focusing problems with the soft-limits.



If the live image is not in focus, select **No, focus is not OK** and click **OK** in the **Focus level** window. You can adjust focus level with z-drive manually in **Free mode**.



Next is the brightness adjustment verification. After the focus is learned, the **Lamp Setup** window (shutter time) and **Macro Wizard** window (image brightness) are shown again for verification. It is the same as described in brightness adjustment (see 3.2.1.1). You may readjust the brightness again if not satisfied with the previous adjustment, otherwise just click **OK** to continue.

3.2.1.3 Angle Adjustment

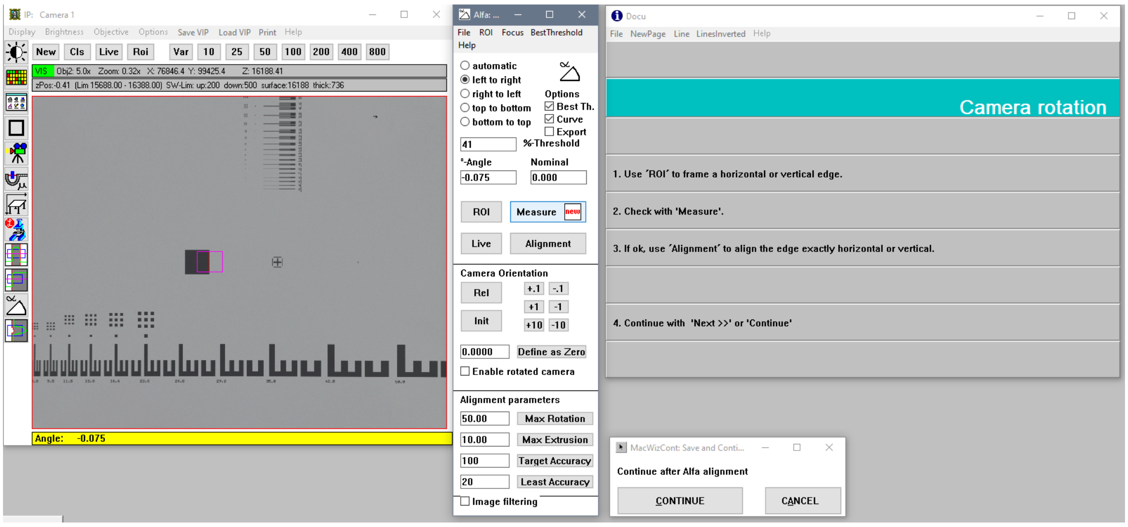
If the checkbox **Use Alfa camera angle alignment** in **Utility Option** window (see 3.1.2.3) is activated, the parameters of the rotating camera can be learned here. The alignment of the camera to the structures on the wafer is supported by the **Alfa** window.

Use the ROI button in the **Alfa** window to define the region of interest either at the vertical or horizontal edge line. In the **Alfa** window, select the direction of the edge detection with the radio button and click **Measure** to evaluate the red line on the edge. The default threshold of 50 % is good for most cases. It may be necessary to adjust the threshold manually or to activate the best threshold option once, especially in a low contrast image. If the correct setting is found (e.g. left to right direction at 40 % threshold), then use this setting. For bad structure edges with noise or ripple (poor edge roughness), the automatic option may result in instability, error messages and never-ending angle alignments. Do not use it for production recipes.

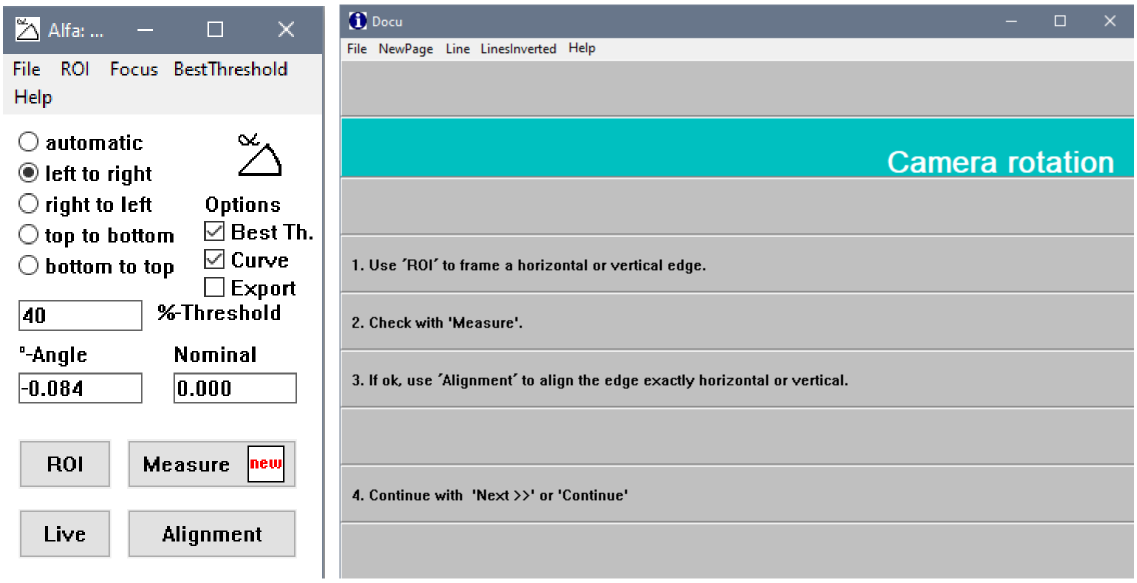
It is recommended to choose the Alfa alignment structure and its position as below:

- Choose a relatively long and straight edge with lesser edge roughness. The Alfa alignment is more precise if the ROI defined edge is longer. But 25% of the screen size is enough.
- Set the direction either **left to right/right to left** for vertical line or **top to bottom/bottom to top** for horizontal edge. It is not advisable to use the automatic option.
- The **Best threshold** option might be nice for detecting threshold % but may not be good during normal job executing. Turn it off after using it once. Typically, a good threshold is around 30-50 %.

If the Alfa setup is correct, click the **Alignment** button to align the edge exactly to a horizontal or vertical line. In the **MacWizCont** window, click **Continue** to continue. This stores the settings of the Alfa window.



The **Docu** window displays information and the guideline for angle adjustment.



Selection	Description
ROI	Use this button to define the region of interest. The angle is adjusted based on a straight line in this region.
Measure	The angle of the edge is measured and displayed in the Angle box.
Alignment	The Alfa camera is adjusted to align the edge exactly to horizontal or vertical line.
CONTINUE	Click this button to continue with finder.

CANCEL	To interrupt the job learning. The Wizard stops.

If the message **Alfa: not enough data for regression** is shown in the **Message** window frequently during test run or job executing, the threshold and ROI position need to be re-checked. The threshold setting maybe is wrong, and ROI at wrong position where the edge is not inside the ROI or the ROI is too small.

If the message **Alfa: Best Threshold: no best threshold found i**s shown in **Message** window frequently during test run or job executing, deactivate the **Best Threshold** option for this job if not due to bad focussing. Set a good threshold in the entry field manually.

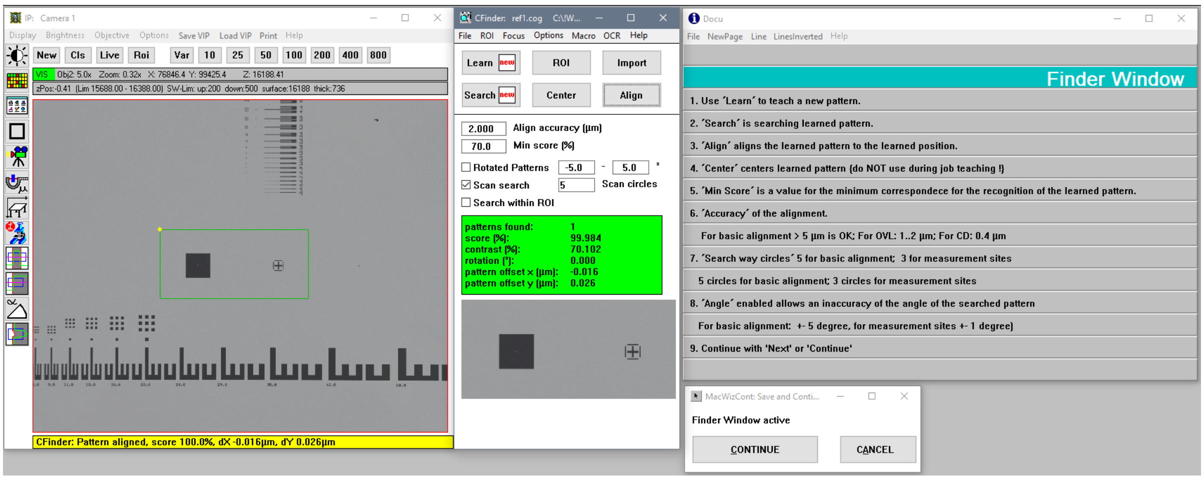
3.2.1.4 CFinder at Alignment Site

After focus, brightness and angle of first alignment structure are well adjusted, next is to define the parameters for **CFinder** window. The **CFinder** is used for all alignment purpose within the Wizard. The **CFinder** performs pattern recognition and alignment to the position as it was learned.

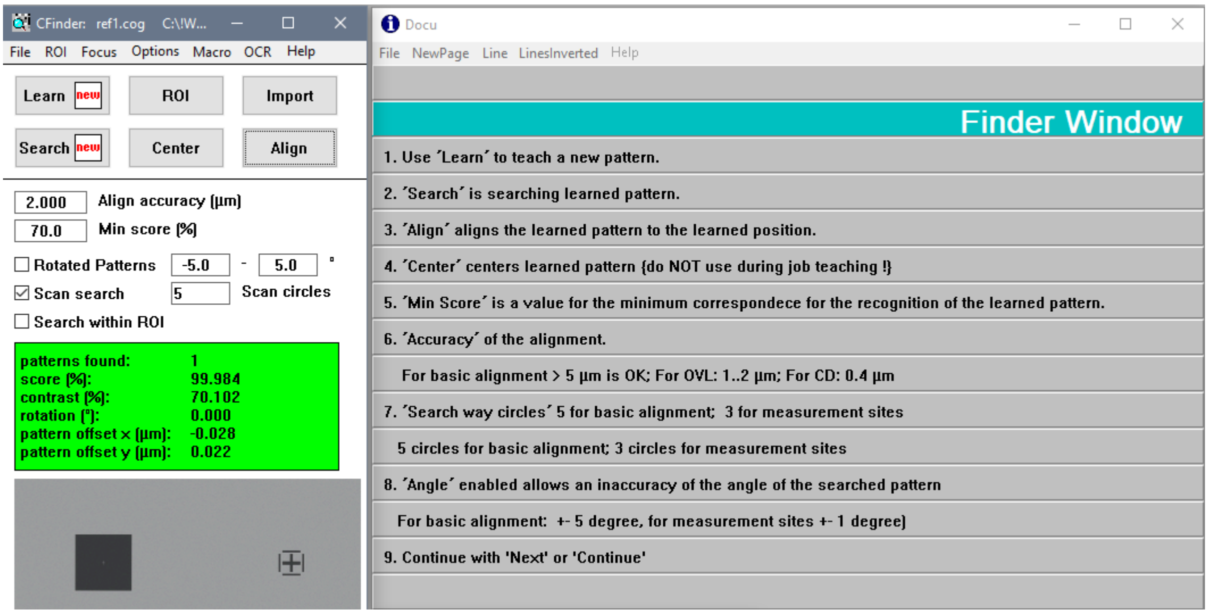
Click the **Learn** button to define an alignment structure in the **IP** window as reference image for pattern recognition. Subsequently, click the **Search** button to search for the learned pattern. During searching, observe the pattern recognition process in **IP** window and the obtained **score%** in the **CFinder** window. It may be necessary to adjust the **Min score (%)** or relearn the reference image with better structure. The **Min score%** must be at least 60% or more. If the pattern is found with high score, click the **Align** button to align the learned pattern to the learned position. ROI can be defined at the exact region of the image that is to be recognized and aligned by clicking the **ROI** button. It is advised to place the structure before defining the ROI exactly at the position on the **IP** window. This is currently only used for overlay-measurements with identical overlay boxes close to each other.

It is recommended to choose the finder pattern and its position as follows:

- The finder pattern for measurement position is simple but unique.
- Do not have similar structure nearby.
- The finder pattern should not be too large, about 25-30 % of the **IP** window. If the learned pattern is larger than 50 %, the structure must contain horizontal and vertical edges.
- Do not include too much empty space in the finder pattern.
- The laser spot must not stay on any structure edge.
- The learned pattern must not touch or be close to the border of the entire image, but be away from the border at least 10-20 % of the screen area. Otherwise, the finder becomes unstable.



The **Docu** window displays information and the guideline for the **CFinder** window.



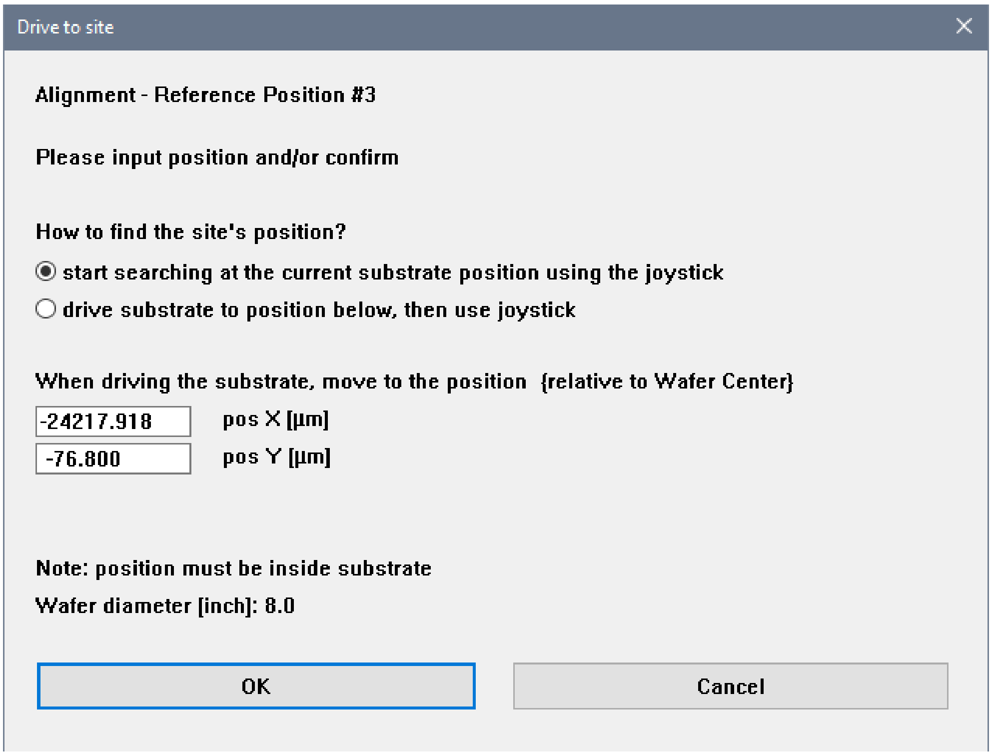
Selection	Description
Learn	To learn a new pattern as reference image
ROI	To define the region of interest for pattern searching. Pattern is searched only within this ROI.
Import	To import a reference image
Search	To search for the learned pattern
Center	To center the learned pattern to the crosshairs (not used for the Wizard)
Align	To align the learned pattern to the learned position (used for the Wizard)
Rotated Patterns	By activating this function, the finder recognizes a rotated pattern within the angle entered in the entry field.
Scan search	To scan the searched pattern in a spiral circle if the pattern is not found at the finder position. The number of scan circles and scan steps can be defined in the entry field of **CFinder option** window.
Search within ROI	Used only for alignment with two identical structures in IP window at the same time. Differentiate the two structure to left-right or top-bottom. The ROI is defined only at one of the structures. Do not activate this function for normal jobs.
Continue	To continue with second alignment
Cancel	To interrupt the job learning. The Wizard stops.

By activating the **Scan search** function, the stage will move in a spiral (scan circles) from its nominal position if the searched pattern is not found at the learned finder position, In the menu of the **CFinder** window, click **Options-Search** to define the scan step. It is advisable to set the scan step to 30-50 %, depending on the size of the learned pattern. The pattern may still not be found if the scan step is too large. Typically, the pattern not found problem is due to unsharp focusing. During scan searching, the focus will change according to the stage movement and may help to find a sharp image for recognizing the pattern. In the **MacWizCont** window, click **Continue** to continue with the second alignment.

3.2.2 Second and Third Alignment

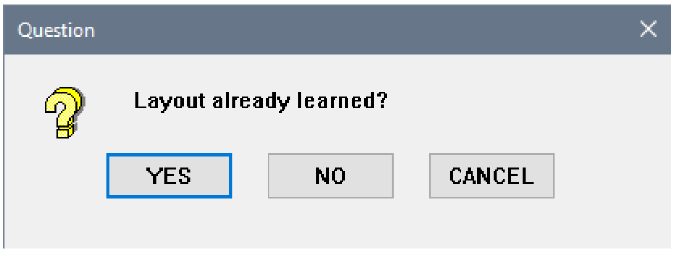
The learning of the second alignment position is the same as the first alignment position (see 3.2.1). It is important the second alignment structure to be same as the first alignment structure. To avoid alignment error, the second alignment position has also to be on the same nominal X or Y coordinate as the first alignment position.

After driving to the second position, check and correct the position with joystick or mouse in the live window. Click **OK** to continue driving to the third position if a third alignment is configured. A third alignment position is helpful for manual loading but can usually be skipped for robot loading.



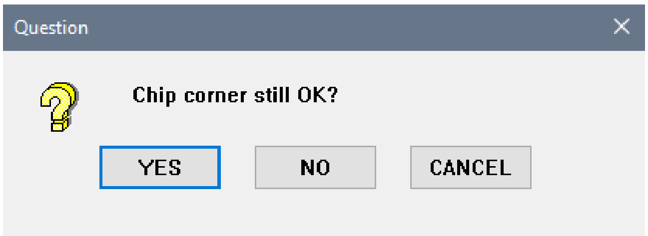
3.3 Wafer Layout and Measurement Way

If the job is copied based on an existing job or to modify the existing job, the **Question** window is shown and asking if the layout is already learned.



Selection	Description
YES	To skip layout learning and use the layout and way according to existing parameter
NO	To continue with layout learning, starting from Chip Index definition
CANCEL	To interrupt the job learning. The Wizard stops.

If the layout is already learned, the stage moves to the existing chip corner position. Subsequently, the next **Question** window is shown and asking if chip corner is still OK. Look at the chip corner shown in the **IP** window, evaluate the chip corner is aligned at the cross-hair center. If the chip corner is not centered, click **NO** button and center it in **Free mode** (see 3.3.1.3).



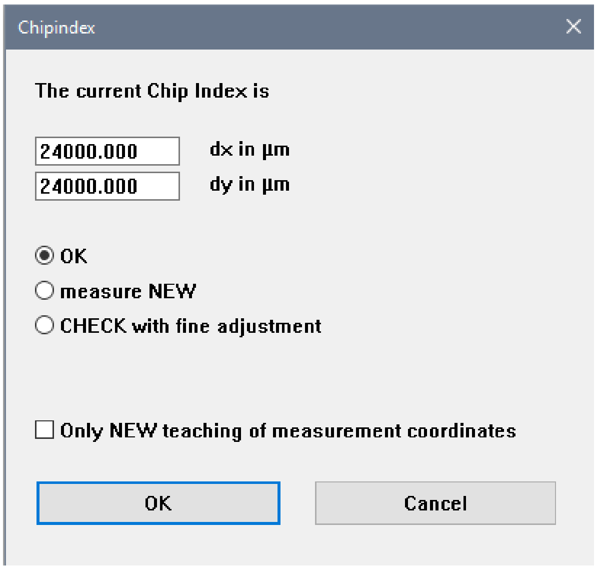
Selection	Description
YES	The existing chip corner position is at center of the cross hair.
NO	To move the stage to a chip corner with joystick and center it in **Free mode** if the chip corner is not at center
CANCEL	To interrupt the job learning. The Wizard stops.

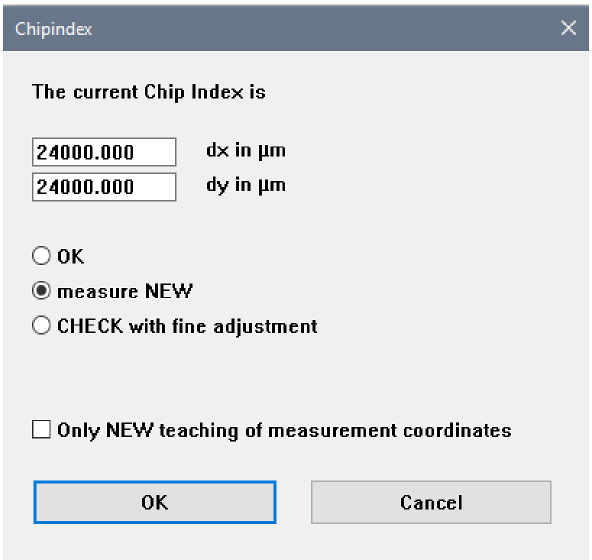
After the layout is shown in the **StageDsp** window, user can have a check on the substrate layout or chip layout by clicking the **Layout** in the menu. If there are changes in the **Substrate layout** window and you clicked **Calculate new maps** button, the new layout will be displayed in the **StageDsp** window according to the new parameters. If the layout is correct, click the **Accept** button to continue (see 3.3.1.4).

3.3.1 Wafer Layout

The wizard uses the wafer layout to define the measurement positions. The layout is based on the wafer size and chip dimension. To calculate the wafer layout, it is necessary to know the chip size accurately.

If the chip size (Chip Index) is known, the value can be entered in the **dx** and **dy** entry fields of the **Chip Index** window. Select **OK** radio button and continue with **OK** if the current chip index shown in **dx** and **dy** are correct or updated. The **Chip Index** is in micrometer, um.

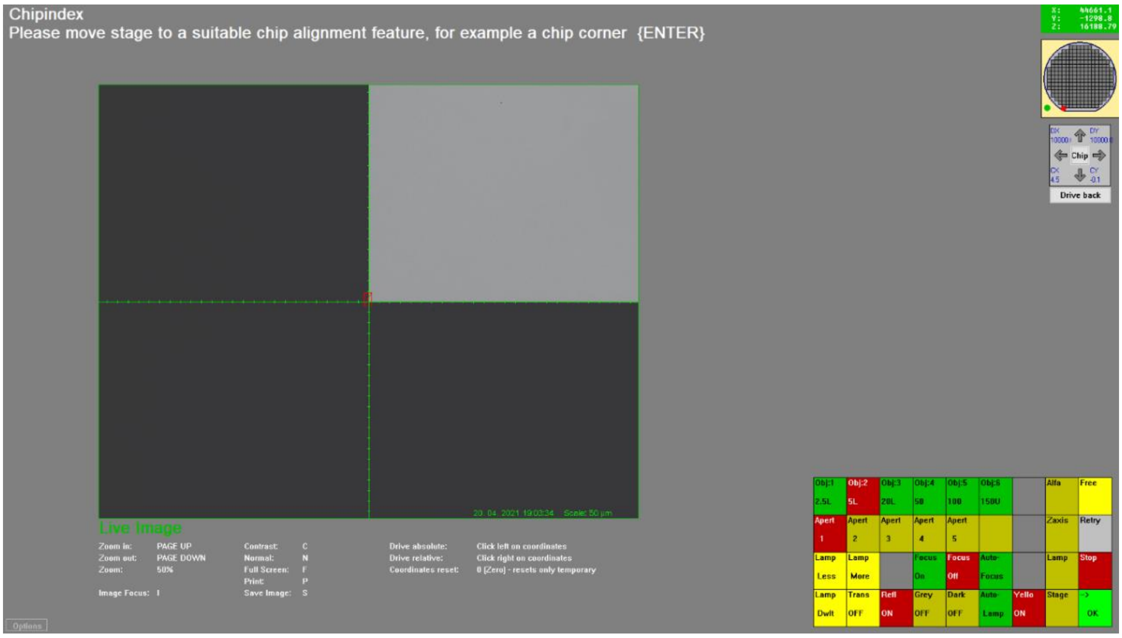




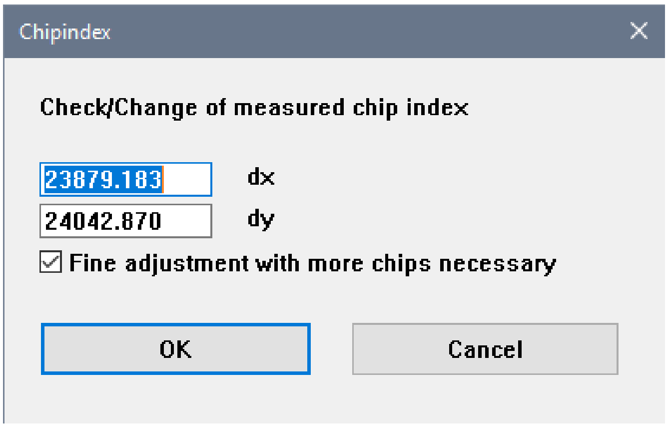
Selection	Description
OK	To accept the current Chip Index shown in dx and dy. The whole chip size measuring procedure will be omitted. The job continues with defining the offset of a chip corner to the center of wafer.
Measure NEW	To measure new chip size for the current chips
CHECK with fine adjustment	To check the chip size with higher magnifying objective.
Only NEW teaching of measurement coordinates	To define the exact measurement coordinates in the chip. This might differ and should be proofed during job learning.
OK/NEXT	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.3.1.1 Measure NEW

If the chip size is unknown, select **measure NEW** radio button and click **OK** button to measure the rough chip size. In **Free mode**, use an objective of relatively small magnification but large field of view to find a chip corner. Drive the stage to a suitable chip alignment feature, center any chip corner with the joystick and click **OK**. Subsequently, move the stage to an identical corner of the neighbouring chip that is diagonal to the first chip corner, center it and click **OK** button.



The measured chip index is shown in the **Chip Index** window. The value can be verified again using a larger magnifying objective and covering greater distance between the first and second chip corner position.

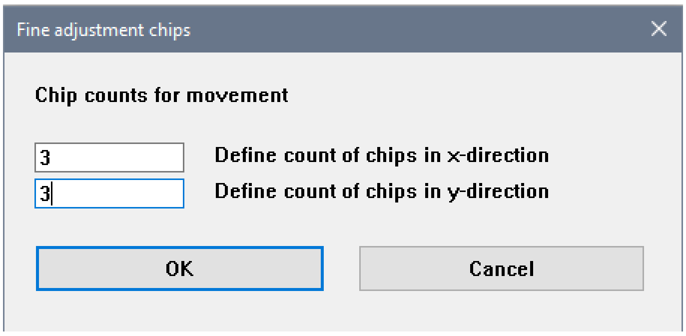


Selection	Description
Fine adjustment with more chips necessary	To measure the chip size more precise with higher magnification and multiple chips apart
OK	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

If no fine adjustment is necessary, deactivate the **Fine adjustment with more chips necessary** checkbox, the job will continue with defining offset of a chip corner (see 3.3.1.3).

3.3.1.2 Fine Adjustment

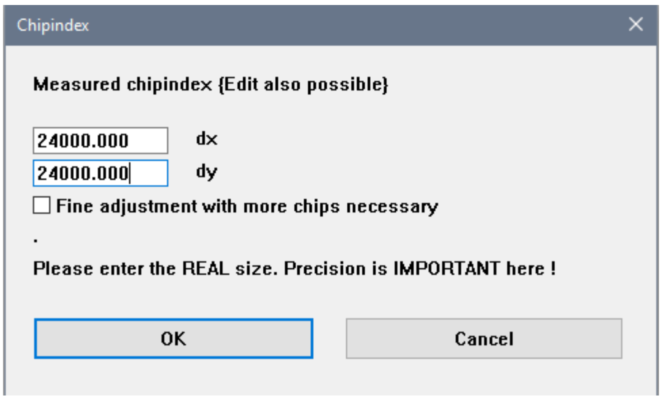
If the **Fine adjustment with more chips necessary** checkbox is activated, the first chip corner with higher magnification will be shown in **Free mode** for manual refining the chip corner centering and click **OK** button. Then, the **Fine adjustment chips** window is shown and the user enters the **count of chips in x and y direction** to move to another chip corner. Chip size measurement over few more chips allows more precise calculation. The further the distance, the smaller the relative error of measurement. It is advisable to choose more chips (e.g. 50 chips) to travel over half of the wafer size.



After entry of the chip count, click **OK** button. Based on the rough chip size measured in **measure NEW**, the stage drives to the chip corner of the defined chip counts, which is diagonally apart to the first chip. In **Free mode**, refine the chip corner to center with the joystick and click **OK** button.

The refined chip index is shown in the **Chip Index** window. It is advisable to align the chip corner precisely and accurately, especially for wafers with high quantity of chips. For example, with the different of 0.28 µm per chip and travelling over 100 chips, the positioning error will be 28 µm. This is a lot for higher magnification objectives and the **CFinder** may not locate the learned pattern at wrong position. This may also cause focussing on a wrong position (edges or other areas with more structures), especially during overlay measurement. Thus, always use the correct chip size. Otherwise repeat the fine adjustment again if the result seems to be unacceptable.

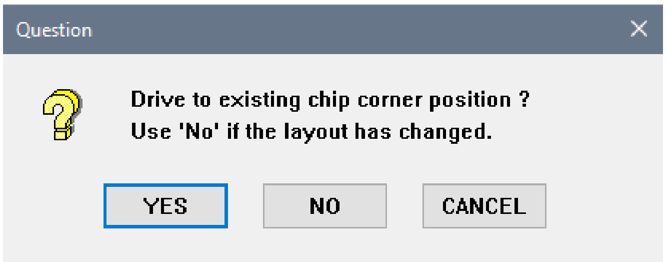
If a chip index of 5901.45 is shown and user know that it must be 5900.0 then enter the correct value. If the deviation is always high (> 5 um) then check the stage correction or call service.



If the **Fine adjustment** checkbox is deactivated and the measured chip index is acceptable, click **OK** to continue.

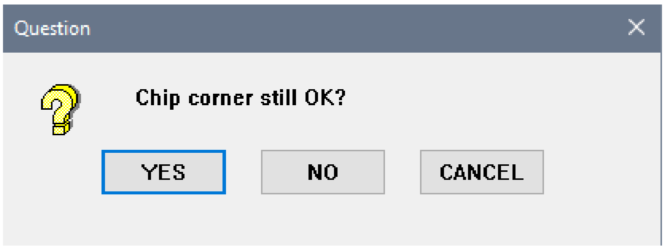
3.3.1.3 Defining Offset of a Chip Corner to the Wafer Center

To calculate an exact layout, the chip arrangement is needed to know by defining an offset of a chip corner to the wafer center. The **Question** window is shown and askes for driving to existing chip corner position.



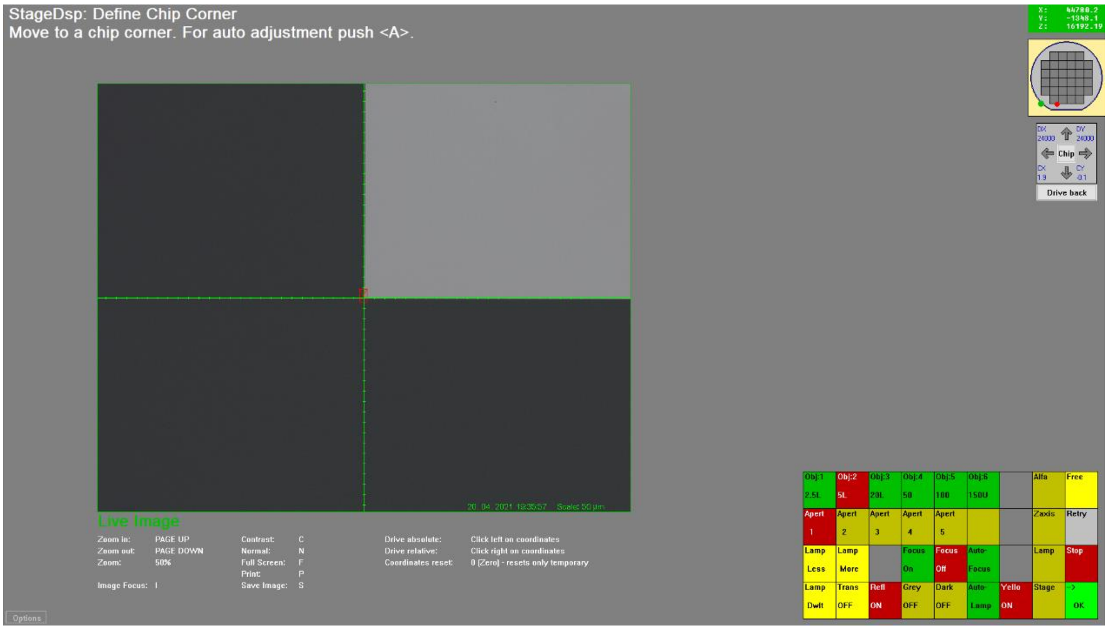
Selection	Description
YES	To drive to existing chip corner position if the layout is still valid
NO	To move the stage to a chip corner with the joystick and center it in **Free mode** if the layout has changed or is a different layout
CANCEL	To interrupt the job learning. The Wizard stops.

If you wish to drive to an existing chip corner position by clicking **YES**, the stage is moved to the existing chip corner position. Subsequently, the next **Question** window is shown and asks if the chip corner still OK. Look at the chip corner shown in the **IP** window and evaluate if the chip corner is aligned at the cross-hair center. Click **YES** to accept the chip corner alignment.



In both **Question** windows, click **NO** to define a new zero position at a chip corner or further fine tune the chip corner. In **Free mode**, move the stage to a chip corner with the joystick and place the chip corner at the cross-hair center as precise as possible. This measurement is the base for calculating the wafer layout. The layout should be real because the measurement positions are based on this layout and calculated relatively to this chip corner. If the alignment structure is moved and the chip corner is reset properly, then all the measurement positions are still valid.

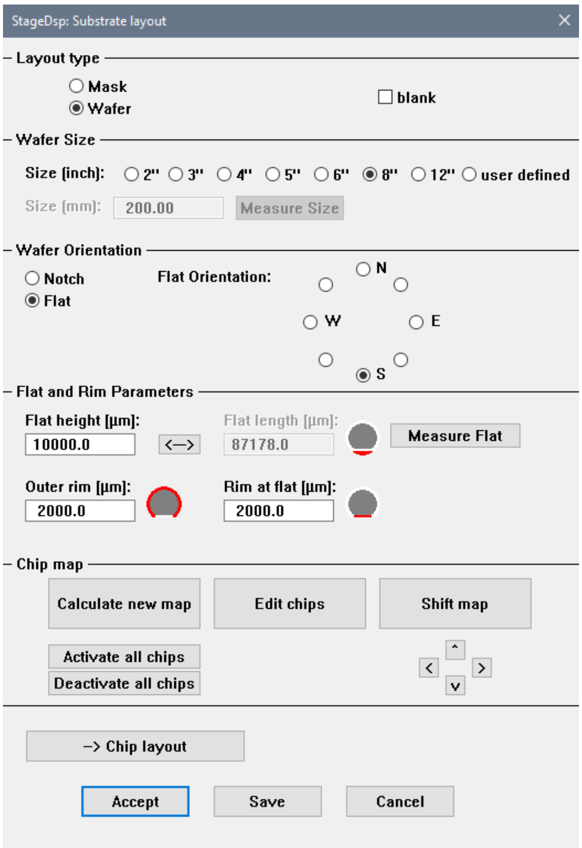
The chip corner should be set at the lower left corner of a chip. In **Free mode**, the upper right quarter is the measurement chip. If the measurement position is at the left or lower quarter in **Free mode**, then it is the neighboring chip and the chip number is wrong. The Wizard will warn that the measurement position is outside of the current chip. Click **OK** on the keypad to continue.

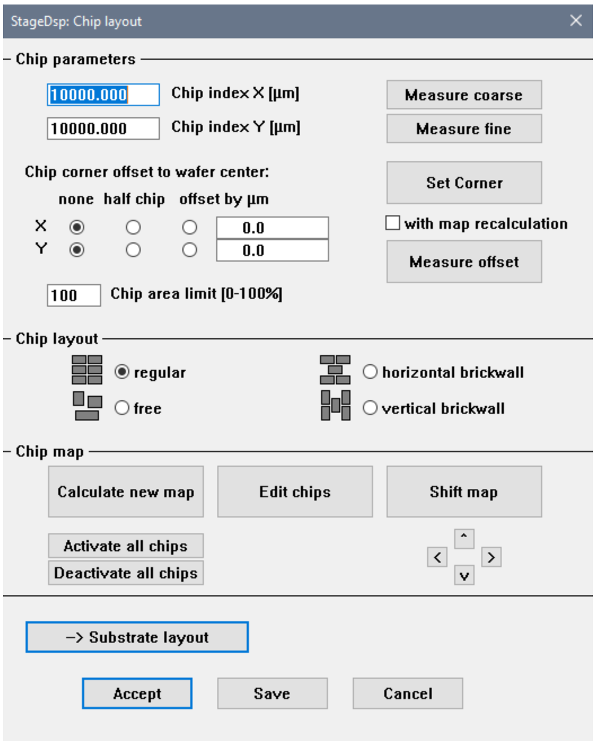


3.3.1.4 *Substrate and Chip Layout*

For CD/OVL measurement, the **Substrate layout** window is shown in the first place. You can define the **Layout Type**, **Wafer Size** and **Wafer Orientation** ahead.

By clicking the **Chip layout** button in the **Substrate layout** window, you can also pre-define the known chip index. If the chip index is unknown, just leave it and the chip index will be learned later. Click **Accept** or **Substrate layout** button to go back to **Substrate layout** window. Click **Accept** again to continue the job learning with the **Job setup** window.

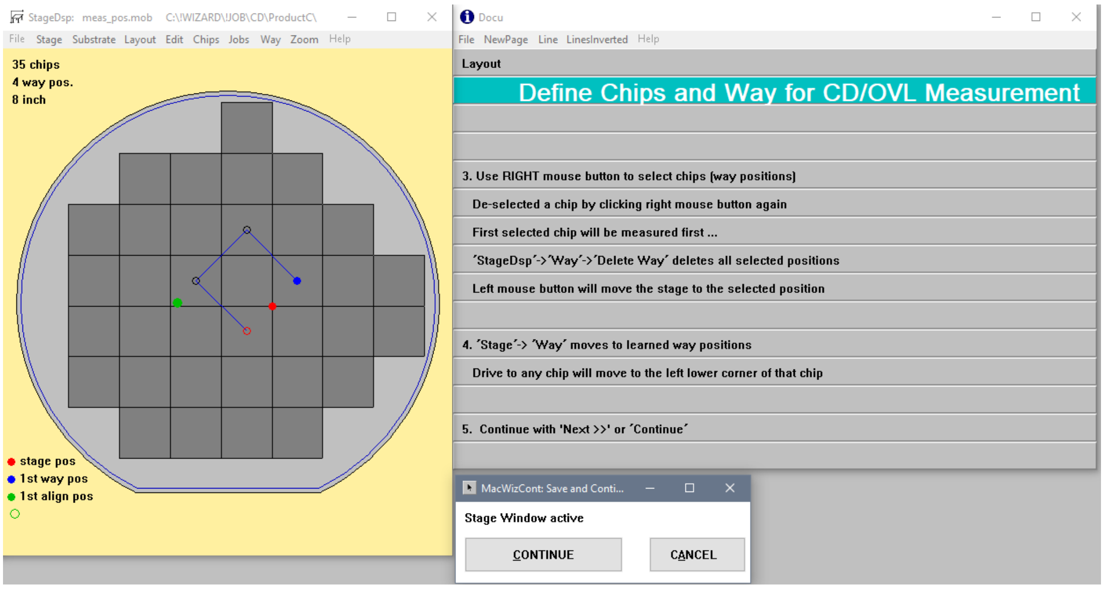




3.3.2 Defining CD/OVL/FT Measurement Chips and Way

After the new layout is shown in the **StageDsp** window, you can define the way by right-click to select the measurement chips one by one, following measurement sequence. This sequence or way will be displayed by small blue circles connected with blue line and the current measurement position is marked with red circle. In the **MacWizCont** window, click **Continue** to continue learning.

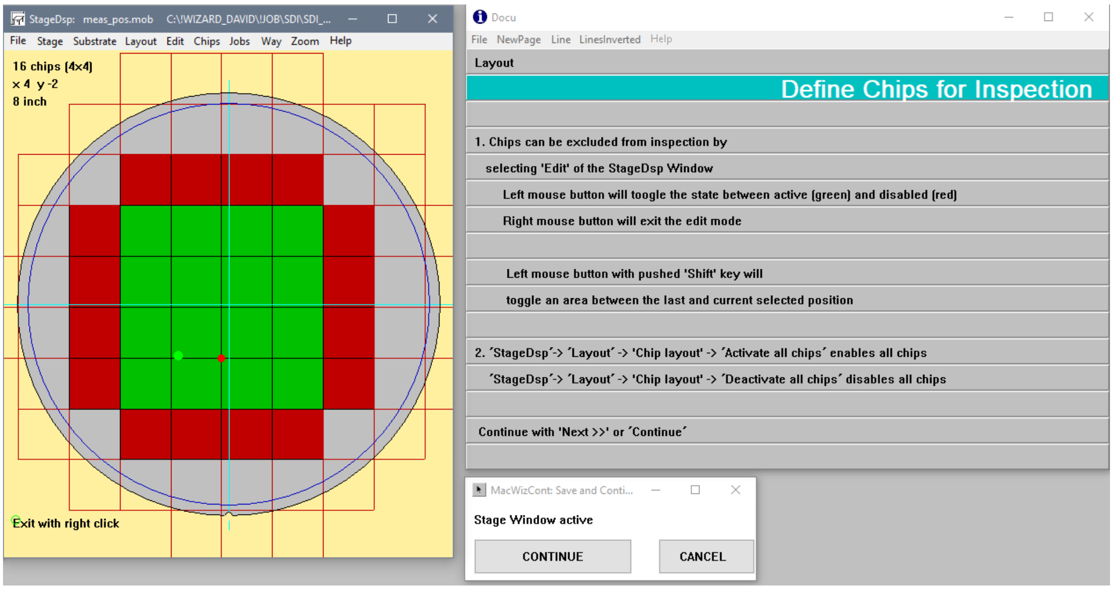
The **Docu** window displays information and the guideline for Defining Chips and Way for CD/OVL Measurement.



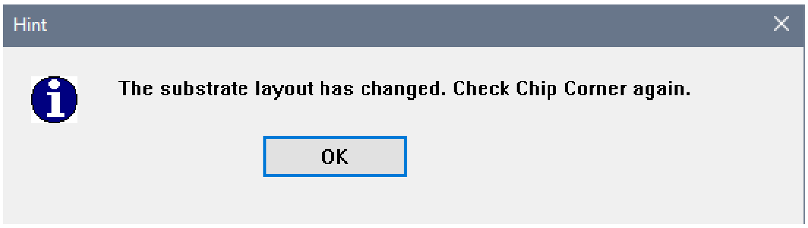
3.3.3 Defining Device Inspection Chips

Correctly defined wafer layout is the basic requirement for a wafer scan. Define it as exact and as precise as possible. The layout is displayed in the **StageDsp** window. If you wish to edit the layout, this can be done by clicking **Edit** in the menu of the **StageDsp** window. Left-click on the chip to toggle the state between active (green) and disabled (red). Right-click to exit the edit mode. In the **MacWizCont** window, click **Continue** to continue the job learning.

The **Docu** window displays information and the guideline for Defining Chips for Inspection



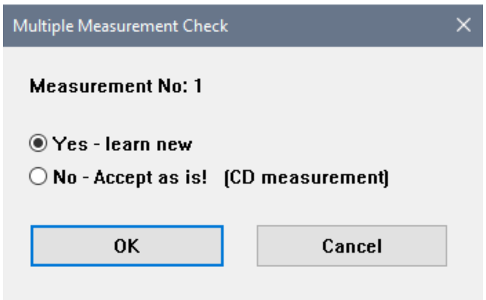
If the layout is changed, the **hint** window is shown and askes to check the chip corner again. By clicking **OK**, check & adjust the chip corner in fee mode if necessary. If the chip corner is well aligned, click **OK** on keypad or **ENTER** to continue.



3.4 Learning Measurement Parameter

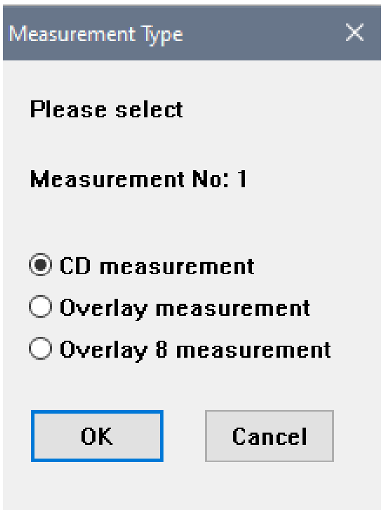
3.4.1 First Measurement

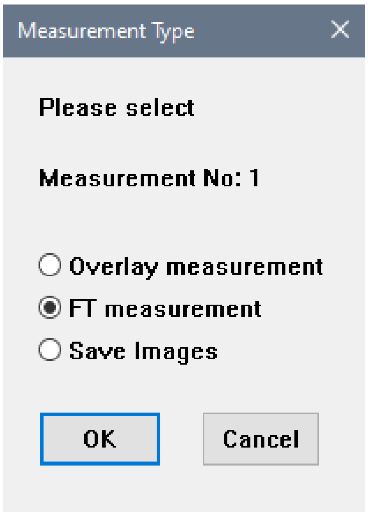
After wafer layout and measurement chips and way are learned, the next to learn is the first measurement position. In the **Multiple Measurement Check** window, select the radio button **Yes** to learn new measurement parameter. Select **No** if it is already learned previously and directly go to next measurement position.



Selection	Description
Yes – learn new	To learn a new measurement parameters.
No – Accept as is!	To accept the existing measurement parameter. By clicking **OK**, the job will skip the learning of measurement parameters.
OK	To continue with job learning
CANCEL	To interrupt the job learning. The Wizard stops.

The **Measurement Type** window is shown if **Yes - learn new** is selected. The exact content depends on the wizard configuration. Select the measurement type that you wish to learn for the first measurement.





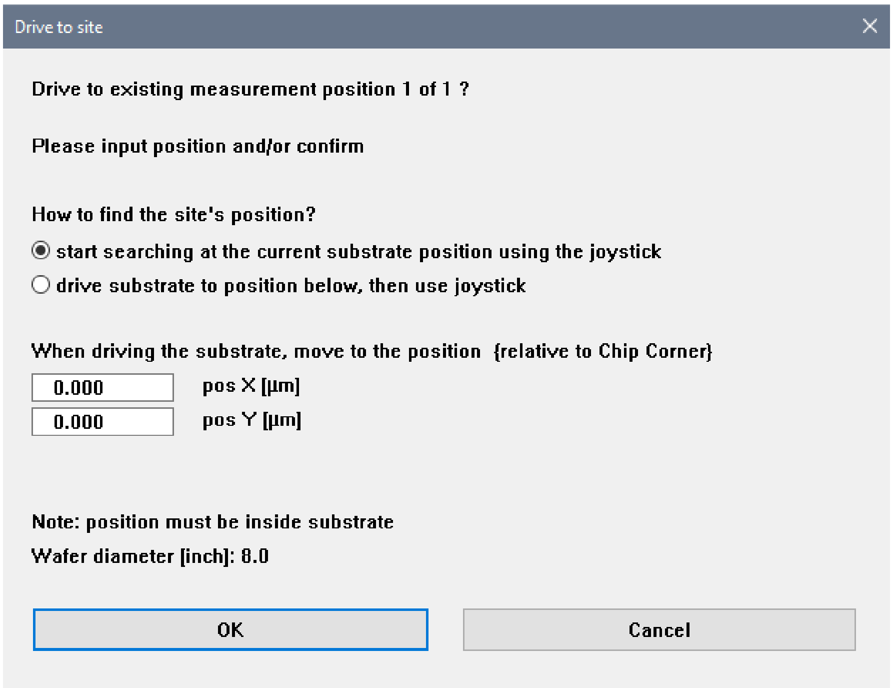
3.4.1.1 Finder Positions

At the defined measurement chip of the way, a so-called micro way within a chip is needed to be defined as well. Multiple measurements in one chip are possible.

Similar to the stage alignment position, it is recommended to choose the finder pattern and its position as follows:

- The finder pattern for alignment is simple but unique.
- Do not have a similar pattern nearby.
- Preferable is a pattern that contains both horizontal and vertical edges, if possible.
- The pattern can be part of the CD/OVL structure or a structure nearby.
- The same pattern is available at every measurement position.
- The structure does not change its appearance with process variations between wafers.
- For teaching, the first chip in the specified way (in the layout) is used.
- For measurements in a line field with only one type of line (only horizontal or only vertical lines) use a feature outside the field. Then move with the measurement xy offset into the field.

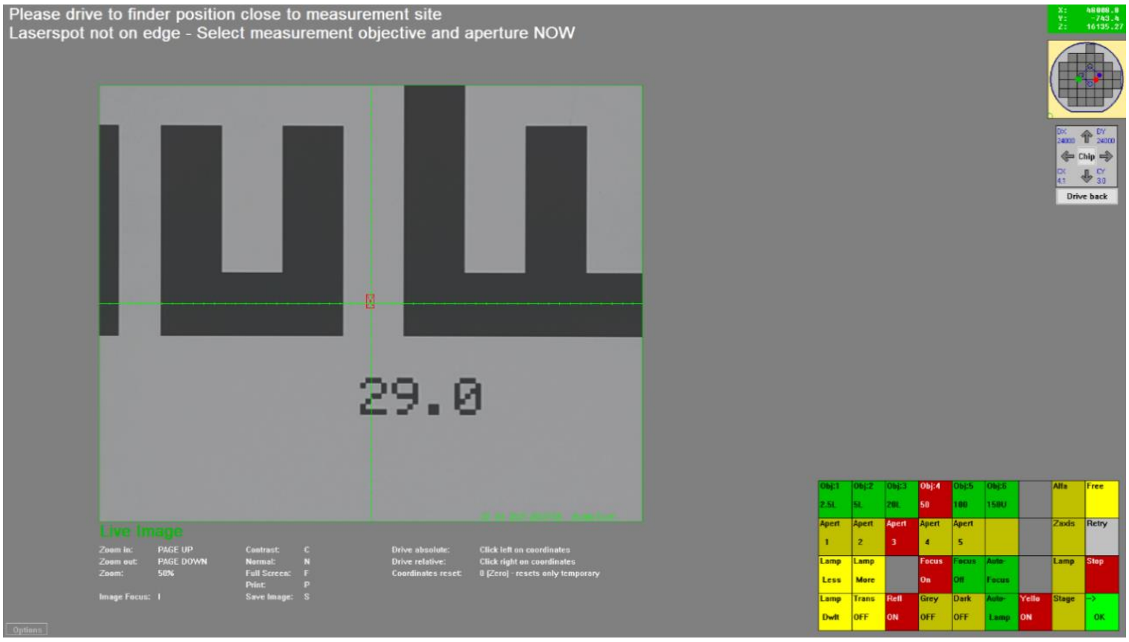
The first measurement position is needed to be set with the guidance in the **Drive to site** window.



After clicking **OK** button, the user will be guided to **Free mode**, to drive to the finder position with the joystick. The measurement objective and aperture need to be defined using the keypad. Also the illumination type (reflected or transmitted light, if available) is selected here. At measurement site, it is important that the LFS spot must not stay on or close to any edges, leading to focusing error. Focus on a plane area where the focus can work reliably.

For overlay structure, it is recommended to focus on the center of the overlay structure if it is large enough. Otherwise, position the entire overlay structure to the left or right from the center vertical line. Maybe it is necessary to adjust the light with the **DWIT** button on the keypad if the brightness is not in good condition. Click **OK** on the keypad to continue.

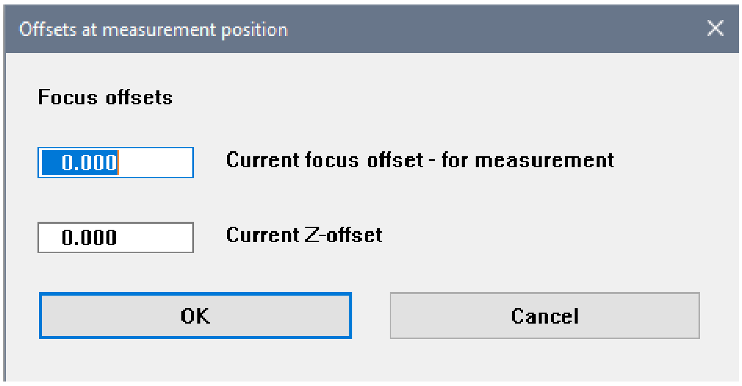
Alternatively, you can also use the mouse pointer and click on the **IP** window to move the stage exactly to the desired position. The clicked position will be placed at the screen center. With this method, it is much more precise to drive the stage than by using joystick, especially when you need to move a very short distance.



3.4.1.2 Focus Offset

The **Offsets at measurement position** window is shown and the current focus offset for measurement and current Z-offset are displayed. Click **OK** to continue with brightness adjustment.

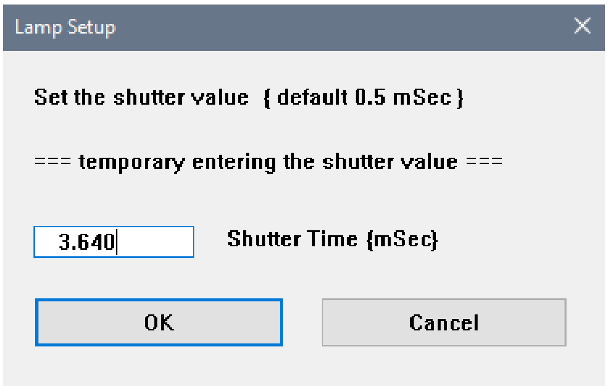
The **Z-Offset** is used if the focusing range is too small to get a sharp image. The 100x lens can focus within a range of +/- 5 µm from the reflective material. It is activated in the job options.



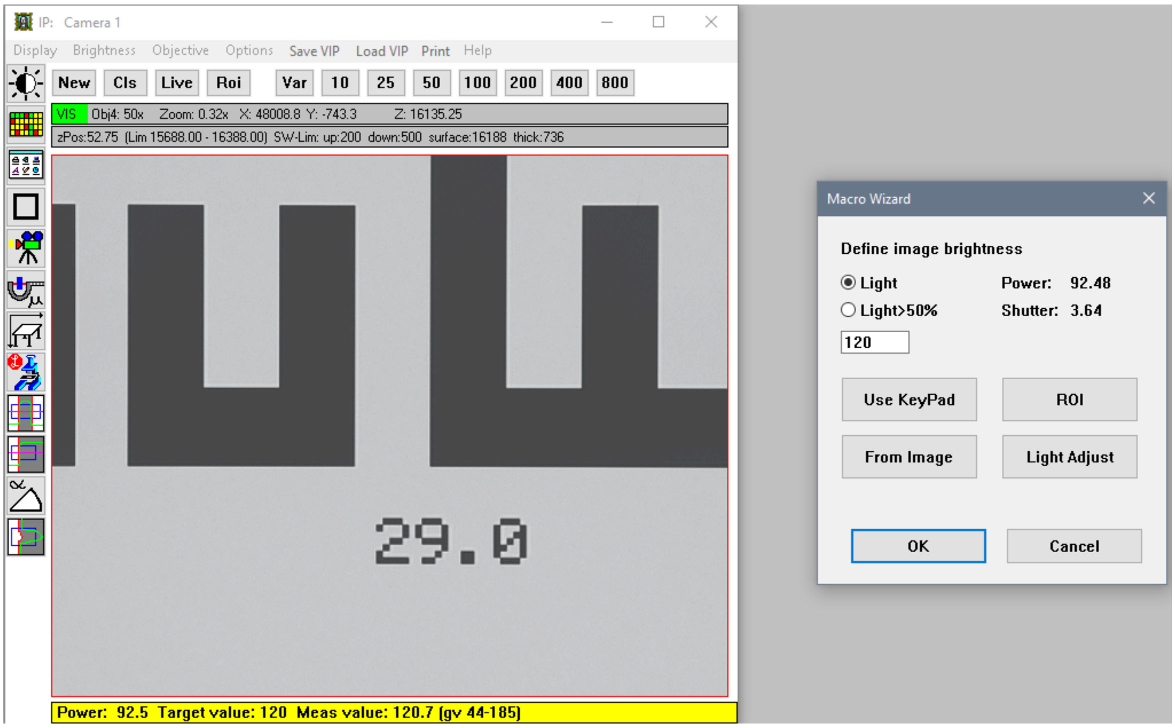
Selection	Description
Current focus offset	The current focus offset at the measurement site with focus compensation
Current Z-offset	The current Z-offset with stage compensation
OK	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.4.1.3 Brightness Adjustment

The procedure for brightness adjustment at the measurement site is the same as described for the stage alignment position (see 3.2.1.1). The **Lamp Setup** window is shown and you set the shutter value with the suggested shutter time by clicking **OK**.

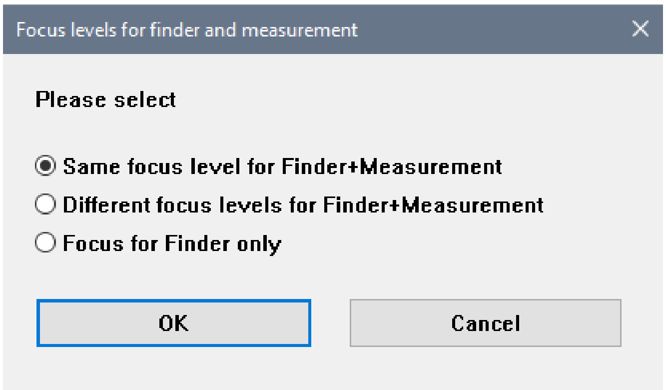


Next, the **Macro Wizard** window is shown for defining the image brightness. The brightness must be set carefully to avoid over-brightness at the measurement structure. Be reminded do not use **Light>50%** for wafer with many different gray levels. It is recommended to define a **ROI** in a homogeneous background on the wafer. For wafers, select the **Light** radio button and set a constant value in between 80-150 in the entry field, so that the measurement structure is around 120-150 in the **IP** window. If there are many thin lines everywhere in the **IP** window, a large ROI with **Light** setting is preferable for a good average intensity.



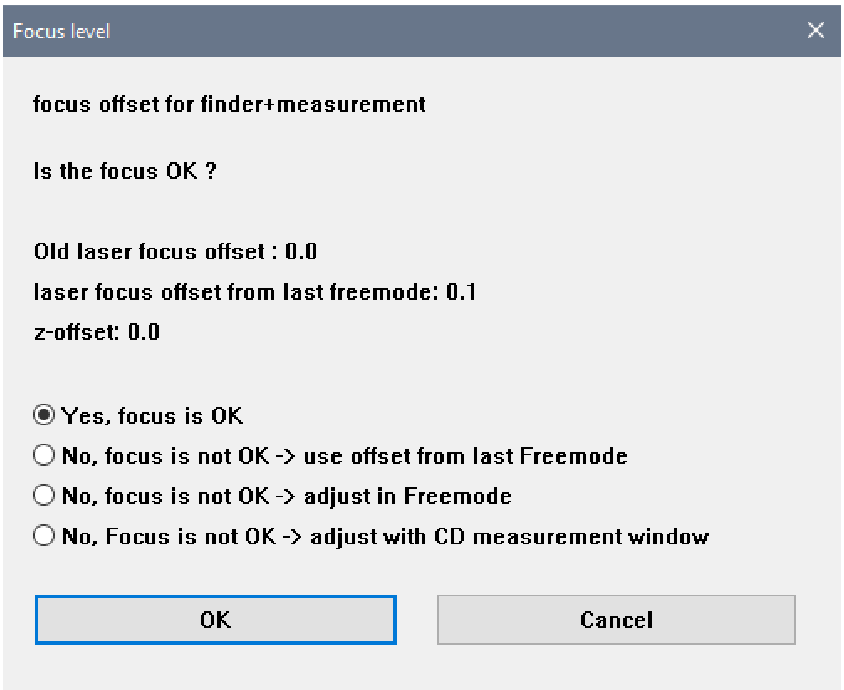
3.4.1.4 *Focus Adjustment*

After brightness adjustment is learned, you define the focus level for finder and measurement. Typically, select the same focus level for finder and measurement if the finder alignment structure and measurement structure are at the same or very near position where the focus level should be the same. Different levels also need execution time and lifetime.



Selection	Description
Same focus level for Finder+Measurement	Use this radio button if the finder and measurement structure are on the same focus level
Different focus levels for Finder+Measurement	Use this radio button if the finder and measurement structure height are on the different focus level. By focusing separately, either focus on top or bottom of the structure, would be helpful.
Focus for Finder only	Use this radio button to learn the focus of finder structure only
OK	To continue with job learning.
Cancel	To interrupt the job learning. The Wizard stops.

The procedure for focus adjustment at the measurement site is the same as described at stage alignment position (see 3.2.1.2). The live image is displayed in the **IP** window and **Focus level** window is shown. Evaluate the sharpness of the live image and response.

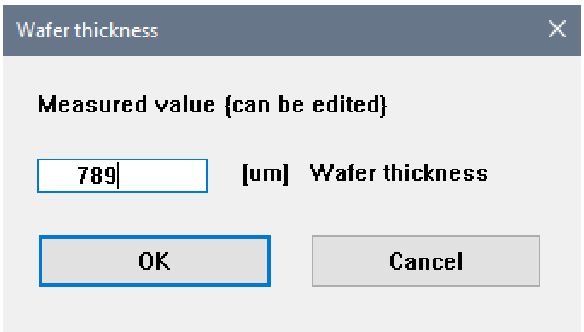


Selection	Description
Yes, focus is OK	The current focus level is acceptable. Wafer thickness is displayed.
No, focus is not OK → use offset from last Free mode	The current focus level is not acceptable and you wish to use the offset from last **Free mode**. Last **Free mode** offset applied and wafer thickness is displayed
No, focus is not OK → adjust in Free mode	The current focus level is not acceptable and you wish to adjust manually in **Free mode**. New offset applied and wafer thickness is displayed.
No, focus is not OK → adjust with CD measurement window	The current focus level is not acceptable and you wish to use the function of the **CD-Meas** window, e.g. focus series.
OK	If yes, the job will continue with displaying the wafer thickness If no, the job will go to **Free mode** for manual focus readjustment.
CANCEL	To interrupt the job learning. The Wizard stops.

If the live image is not in focus, select **No, focus is not OK** with other options and click **OK** in the **Focus level** window. You can either use the offset from last **Free mode**, adjust focus level manually in **Free mode** or use the focus function of the **CD-Meas** window.

If the live image is in focus, select **Yes, focus is OK** and click **OK** in the **Focus level** window. Based on the z-height corresponding to the focused images, the wafer thickness is calculated and displayed in the **Wafer thickness** window.

If the measured wafer thickness differs too much from the nominal value, stop learning and find the reason for the difference. Maybe it is a wrong wafer or the tool parameters changed or another possible reason. Please do not ignore a false value, it can cause damage. If the displayed value is not acceptable, please edit the normal thickness before clicking **OK**. If your wafers have different thickness (due to re-use) then enter a normal median value.



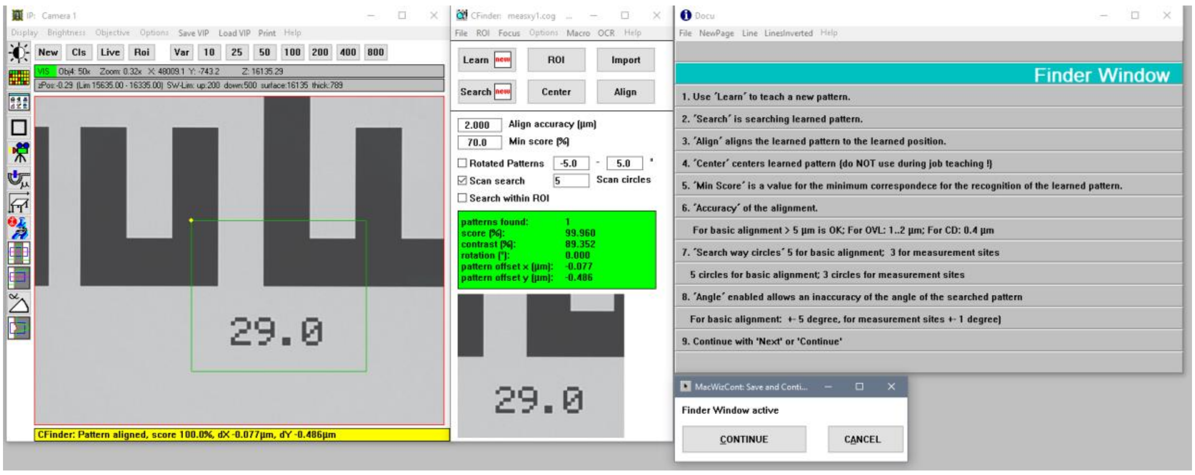
3.4.1.5 CFinder at Measurement Site

The procedure of the **CFinder** at the measurement site is the same as described for the stage alignment position (see 3.2.1.5).

Again, it is recommended to choose the finder pattern and its position as follows:

- The finder pattern for measurement position is simple but unique.
- Do not have similar structures nearby.
- The finder pattern should not be too large, about 25-30 % of the **IP** window. If the learned pattern is larger than 50 %, the structure must contain horizontal and vertical edges.
- Do not include too much empty space in the finder pattern.
- The laser spot should not stay on any structure edge.
- The learned pattern must not touch or be close to the border of the image, but be away from the border at least 10-20 % of the screen area. Otherwise, the finder becomes unstable.

The finder position does not necessarily have to use the measurement structure. It can be any structure that is near to the measurement structure. Finding a good finder pattern at measurement site in the same field of view will help to save much time. For high alignment precision, a smaller value of alignment accuracy can be defined (</= 1 µm) if the focus spot is at empty space without any structure nearby. A precise alignment helps to improve the repeatability of the measurements. However, a too high demand for the alignment accuracy may end up causing the stage oscillating and interrupting the job. Depending on the stage type, 0.5 µm is a good value.

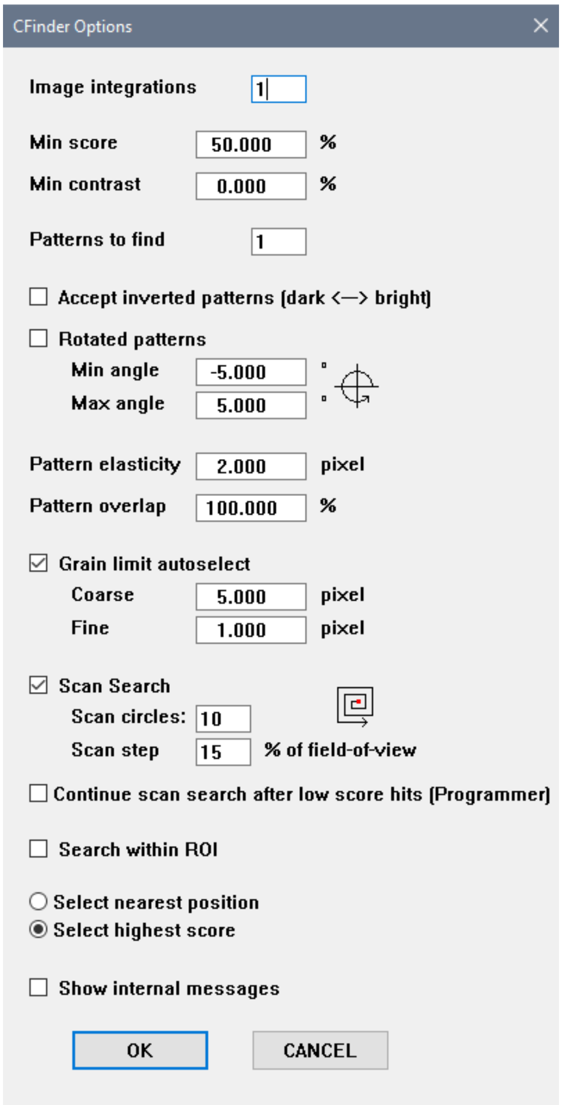


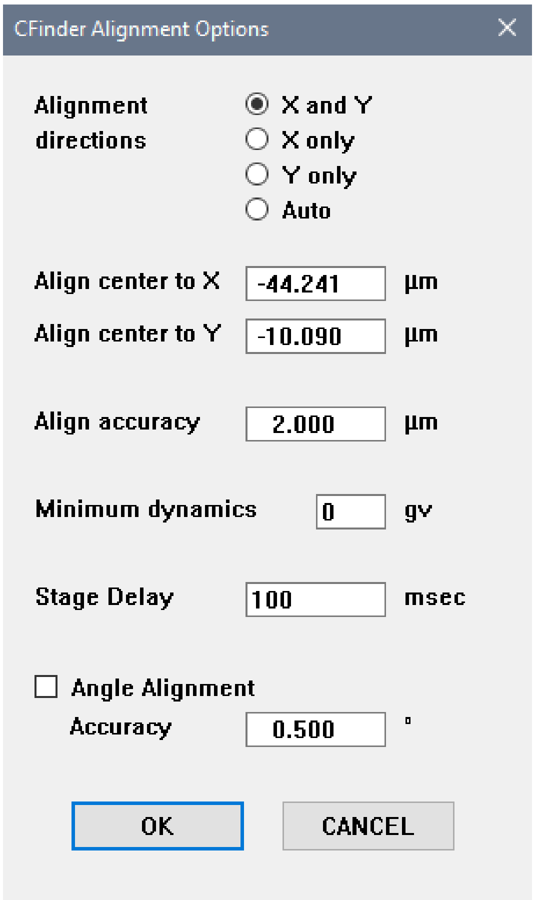
In the menu of **CFinder** window, select **Options-Search** and the **CFinder Options** is shown.

It is recommended to define the **CFinder Options** as follows:

- The finder **Min score%** must be at least 60 % or more.
- Do not activate **Accept inverted patterns (dark < - > bright)**.
- Do not activate **Rotation Patterns**. This is only for manually loaded substrates and first alignment position.
- The **CFinder** options needs a **Pattern elasticity** of 2-5 pixels or even 10 pixels for strongly deviating structures.
- Define a scan step of 10-20 % (or even 5 %) of field-of-view and 10 search circles, if necessary.
- Typically, select the **Select highest score** radio button. Only use the **Select nearest position** radio button when there are simlar finder patterns in the field of view and **search within ROI** is activated.

In the menu of **CFinder** window, select **Options-Alignment** and the **CFinder Alignment Options** is shown. If the measurement site is having hilly topography or full of different height structure, then the focus might take some time to correct the z-level. In this case, the **Stage Delay** after each scan step should be set longer than usual. The typical value of 100 ms can be set to 200-500 ms for better focusing.





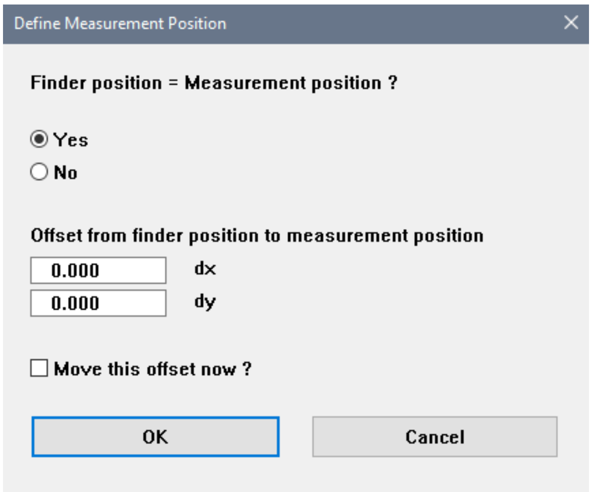
3.4.1.6 Measurement Position

Subsequently, the **Define Measurement Position** window is shown where you define the measurement position as at the same or a different position as the finder.

It is helpful to move an offset between finder position and measurement position for these cases:

- No vertical and horizontal edges for XY alignment at measurement site available (field of lines)
- Focussing not reliably close to the measurement structure (holes, edges, obstacles nearby)

Please keep in mind that this movement is done with stage precision (1-2 µm) over small distances. So the ROIs in the CD/OVL measurements must be large enough to cover the deviation.



Selection	Description
Yes	Use this radio button if the measurement position is the same as the finder position.
No	Use this radio button if the measurement position is having offset from finder position. If there is any predefined offset, the stage will move to the measurement position as stated in **dx** and **dy** (typically the existing offset is from base job). You can further adjust to make necessary corrections of the position.
Move this offset now?	Activate this checkbox to move to measurement position with the **dx** and **dy** offset.
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

3.4.2 Learning Critical Dimension (CD) Measurement

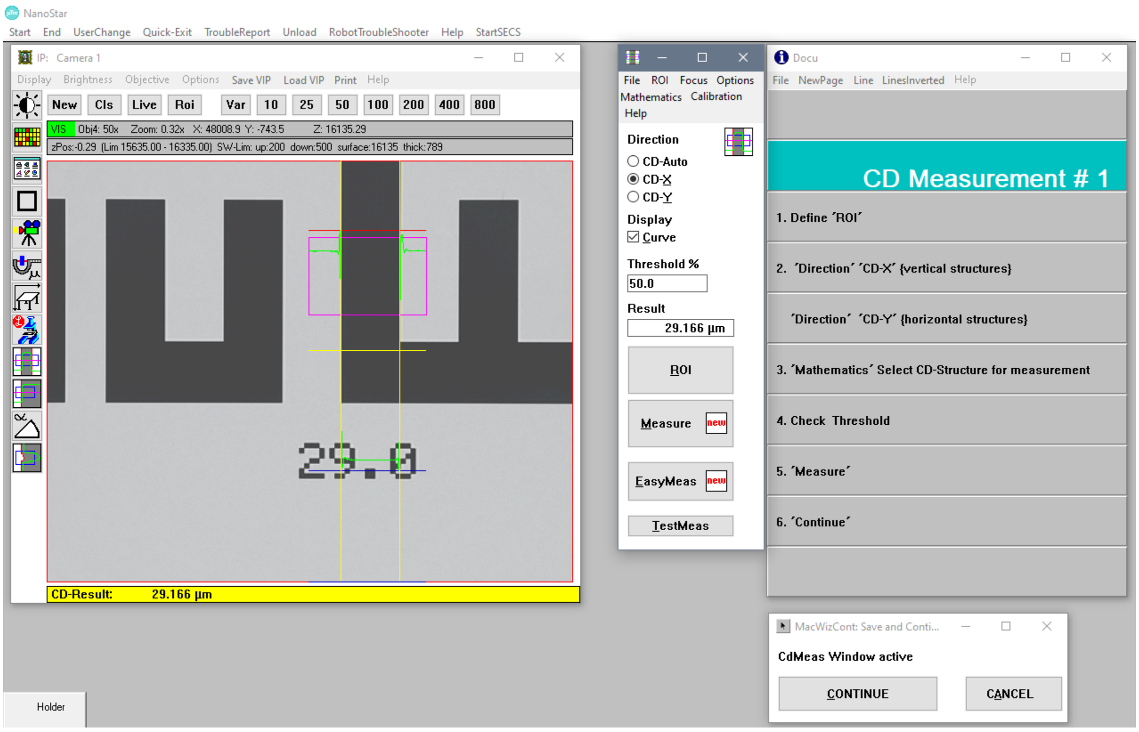
Define all necessary parameters in the **CD-Meas** window to get good repeatability:

Click the **ROI** button and define the ROI rectangle (purple colour) around the line to be measured in the **IP** window with the mouse.

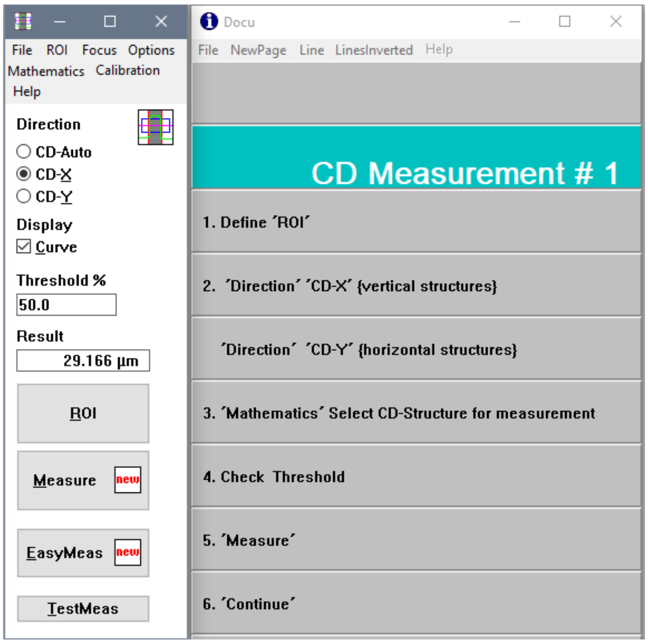
Instead of setting the CD measurement direction as CD-Auto, it is sometimes preferable to set it as CD-X or CD-Y according to the measurement direction of CD structure (X for vertical lines, Y for horizontal lines). For best results, the measurement should be done near the center of the camera image, if possible.

If there is more than one line or structure in the ROI, it is necessary to define the line to measure in the **CD-Mathematics** window.

Adjust the brightness threshold accordingly (0 % is the darkest, 100 % is brightest part of the structure). Typically, a threshold of 50 % gives the best result in term of stability and focus deviation. A threshold of 40-42 % is often very close to the real and physical width. For photomasks, the absolute width is often met at 30-34 %. Avoid thresholds which are very close to a peak or a valley in the brightness curve.



The **Docu** window displays some helpful information and the guideline forthe **CD-Meas** window.



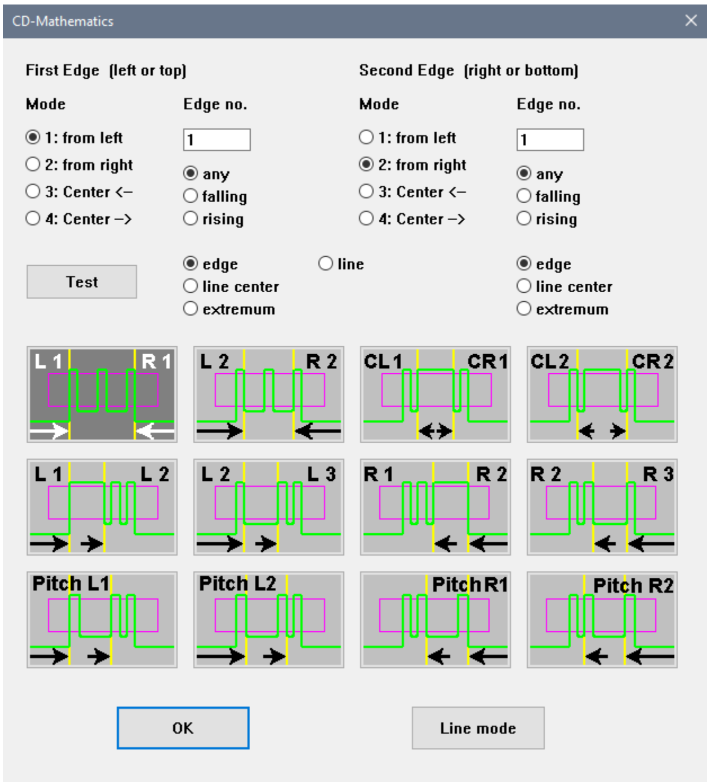
Selection	Description
CD-Auto	The direction of the CD measurement is defined automatically.
CD-X	CD measurement of the structure width in X direction
CD-Y	CD measurement of the structure width in Y direction
Display Curve	To display the brightness curve and the CD measurement result in the IP window
Threshold%	The brightness cut threshold used for the CD measurement (0-100 %)
Result	Value of the CD measurement result
ROI	Defines the region of interest with the mouse.
Measure	Performs the CD measurement.

In the menu of the **CD-Meas** window, click **Mathematics** and the **CD-Mathematics** window is shown. This window selects the structure to measure and the characteristics of its both edges. To get better measurement repeatability, it is recommended to specify criteria to describe both edges as precisely as possible. For both edges, select the direction from where to count the number of edges in the defined ROI ("left" and "right" means "top" and "bottom" when measuring horizontal lines). Enter the exact edge number to be measured (counted is each edge which cuts the threshold) and whether it is a falling or a rising edge, as seen coming from the direction selected for this edge.

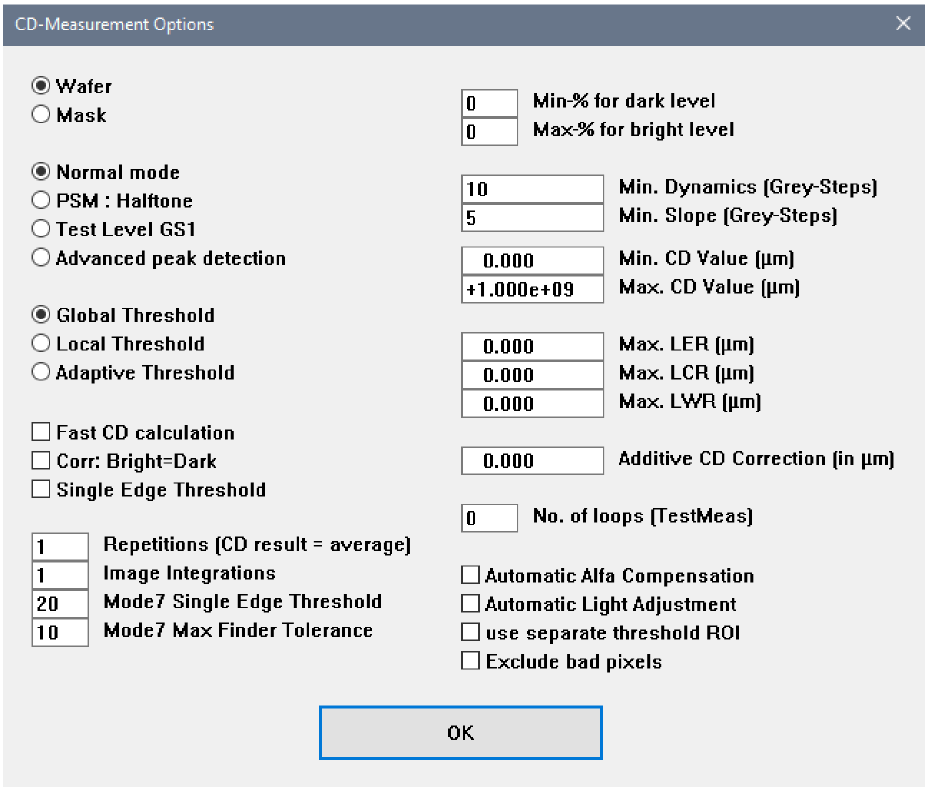
Alternatively, the user can simply select one of the pictograms below which represent some common measurement tasks.

For photoresist CD, the edges usually appear as rather thin black lines. It is recommended to use in this case the **line center** option for those weak and narrow edge lines. With the **line center** option, the rising and falling edges of the black lines are measured at the given threshold and the arithmetic mean value will be used as edge position. When using the **line center** option, the threshold should be set rather close to the line minimum (e.g. 10 % for thin black lines), especially for asymmetric lines. In very rare cases where the structure has strongly asymmetric rising and falling edges, it might be better to use the slightly less precise **extremum** option. In this case, the edge positions will be defined by the minima (or maxima) in the intensity curve.

If the ROI is defined, the threshold is set, and the criteria for edge detection are selected. Test and check your selection by clicking the **Test** button.



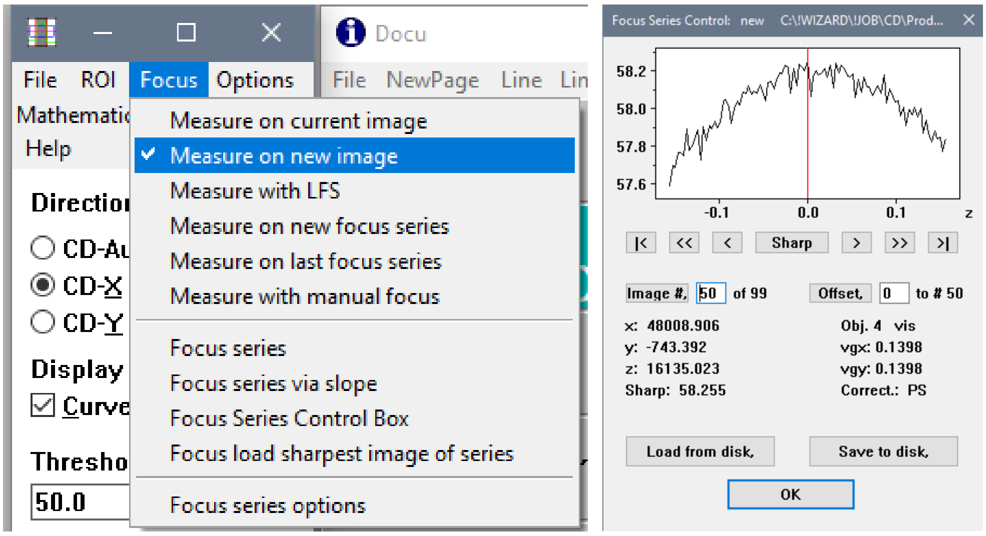
In the menu of the **CD-Meas** window, select **Options-Measurement** and the **CD-Measurement Options** window is shown. For **measurement with LFS**, you may define more **Repetitions**, i.e. the measurements are repeated and the result is calculated as the average of those measurements. This can be useful to improve the repeatability. However, for **measurement with Focus Series**, the repetitions must be set to 1. The Focus Series is already using about 30 images for evaluation, so repetitions are not required here. Other optional parameters are described below.



Selection	Description
Wafer Or Mask	Defines CD measurement on wafer or mask. Activates slightly different algorithms.
Normal mode	Standard evaluation of intensity profiles for CD measurement
PSM: Halftone	Special evaluation of intensity profiles for CD measurement on Phase Shift Masks halftone type
Global Threshold	Measure at the threshold defined by using the minimum and maximum intensity values of all structures in the ROI
Local Threshold	Measure at the threshold defined by using the minimum and maximum intensity values of the measured structure only. Useful when the ROI encloses several structures of different intensity profiles
Fast Calculation	Activate this option if the structure edge is very steep, if measuring on thin photoresist or if the warning message of "CD-Curve both rising and falling near cut" is shown.
Corr: Bright=Dark	This option concerns the **Additive correction** or the **multipoint calibration** file. If active, the bright and dark structures are corrected by the same value. If inactive, the **additive correction** is applied with an inverted sign for dark structures, or the specific entries for dark and bright structures within the **multipoint calibration** file are used.
Repetitions	Defines the number of subsequent measurements done and averaged for the result. For **measure with Focus Series**, use 1.
Image Integrations	Number of intensity curves to be integrated before measurement, helpful for noisy images. A value of 0 or 1 means integration is off. Values of 2,3,4,5… mean 2,4,8,16… images are taken and their intensity curves are averaged before measurement.

Min% for dark level Max% for bright level	Enter the trimming percentage for the dark and bright level, to trim the minimum and maximum values of the intensity curve. The threshold is calculated from these trimmed levels. Increases reliability for images with an uneven or unstable brightness curve behaviour near the minimum or maximum.
Min Dynamics	The range between minimum and maximum grey levesl of the measured edge needs to be larger than Min Dynamics. The default is 10 grey values. Use this to ignore very weak features.
Min Slope (Grey Steps)	The minimum value of the slope of the intensity curve at the intersection with the threshold. The standard value is 5 grey values per pixel. Use this to ignore very shallow features.
Min/Max CD value (um)	Minimum and maximum tolerated CD value. Measurement results outside this range will result in a error message and will not be stored. Values of zero mean no checking.
Additive CD Correction	To match the measurement result to a standard. Add the correction to bright structure CD results and subtract it from dark structure CD results
No. of loops (TestMeas)	Number of measurement loops executed when using the **TestMeas** button in the **CDMeas** window. A value of Zero means an endless loop, which can be stopped by either the **ESC** or the **F12** keys.

In the menu of the **CD-Meas** window, select **Focus** to determine the focus mode. Using short working distance objectives, the image may not be in focus immediately after moving to the structure due to the limited focal depth. Thus, the **Measure on current image** and **Measure on new image** options should not be used. Use **Measure with LFS** (before measurement, enable the laser focus until the best sharpness is achieved) or **Measure on new focus series** (obtaining the best image sharpness from evaluating several through-z images) instead.



Selection	Description
Measure on new image	The LFS is active all the time, so this is the fastest.
Measure with LFS	The LFS will be switched on and off for each measurement, so this uses up more time than a continuously active LFS.
Measure on new focus series	The z level of the wafer is varied around the nominal level and the camera images are recorded continuously. The sharpest image from the recorded image series is determined and thus the optimal focus level is calculated. The CD result is interpolated using the images around the best focus. This option is especially recommended when using short-working distance objectives, but needs considerably more execution time
Focus Series	Executes a focus series to get the best sharpness image.
Focus Series Control Box	The Focus series control box displays the image sharpness values and other information for a recorded focus series.
Focus series options	Definition of the focus series parameters

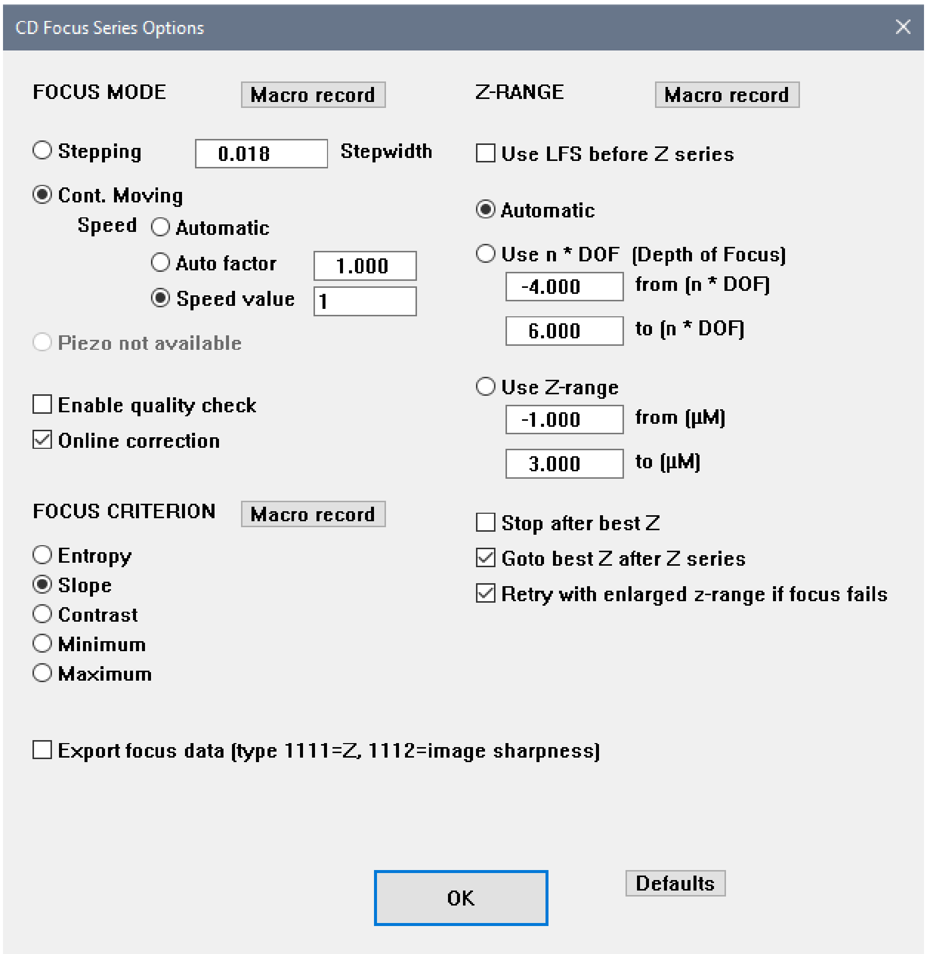
When executing a Focus series, multiple images are grabbed continuously while the stage is travelling in Z. The speed and Z-range must be matched in the way that around 80-100 images are grabbed. Focus series store only the best 100 images and ignore the rest. Only the 30 images around the best sharpness are used for the actual CD calculation, the rest of the images are ignored for the measurement as well.

In the menu of **CD-Meas** window, select **Focus-Focus series options** to optimization focus series parameters (e.g., the z-range and the speed of moving in z). Define the z range in order to have a pronounced maximum of the sharpness criterium by checking the graph in the **Focus Series Control box**. Visually, the image should be noticeably unsharp at the beginning, sharp around the middle, and unsharp again at the end of the focus series. Select **Con. Moving** to drive the z-axis in a continuous movement, and enter the **Speed value** in the entry field. The greater the magnification (i.e. the smaller the focal depth) of the objective, the smaller the range should be. For 100x or 150x magnification, the z-range should be not more than a few microns. After choosing the z-range, adjust the focus speed to be fast enough so that not more than 30-100 images are grabbed during the focus series (check the yellow status bar at the bottom of the Nanostar screen) to avoid excessive execution times.

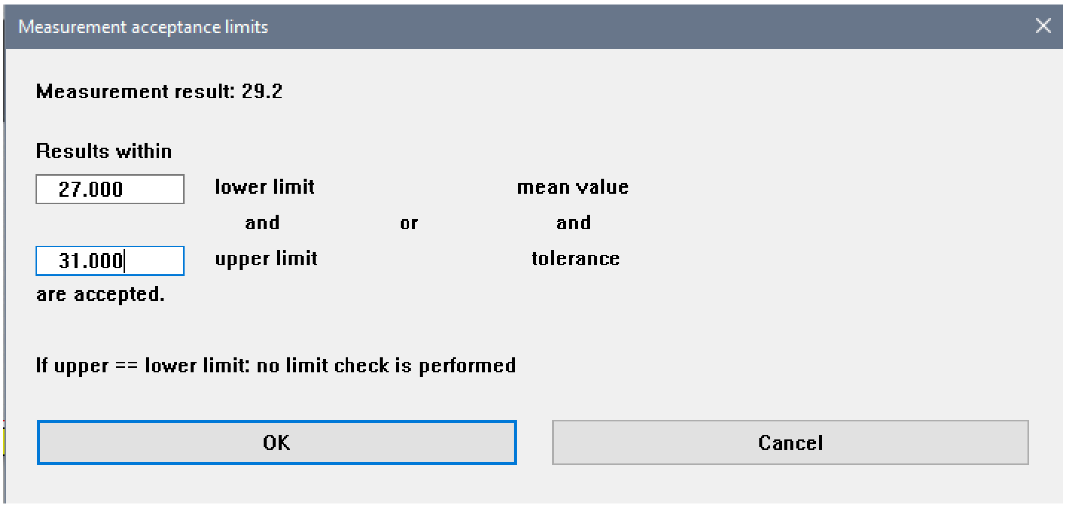
The time-saving option of **Stop after best Z** (stop the series after the sharpness maximum, when images are getting unsharp again) is faster, but in some cases, it may not find the real best focus due to multiple sharpness maxima. If this happens, deactivate this option and the real best Z position will be determined using the entire z-range. Optionally activate **Goto best Z after Z series** and **Retry with enlarged z-range if focus fails**. When activating these options, the z-range will be automatically enlarged and the focus series will be repeated with higher speed, if the defined z-range turns out to be too small (no sharpness maximum found, or the maximum is too close to the begin or end of the focus series).

Confirm with **OK** and click the **Measure** button in the **CD-Meas** window to test the setting. During measurement with focus series, the image in the **IP** window will be defocused in the beginning, slowly getting into focus as the series progresses, and getting defocused again at the end.

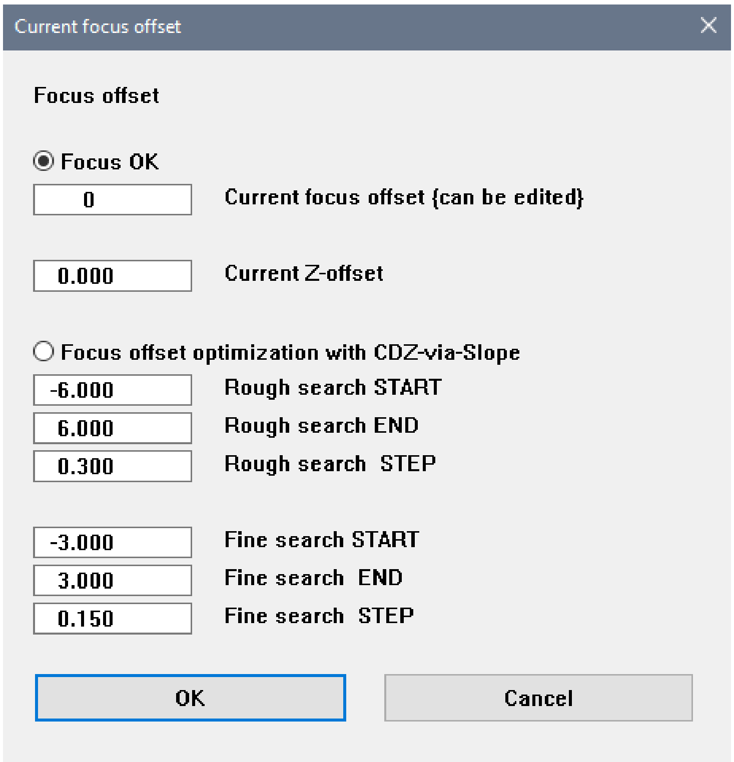
If the CD measurement result is acceptable, click **Continue** in the **MacWizCont** window.



Next, define the limits to accept or reject the measured result in the **Measurement acceptance limits** window. If both the lower and upper limits are set to zero, no limit check is performed. If a measurement is rejected, the measurement result will be stored in the results but flagged as a failed measurement.

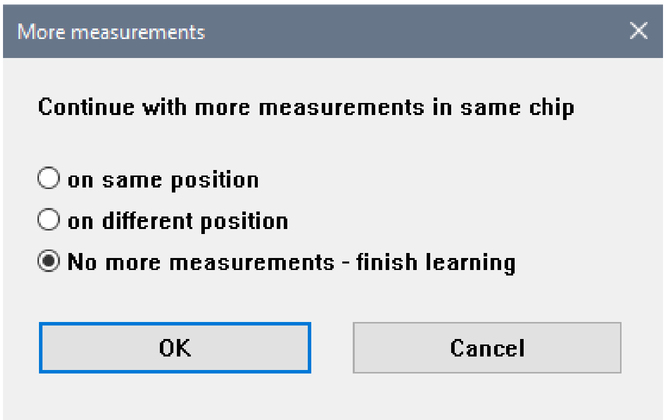


The current focus z-offset (relative to the tool setting for the laser focus) is displayed in the **Current focus offset** window. Click **OK** if the current focus offset is accepted.



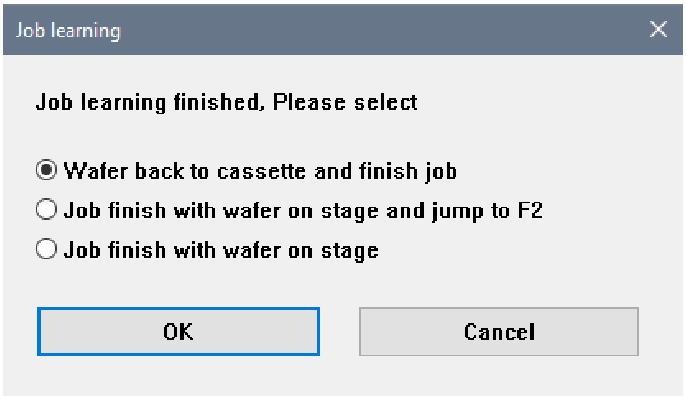
Selection	Description
Focus OK	If the current focus offset is acceptable
Focus offset optimization with CDZ-via-Slop	To proceed with a test measurement using a focus series. The stage will move to the Z-position of the best focus and the new offset is displayed.
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

After defining all the parameters for the first measurement, user may decide to continue with more measurement in the same chip in the **More measurements** window. If you plan to continue with more measurements, either on the same or on a different position of the chip, then the whole learning cycle is repeated for second measurement and so on (see 3.4.1).



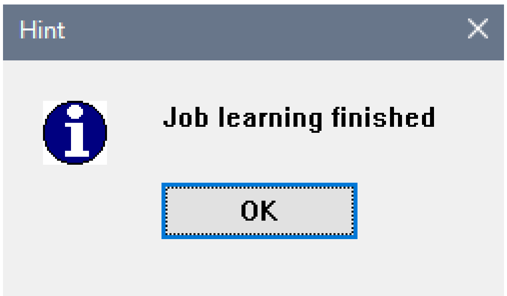
Selection	Description
on same position	To learn another measurement on the same position in the same chip
on different position	To learn another measurement on a different position in the same chip
No more measurements – finish learning	If there are no more measurement to learn in the same chip
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

If no other measurements are to be learned on the same chip, the **Job learning** window is shown and the user may decide to handle the wafer back to cassette or keep the wafer on stage.



Selection	Description
Wafer back to cassette and finish job	To handle the wafer back to cassette and finish the job learning
Job finish with wafer on stage and jump to F2	To keep the wafer on stage and finish the job learning. The **F2** function is called and the **Start From** window is shown for the user to continue immediately with job execution.
Job finish with wafer on stage	To keep the wafer on stage and finish the job learning
OK	To continue with the selected option
CANCEL	To Interrupt the job learning. The Wizard stops.

At last, the **Learning finished!** message is shown in the **Hint** window. Click **OK** to end the job leaning.



It is advisable to test run the newly learned job both with the wafer used for learning and with other wafers. There are variations among wafers. The job may be perfect for the learned wafer but may not run well for other wafers. After test runs, fine tuning may be required for a more stable job.

3.4.3 Learning Overlay Measurement

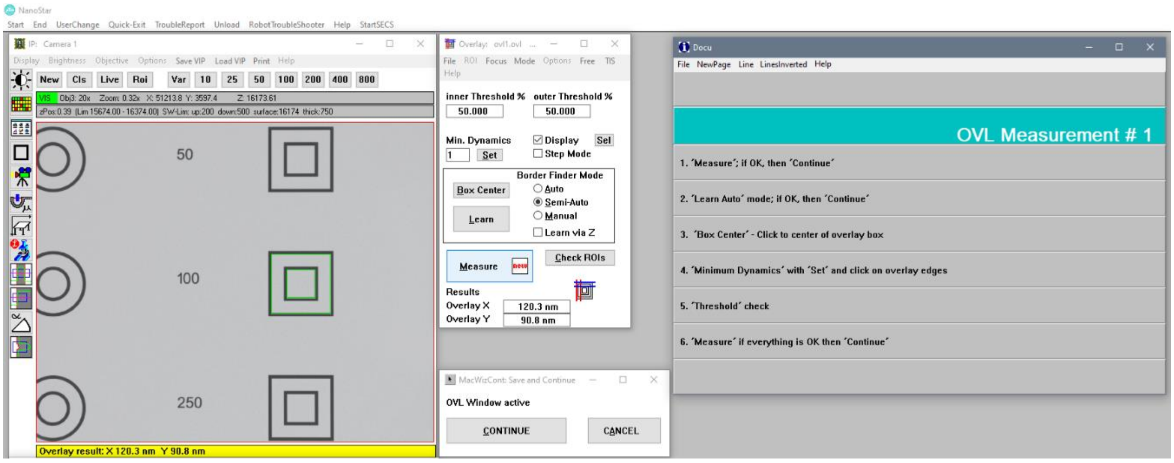
The **Overlay** window is used for the overlay measurement of overlay structures. Carefully adjusting all necessary parameters is required to get good repeatability. Experience in working with overlay structures helps finding the best parameters. See also the NanoStar User Manual and the overlay online help for more detailed information.

Before setting any other parameters, mark the approximate box center of the inner overlay structure. Do this by clicking the **Box Center** button in the **Overlay** window and subsequently click on the center of the inner overlay structure in the **IP** window. The overlay measurement will use this box center as starting point for finding the overlay structure features.

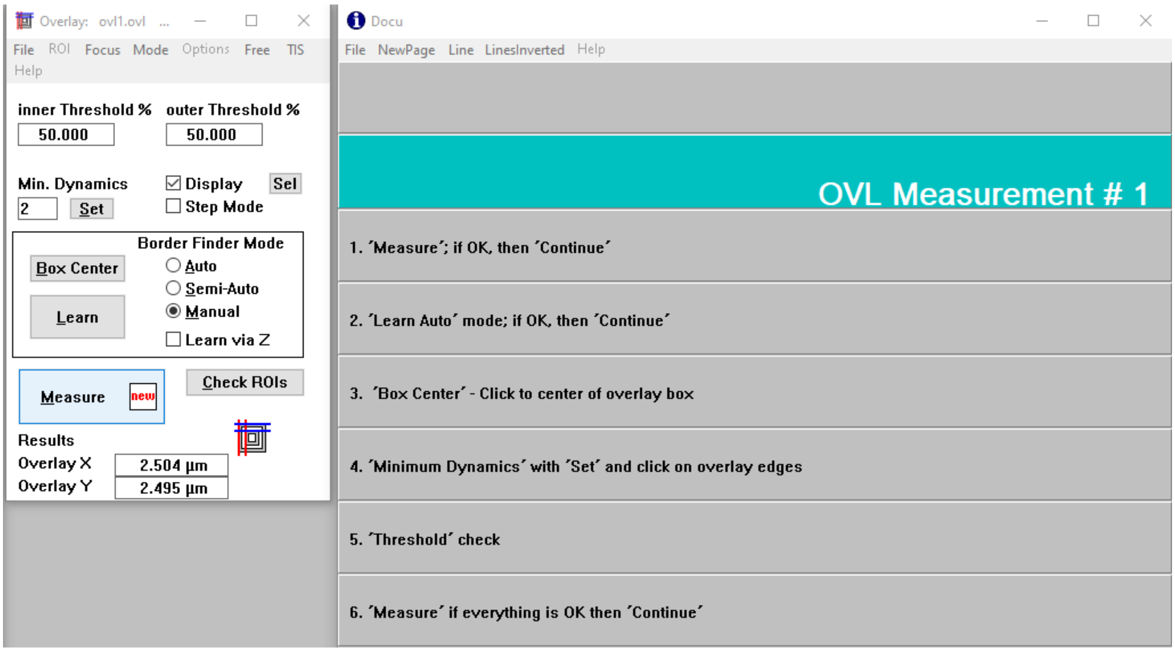
Next, set the **Border Finder Mode** to **Auto** and click the **Learn** button. The **Auto** mode creates overlay measurement parameters reliably and successfully for most of the common overlay box-in-box structures by automatically identifying the approximate position and appearance of the structure edges to measure. If problems emerge in the **Auto** mode, **Semi Auto** or **Manual** modes can be tried, helping the automatic algorithm by identifying some or all edges manually.

The **Min. Dynamics** value in units of grey values is used to differentiate real structure edges from noise or secondary features.

The algorithm also tries to find useable measuring thresholds for the edges, but these should always be checked and adjusted if necessary. Some edges might be weaker depending on process variation, so the **Inner** or **Outer threshold** can be defined at different levels.

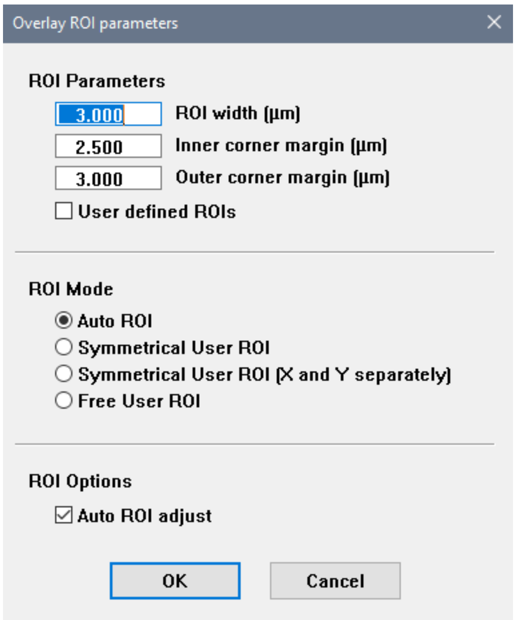


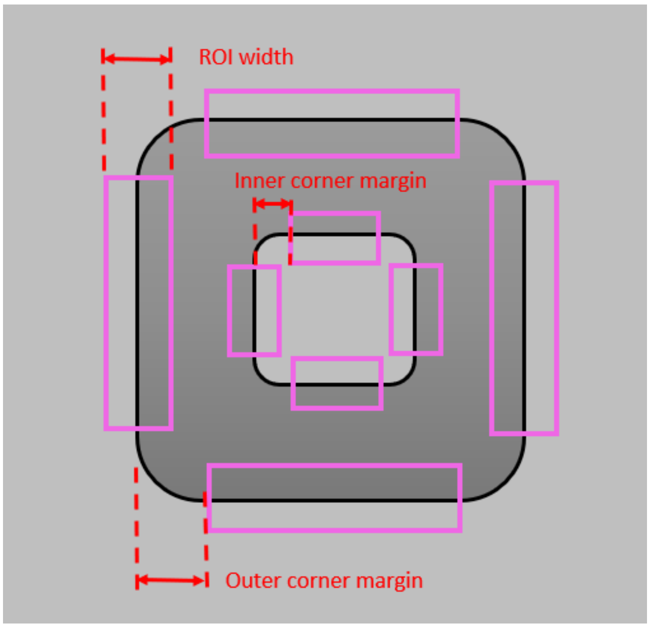
The **Docu** window displays some helpful information and the guideline for the **Overlay** window.



Selection	Description
Inner Threshold%	The threshold of the inner box on the overlay structure
Outer Threshold%	The threshold of the outer box on the overlay structure
Min Dynamics	The minimum height of all edges or line borders to be measured. The value must be lower than the height of both edges of all borders but higher than any other noise. **Set** allows to click on any edge (use the weakest edge) and set the minimum dynamics accordingly.
Box Center	To mark the box center of the overlay structure.
Learn	To learn the box-in-box structure with the selected border finder mode. **Learn** sets the ROIs automatically where required and saves them later to the overlay parameter file.
Border Finder Mode	Define the Border Finder mode for **Learn**, either **Auto**, **Semi-Auto** or **Manual**
Measure	Use this button to perform the overlay measurement
Check ROIs	Show and the found regions of interest for checking
Result	Display of the overlay X and Y result

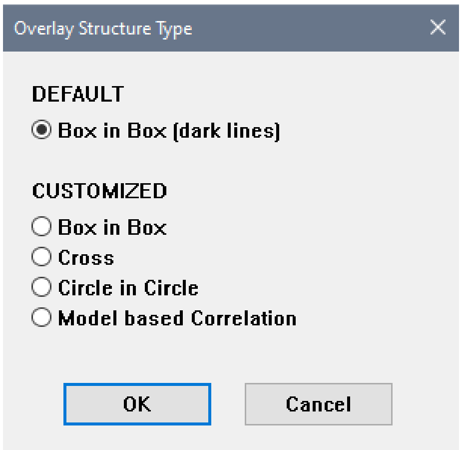
In the menu of **Overlay** window, select **Options-ROI parameters** and the **Overlay ROI parameters** window is shown. Select **Auto ROI** and activate **Auto ROI adjust** for automatic ROI.





Selection	Description
ROI width (µm)	The width of the ROI perpendicular to the line or edge. Do not set it too large to leave some space between the inner and outer ROIs.
Inner corner margin (µm)	A measure of the shortening of the inner ROIs along the border to exclude the non-linear sections of the border. The value must be larger than the radius of the rounded corner to avoid inaccurate measurements by including nonlinear sections in the edge measurement.
Outer corner margin (µm)	A measure of the shortening of the outer ROIs along the border to exclude the non-linear sections of the border. The value must be larger than the radius of the rounded corners.
User defined ROIs	Checkbox for user defined ROIs, allowing to set each ROI manually
Auto ROI	All the ROIs are automatically defined.
Symmetrical user ROI	Define an outer and an inner ROI manually. The remaining ROIs will be set automatically and symmetrically with respect to the box center.
Symmetrical user ROI (X and Y separately)	Define two ROIs for the outer structure, one in X and one in Y, and also two ROIs for the inner structure manually. The remaining ROIs will be set automatically and symmetrically with respect to the box center.
Free User ROIs	All the ROIs are manually defined by the user.
Auto ROI adjust	By activating this checkbox, the ROIs are relocated during measurement, if the overlay of the measured structure is different from the overlay of the structure used for learning. The four inner ROIs are moved as a group and the four outer ROIs are moved as another group.

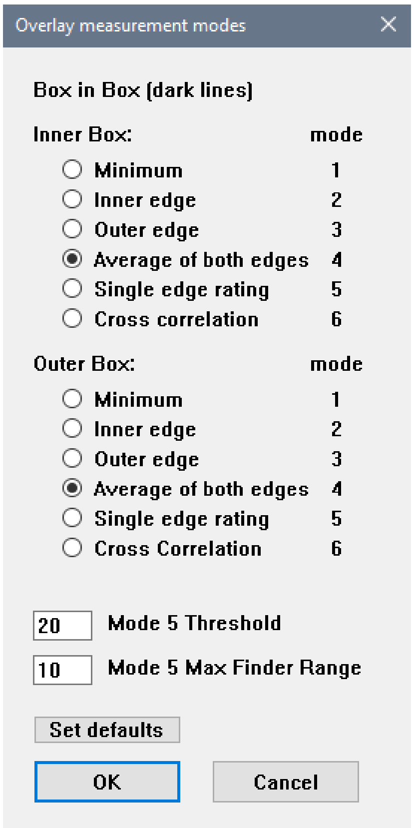
In the menu of **Overlay** window, select **Options-Structure type** and the **Overlay Structure Type** window is shown. Select **Box in Box (dark lines)** if the overlay structure is box-in-box with dark lines. Other options of customized overlay structures can be selected as well. See the manual or online help for details. Click the **OK** button to continue.

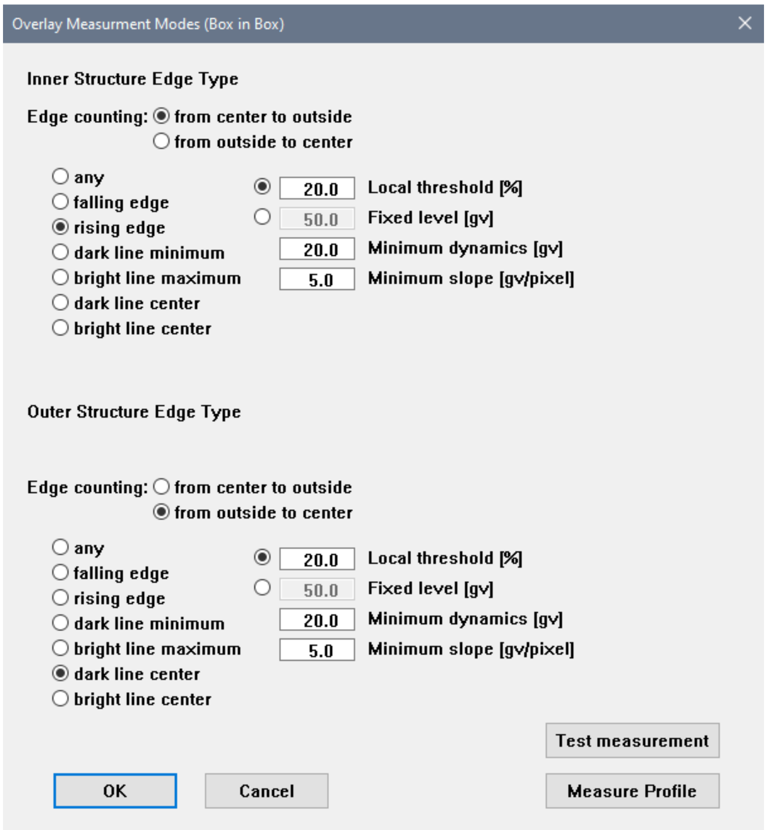


In the menu of **Overlay** window, select **Mode** and the **Overlay measurement modes** window is shown. If the **Box in Box (dark lines)** is selected, the overlay measurement modes of the **Inner Box** and **Outer Box** can be selected from 6 pre-defined modes. If any of the **Customized** structure types was chosen in the

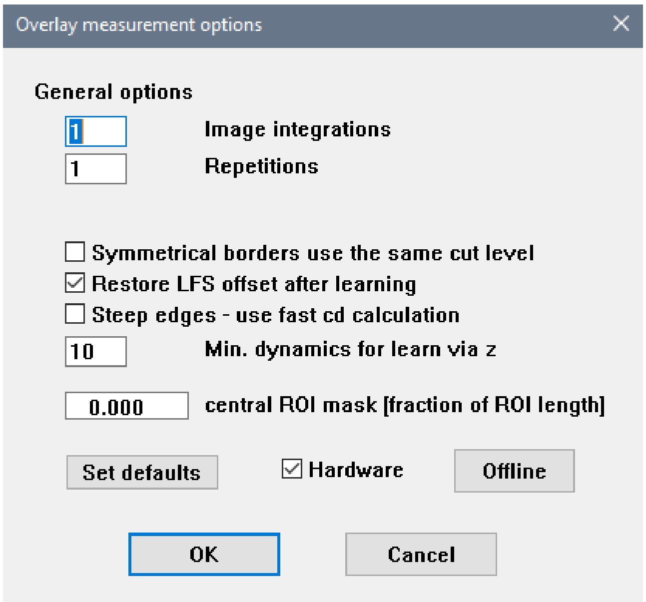
Overlay structure type window, more parameters are available to fine-tune the measurement mode. Most of these parameters work quite similar to those for the CD measurement.

Click the **OK** button to continue.





In the menu of **Overlay** window, select **Options-Measurement** and the **Overlay measurement options** window is shown.



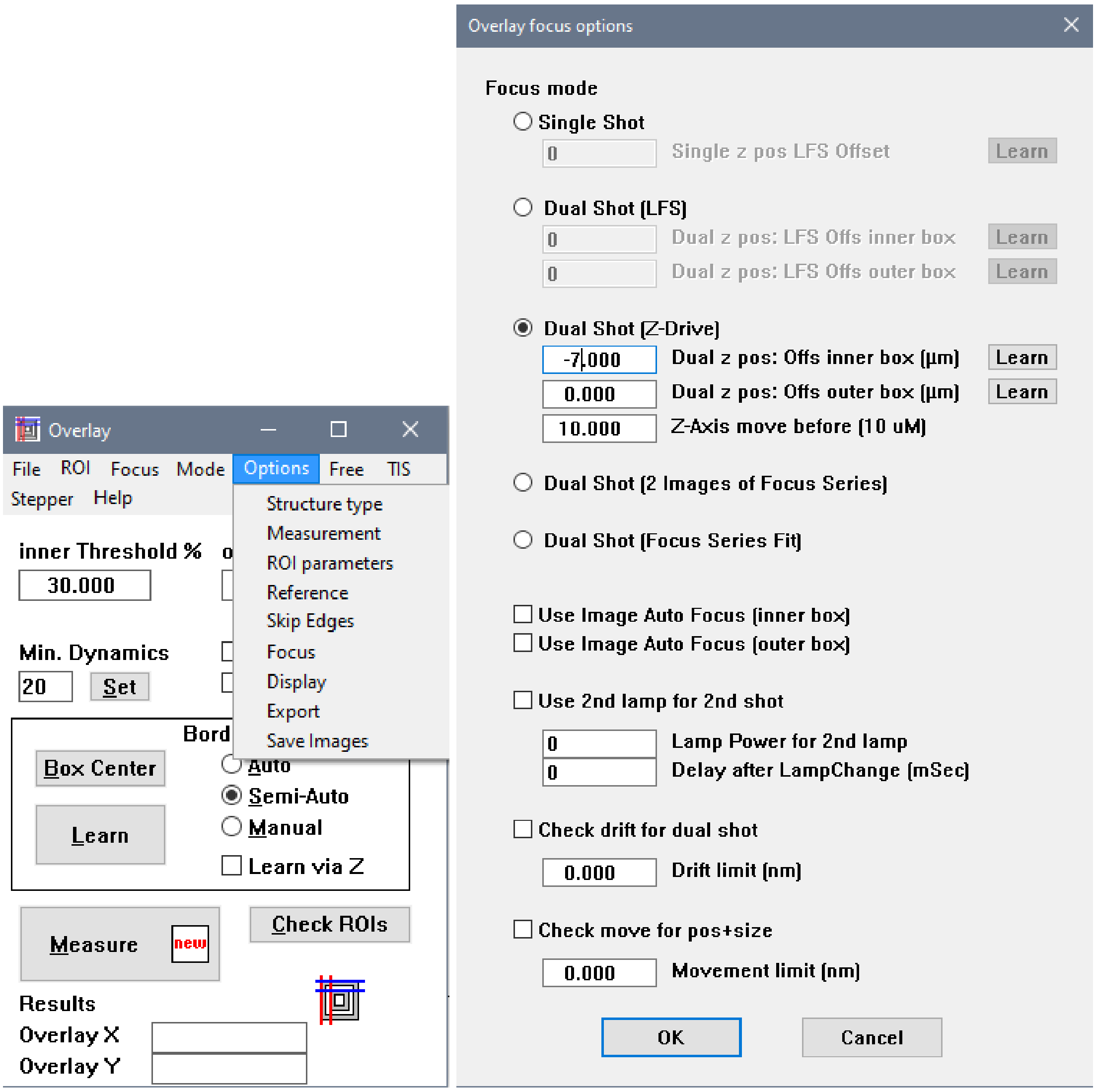
Selection	Description
Image integrations	Number of images to be integrated before measurement
Repetitions	Number of measurements done on different images. Results are averaged.
Symmetrical borders use the same cut level	Activate this button only for very regular and symmetric overlay structure. The same absolute intensity level is used for the two symmetrical edges.
Restore LFS offset after learning	Activate this button to restore the LFS offset if the box parameters were **learned via z** in Overlay window. Possibly the laser focus offset was changed during learning. As default, it should be activated.
Central ROI mask	Excludes a defined fraction of the ROI length from the calculation. Useful for cross-like overlay structures.
Steep edges – use fast cd calculation	Activate this button if the structure edge is very steep or if measuring on thin photoresist.
Min. dynamic for learn via z	Defines the min dynamic to help the border detection in out-of-focus images during learn via z
Set defaults	Resets all parameters of this option window to their default values.

Subsequently, the procedure to define the focus type, acceptance limit and focus offset are the same as described in the CD measurement until the job is finished.

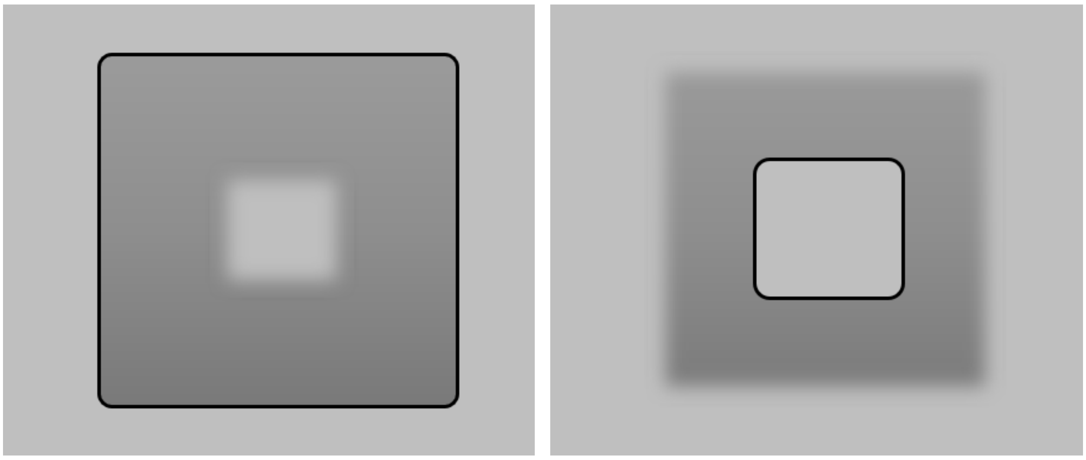
3.4.3.1 Dual Shot

If the inner and outer box of an overlay structure are not in the same focus plane, dual shot focus mode is used for the overlay measurement. In the menu of **Overlay** window, select **Options-Focus** and the **Overlay focus options** window is shown. Select **Dual Shot (Z-Drive)** and click **OK** button to go back to the main **Overlay** window.

Like when using single shot, click the **Box Center** button to mark the center of the overlay structure in the **Overlay** window.



After pressing the **Learn** button in the **Overlay** window, focus the outer structure manually and press **ENTER** in **Free mode**. The z-offset of the outer box will be saved as **Dual z-pos: Offs outer box** in the **Overlay focus options** window.



Subsequently, repeat focussing the inner structure manually and press **ENTER** in **Free mode**. The z-offset of the inner box will be saved as **Dual z-pos: Offs Inner box** in the **Overlay focus options** window.

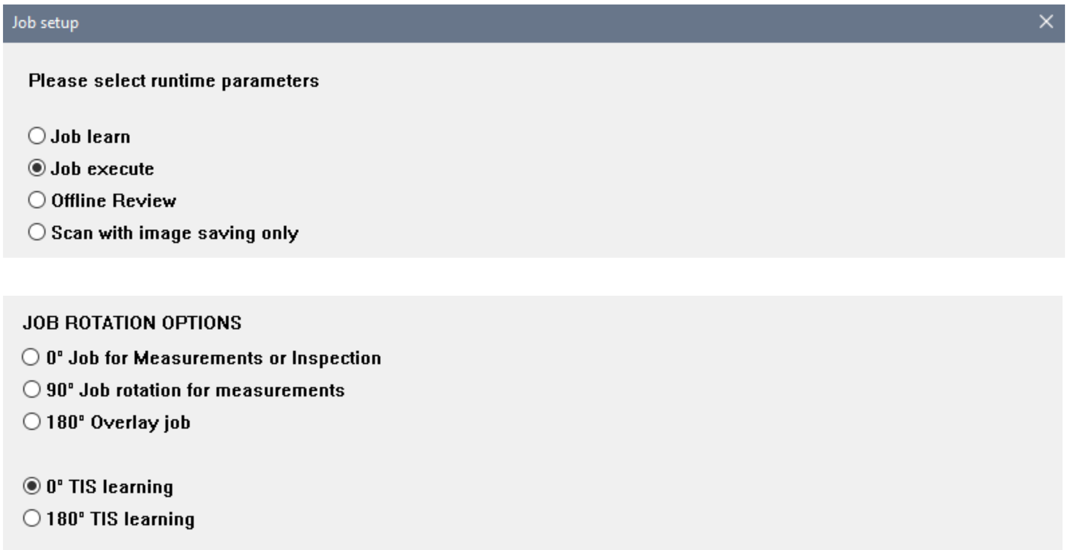
In the **Overlay focus options** window, **Dual z-pos: Offs outer box** and **Dual z-pos: Offs Inner box** can be redefined or fine-tuned by clicking the **Learn** button.

The procedure of defining the ROIs is the same as described as for single shot, just done on two separate images for the two focus planes.

3.4.3.2 Tool Induced Shift (TIS)

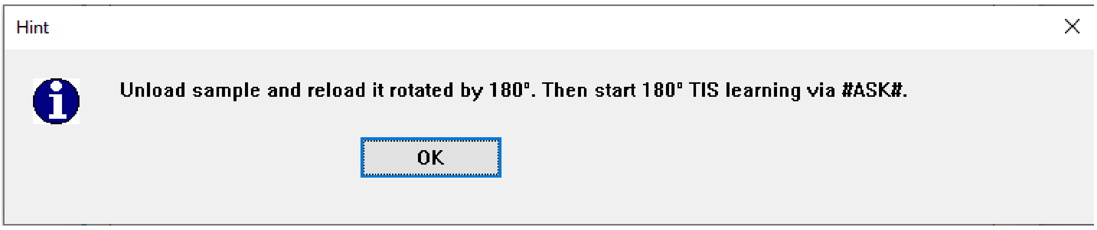
Before performing overlay measurement with a Tool Induced Shift (TIS) correction file, the TIS correction needs to be learned as follows:

- Learn an overlay job with the wafer without activating **use TIS calibration** in the CD/Overlay Options and test run it.
- Restart the job by selecting **# ASK #** to execute the job with the **0° TIS learning** radio button.



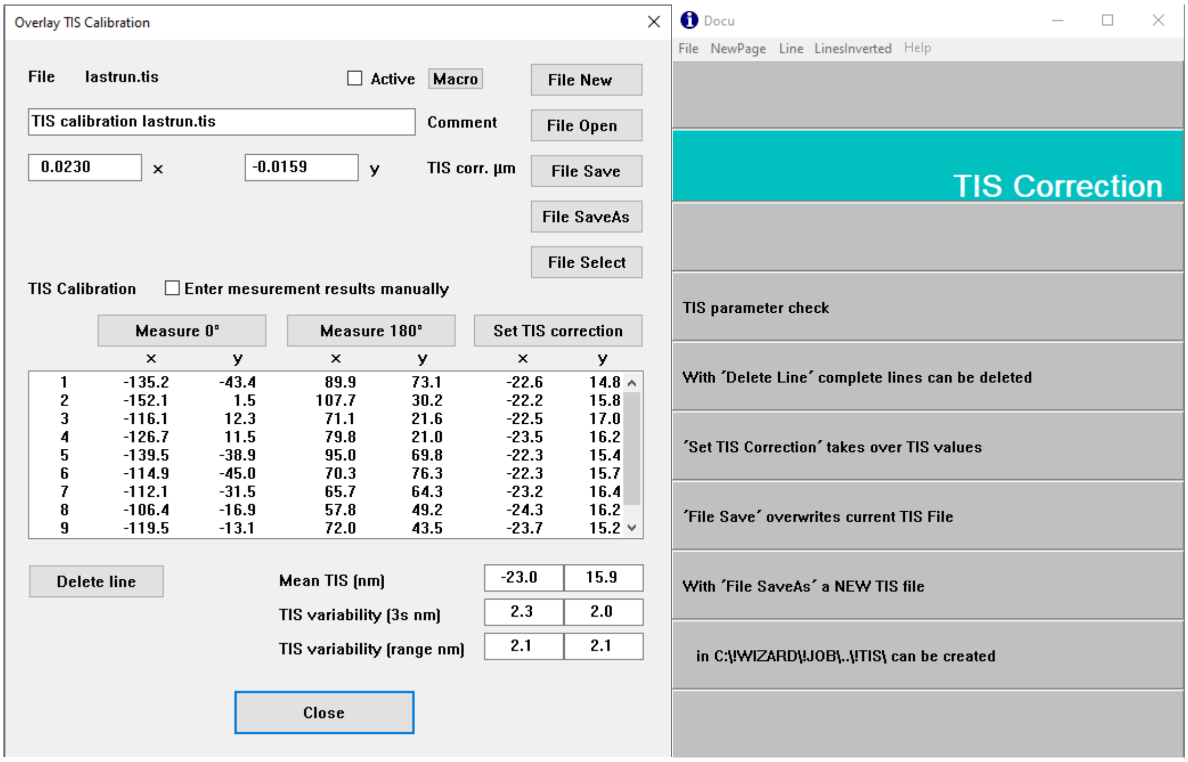
After finished **0° TIS learning**, the **Measure 0°** result will be stored in the TIS file.

- Unload the wafer and reload it by rotating to 180° orientation.



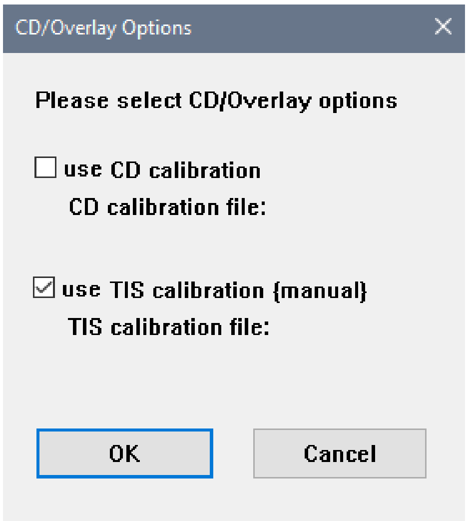
- Restart the job by selecting **# ASK #** to execute the job with the **180° TIS learning** radio button active.
- After the **180° TIS learning** is finished, the **Overlay TIS Correction** window will be shown for reviewing. Each measured overlay structure is represented with a line showing both the 0° and 180° results and also the estimated TIS based on this overlay structure.
- Before clicking **Set TIS correction** button, any mis-measurement line can be deleted by selecting the line and clicking **Delete line** button in the **Overlay TIS Correction** window .
- Enter a comment in the **Comment** line as user preference.
- The **Set TIS correction** button averages the TIS results of all lines/measured structures to be used as TIS correction values. Save the file by clicking **File SaveAs** button. It is recommended to save all the TIS files at the same TIS folder in the directory of *C:\!Wizard\!TIS*.

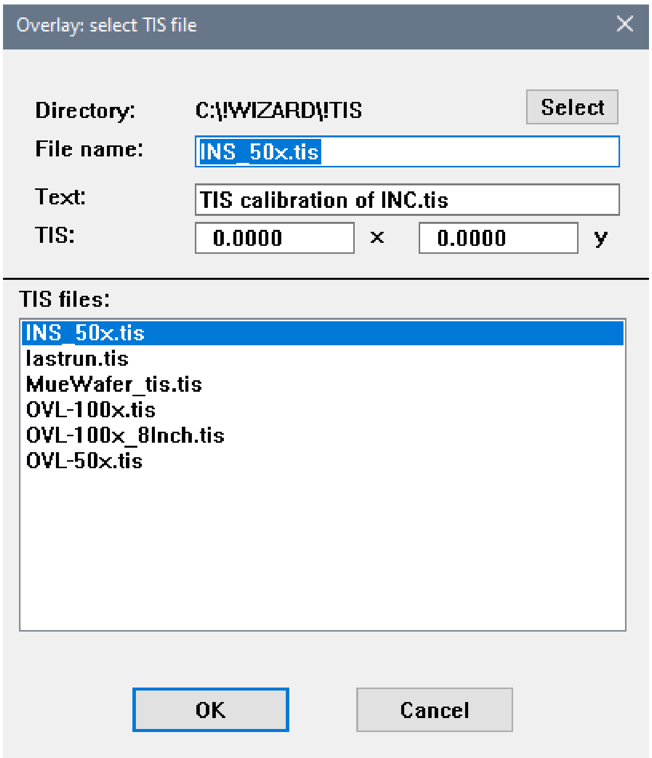
- It is recommended to learn TIS corrections separately for different types of overlay structures, different layers, or different process stages and save it in separate TIS files.



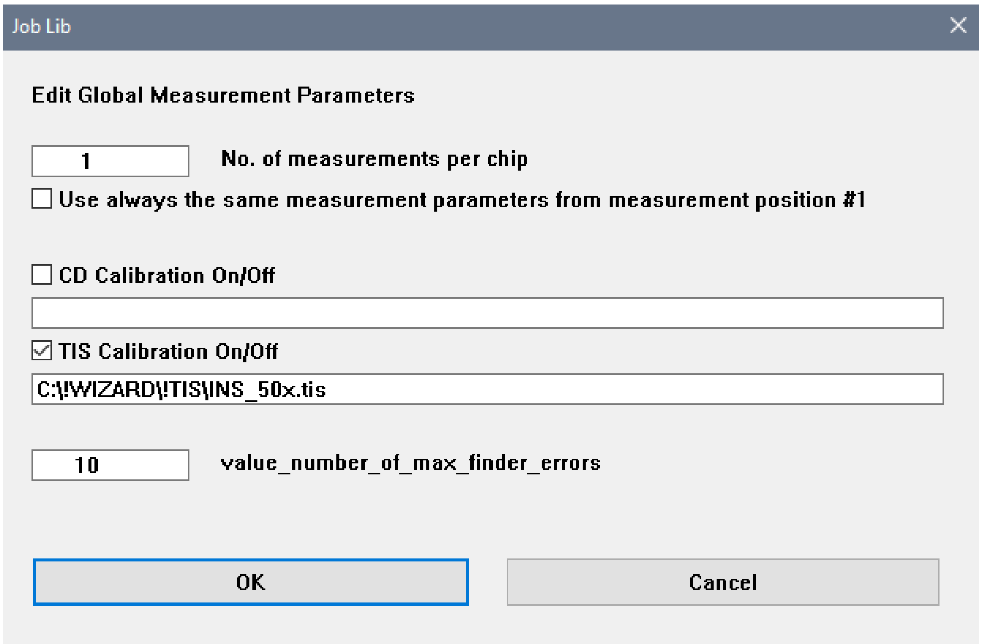
There are two ways to activate the TIS calibration for the overlay job:

- During the overlay job learning, activate the **use TIS calibration** in the CD/Overlay Options (see 3.1.2.2) for automatic application of a TIS correction to the overlay measurement. After the **Measurement Site Options** window, the **Overlay: select TIS file** window is shown for TIS correction file selection. Select the appropriate TIS correction file and click the **OK** button.



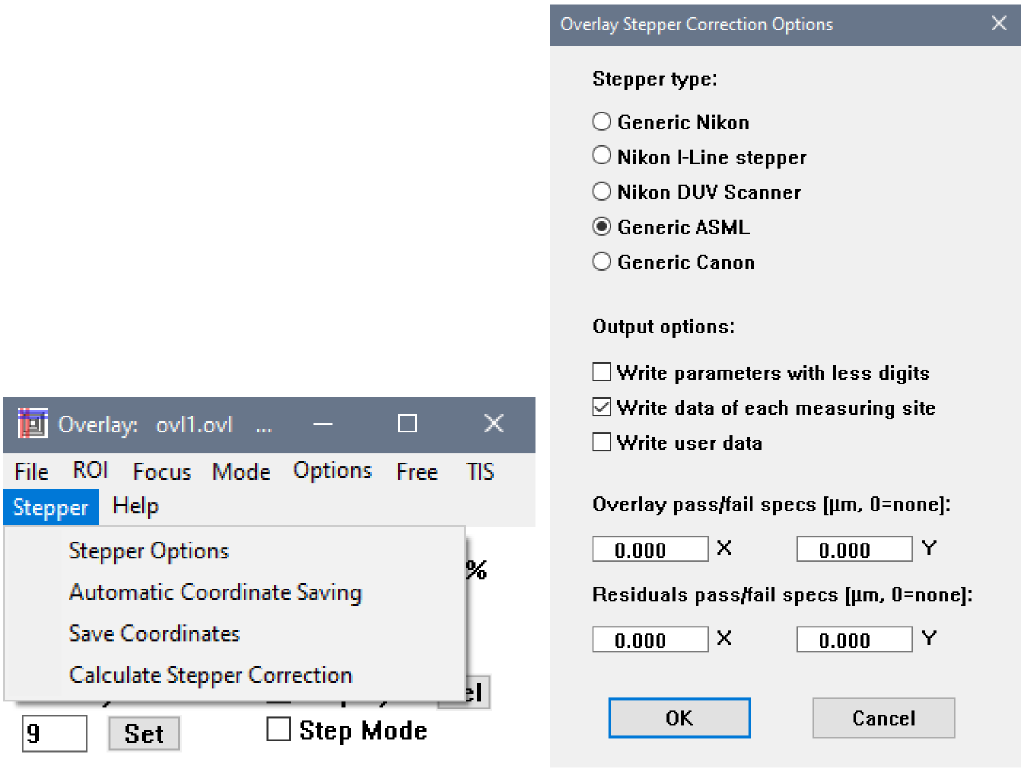


- With the Overlay recipe loaded in the **Applic** window, press the **F9** function key and click the **OK** button two times until the third **Job Lib** window is shown for editing the **Global Measurement Parameters.** Activate **TIS Calibration On/Off** and enter the path and filename of the appropriate TIS calibration file. Click the **OK** button.



3.4.3.3 *Stepper Correction*

To automatically generate a file with stepper correction parameters from the overlay results for stepper use, select **Stepper-Stepper Options** in the menu of **Overlay** window and the **Overlay Stepper Correction Options** window is shown.



By selecting the stepper type, the stepper correction file is generated with a format matching the correction parameter set and parameter format of the selected stepper type. The **OK** button saves the stepper type selection to be used for all subsequent overlay measurements on the tool.

Below is the example of a correction file for Generic ASML stepper type.

File Edit Format View Help

Interfield Corrections

Translation X	[µm]	0.426
Translation Y	[µm]	0.126
Rotation	[µrad]	-23.51
Non-orthogonality	[µrad]	-1.22
Expansion X	[ppm]	-0.657
Expansion Y	[ppm]	-0.507

Intrafield Corrections

Translation X	[µm]	0.000
Translation Y	[µm]	0.000
Rotation	[µrad]	-23.38
Magnification	[ppm]	-34.15

Measurem.	Die_X	Die_Y	Waf_X mm	Waf_Y mm	Fld_X mm	Fld_Y mm	Ovl_X µm	Ovl_Y µm	Res_X_µm	Res_Y_µm
1	8	8	1.333349	-0.418744	-4.372858	-4.140734	0.483	0.372	0.016	0.033
2	8	8	1.333349	-0.418744	-4.373159	4.274246	0.605	0.054	-0.059	0.003
3	8	8	1.333349	-0.418744	4.042709	4.274329	0.358	-0.144	-0.019	0.001
4	8	8	1.333349	-0.418744	4.042898	-4.139994	0.170	0.114	-0.010	-0.029
5	13	8	47.233349	-0.418744	-4.373953	-4.139262	0.441	-0.711	0.004	0.029
6	13	8	47.233349	-0.418744	-4.374125	4.274387	0.567	-1.030	-0.066	-0.003
7	13	8	47.233349	-0.418744	4.040731	4.274347	0.330	-1.245	-0.017	-0.021
8	13	8	47.233349	-0.418744	4.041513	-4.139306	0.133	-0.968	-0.016	-0.032
9	8	4	1.333349	-37.138744	-4.372094	-4.141881	-0.376	0.407	0.065	0.050
10	8	4	1.333349	-37.138744	-4.372643	4.272757	-0.246	0.069	-0.001	-0.002
11	8	4	1.333349	-37.138744	4.042926	4.273240	-0.498	-0.138	0.034	-0.012
12	8	4	1.333349	-37.138744	4.043432	-4.141917	-0.695	0.146	0.034	-0.015
13	3	8	-44.566651	-0.418744	-4.371827	-4.143574	0.504	1.453	0.007	0.035
14	3	8	-44.566651	-0.418744	-4.372640	4.271228	0.632	1.120	-0.062	-0.011
15	3	8	-44.566651	-0.418744	4.043218	4.271972	0.383	0.916	-0.023	-0.018
16	3	8	-44.566651	-0.418744	4.043470	-4.143019	0.193	1.190	-0.017	-0.032
17	8	12	1.333349	36.301256	-4.374211	-4.142040	1.439	0.354	0.064	0.034
18	8	12	1.333349	36.301256	-4.374030	4.273901	1.558	0.051	-0.015	0.018
19	8	12	1.333349	36.301256	4.040916	4.274487	1.328	-0.167	0.043	-0.003
20	8	12	1.333349	36.301256	4.041063	-4.141285	1.128	0.097	0.039	-0.027

Below is the example of a correction file for Generic Nikon stepper type.

File Edit Format View Help

Stepper wafer offset X	0.428 µm
Stepper wafer offset Y	0.126 µm
Stepper wafer scale error X	-0.632 ppm
Stepper wafer scale error Y	-0.585 ppm
Stepper wafer rotation X	24.735 ppm
Stepper wafer rotation Y	-23.522 ppm
Stepper field scale error	-34.080 ppm
Stepper field rotation	-23.362 ppm

Measurem.	Die_X	Die_Y	Waf_X µm	Waf_Y µm	Fld_X µm	Fld_Y µm	Ovl_X µm	Ovl_Y µm	Res_X_µm	Res_Y_µm
1	8	8	1333.349	-418.744	-4372.573	-4141.229	0.473	0.384	0.004	0.046
2	8	8	1333.349	-418.744	-4372.752	4273.058	0.609	0.050	-0.056	-0.001
3	8	8	1333.349	-418.744	4042.949	4273.073	0.362	-0.153	-0.016	-0.008
4	8	8	1333.349	-418.744	4043.289	-4140.844	0.172	0.114	-0.010	-0.027
5	13	8	47233.349	-418.744	-4373.771	-4140.133	0.448	-0.715	0.008	0.027
6	13	8	47233.349	-418.744	-4373.701	4272.970	0.565	-1.045	-0.071	-0.016
7	13	8	47233.349	-418.744	4041.121	4273.151	0.337	-1.244	-0.013	-0.019
8	13	8	47233.349	-418.744	4041.471	-4140.316	0.136	-0.969	-0.017	-0.030
9	8	4	1333.349	-37138.744	-4371.576	-4142.123	-0.374	0.410	0.066	0.051
10	8	4	1333.349	-37138.744	-4372.207	4272.114	-0.248	0.074	-0.004	0.002
11	8	4	1333.349	-37138.744	4043.358	4273.070	-0.492	-0.127	0.037	-0.003
12	8	4	1333.349	-37138.744	4043.751	-4142.630	-0.693	0.147	0.034	-0.016
13	3	8	-44566.651	-418.744	-4371.211	-4143.608	0.506	1.451	0.008	0.033
14	3	8	-44566.651	-418.744	-4371.697	4271.113	0.634	1.113	-0.060	-0.018
15	3	8	-44566.651	-418.744	4043.942	4272.077	0.380	0.914	-0.027	-0.020
16	3	8	-44566.651	-418.744	4043.983	-4142.971	0.197	1.187	-0.014	-0.034
17	8	12	1333.349	36301.256	-4373.511	-4142.479	1.439	0.356	0.062	0.040
18	8	12	1333.349	36301.256	-4373.205	4273.239	1.556	0.052	-0.018	0.023
19	8	12	1333.349	36301.256	4041.785	4274.081	1.329	-0.167	0.042	-0.000
20	8	12	1333.349	36301.256	4042.459	-4141.835	1.136	0.092	0.046	-0.028

3.4.4 Learning Film Thickness Measurement

The Film Thickness (FT) window is used for the thickness measurement of optically transparent films, based on the evaluation of the spectrum of light reflected on the film, and comparing it to calculated model spectra.

Before learning a film thickness measurement job with the wizard, the **Optical Parameter** of all materials to be measured, containing the optical characteristics of this material, must be available. The optical parameters can be obtained from the material supplier, manufacturer data sheets or published research data.

The optical parameters of a new material can be entered in the **FT: Optical parameter editor** window. In the **FT** window, select **Parameter-Optical parameter editor**. For a transparent material, the **refraction index (n)** of the material within the wavelength range of 400 nm to 800 nm is required. If the material is opaque, both the **refraction index (n)** and the **wavelength dependent absorption coefficient (k)** are required. The number of n,k values within the 400-80 0nm range is dependent on the wavelength-dependent behaviour of the parameters, as seen in the shape of those curves. For most common materials, either the use of a describing formula (**Cauchy**, **Sellmeier**) or the input of n or k values spaced by about 50 nm should suffice. Materials with complex optical behaviour, i.e. sharp features in the n,k curve might require more closely spaced parameter points. n or k values and the associated wavelength might be entered in any wavelength order. Just remember to click the **Recalc** button to sort your entries according to wavelength and update the shown n or k curves before saving.

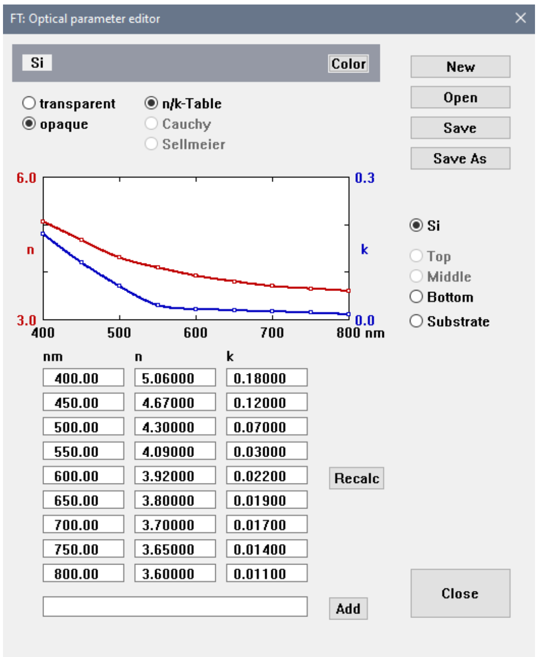
The n,k data points for different wavelengths can be added by entering in the entry field and clicking the **Add** button. For transparent layers alternatively either the **Cauchy** or the **Sellmeier** calculator can be used to calculate the **refraction index (n)** values by entering the coefficients and clicking the **Calculate** button. Cauchy coefficients are commonly used for characterization of photoresist and are provided by resist manufacturers. Please make sure that the coefficients are expressed in the required units (Cauchy: none, µm^2, µm^4).

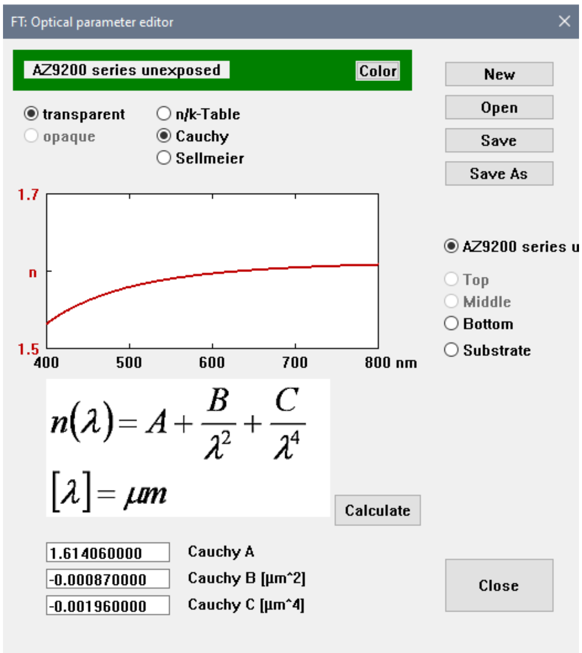
You can also assign a specific **Color** to each material. Just for display purposes, the material will be shown with this color in the **FT** window for easy identification.

Click the **Save As** button to save as a new optical parameter file (*nf*). The files should be saved either into the substrate directory (usually *c:\Wizard\ft.app\substrate*) or into the layer directory (usually *C:\Wizard\ft.app\Layer*).

Sometimes n and k curves are provided in tabulated form. As long as the values are tabulated in the order wavelength, n, k, they can also easily be entered by opening the *nf* files with any text editor and paste the tables directly into the *nf* file. Any commonly used separator (space, comma, semicolon, Tab, …) between wavelength, n and k will do. For transparent materials, k is not needed. You may need to edit the number of data points included in the file also (nk_npts = …).

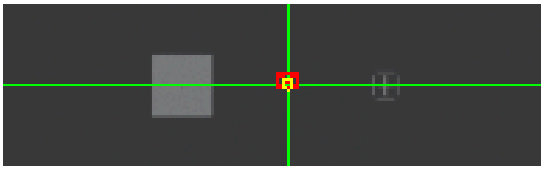
The preparation of the optical parameter files must only be done once. Once a file exists in the FT.app library directories, it is available for all future recipes.





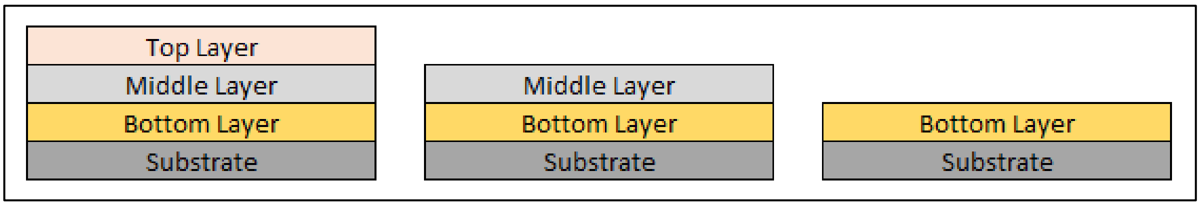
For film thickness measurement on a structured wafer, the alignments and settings are identical to the CD/OVL measurement job. For the **Multiple Measurement Check** and **Measurement Type** windows, the procedures are the same as described in learning CD measurement (see 3.4.1). Note that during the FT measurements itself the wizard will temporarily set the lamp power to a pre-defined high value (usually 255) during the spectrum integration, so lamp brightness adjustment is only required for aligning the measurement sites.

While defining the measurement position in **Free mode**, objectives with long working distance (5x or 10x) are typically selected for FT measurement. Higher magnifications yield less precise results. The FT measurement spot is always at the center of the cross hair, its size and position is indicated as a yellow circle in the **IP** window. Note that the physical size of the FT measurement spot (in µm on the wafer) depends on the objective magnification. Higher objective magnifications result in smaller spots.

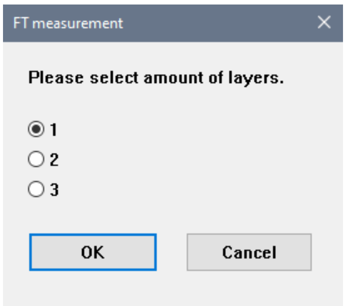


For FT measurement, the common term for layers and substrate are as below.

Selection	Description
Layers	The film layers to be measured (Bottom + Middle + Top), starting with bottom layer
Substrate	The base material underneath the Bottom layer

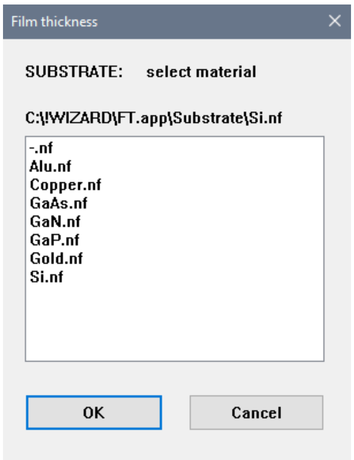


In the **FT measurement** window, select the number of layers to be measured. Up to three film layers (**Bottom**, **Middle** and **Top)** on an opaque substrate can be defined. Click **OK** to continue.

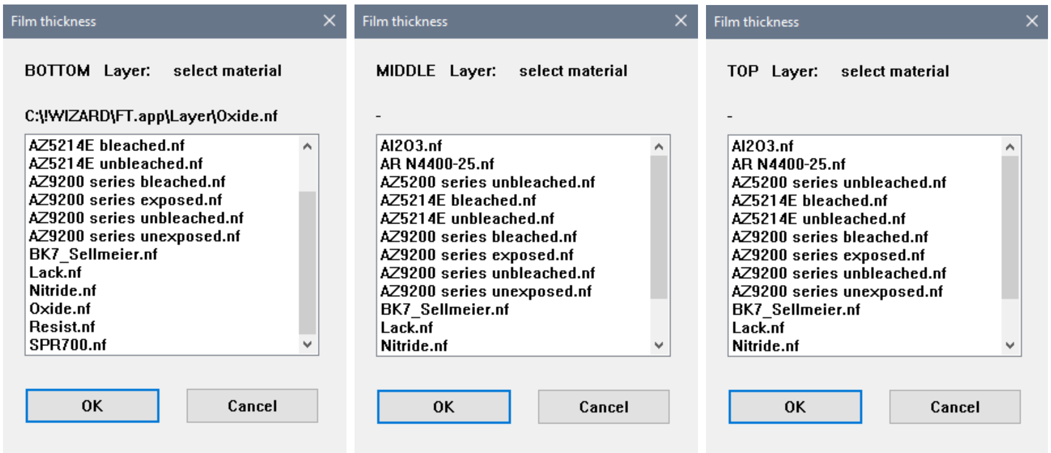


Selection	Description
1	Bottom Layer
2	Bottom Layer + Middle Layer
3	Bottom Layer + Middle Layer + Top Layer

Subsequently, the **Film Thickness** window is shown. Select the optical parameter file of the material for the substrate. The number and types of substrate material in the selection list represent the existent optical parameter files and are specific to the customer. Click **OK** to continue.

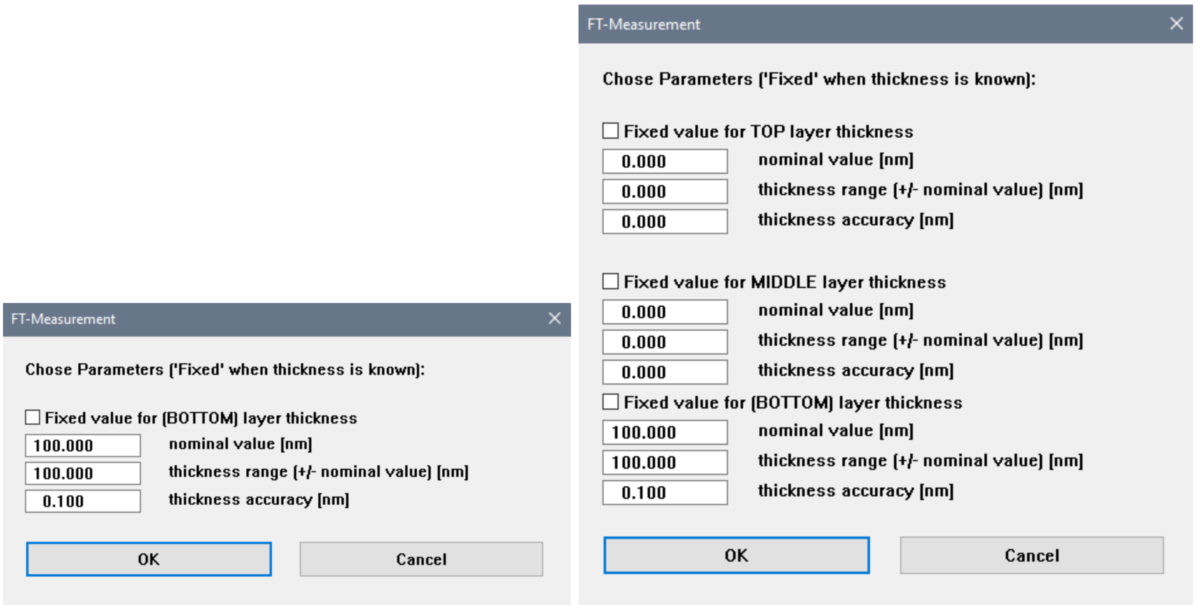


After the substrate material is selected, for each layer another **Film Thickness** window is shown for selecting the layer material. Select the optical parameter file of the material for the layer and click **OK** to continue.

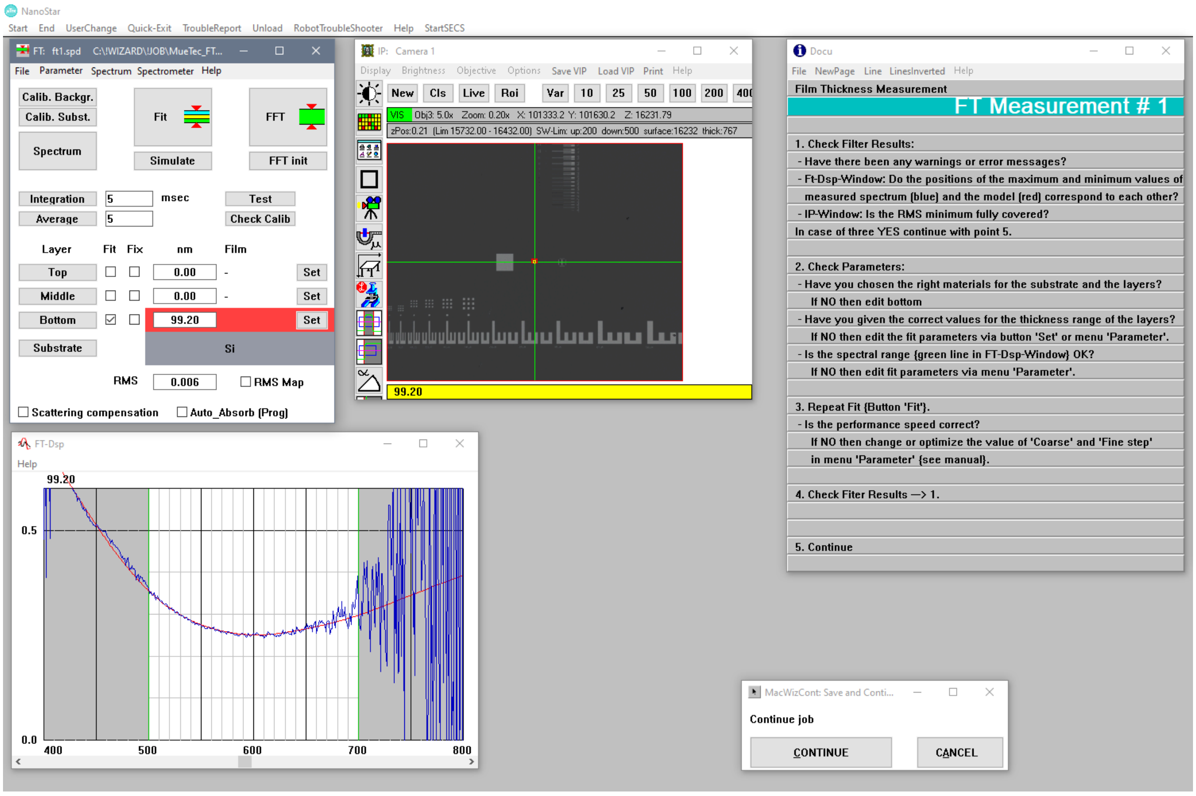


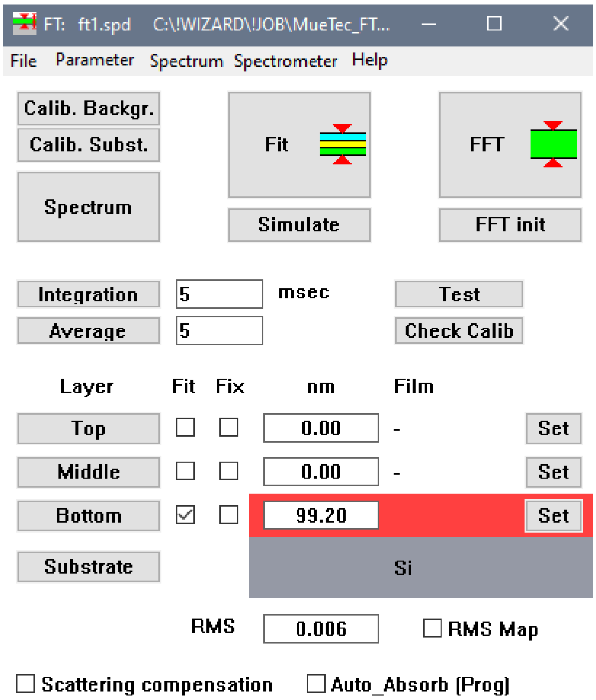
In the **FT-Measurement** window, define the nominal thickness (expected target), the expected thickness range and the accuracy for each layer. Activate **Fixed value for layer thickness** if the thickness is known and fixed.

The larger the thickness range and the better the accuracy is set, the longer the execution time of the model fit will be. This is usually no problem with one-layer or two-layer fits. For three-layer fits either the ranges should be reduced or the required accuracies should be set to higher values. For three-layer fits it is generally recommended to define the best-known layer as **Fixed** and fit only the two other layers.



The **FT** window is shown together with the **FT-Dsp** window and the **IP** window. The **Docu** window displays information and the guideline for the FT Measurement.


NanoStar
Start End UserChange Quick-Exit TroubleReport Unload RobotTroubleShooter Help StartSECS
FT: ft1.spd C:\WIZARD\JOB\MueTec_FT...
Camera 1
Docu
Film Thickness Measurement
FT Measurement # 1
1. Check Filter Results:
- Have there been any warnings or error messages?
- Ft-Dsp-Window: Do the positions of the maximum and minimum values of measured spectrum (blue) and the model (red) correspond to each other?
- IP-Window: Is the RMS minimum fully covered?
In case of three YES continue with point 5.
2. Check Parameters:
- Have you chosen the right materials for the substrate and the layers?
If NO then edit bottom
- Have you given the correct values for the thickness range of the layers?
If NO then edit the fit parameters via button 'Set' or menu 'Parameter'.
- Is the spectral range (green line in FT-Dsp-Window) OK?
If NO then edit fit parameters via menu 'Parameter'.
3. Repeat Fit (Button 'Fit').
- Is the performance speed correct?
If NO then change or optimize the value of 'Coarse' and 'Fine step' in menu 'Parameter' (see manual).
4. Check Fiter Results —> 1.
5. Continue
FT-Dsp
99.20
MacWizCont: Save and Conti...
Continue job
CONTINUE
CANCEL


FT: ft1.spd C:\WIZARD\JOB\MueTec_FT...
File Parameter Spectrum Spectrometer Help
Calib. Backgr.
Calib. Subst.
Spectrum
Fit
Simulate
FFT
FFT init
Integration 5 msec
Average 5
Test
Check Calib
Layer Fit Fix nm Film
Top 0.00 - Set
Middle 0.00 - Set
Bottom 99.20 Set
Substrate Si
RMS 0.006
RMS Map
Scattering compensation
Auto_Absorb (Prog)

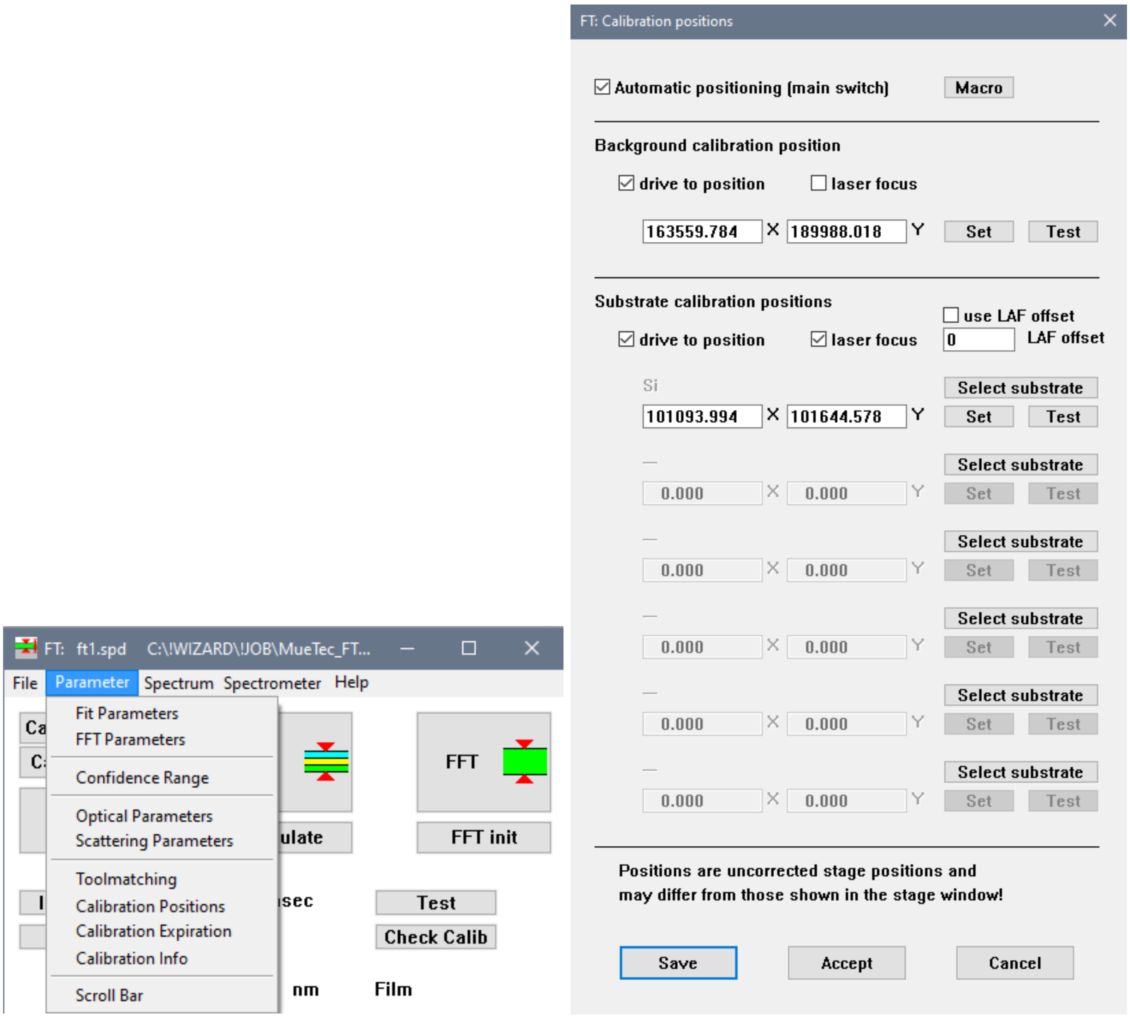
Selection	Description
Calib. Backgr.	Performs the background calibration, preferably on a dark non-reflective surface, like the black surface of holder nearby a mini chuck (With the lamp power set to 0 in **Lamp** window)
Calib. Subst.	Focus and performs the substrate calibration on a blank substrate without any layer on top of it (With the lamp power set to 255 in **Lamp** window)
Spectrum	Records a spectrum (or more if Average is >1) and calibrates the spectrum by subtracting the background spectrum and dividing the substrate spectrum to obtain the reflectance spectrum. The calibrated spectrum is shown in the **FT-Dsp** window.
Fit button	Calculates the film thickness using the Fit method for thin film up to 20 µm and up to three layers. The measured spectrum (blue), the fitted spectrum (red) and the results are displayed in the **FT-Dsp** window.
Integration	Sets the integration time of the spectrometer. Reduce it, if the top of features in the spectrum is clipped (spectrometer overexposure).
Average	Sets the number of averaged spectra. Increase to reduce spectral noise.
Top/Middle/Bottom	Select the optical parameter file of the layer material to be measured. Stored in the directory *C:\!Wizard\!FT.app\Layer.*
Substrate	Select the optical parameter file of the substrate material. Stored in the directory *C:\!Wizard\!FT.app\Substrate.*
Fit Checkbox	Activate this checkbox to include the layer into the model fit. The film thickness result is displayed in the entry box of the layer respectively. Only applicable for Fit method.
Fix Checkbox	Activate this checkbox if the layer thickness is known and should be included with a fixed thickness in the model fit. Only applicable for Fit method.
Set	Define nominal thickness (expected target), thickness range and fit accuracy for the layer. Alternative to the **Fit Parameters** window (see below). Only applicable for Fit method.
RMS	The root-mean-square value of the deviation between measured spectrum and model is displayed as a measure of fit quality. RMS value of 0 indicates perfect fit whereas 1 indicates no match at all. Acceptable values may depend on the layer and fit parameters
Scattering compensation	Activate this checkbox to improve the fit if measuring on rough or scattering surface.

In the **FT** window, select **Parameter-Calibration positions** to check or define the background and substrate calibration positions. In the **FT: Calibration positions** window, activate both the **Automatic positioning** and **drive to position** check boxes to automatically drive the stage to defined calibration position during the job executing or when the **Calib. Backgr** and **Calib. Subst** buttons are clicked.

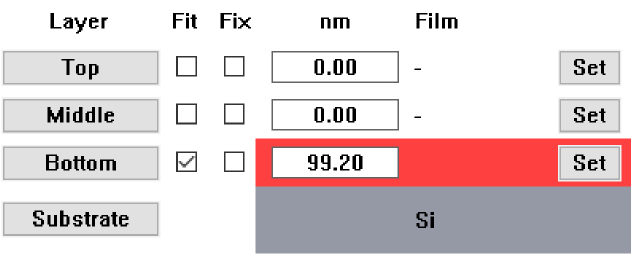
For the **background calibration position**, do not activate the **laser focus** checkbox. Click the **Test** button to move the stage to the defined calibration position. To change the calibration position, drive the stage manually with joystick to the desired position (dark and non-reflective surface) and click **Set** button to set the current stage position as background calibration position. The X and Y coordinates will be recorded in the two entry fields.

For **substrate calibration position**, activate the **laser focus** checkbox to automatically focus before calibration. By clicking the **Select substrate** button, define the substrate material by selecting the corresponding optical parameter file. To define the substrate calibration position, drive the stage to the calibration substrate on the mini chuck and click the **Set** button. Ensure that there is no particle or scratch on the measurement spot. Calibration positions can be defined for up to 6 different substrate materials.

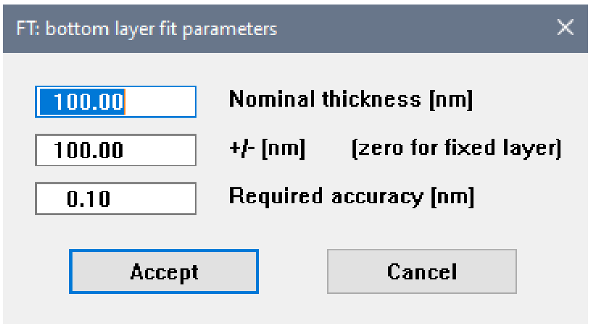
Click **Save** button to save the defined calibration positions in WIN.ini file. Note, that the definition of the **Calibration positions** has only to be done once. After saving they are valid for all further jobs.

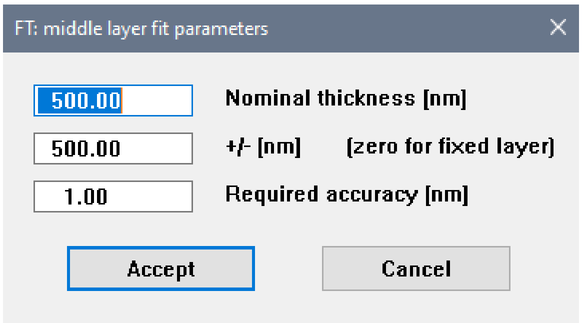


Check if the correct materials are selected for the layers and substrate. The material can be redefined by clicking either the **Top**, **Middle**, **Bottom** or the **Substrate** button, and selecting the appropriate optical parameter file for the requested material.



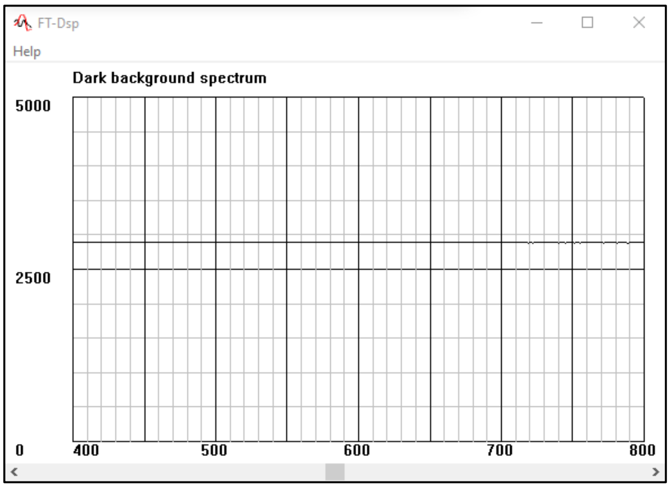
By clicking the **Set** button of each layer, the layer parameter window is shown and the fit parameters can be redefined.

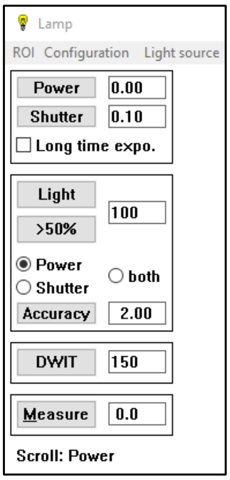




Calibration must be performed after the change of substrate material, objective, aperture, lamp power setting or after a specified time interval. Usually the wizard cares automatically for that.

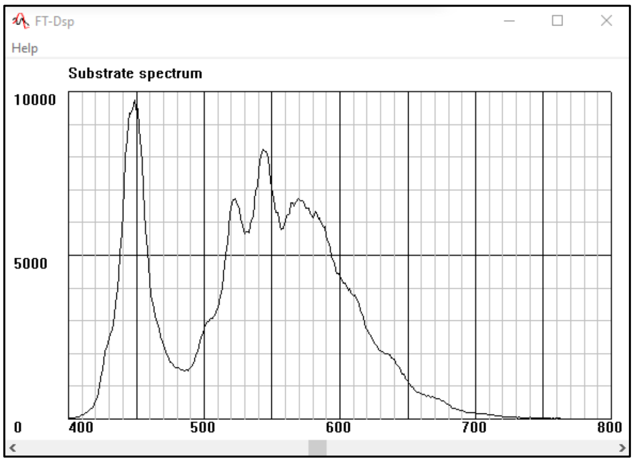
For background calibration, the integration time must be the same as the later thickness measurements. The lamp power must be zero. In the **Lamp** window, switch off the light by entering 0 in the entry field and click the **Power** button to set it to zero. In the **FT** window, click **Calib. Backgr.** button. The stage will move to the defined background calibration position to perform the background calibration and drive back to current position. The dark background spectrum is displayed in the **FT-Dsp** window. It should show a constant level. Any features in it may be due to stray light which should be avoided.

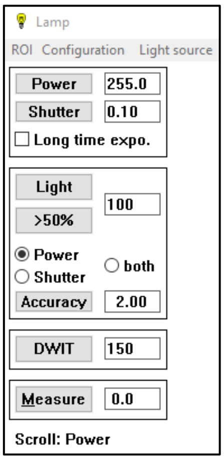




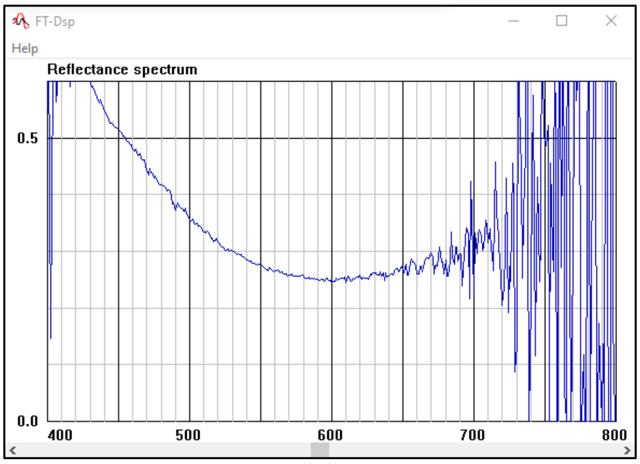
For substrate calibration, the integration time, objective, aperture, yellow filter and lamp power must be the same as for later film thickness measurements. In the **Lamp** window, switch on the light to full power by entering 255 in the entry field and click the **Power** button. Similarly, by clicking **Calib. Subst.** button, the stage moves to the defined substrate calibration position (mini chuck) and focusses on the blank substrate without any layers on top of it to perform the substrate calibration. The substrate spectrum is displayed in the **FT-Dsp** window.

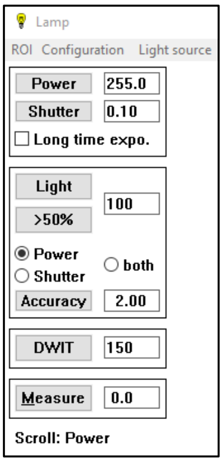
Repeat the background and substrate calibration if the substrate spectrum looks noisy. Increase the integration time if the signal looks small. Decrease the integration time if the highest peak signals look clipped due to overexposure. A peak height of the tallest structure in the spectrum of about 10000 to 30000 is recommended. Use exactly the same integration time also for later measurements.



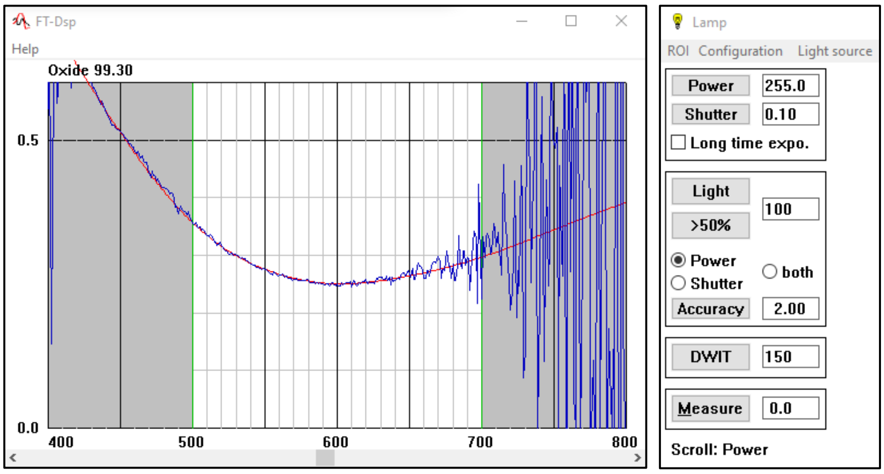


At the current measurement location, click the **Spectrum** button to record and calibrate a spectrum. The calibrated reflectance spectrum is displayed in blue color in the **FT-Dsp** window and can be used for subsequent layer thickness calculation by the Fit method.





Click the **Fit** button to calculate the film thickness by the Fit method. The best fitting model spectrum is displayed in red in the same **FT-Dsp** window. The film thicknesses of the model spectrum are displayed at the top.

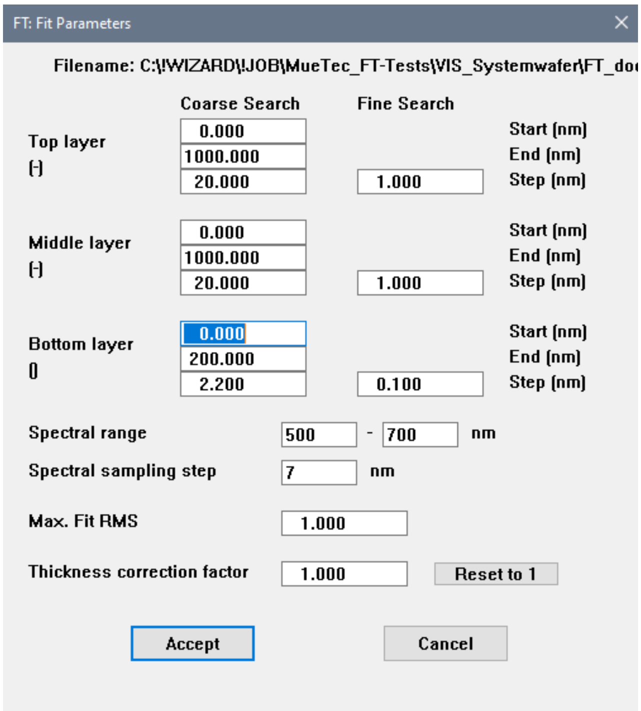


Referring to the curves in the **FT-Dsp** window, the layer thicknesses of the model spectrum giving the best match to the measured spectrum are calculated and shown here. The RMS value in the **FT** window (Root-Mean-Square deviation between model and measured spectrum) is used as a measure of the fit quality. **Max. fit RMS** can be defined to sort out bad measurements.

It is advisable to use only the good part of the spectrum for the fit simulation. The wavelength range can be confined to exclude noise or bad signal at the beginning and end of the bandwidth. In the **FT: Fit parameters** window, optimize the **Spectral range** in the entry field after reviewing the measured and fitted spectrum in the **FT-Dsp** window. The spectral range limits are indicated by green lines in the **FT-Dsp** window, the excluded wavelengths are shown as grey areas.

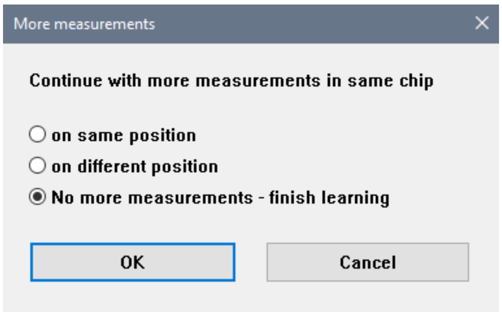
In the **FT: Fit parameters** window, the value of the **coarse** and **fine search step** can be optimized to improve the calculation speed performance. The coarse search calculates spectral models in a rather coarse thickness step width across the expected thickness range. After the interesting thickness values are found, the fine search refines the thickness measurement to the required accuracy.

Normally the steps of the **Coarse** and the **Fine search** are already optimized automatically for best calculation speed performance when using the **Set** button in the **FT** window. If a thickness measurement seems to yield improbable, unreliable or unstable results (this can happen for very thick layers), try decreasing the **Coarse search step** here. Be aware, that this will significantly increase the calculation times especially for multi-layer fits.



After all the FT parameters are defined, click the **Continue** button in the **MacWizCont** window.

At last, the **More measurements** window is shown and the procedures are the same as described in the CD measurement until the job is finished.

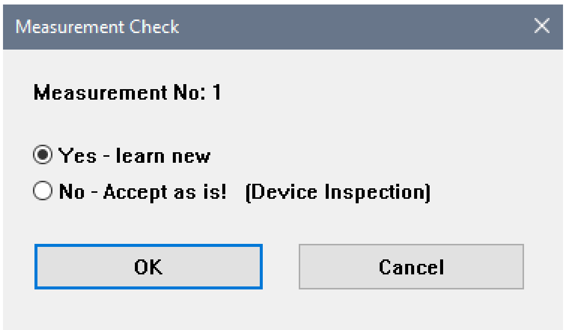


3.5 Learning Inspection Parameters

For **Device Inspection**, the wafer is scanned line by line either horizontally or vertically. The scanned images are joined by software and create the so-called **Composite Map**. One chip may contain several scan lines and several pictures for each scan line if it is larger than the field of view. For **Sealing Inspection**, it is essential that the chip completely fits into the field of view, thus lower magnification of the inspection objective is usually used.

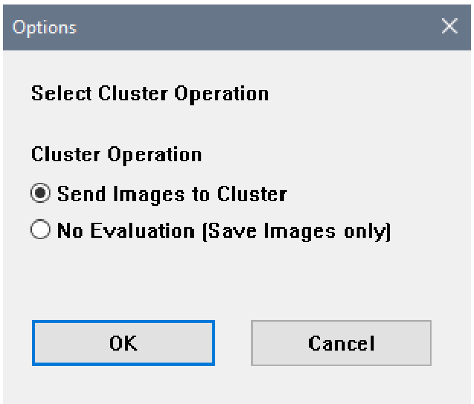
3.5.1 Learning Device Inspection (SDI)

In the **Measurement Check** window, select the radio button **Yes** to learn new inspection parameters. Select **No** if they are already learned previously and proceed with inspection scan. It is same as described for CD/OVL measurement (see 3.4.1).



Selection	Description
Yes – learn new	To learn a new inspection parameter
No – Accept as is!	To accept the existing measurement parameter. After clicking **OK**, the job will skip the learning of inspection parameters.
OK	To continue with job learning
CANCEL	To interrupt the job learning. The Wizard stops.

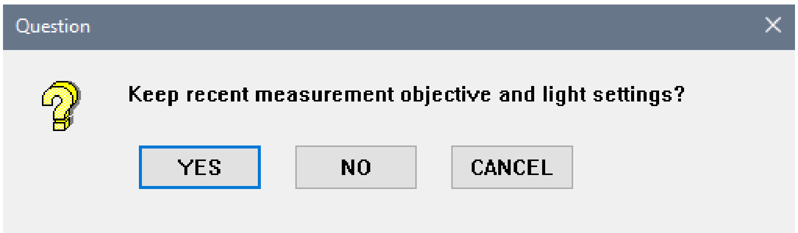
If **Yes - learn new** is selected, the **Options** window is shown to select the cluster operation.



Selection	Description
Send Images to Cluster	To send images to cluster for evaluation
No Evaluation	To save images only, without performing any evaluation. This is for jobs that shall only create composite maps.
OK	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

3.5.1.1 Objective and Illumination

The Question asks whether to keep the recent measurement objective and light settings.



Selection	Description
YES	To keep recent measurement objective and light setting
NO	To check or adjust the brightness and shutter time
CANCEL	To interrupt the job learning. The Wizard stops.

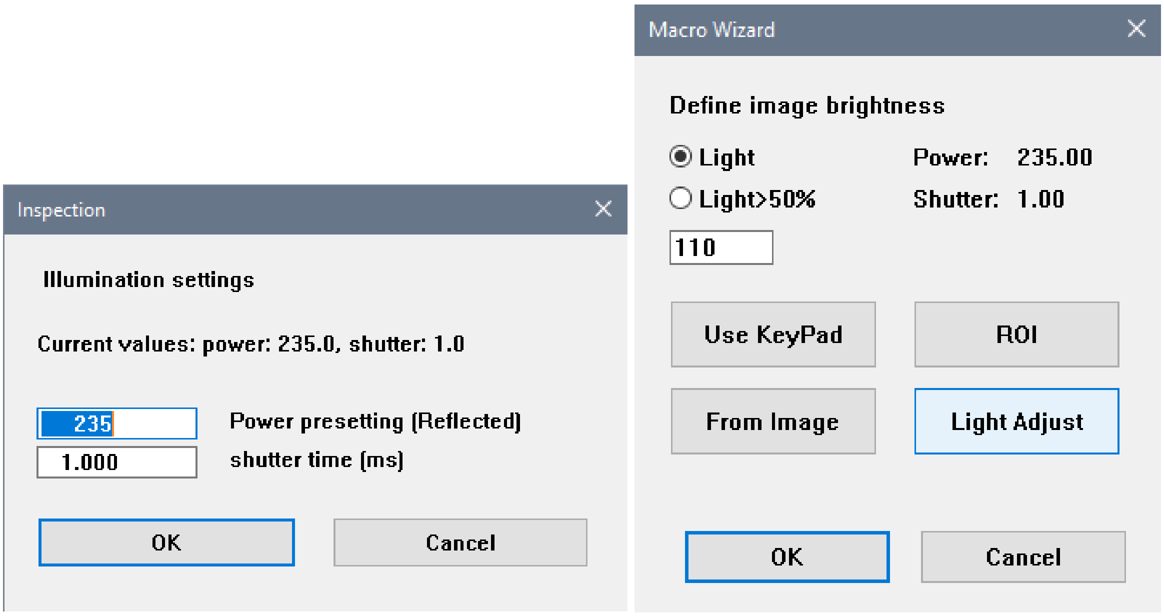
If you wish to redefine the measurement objective and light setting by answering **NO** to the previous question, the next **Question** window asks whether to drive to existing position for brightness adjustment.

Selection	Description
YES	To drive to existing position for objective selection and brightness adjustment
NO	To search for suitable position for brightness adjustment in **Free mode**
CANCEL	To interrupt the job learning. The Wizard stops.

In **Free mode**, select the objective with keypad and drive to the position where the brightness should be adjusted. Focus with z-wheel or keypad to get a sharp image. Click **OK** or **ENTER** to continue.

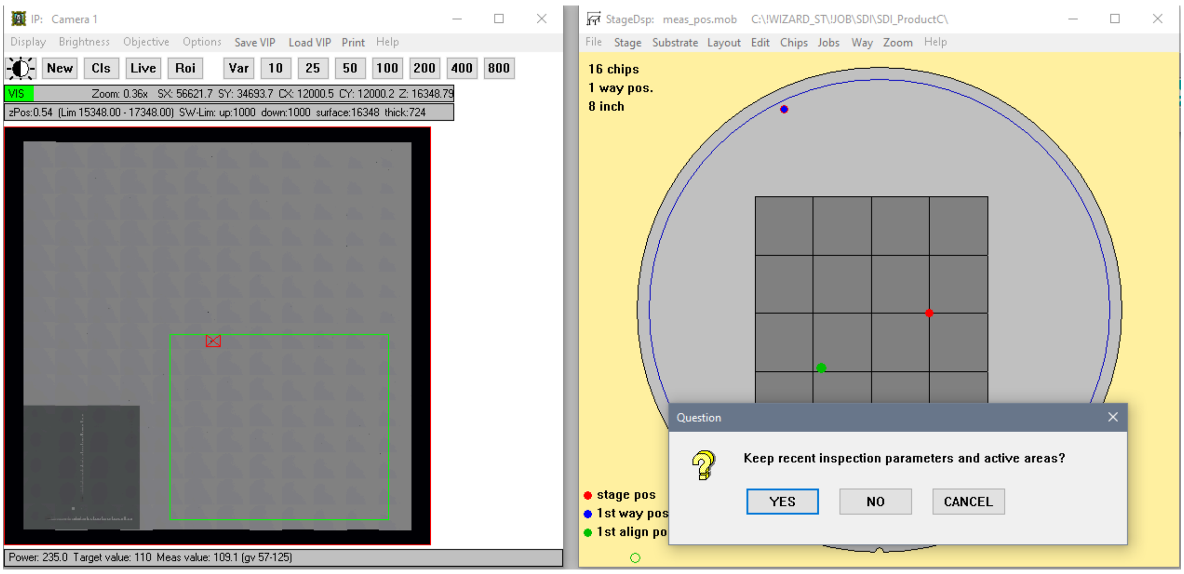
Next, the **Inspection** window is shown to check the current illumination setting. It is advised to set the power with at least a distance of 10 to maximum value (256) to allow light adjustment and the shutter time is advised to set around 1ms or below. Larger shutter times require slower scanning speed due to the avoidance of movement blurring and decreases the throughput, so shorter shutter times should be preferred. Please note that for defect inspection the light adjustment is done by the regulation of lamp power using the predefined shutter time.

Click **OK** to continue to check or adjust the image brightness. Perform the brightness adjustment as described in alignment and measurement position (see **Error! Reference source not found.** and 3.4.1.3). Click **OK** to continue.



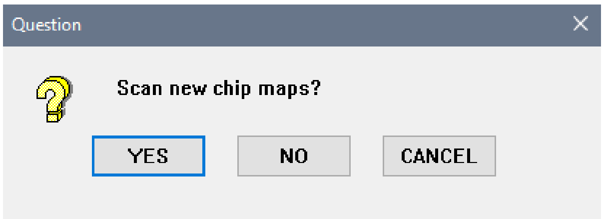
3.5.1.2 Inspection Parameter and Active Area

The **Question** window asks for keeping recent inspection parameters and active areas.



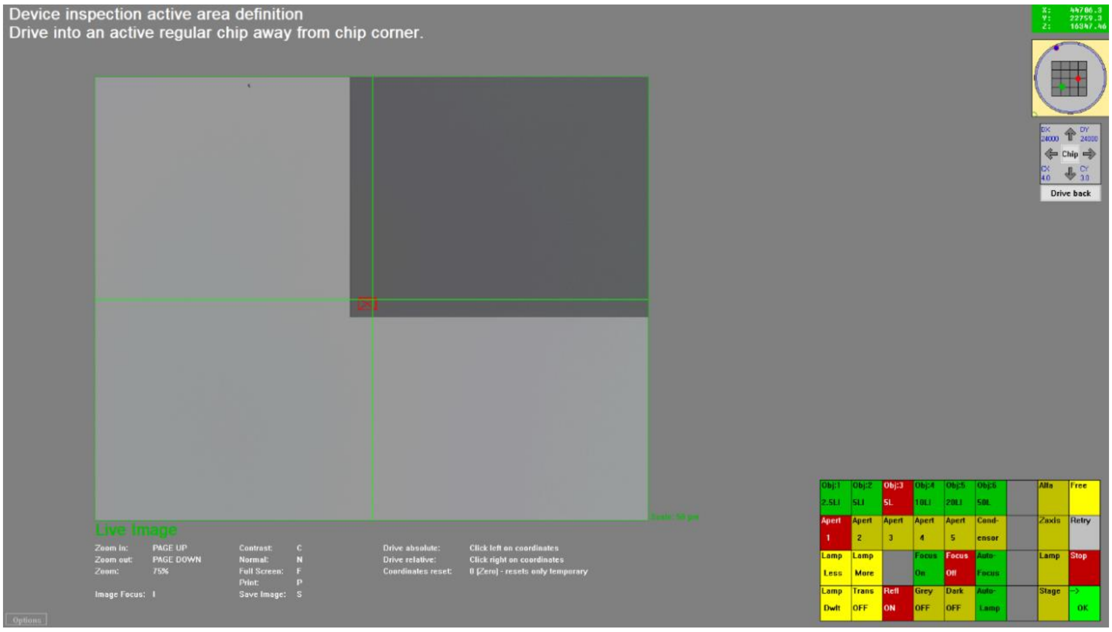
Selection	Description
YES	To keep recent inspection parameters and active area
NO	To rescan a new chip map and define inspection parameter setting
CANCEL	To interrupt the job learning. The Wizard stops.

To modify the inspection parameters, a scan of a single chip is recommended. To redefine inspection parameters and active area, select **NO**. The next **Question** window asks whether to scan new chip maps.



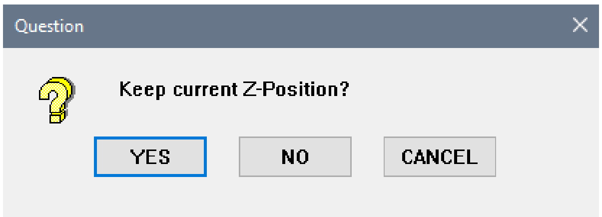
Selection	Description
YES	To scan new chip map
NO	To skip the scanning of new chip map
CANCEL	To interrupt the job learning. The Wizard stops.

After **YES**, drive the stage from chip corner to an active regular chip in **Free mode**, where the cross hair is on the chip as shown in the image below. Click **OK** or **ENTER** to continue.



3.5.1.3 Focus and Scan Speed

After defining the active chip, the **Question** window asks whether to keep current Z-position.

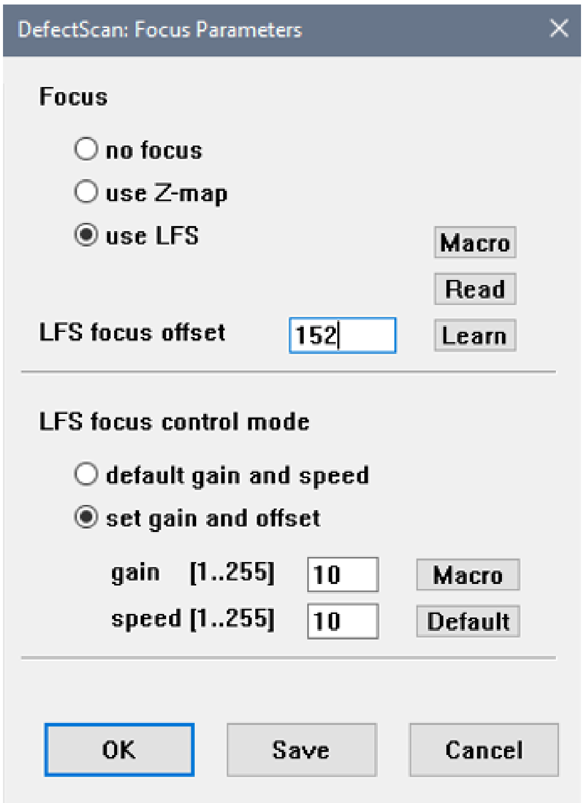


Selection	Description
YES	To use the current Z-position if the focus looks good in IP window
NO	To readjust focus in **Free mode**
CANCEL	To interrupt the job learning. The Wizard stops.

If you wish to keep the current Z-position by answering **YES** to the previous question, the **DefectScan: Focus Parameters** window is shown for defining focus parameters. The **Z-Map** function is intended for wafers with focussing problems that prevent the usage of the laser focus. Do not use this option for other purposes. It is advised to select **use LFS** radio button for focussing. You may relearn the **LFS focus offset** by clicking the **Learn** button.

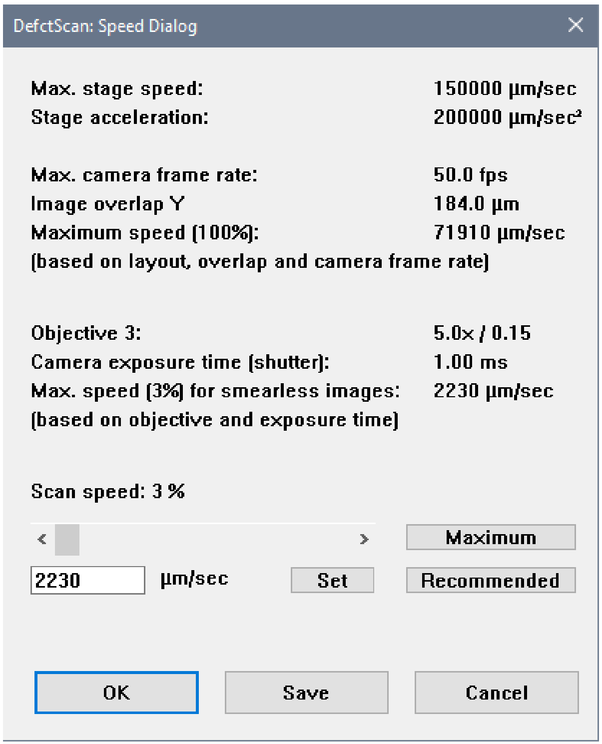
For the **LFS focus control mode**, the **default gain and speed** selection are for CD/OVL where the focus is moving fast but sensitive to edges. It is advisable to select the **set gain and offset** for scanning with the setting of 5-10. You must not press the **Default** button because this will set the default values that are optimized for CD/OVL and not for scanning.

Click **OK** to accept the parameters setting.



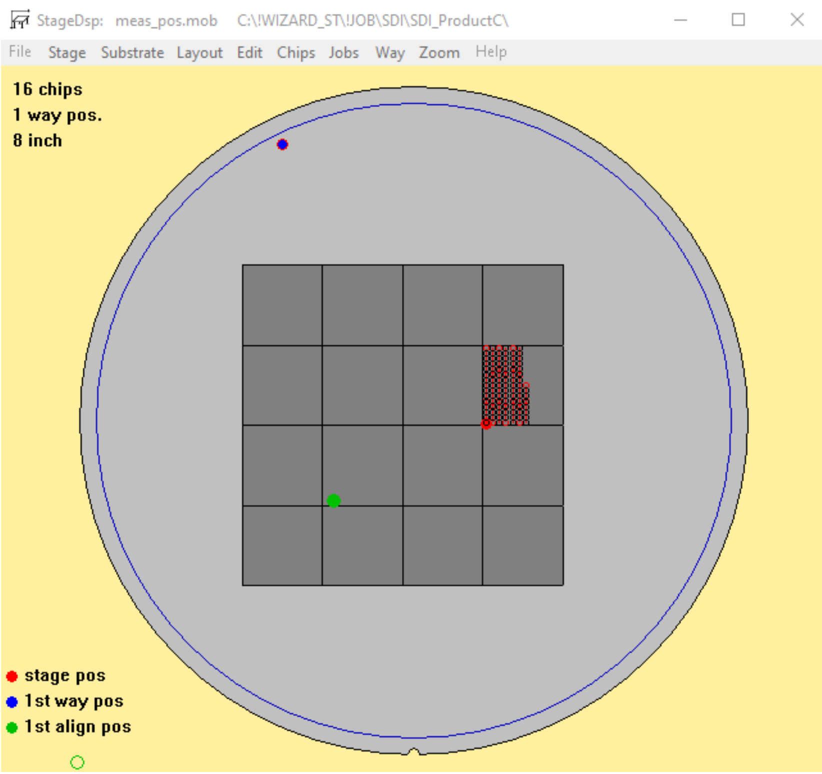
After defined the focus parameters, the **DefectScan: Speed Dialog** window is shown. It is advised to click **Recommend** and the suggested scan speed is displayed in the entry field. The slider can be adjusted to increase or reduce the scan speed. It is also possible to edit the scan speed in the entry field and click **Set**. Please note that a too fast scanning speed leads to blurry images and bad inspection results.

Click **OK** to accept the scan speed and the scan of one chip is executed.

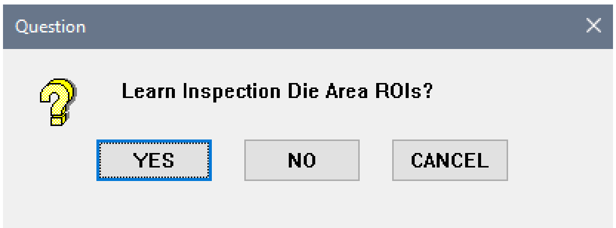


3.5.1.4 Scan One Chip and Define Active Area

The scanning of one chip is started as shown in this image:

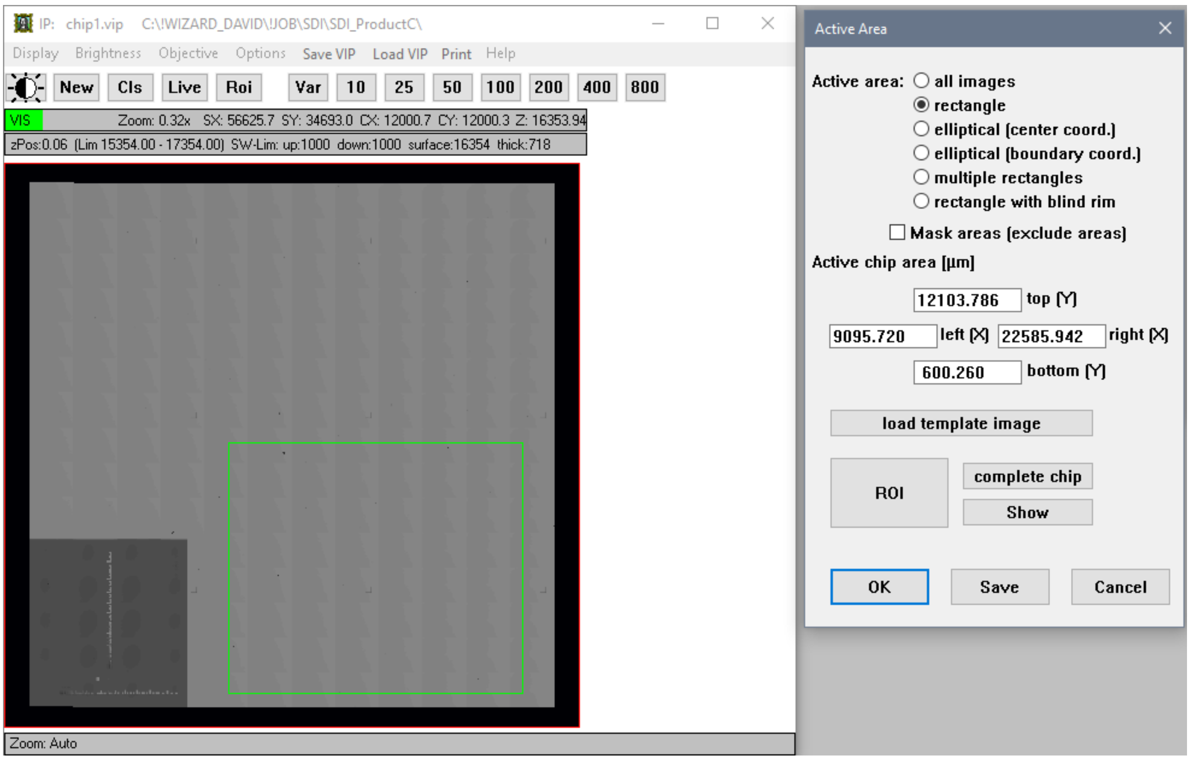


After the scan is completed, the **Question** window asks whether to learn the inspection die area ROIs.

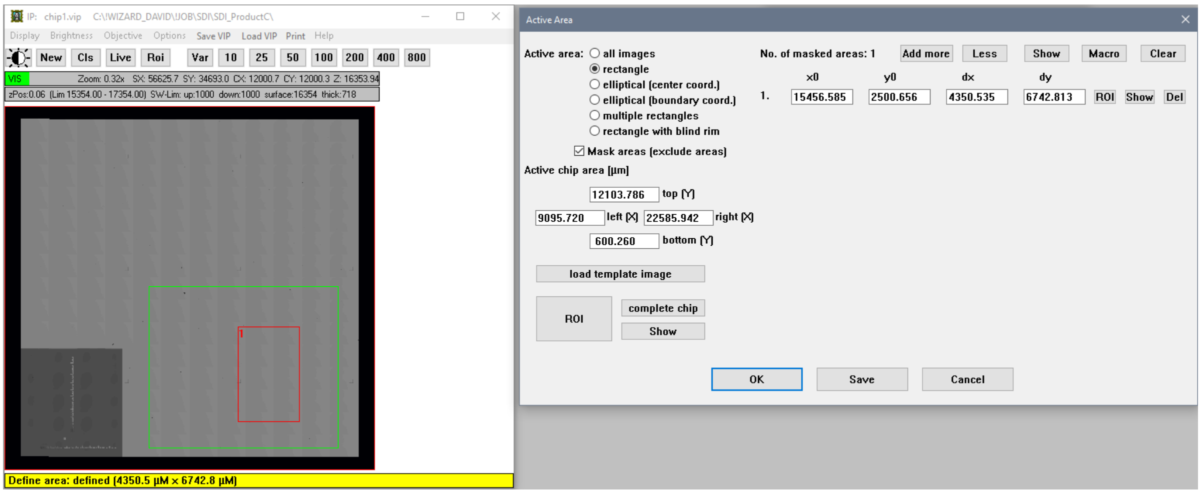


Selection	Description
YES	Click this button to learn the inspection die area ROIs
NO	Click this button to skip the learning of inspection die area ROIs
CANCEL	To interrupt the job learning. The Wizard stops.

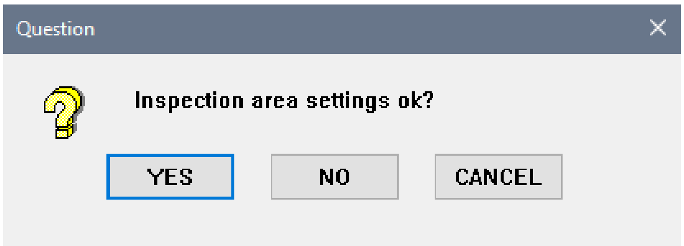
By answering **YES**, the composite map of one chip (the sum of all images that belong to the chip) is displayed in the **IP** window. The **Active Area** window is also shown for user to define the active area in the scanned chip. The ROI at a certain region in a chip or the whole chip can be defined for defect inspection. By selecting the **rectangle** radio button and clicking the **ROI** button in the **Active Area** window, define the inspection ROI rectangle in **IP** window to contain the active area. The **Active chip area** result will be displayed in the **Active Area** window and it is editable numerically. To perform inspection for whole chip, click the **complete chip** button in the **Active Area** window.



Activate the **Mask areas** to exclude defect inspection at certain region within the inspection area. Click **Add more** button and **ROI** button at the upper right of the **Active Area** window as below image. Similarly, define the ROI rectangle in the **IP** window to exclude the region from performing defect inspection. Click **OK** in the **Active Area** window to accept the parameter setting.



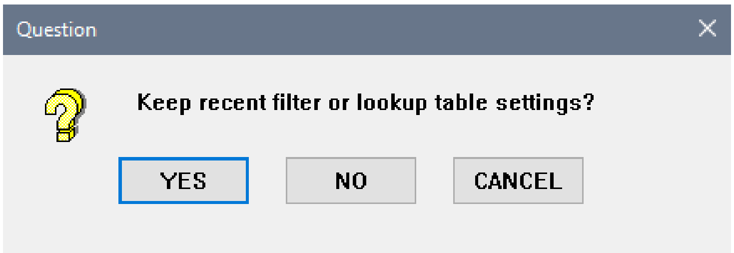
The **Question** window asks to verify if the inspection area setting are ok. Click **YES** to accept the inspection area setting.



Selection	Description
YES	To accept the inspection area setting
NO	To reset the inspection area setting. The **Active Area** window is shown again
CANCEL	To interrupt the job learning. The Wizard stops.

3.5.1.5 Filter and Contrast

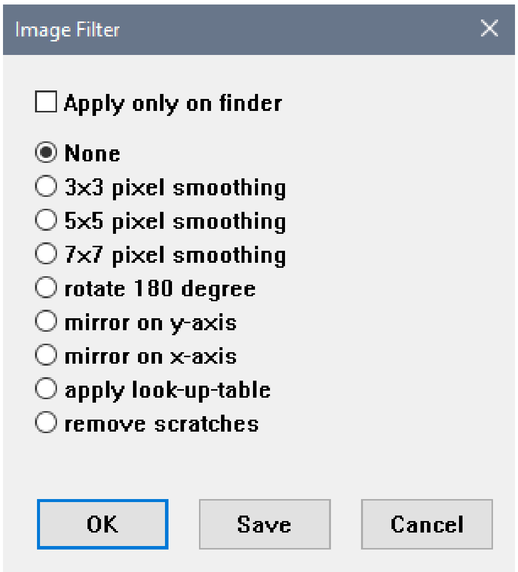
The **Question** window asks to keep the recent filter or lookup table settings.



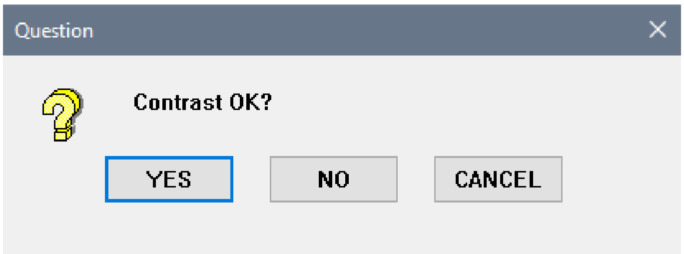
Selection	Description
YES	To keep recent filter and proceed to contrast setting
NO	To redefine the filter in Image Filter window
CANCEL	To interrupt the job learning. The Wizard stops.

If answering **NO** to the previous question, the **Image Filter** window is shown and the filter setting can be selected. Filters are used for very special purposes and need a special introduction for usage. Usually this option is disabled (None).

Click **OK** to continue with contrast setting.

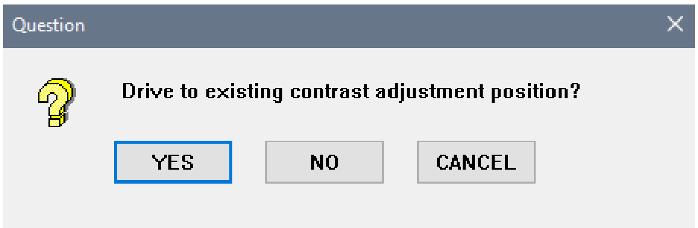


Next the **Question** asks if the contrast is ok.



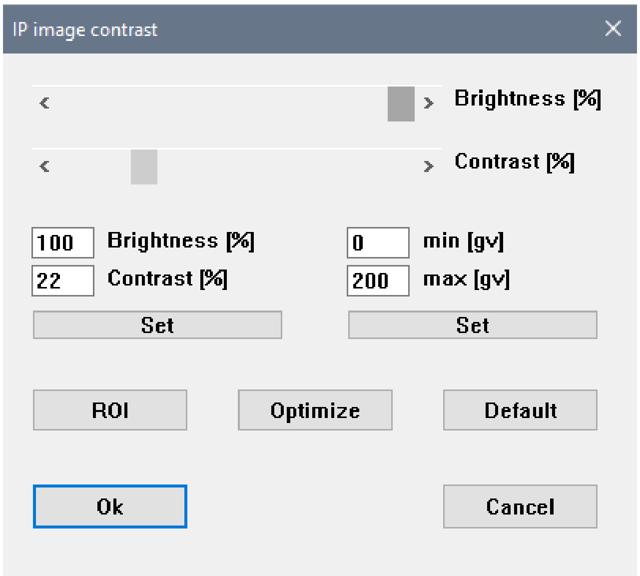
Selection	Description
YES	To accept the current contrast
NO	To readjust the contrast
CANCEL	To interrupt the job learning. The Wizard stops.

If wish to adjust the contrast, click **NO** and drive to existing adjustment position or look for other suitable position for contrast adjustment.



Selection	Description
YES	To drive to the existing contrast adjustment position
NO	To search for a suitable position for contrast adjustment in **Free mode**
CANCEL	To interrupt the job learning. The Wizard stops.

The **IP Image contrast** window is shown for optimizing the contrast with sliders. The defect inspection can be done on a contrast image, that is an expanded gray value cutout of the original image. This makes only sense for low contrast infrared images. The contrast enhancement increases gray value variations, therefore, if ever possible, use default contrast. Click **OK** after done adjusting the contrast in the **IP Image contrast** window.



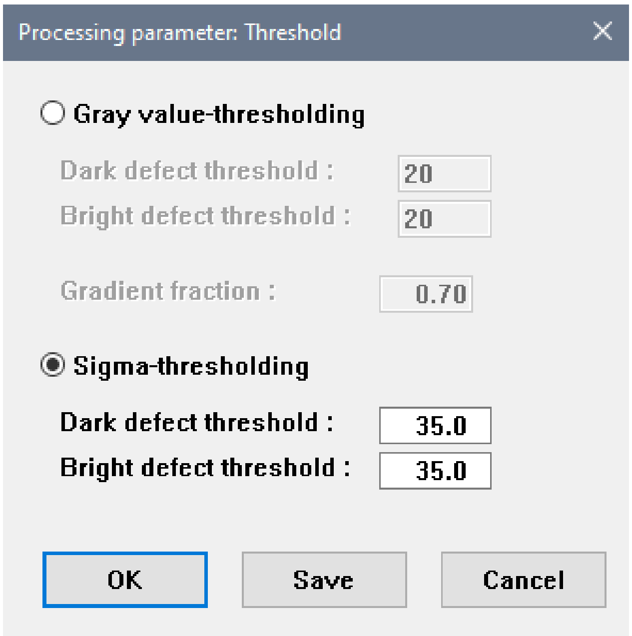
3.5.1.6 Processing Parameters

In the **Processing parameter: Threshold** window, there are options of **Gray value-thresholding** or **Sigma-thresholding** for **dark** and **bright defect threshold** settings. For **Gray value-thresholding** the defect criterium is the difference of gray values between golden image and recent image. The gray value difference can be suppressed at structure edges by scaling the gradient fraction from 0 (disabled) to 1 (maximum) to avoid false detections at edges. The base of edge suppression is a **gradient image** that is calculated from the golden image. This mode is especially usable on reticles but also on some wafers.

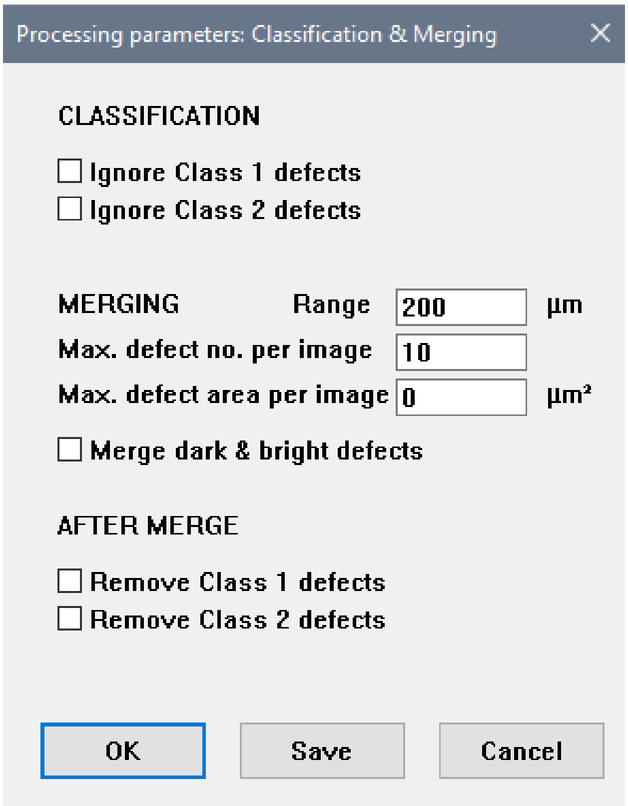
For **Sigma-thresholding** the defect criterium is a multiple of the standard deviation of gray values for each pixel. The base of sigma thresholding is a **sigma image** that is calculated from all images that contribute to the golden image. The sigma image shows how the gray value of each pixel in the image varies comparing all contributing images. The sigma value is small for homogeneous areas and larger for inhomogeneous areas such as metal layers. In this way it is possible to have dynamically adapted sensitivities on a single image depending on local homogeneity.

Example: Let the defect threshold be 10 sigma. On a quite homogeneous area the standard deviation (sigma) may be 1 gray value. In this case the final defect threshold is 1x10=10 gray values. In another area of the image the gray value has a larger variation, sigma is for example 3. Here the final defect threshold is 3x10=30 gray values, so in inhomogeneous areas it is less sensitive than in homogeneous areas. This mode is especially useful on many wafers.

Enter both thresholds in the entry field and click **OK** button to accept the threshold setting.



In the **Processing parameter: Classification & Merging** window, inspected defects within defined range can be merged. Click **OK** after all the parameters are defined.

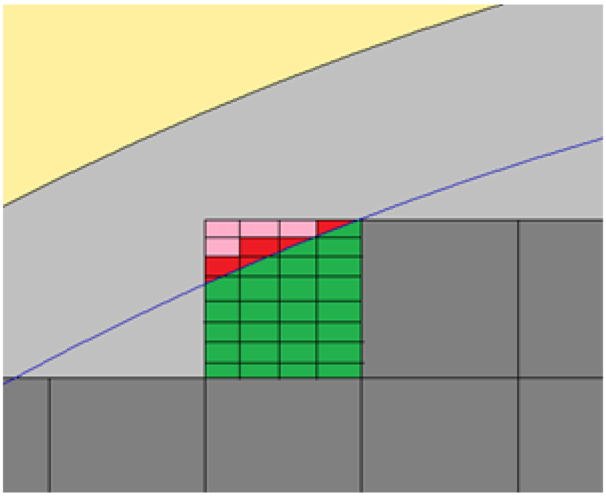


In the **Processing parameter: Golden Image** window, the **Golden Image Averaging** setting is typically used to define how the golden image is calculated. It can be defined how many images contribute to the golden image. The **n** sharpest images from total **m** images are taken and **p** outlier pixels are excluded. The number of discarded outlier pixels should be at least 1 to avoid that defects contribute to the golden image. Using this setting the golden image calculation can start after **m** chips are scanned. Golden and sigma image are calculated with **n-p** pixels that must be at least 3. With **n-p**=2 there is no decision possible what is the defect and what is good.

If a golden image is available on disk, the option of **Use golden images** can be activated. For this function, the system uses the existing golden image for evaluation instead of calculation one during runtime. Select the path of the existing golden image by clicking the **Directory** button.

Mask to Mask is a special mode for mask inspection. The mask area is scanned completely without attention to any chip layout and is compared to stored images.

Skip rim defines how images outside the defined wafer rim are handled. Depending on layout setting chips can partially go into the rim exclusion area (red/pink chip area). Let each chip consists of 4 scan lines in Y and 8 images each scan line. With standard setting the complete chip is scanned. With active **Skip rim** option only green/red marked images are scanned, images that are completely part of the rim exclusion area (pink) are not scanned.



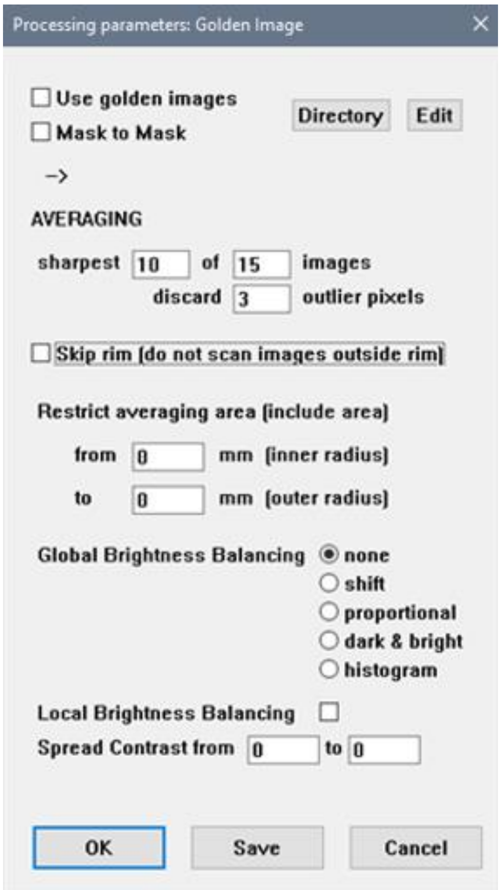
Restrict averaging area: Because the wafer edge and sometimes the wafer center are critical areas it is possible to exclude these areas from collecting images for golden image calculation. Your can define 2 radii (inner and outer) that define a inner circle (if the inner radius is 0) or a ring as a valid area for golden image determination. If inner radius >= outer radius this option is disabled.

Global brightness balancing: Some wafers show a radius dependent brightness variation that should not be detected as defect. In this case the **Global brightness balancing** can be activated. The brightness of each image is adapted before evaluation compared to a reference image. There are several modes:

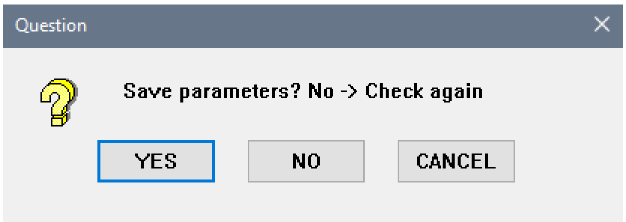
- **shift:** An additive constant is estimated to minimize mean gray value difference.
- **Proportional:** a constant factor is estimated to minimize mean gray value difference
- **dark & bright:** Special mode for binary masks, handle dark and bright areas separately
- **histogram:** Special mode for binary masks based on histogram evaluation

Local Brightness Balancing: This is a very special option that corrects unbalanced brightness within each single image by bilinear.

Click **OK** to continue after all the parameters are defined.



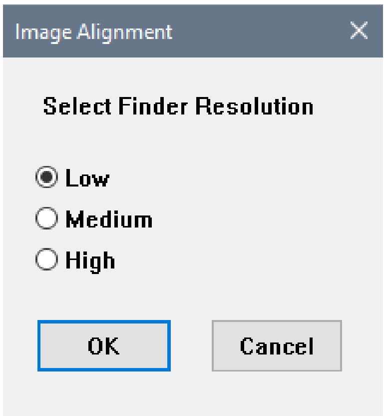
Subsequently, the **Question** window asks for saving the parameters.



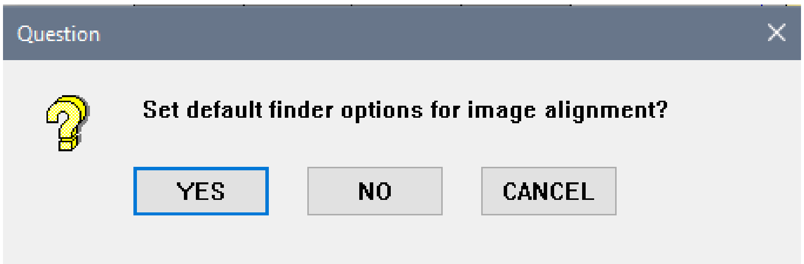
Selection	Description
YES	To save the golden image parameters
NO	To redefine the golden image parameters
CANCEL	To interrupt the job learning. The Wizard stops.

3.5.1.7 Finder

The finder is a very essential part of the defect inspection. It is responsible for image alignment before golden image and difference image calculation. Misalignments would result in false defects. The are 3 resolution steps available. In the **Image Alignment** window, it is advised to select **Low** for the **Finder Resolution**. By setting to **Medium** or **High** finder resolution, longer time will be taken for inspection. Increasing resolution by one step will enlarge alignment time by a factor of 4. Low resolution is sufficient if there is no danger of misalignment due to periodical structures in the image. If you face misalignments due to periodical structures, you should increase the finder resolution by one step.



Next, the **Question** window ask whether to set default finder options for image alignment.

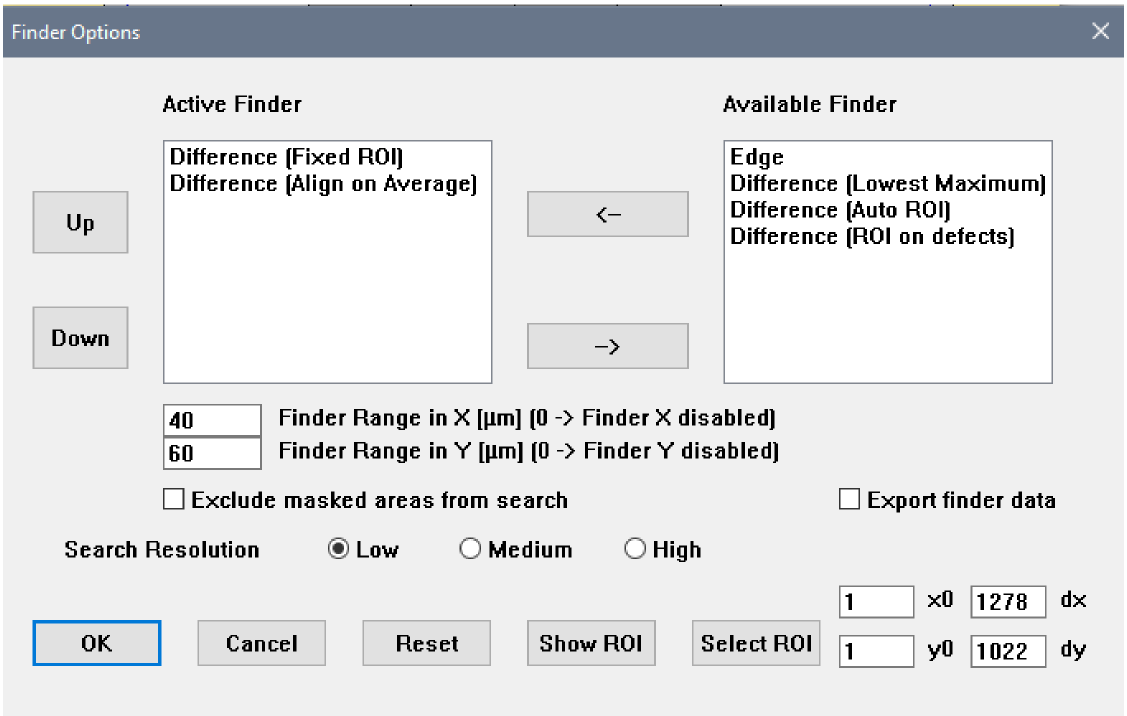


Selection	Description
YES	To set default finder option for image alignment. The object parameter window will be shown.
NO	To select the finder in the **Finder Options** window
CANCEL	To interrupt the job learning. The Wizard stops.

If you clicked **NO**, the **Finder Options** window is shown and the desired finder can be moved from **Available Finder** to **Active Finder** for image alignment. The **Finder range in X and Y** are also can be defined. Click **OK** after defining the finder and proceed to object parameters setting.

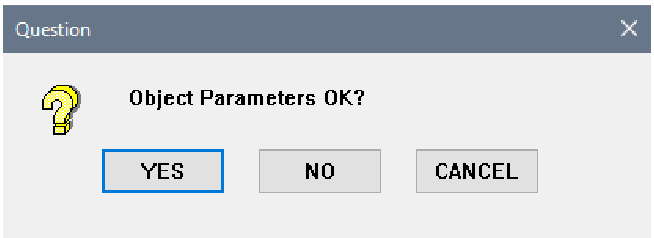
There are different finder types for image alignment available that are executed in the set order from top to bottom as long as defects are found.

- **Fixed ROI:** This finder is preset to first position. It works on a fixed ROI that is defined in the lower right corner of the **Finder Options** edit box. Usually the fixed ROI includes the entire image with one pixel margin. Reference image is the first image that is used for golden image calculation.
- **Align on Average:** Same as Fixed ROI but uses the golden image as reference
- **Auto ROI:** Reduces the ROI automatically to a feature that might be very suitable for alignment.
- **ROI on defects:** Reduces the ROI automatically to a region around the strongest defect
- **Lowest Maximum:** Same as Align on Average but tries to minimize the maximum in the difference image. Should be the last finder in the application chain.
- **Edge:** a very special algorithm based on edge detection.



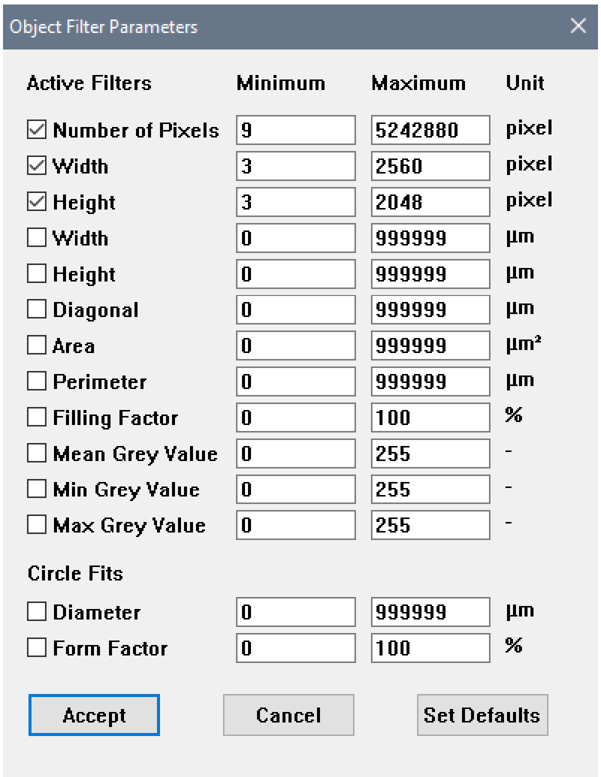
3.5.1.8 Object Parameters

The **Question** window asks if the object parameters are ok.



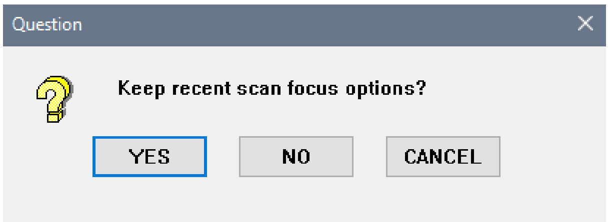
Selections	Descriptions
YES	To accept the current object parameters
NO	To define the object filter in the **Object Filter Parameter** window
CANCEL	To interrupt the job learning. The Wizard stops.

In the **Object Filter Parameters** window, the object filters are used to define the condition of defect selection. These object filter parameters are used to restrict the defect found based on the settings. Different kind of object filters can be activated as user preference. The minimum and maximum values can be entered to the object filter as well. For example, by number of pixels, width, height and area which are more commonly used. With these object filters, the type of defects can be narrowed down in the result list. The defect that filtered out here will not appear in the result list after scan completed. If user prefer to filter the result later, it is advised to use the filter in the **InspReview** window. Click the **Accept** button to continue after defining the object filter parameters.



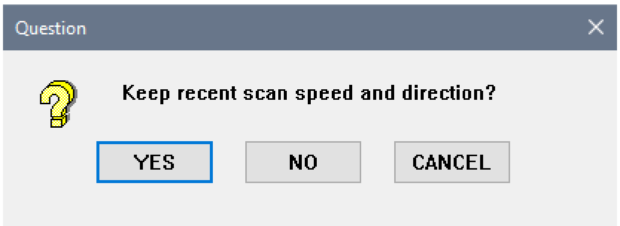
3.5.1.9 Focus and Scan Speed Check

The **Question** window asks for keeping the recent scan focus option. The procedure is the same as described in focus for scanning new chip map (see 3.5.1.3).



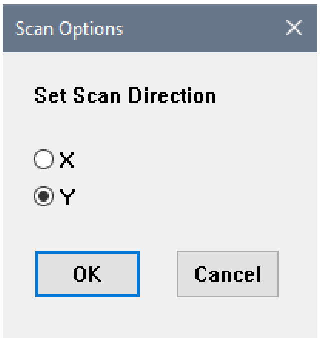
Selection	Description
YES	To accept the current focus options.
NO	To readjust focus in **Free mode**.
CANCEL	To interrupt the job learning. The Wizard stops.

Subsequently, the next **Question** window asks whether to keep the recent scan speed and direction. The procedure is the same as described in scan speed for scanning new chip map (see 3.5.1.3).



Selection	Description
YES	To accept the recent scan speed and direction
NO	To redefine the scan speed and direction
CANCEL	To interrupt the job learning. The Wizard stops.

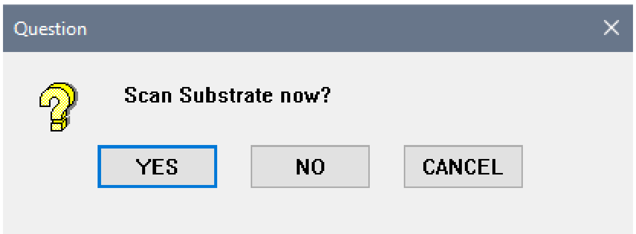
In the **Scan Options** window, define the scan direction. It is advised to perform the scan in Y direction.



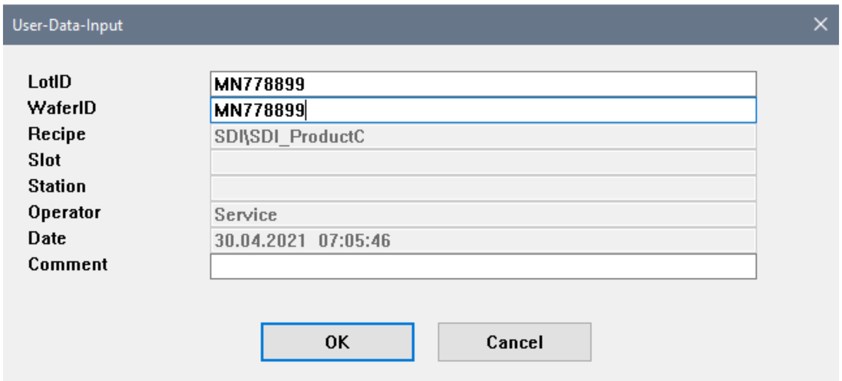
All defect inspection parameters are now defined and you can continue with a test scan.

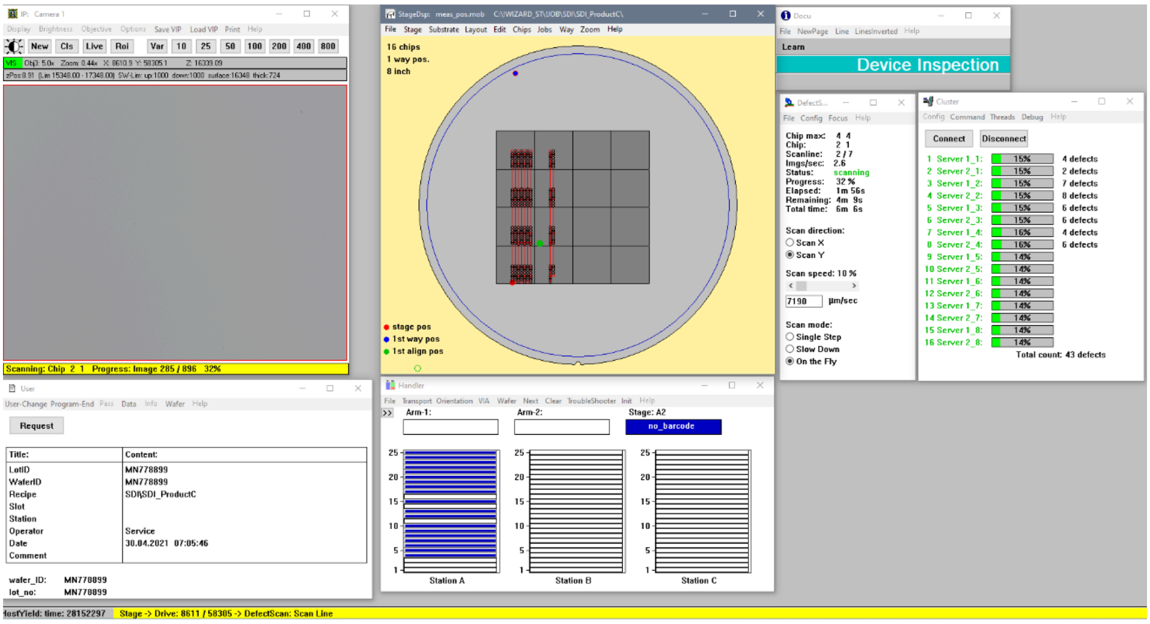
3.5.1.10 Scanning

After all the parameters are defined, the **Question** window asks for scanning the substrate now. The wafer scan for defect inspection and the option of offline review are the same as described in inspection job executing (see 2.3.1 and 2.6.1).

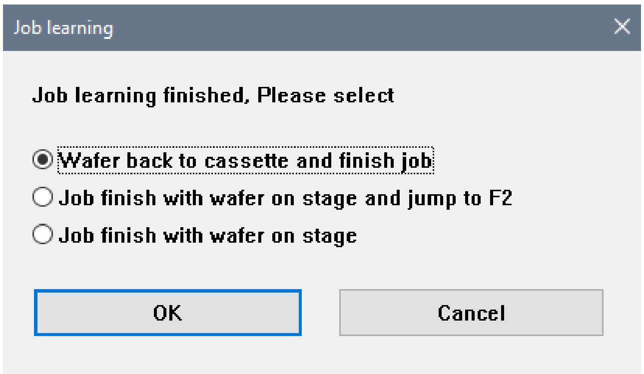


Selection	Description
YES	To start the learned defect inspection job
NO	To skip the defect inspection job and job learning completed
CANCEL	To interrupt the job learning. The Wizard stops.





After inspection is completed, the same **Job learning** window is shown. Job learning is finished and user may decide to handle the wafer back to cassette or leave the wafer on stage. Same as described in the last part of CD/OVL job learning.

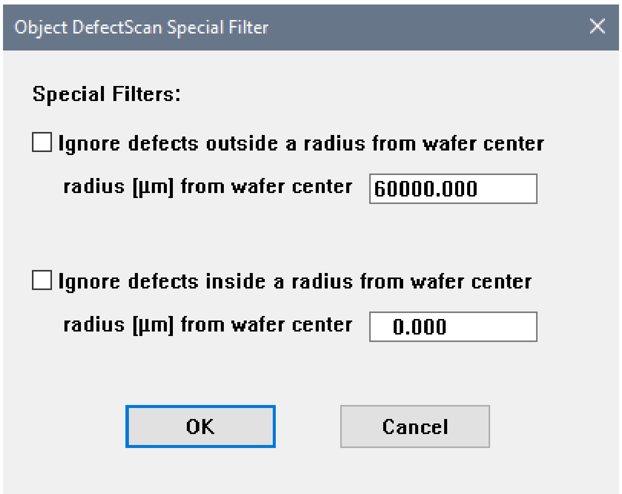


3.5.2 Learning Blank Inspection

For blank wafer inspection, the job learning is much more simple compared to the device inspection. In the Job Parameter window, define the measurement type as blank inspection (see 3.1.2.1). The first and second alignment is skipped, as well as the chip layout. It also requires lesser setting for inspection parameters.

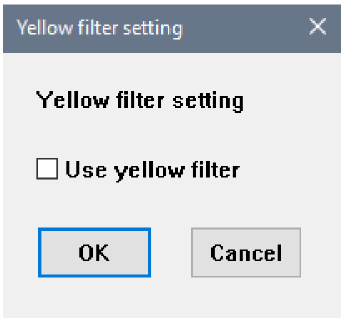
3.5.2.1 Object Defect Scan Special Filter and Yellow Filter Setting

At the beginning, the **Object DefectScan Special Filter** window is shown. If blank inspection is defined and the layout as a wafer, these special filters can be applied to exclude the defects from evaluation by defining the radius from the wafer center. The filter can be activated within the defined radius or away from the defined radius. However, the **Edge Exclusion** setting later will override this special filter. Typically, the defect measurement of a blank scan cannot be performed all the way to the edge because the transition from wafer edge to substate would be interpreted as defects.

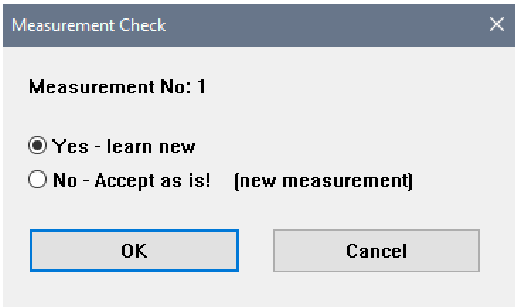


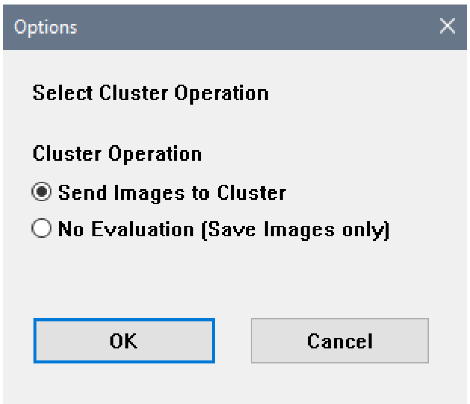
Selection	Description
Ignore defects outside a radius from wafer center	To exclude the defects outside of the defined radius. This setting will be overridden by edge exclusion setting
Ignore defects outside a radius from wafer center	To exclude the defects inside of the defined radius
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

If the blank wafer to be inspected is coated with photoresist, the yellow filter can be activated.



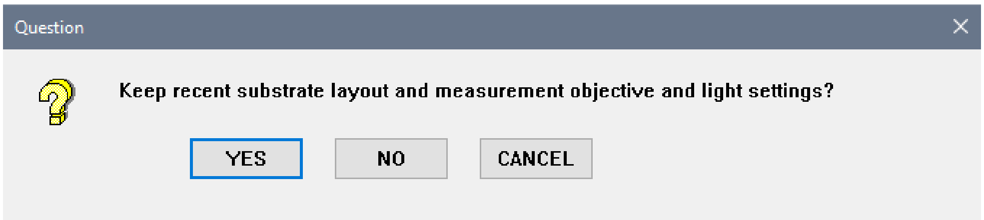
Since there is no alignment and no layout, the **Measurement Check** window is shown and will direct proceed with the inspection setting. The procedure here is the same as described in device inspection (see 3.5.1). In the **Options** window, select the cluster operation either send the images to cluster for evaluation or save the images only without evaluation. Click **OK** to continue the job learning.



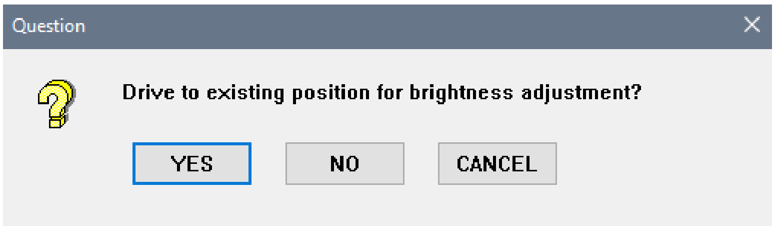


3.5.2.2 Substrate Layout, Objective and Illumination

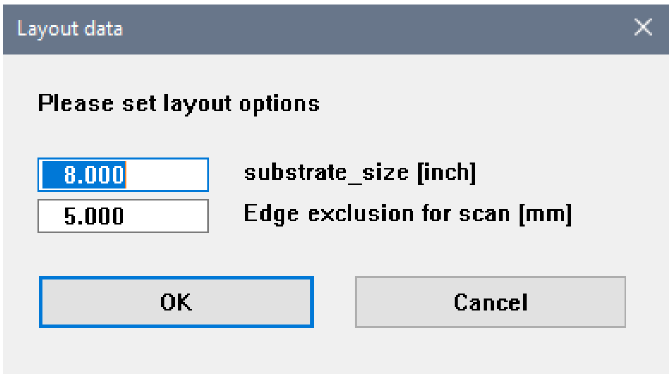
The **Question** window asks for keeping recent substrate layout, measurement objective and light setting. The procedure is same as described in device inspection (see 3.5.1.1.), with additional layout option settings.



If you wish to redefine the measurement objective and light setting by answering **NO** to previous question, the next **Question** window is asks to drive to existing position for brightness adjustment. In **Free mode**, drive to the position where the brightness should be adjusted and select the objective, focus and adjust illumination.

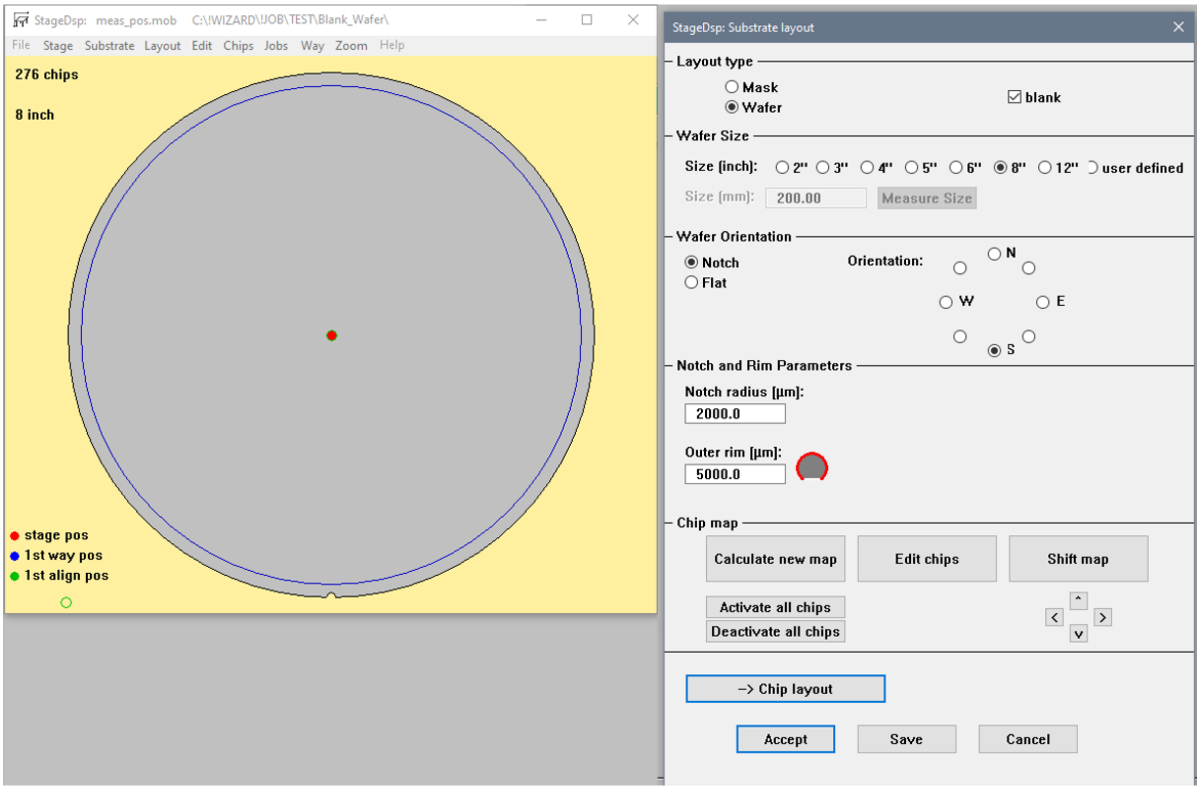


Next, the **Inspection** window is shown for illumination settings. Click **OK** to continue with brightness adjustment. After done with the brightness adjustment, the **Layout data** window is shown. Define the **Substrate size** and **Edge exclusion** to avoid scanning beyond the wafer edge and to minimize the inspection area intentionally.



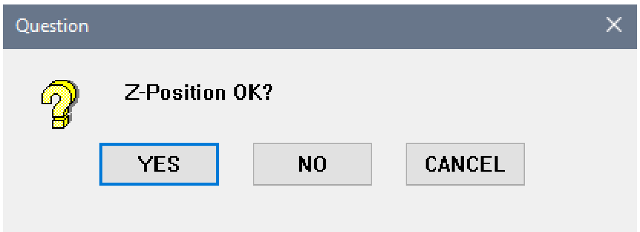
Selection	Description
Substrate Size [inch]	Enter the substrate size or the intentional inspection size.
Edge Exclusion	Enter the edge exclusion from wafer edge.
OK	To continue the job learning
CANCEL	To interrupt the job learning. The Wizard stops.

Check on the **StageDsp** and **Substrate Layout** window and click **Accept** button to confirm.



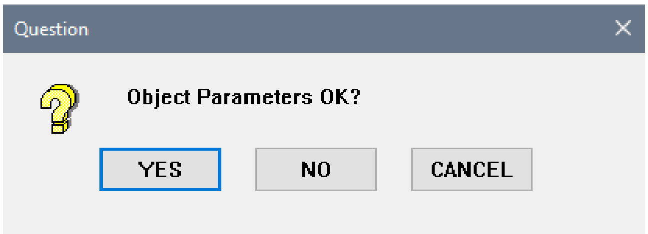
3.5.2.3 Focus

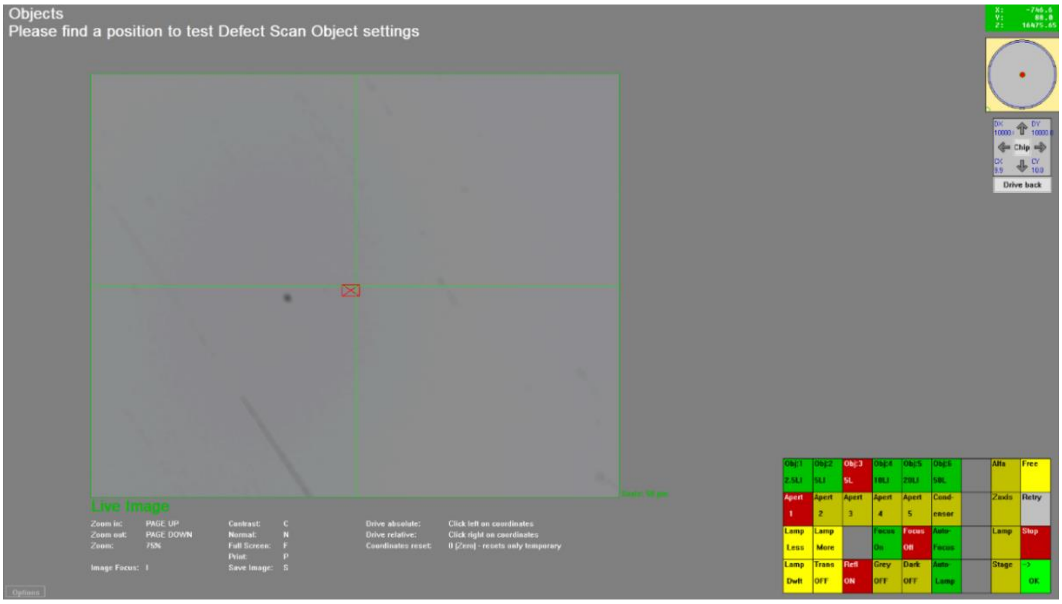
The procedure is the same as described in the device inspection (see 3.5.1.3), check the z-drive position to get sharp images in **IP** window and confirm with **YES**. By clicking **NO**, the focus can be adjusted in **Free mode** by z-wheel. Because of the lack of structures it might be necessary to go to the wafer edge to decide if the z-drive position is ok or not. Be sure that the focus spot is on the wafer if you leave the **Free mode**. Click **OK** or **ENTER** to continue.



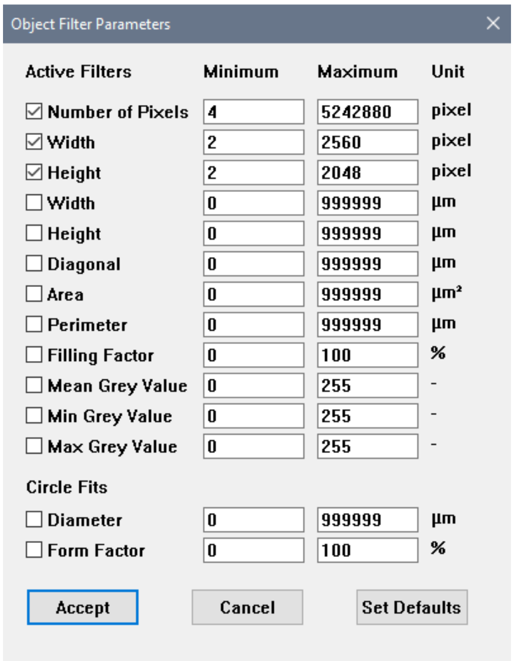
3.5.2.4 Object Parameters

The procedure is same as described in the device inspection (see 3.5.1.8). The **Question** window asks if the object parameters are ok. Click **NO** to find a position with known defect in **Free mode** to review the object filter setting.





Same as in device inspection, the defects that are filtered out here will not appear in the result later. Alternatively, filter the defects in the **InspReview** window after evaluation. Click **Accept** after defining the object filter.



3.5.2.5 Defect Scan Options

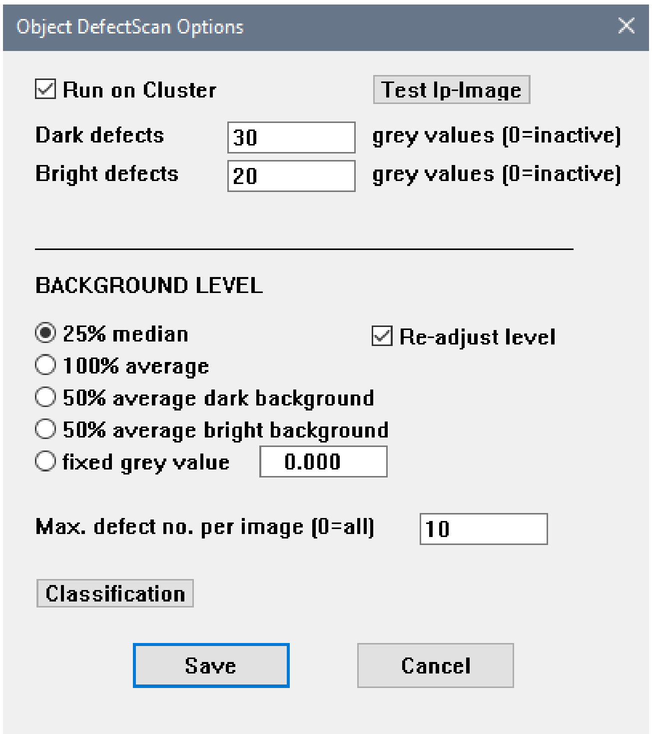
In the **Object DefectScan Options** window, the **threshold of dark and bright defects** can be defined with the grey value separately (range 1-255, usually the setting is between 15-40, 0 = ignore). Click the **Classification** button to define the classification. The **Defect Classification** window is the same as shown in defect review for inspection measurement (see 2.6.1).

There are different kind of background gray value level estimation to identify defects:

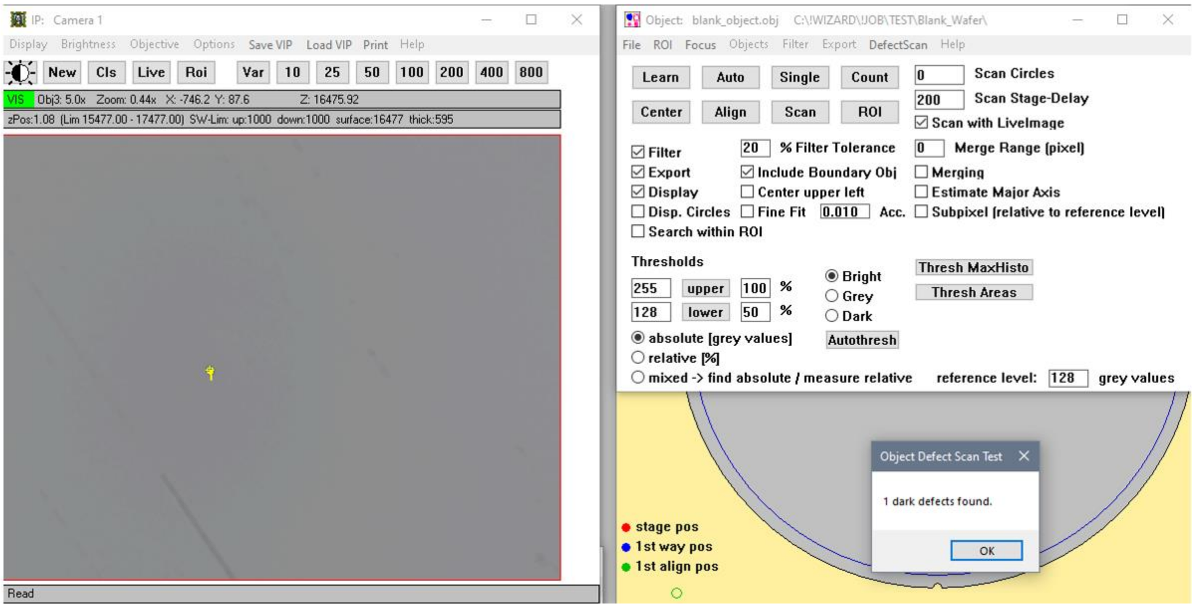
- **25 % median:** Median of all gray values in the histogram that are within 25% around the histogram peak level.
- **100% average:** Average of the entire image.
- **50% average dark background:** The histogram is divided into 2 parts, the background gray value is the average of the lower (dark) part of the histogram.
- **50% average bright background:** The histogram is divided into 2 parts, the background gray value is the average of the upper (bright) part of the histogram.
- **fixed gray value:** a fixed predefined value.

Re-adjust level: The analysis is performed twice. After the first loop the background level is re-adjusted with the same criterium but with exclusion of the defects found in the first loop.

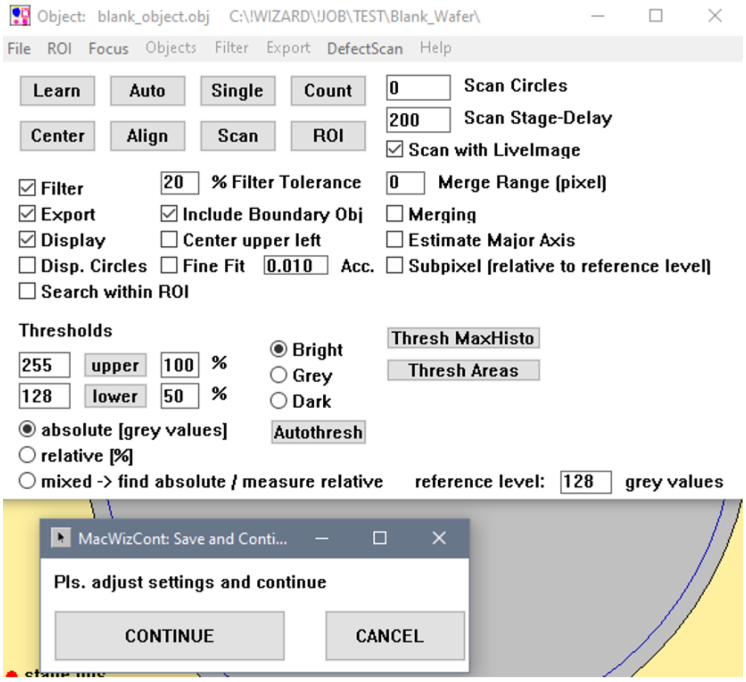
Max. defect no per image: In worst cases the algorithm may find several thousand defects in a single image. Because this can barely be handled reasonably it is possible to restrict the maximum number of exported defects per image.



Click the **Test Ip-Image** button to test the defects in the **IP** window can be found or been filtered. Click **OK** and review the parameters in the **Object** window.

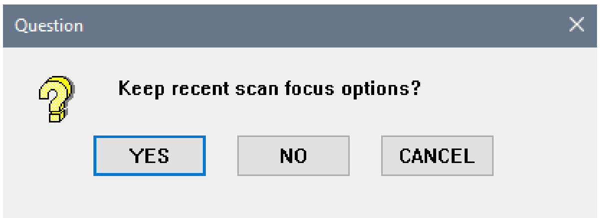


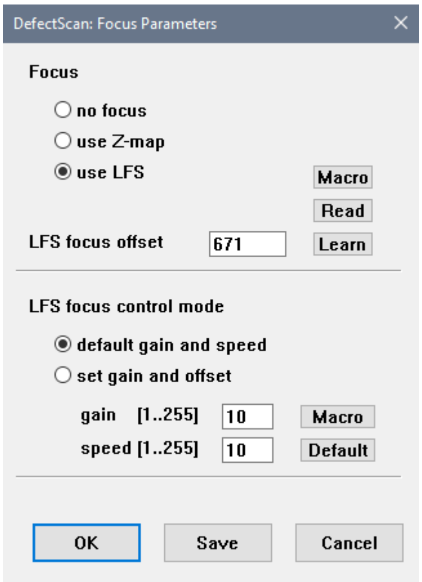
In the **MacWizCont** window, click **Continue** to continue the job learning .

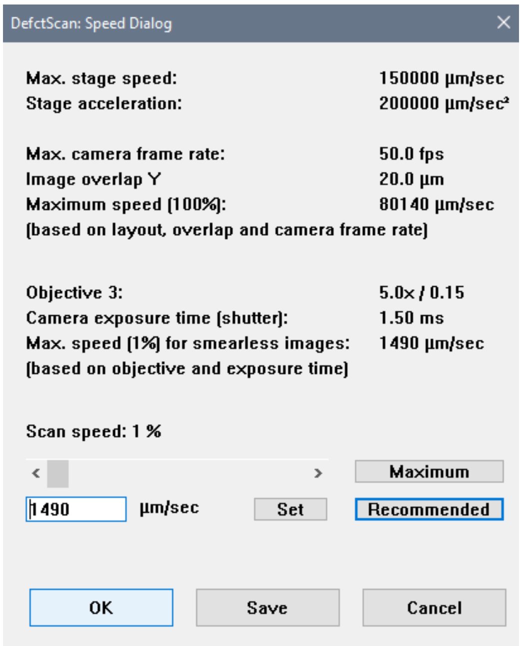


3.5.2.6 Focus and Scan Speed

The procedure of the focus and scan speed setting is the same as described in device inspection (see 3.5.1.3. and 3.5.1.9).

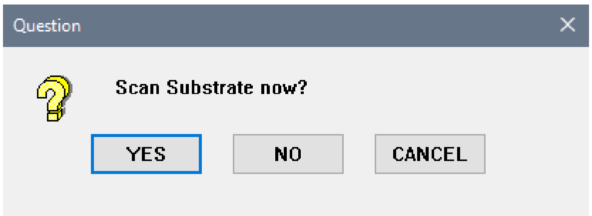






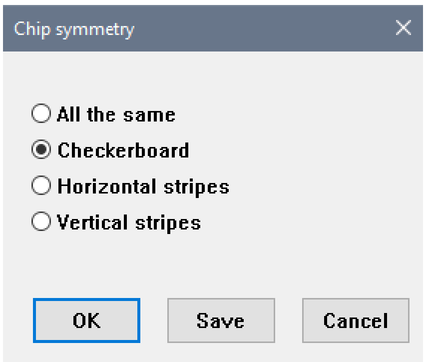
3.5.2.7 Scanning

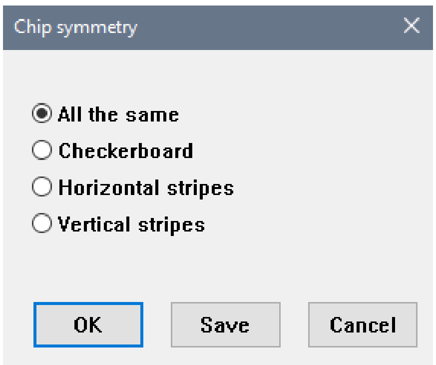
After all the parameters are defined, the **Question** window asks whether to scan substrate now. The wafer scan is the same as device inspection (see 3.1.5.10) and the options of offline review are the same as described in measurement job (see 2.6.1).



3.5.3 Learning Sealing Inspection

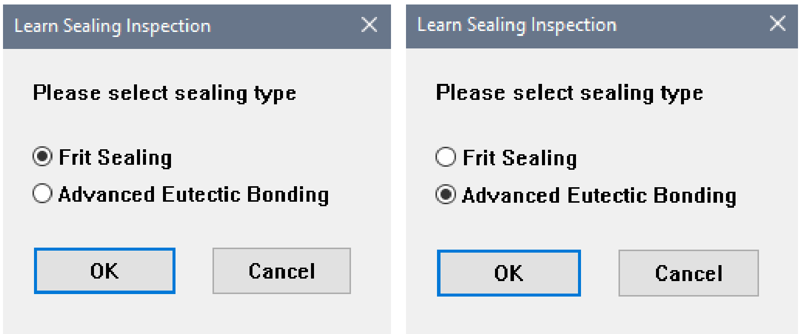
For sealing inspection, the alignments and settings are identical to the device inspection job. During layout learning, the **Chip symmetry** window is shown after accepted the chip corner.





Selection	Description
All the same	The orientation of chips is arranged in the same way, all are regular chips.
Checkerboard	The orientation of chips is arranged in checkered pattern, consisting of alternating regular and horizontally and vertically rotated chips, .
Horizontal stripes	The orientation of chips is arranged in stripes pattern, consisting of alternating regular and horizontally rotated chips.
Vertical stripes	The orientation of chips is arranged in stripes pattern, consisting of alternating regular and vertically rotated chips.

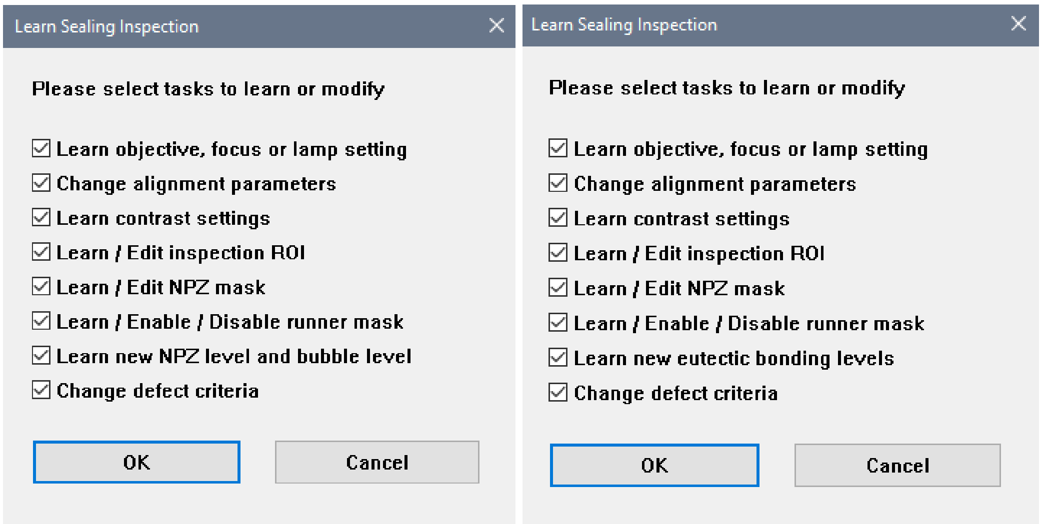
For **Measurement Check** and **Cluster Operation** window, the procedures are the same as described in device inspection (see 3.5.1). In the **Learn Sealing Inspection** window, select the sealing type as **Frit Sealing** or **Advance Eutectic Bonding**. Available options are a matter of purchased configuration and can differ.



Selection	Description
Frit Sealing	For this sealing type, normally the sealing area is in black or darker color. The brighter area is categorized as defect from dark background
Advanced Eutectic Bonding	For this sealing type, normally the sealing area is mixture of black and white color, like salt and pepper texture background.
OK	To continue with learning tasks selection
CANCEL	To interrupt the job learning. The Wizard stops.

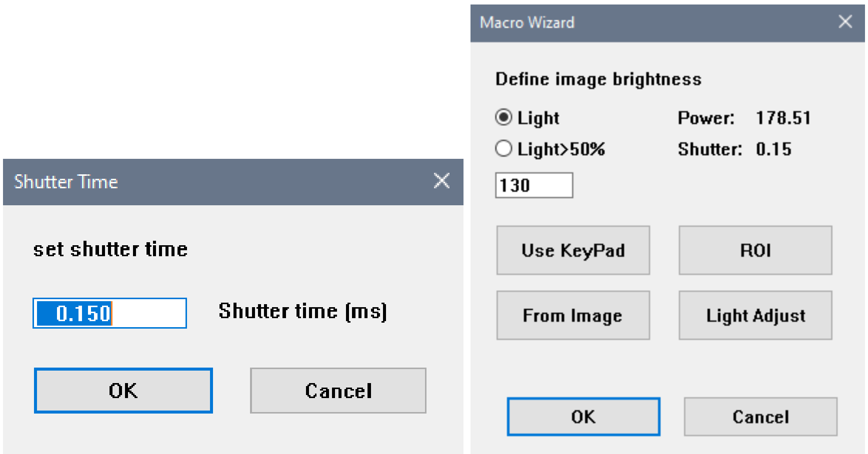
For job modification, activate the necessary checkboxes in the **Learn Sealing Inspection** window. Depending on the selected tasks, different subroutines are executed. For new job learning, all the tasks can be activated. Click **OK** to continue.

The left window is for **Frit Sealing** and right window is for **Advanced Eutectic Bonding**.



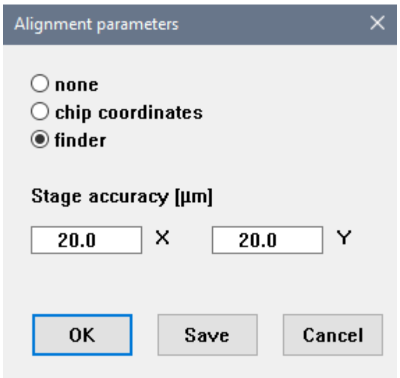
3.5.3.1 *Objective, Focus and Lamp Setting*

The procedures for objective, focus and lamp setting are the same as described in device inspection. (see 3.5.1.1.) It is advisable to define the shutter time at around 0.15 - 0.2 ms. Lower shutter time is better, provided the image brightness is sufficient. Define the ROI to include the sealing area.



3.5.3.2 *Alignment Parameters*

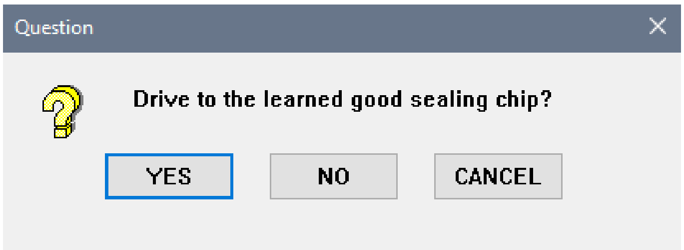
This alignment refers to the feature alignment from chip to chip to compensate the stage inaccuracy during scan in order to have the same ROI location during evaluation as during job learning. In the **Alignment Parameters** window, select **finder** as the alignment mode and define the **Stage accuracy**.



Selection	Description
None	The alignment is deactivated
Chip coordinates	For this option, it is aligned by chip coordinates. The stage measures the current coordinates during scanning and corrects the ROI position in the image.
Finder	Additionally to the chip coordinates alignment function, during evaluation of the scanned images, the finder will compare the scanned images with the reference image. Based on the superposition, the ROI positions are corrected and aligned. Select this alignment mode for most of the jobs
Stage accuracy	Edit the value for stage accuracy a bit larger than they are valid for the tool. This ensures a successful finder search as the finder works inside this defined region only.
OK	To continue with job learning
Cancel	To interrupt the job learning. The Wizard stops.

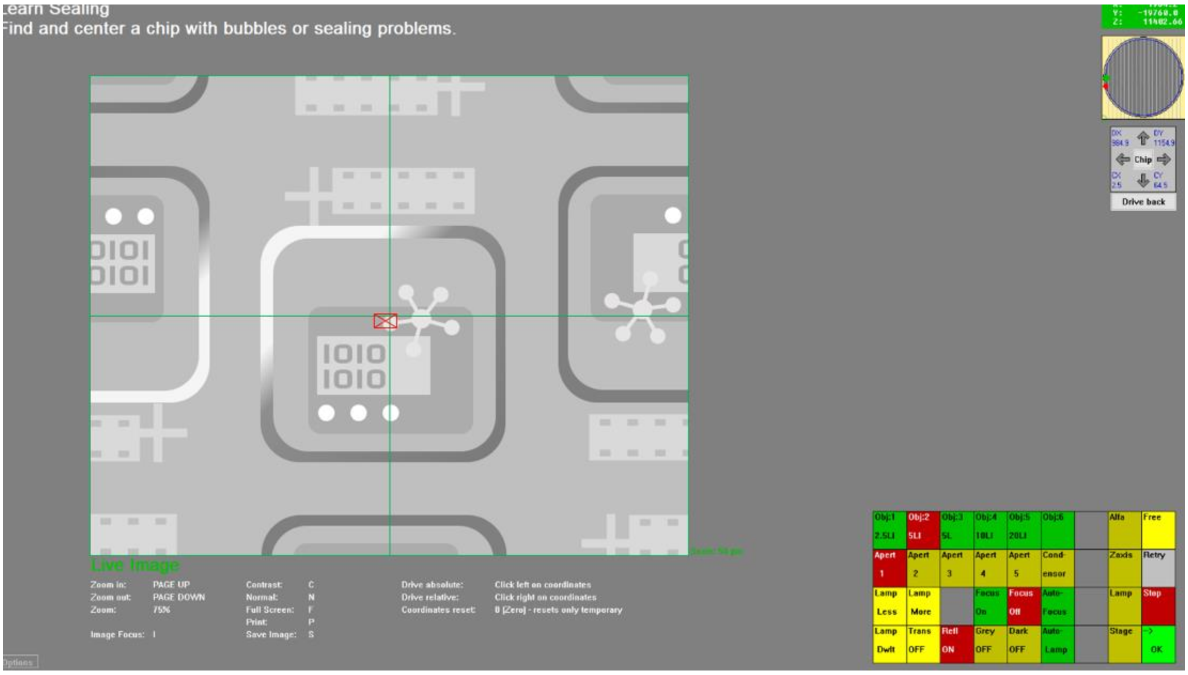
After accepting the alignment parameter with **OK**, the **Question** window is shown. For frit sealing type, the **Question** window asks whether to drive to the learned sealing chip with defects.

For the advanced eutectic bonding type, the **Question** window asks whether to drive to the learned good sealing chip.

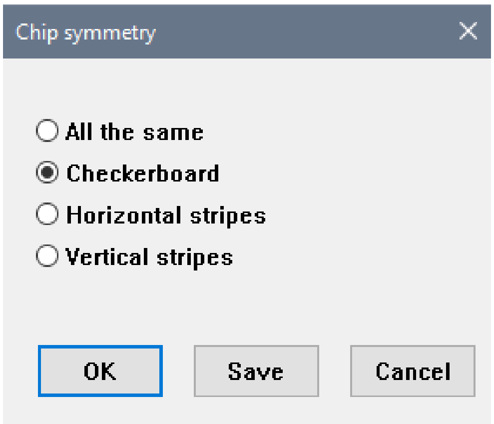


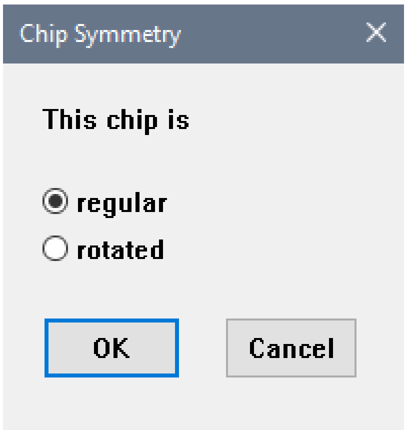
Selection	Description
YES	To drive to the learned sealing chip with defects for **Frit Sealing** or drive to the learned good sealing chip for **Advanced Eutectic Bonding** type.
NO	Click this button to drive the stage to other suitable position in **Free mode**.
CANCEL	To interrupt the job learning. The Wizard stops.

After driving to the test chip by answering **YES**, search and center a chip with bubbles or sealing problems which look brighter than the sealing area for frit sealing. The focus can be optimized, as the contrast may change according to the z-level. For advanced eutectic bonding, find and center a good chip to learn the nominal sealing level. It is essential that the whole chip is visible on the screen. Click **OK** or **ENTER** to continue.



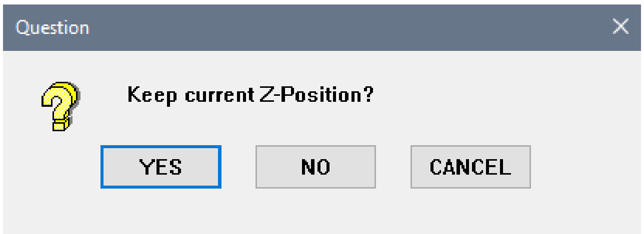
If **Checkerboard, Horizontal or Vertical stripes** is selected in previous **Chip Symmetry** window, this **Chip Symmetry** window will be shown for defining if this chip is either in regular or rotated arrangement. If **All the same** is selected in previous **Chip Symmetry** window, this **Chip Symmetry** window will be skipped and you directly go to the next **Question** window to keep current Z-position. Click **OK** to continue.





Selection	Description
regular	Click this button if this chip is in regular arrangement.
rotated	Click this button if this chip is in rotated arrangement.
OK	To continue with job learning.
CANCEL	To interrupt the job learning. The Wizard stops.

Subsequently, the next **Question** window asks whether to keep current Z-position. The procedure is the same as described in device inspection. (see 3.5.1.3).

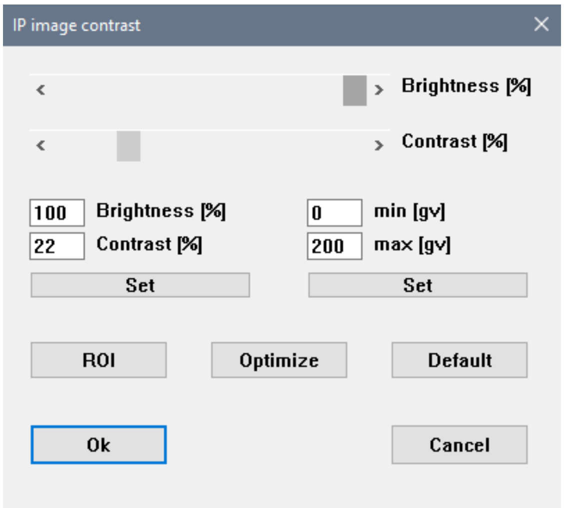


3.5.3.3 Contrast Enhancement

This option is only useful for defect detection in the regions with low contrast. It cuts the narrow range of gray values that represent the image content out and stretch it. Because the contrast enhancement increases the general gray value variation it should only be used if necessary. If ever possible use **Default** contrast.

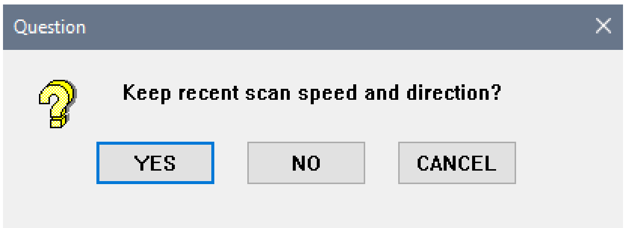
Move the slider for brightness [%] and contrast [%] with the mouse to adjust the contrast if all the bubbles/defects are not clearly recognizable. The effects of these changes are immediately displayed in the **IP** window. This adjustment has no effect on the scanned images. It supports the digital post-processing for the evaluation only.

Typically, there is no need to perform contrast enhancement if the defects are clearly recognizable. Click **OK** to continue. The brightness will be auto adjusted (see 3.5.1.1).

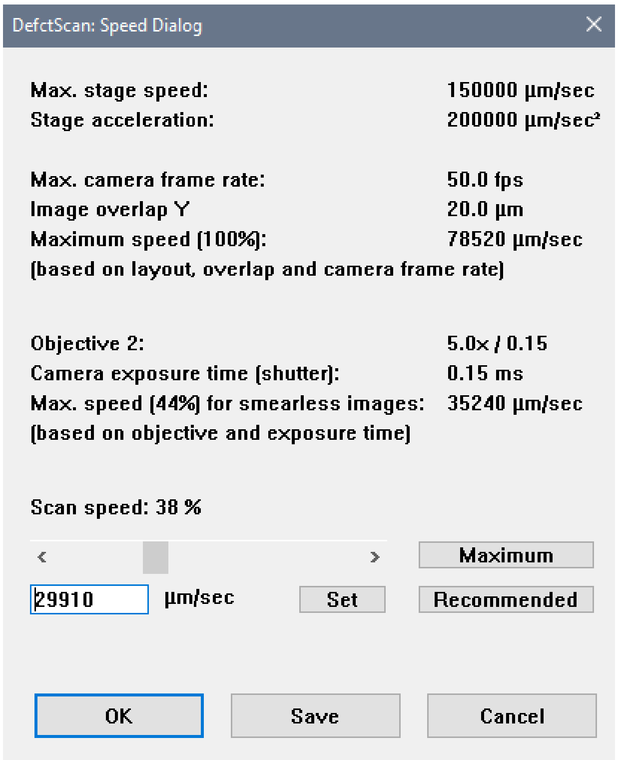


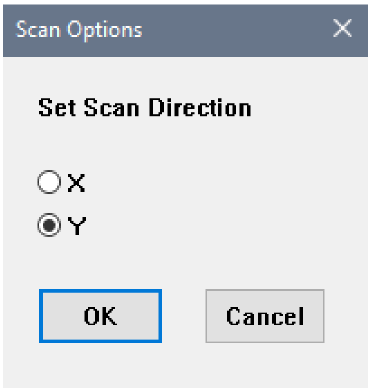
Subsequently, the **Question, Speed Dialog and Scan Options** windows are shown one after another. The procedure for the scan speed and direction are same as described in device inspection (see 3.5.1.9).

The **Question** window asks to keep the recent scan speed and direction.



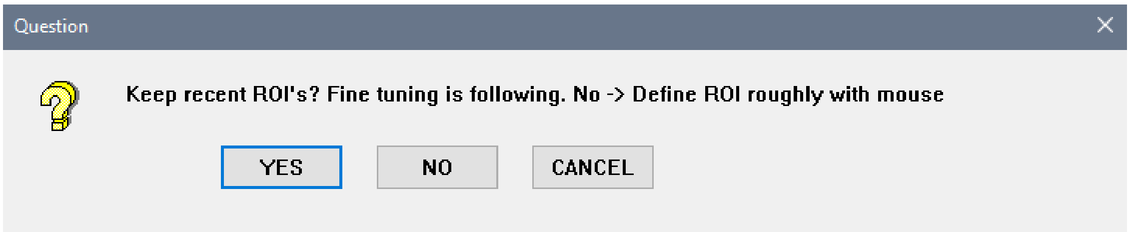
In the **Speed Dialog** window, click the **Recommended** button to get the recommended scan speed. The recommendation depends on many parameters and tries to find out the maximum scan speed to get usable images. However, recommended scan speed may be too fast. In case of blurry images it is advised to define the scan speed lower than the recommended scan speed. The scan speed can be adjusted by moving the slider or edit in the entry field and click **Set** button. Click **OK** to continue.



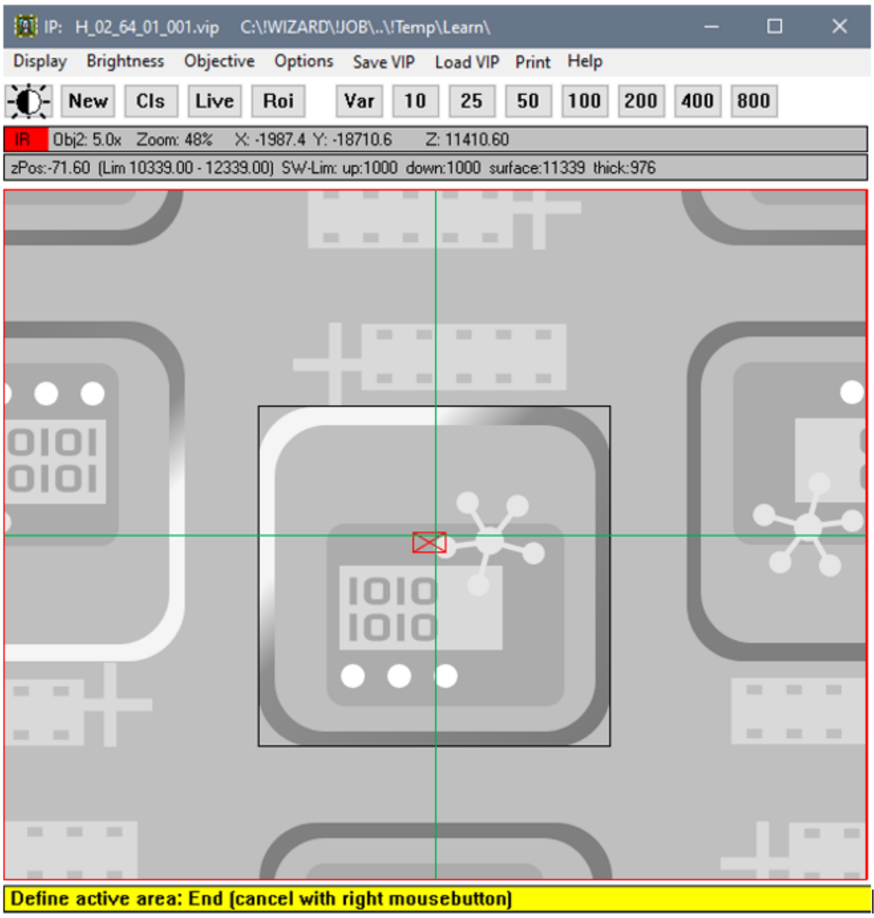


3.5.3.4 Inspection ROI

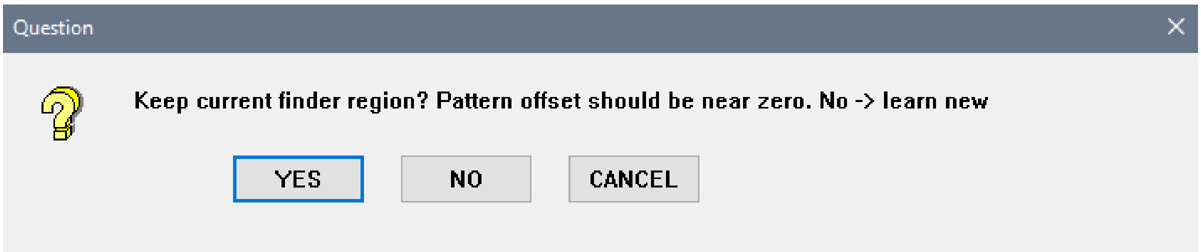
The Inspection ROI defines the active area for the sealing inspection. The image is only processed inside this ROI. The **Question** window asks to keep recent ROI. After **NO**, you redefine the ROI roughly with the mouse in the **IP** window.



Selection	Description
YES	To keep recent ROI, following with the fine tuning in the ROI parameters window.
NO	To redefine new ROI with the mouse
CANCEL	To interrupt the job learning. The Wizard stops.

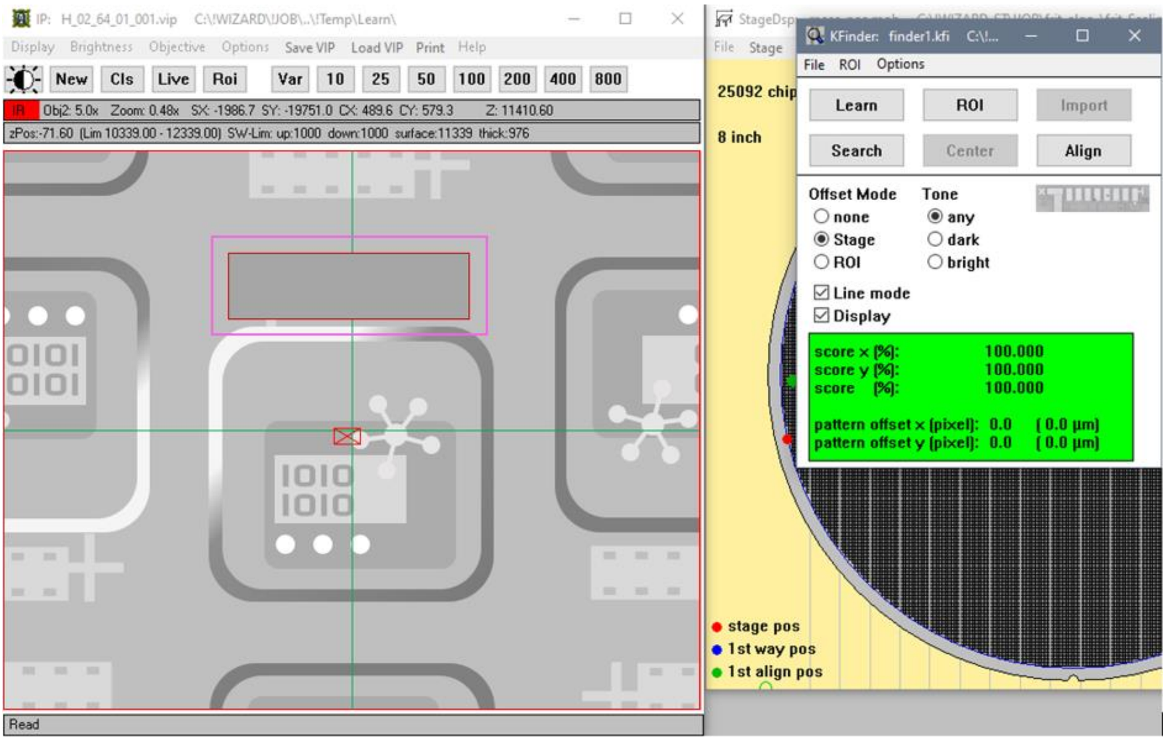


If the radio button **Finder** was selected in the **Alignment Parameter** (3.5.1.2), the **KFinder** window is shown for alignment purpose. It is executed by software, not by hardware stage moving. The **KFinder** executes a search based on the existing parameter files. The difference between the current image and the template is displayed in the **IP** window. The next **Question** window asks whether to keep the current finder region. The pattern offset x and y should be very near to zero or less than 0.5 µm during learning, otherwise the ROI will be shifted later during evaluation.

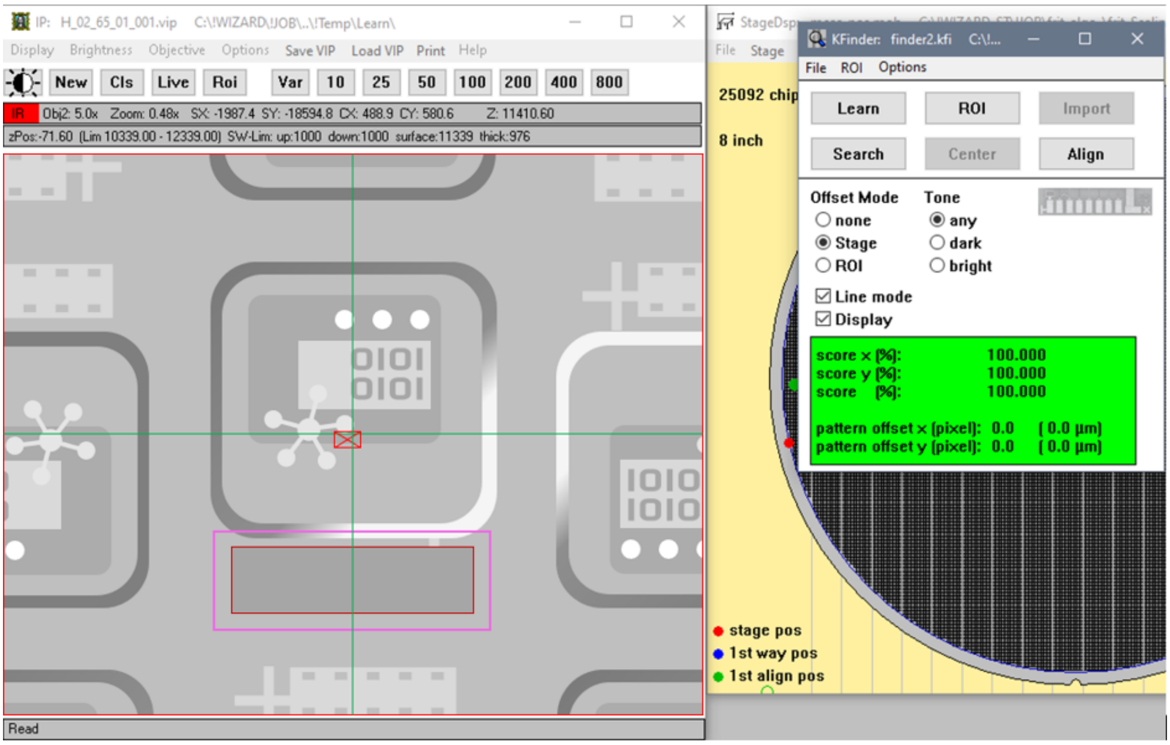


Selection	Description
YES	To keep finder region if the pattern offset x and y are very near to zero
NO	To redefine new finder region if the pattern offset x and y more than 0.5 µm.
CANCEL	To interrupt the job learning. The Wizard stops

By answering **NO** to the previous question, a new finder region can be redefined. In the **KFinder** window. Click the **Learn** button and define a new finder ROI for the regular chip in the **IP** window. Define the region where the area inside is identical in all chips. Click the **Search** button and both the pattern offset x and y will become zero value. Click **Continue** in the **MacWizCont** window.



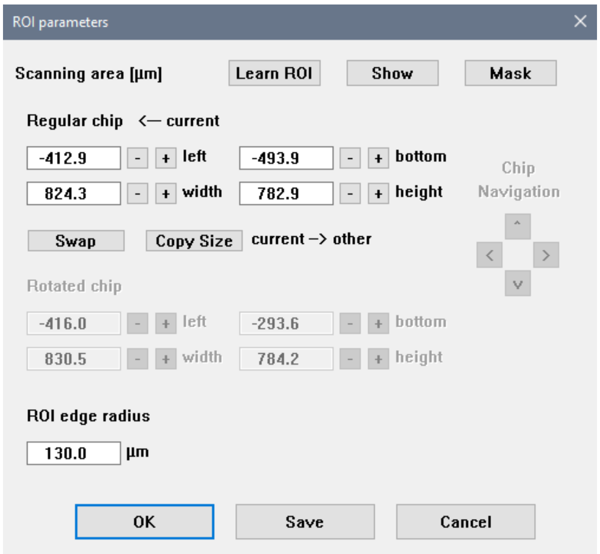
The shown area inside the red ROI is a difference image calculated from the recent image and the finder template. If the **rotated** radio button was selected in the **Chip Symmetry** window previously (see 3.5.3.2), the rotated image will be loaded in the **IP** window. Repeat again the same steps to define ROI and finder region for the rotated chip, as learned for regular chips (see 3.5.3.4).

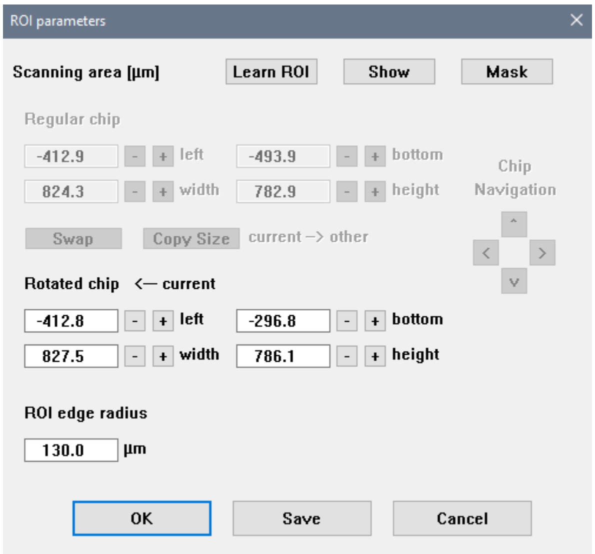


After clicking **Continue** in the **MacWizCont** window, the **ROI parameters** window is shown as an overview for checking and finetuning if needed. Click **Show** to display only the ROI or **Mask** to display the mask in the **IP** window. The size values in the entry field for regular are adjustable by +/- buttons. To redefine the ROI in the **IP** window, click the **Learn ROI** button. By clicking the **Copy Size** button, the size value of a regular chip can be copied to a rotated chip by adapting the right and upper ROI edge.

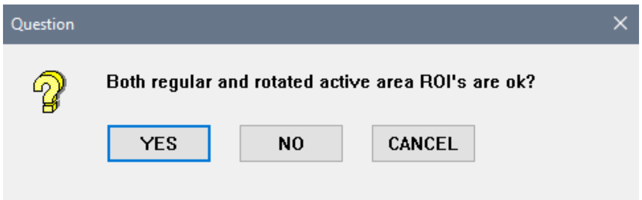
If the sealing area is a rectangle with a radius at the 4 corners, radius value can be entered in the entry field of **ROI edge radius** to mask the corner as well. Click the **Mask** button again to check on the corner masking and fine tune the radius value until fully masks the non-active area at corner. The ROI learning is on the saved scanned image from file, so the **Chip Navigation** is not working here.

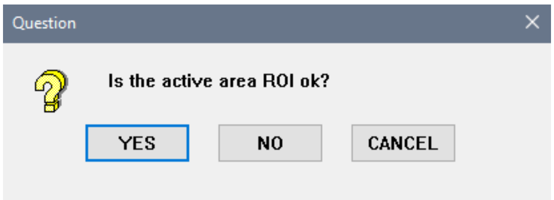
Click **OK** to continue with the rotated chip and the rotated image will be loaded in the **IP** window. Repeat the same steps as for the regular chip to finetune the ROI parameter for the rotated chip.





After clicking **OK** in the **ROI parameters** window, the **Question** window asks if both regular and rotated active area ROIs are OK.

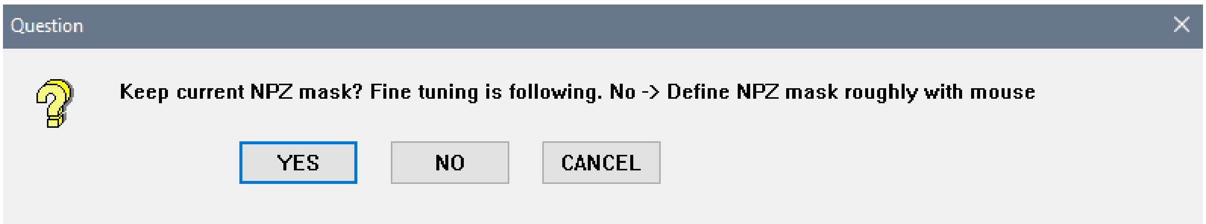




Selection	Description
YES	To accept both regular and rotate active area ROI
NO	To redefine both regular and rotate active area ROI
CANCEL	To interrupt the job learning. The Wizard stops.

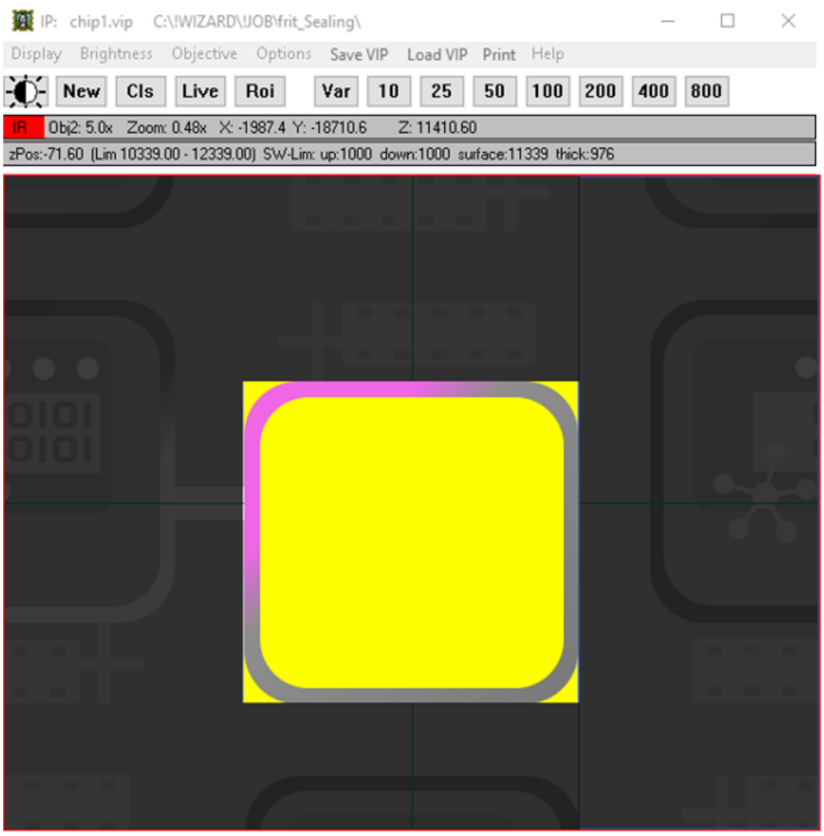
3.5.3.5 Non-Processing Zone (NPZ) Mask

The **Non-Processing-Zone (NPZ)** mask defines the non-processing area in the active area of chip image. The NPZ area is the inner area of the chip where it is non-sealing area. The image inside this masked area is not processed.

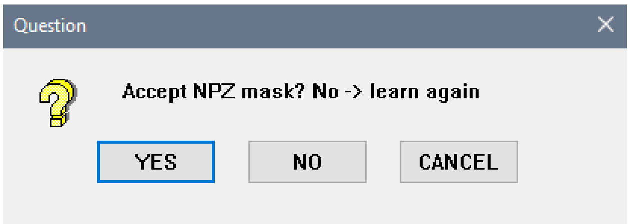


Selection	Description
YES	To keep the current NPZ masks, followed by the fine tuning in the NPZ mask parameters window
NO	To redefine new NPZ masks with the mouse
CANCEL	To interrupt the job learning. The Wizard stops.

The **Question** window asks to accept current NPZ masks. After **NO**, redefine the NPZ mask area by mouse in the **IP** window. The masked area can be further finetuned later in the entry field of **NPZ mask parameters** window.



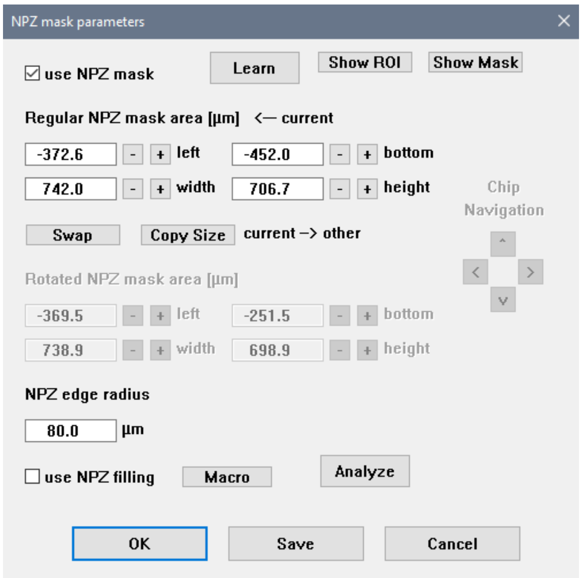
After redefining the NPZ mask area, the next **Question** window asks whether to accept the NPZ mask.

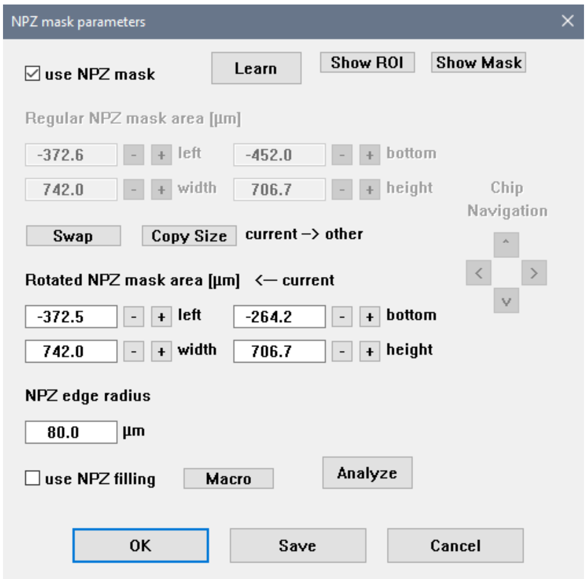


Selection	Description
YES	To keep current NPZ masks
NO	To redefine new NPZ masks
CANCEL	To interrupt the job learning. The Wizard stops.

By answering **YES** to accept the NPZ mask, the **NPZ mask parameters** window is shown. The function in the **NPZ mask parameters** window is very similar to the **ROI parameter** window (see 3.5.3.4). **NPZ edge radius** can be finetuned as the **ROI edge radius.** There are two check boxes for the method selection to define the NPZ mask, either use NPZ mask or NPZ filling.

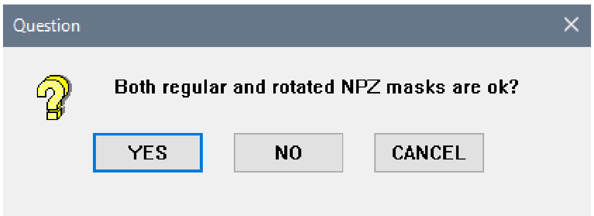
If the **rotated** radio button was selected in the **Chip Symmetry** window previously (see3.5.3.2), the rotated image will be loaded in the **IP** window. Repeat the same steps to define NPZ mask area for the rotated chip, as learned for regular chips (see 3.5.3.5).

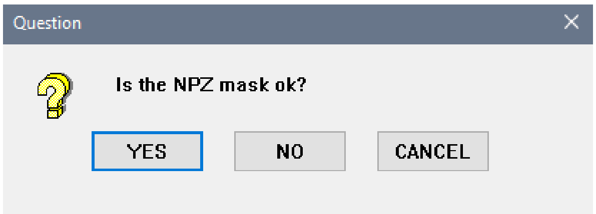




Selection	Description
Use NPZ mask	To use a rectangular ROI which surrounds the NPZ mask area, optionally with rounded corners. Typically this method is used.
Use NPZ filling	To use a software-controlled process that marks automatically the NPZ area using gray value levels in a flood-fill algorithm. Can be combined with NPZ mask. The resulting NPZ may have an irregular shape.
OK	To continue with job learning
Save	To save the NPZ mask parameters
CANCEL	To interrupt the job learning. The Wizard stops.

After clicking **OK** in the **ROI parameters** window, the **Question** window asks if both regular and rotated NPZ masks are OK.



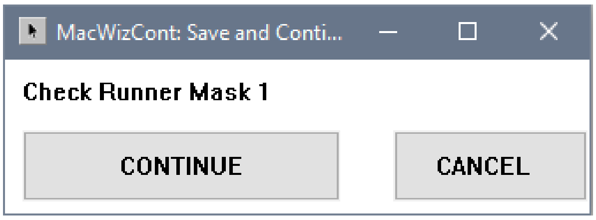


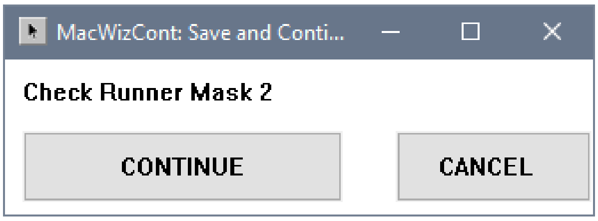
Selection	Description
YES	To accept both regular and rotated NPZ masks
NO	To redefine both regular and rotated NPZ masks
CANCEL	To interrupt the job learning. The Wizard stops.

3.5.3.6 Runner Mask

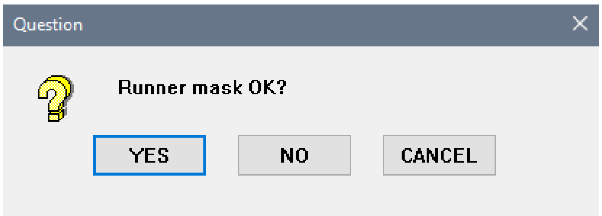
The **Runner Mask** defines the non-processing area in the sealing area of chip image. The masking areas for both regular and rotated chips are painted respectively using the mouse.

The regular chip image is displayed in **IP** window to check runner mask 1. Click **Continue** in the **MacWizCont** window. The rotated chip image is displayed in **IP** window to check runner mask 2. Click **Continue** in the **MacWizCont** window.





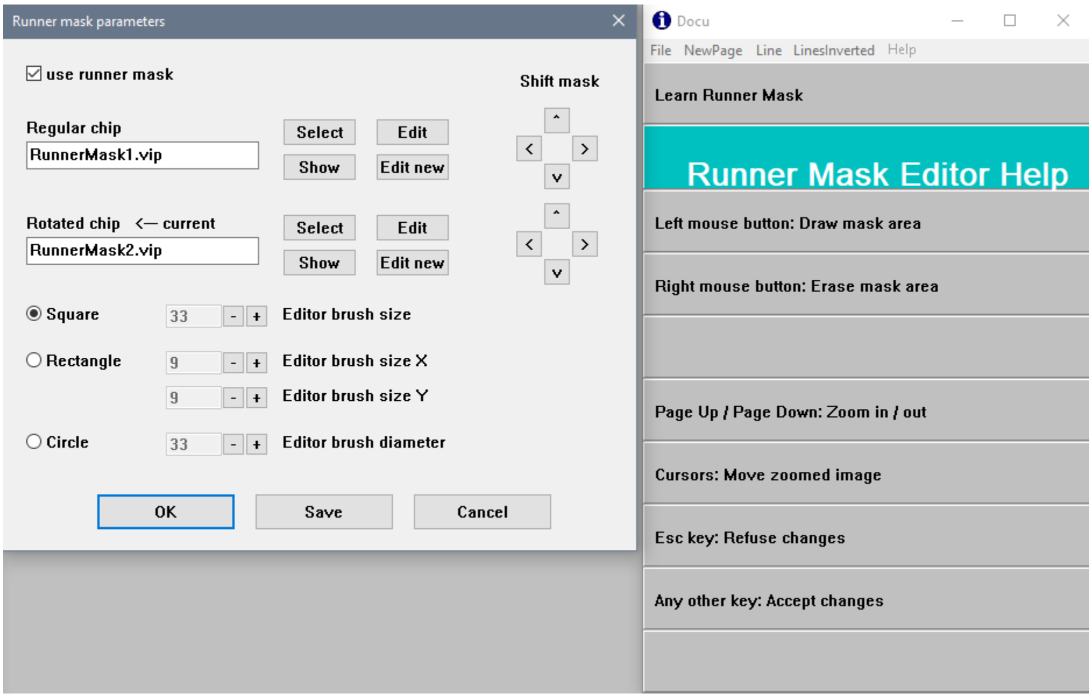
The **Question** window asks if the runner masks are OK. If no runner mask is needed, just click **YES** to continue.



Selection	Description
YES	To accept both runner masks
NO	To edit runner masks
CANCEL	To interrupt the job learning. The Wizard stops.

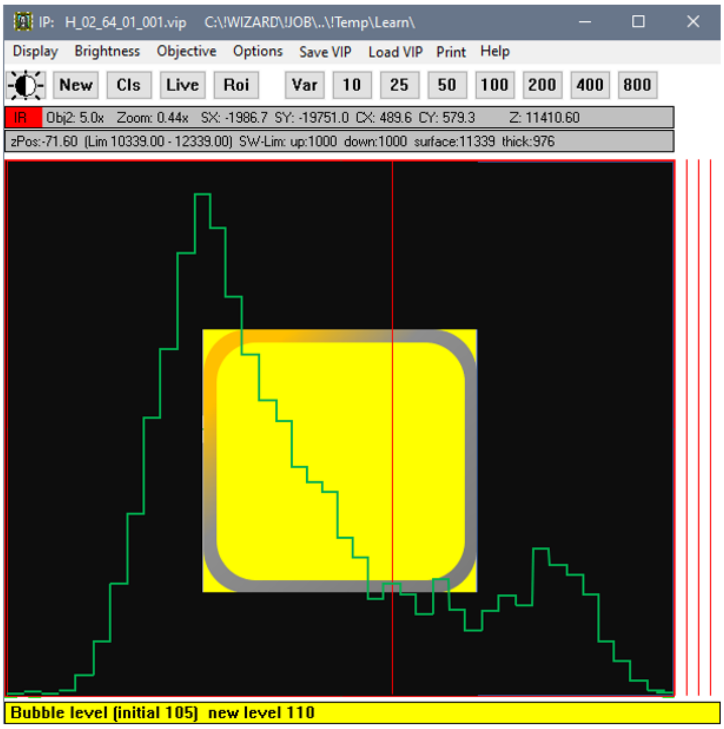
If you want to edit the runner mask, click **NO**. The **Runner mask parameters** window is shown for editing both the runner mask of regular and rotated chip images. By clicking **Edit**, you can paint the runner mask in the IP window. Hold down the left mouse button to paint the mask areas and hold down the right mouse button to erase the mask areas. Activate the **use runner mask** checkbox.

The **Docu** window displays information and the guideline for **Runner Mask Editor Help** window.

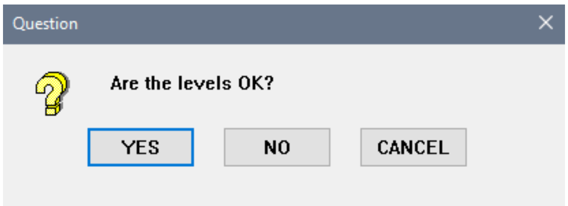


3.5.3.7 New NPZ Level and Bubble Level

The histogram curve in green color is displayed in the **IP** window. Use the mouse to define the detection threshold by moving the vertical red line. The highest peak at the left side is the good sealing and the higher peak at the right side are defects. Depending on the cursor position, the sealing area in the image changes its colours. It is advisable to set the vertical red line at the center between good sealing (left highest peak) and defects (right peak) to define as the detection threshold.

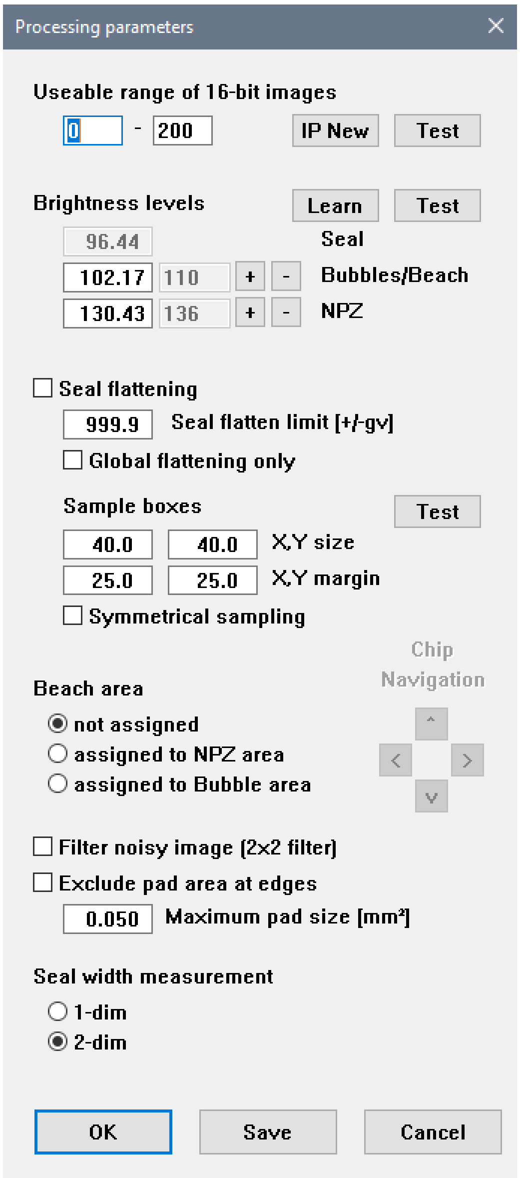


The **Question** window asking if the levels are OK.



Selection	Description
YES	To accept the learned levels
NO	To redefine the levels
CANCEL	To interrupt the job learning. The Wizard stops.

In the **Processing parameters** window, all essential parameters of sealing inspection are listed for overview and editing. Click the **Test** button to check after defining the processing parameters.

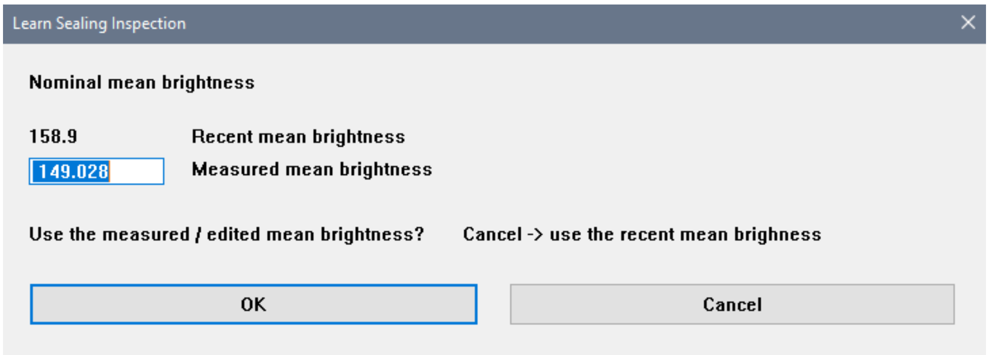


Selection	Description
Usable range	The defined contrast range
Brightness levels - Seal	The average gray value of sealing area
Brightness levels-Bubbles/Beach	The learned gray value threshold used to identify the bubble and the beach areas
Brightness levels - NPZ	To identify the NPZ automatically if NPZ filling is used
Seal flattening	A software procedure that equalizes brightness differences in the seal area either in one chip or from chip to chip in the wafer. Therefore the gray value level is measured in sample boxes in the corners of the sealing area. The allowed gray value range of equalizing and the sample box positions are set here.

Beach area	The intermediate between NPZ area and sealing area. Define if the beach area should be part of the NPZ area, the bubble area, or not assigned. If the NPZ mask directly joins the sealing area there is no beach area.
Seal width measurement 1 dim	The detection of sealing breach across the sealing area in 1 dimension way (a breach is detected only if the breach is perpendicular across the sealing area)
Seal width measurement 2 dim	The detection of sealing breach across the sealing area in 2 dimensions way (a breach is detected with an angle across the sealing area)
OK	To continue with job learning
Save	To save the inspection parameters
CANCEL	To interrupt the job learning. The Wizard stops.

3.5.3.8 New Advanced Eutectic Bonding Level

In the **Sealing inspection** window, recent and measured mean brightness is shown. The mean brightness can be edited in the entry field. Click **OK** to continue. Click **Cancel** if wish to use the recent mean brightness.

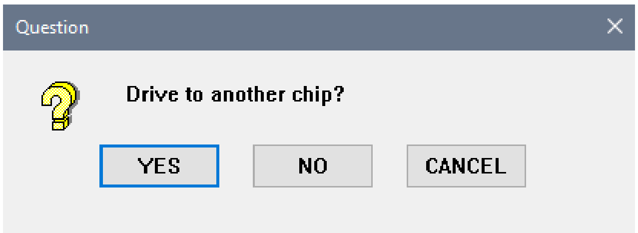


Selection	Description
OK	To use the measured/edited mean brightness
CANCEL	To use the recent mean brightness

The **Question** window asks whether the mean brightness is OK.

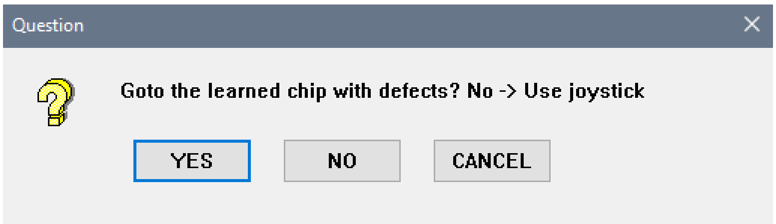
Selection	Description
YES	To accept the mean brightness
NO	To go to another chip and remeasure the mean brightness
CANCEL	To interrupt the job learning. The Wizard stops.

If answered **NO** to the previous question, the next **Question** window asks for driving to another chip. The measurement procedure of the mean brightness is repeated.



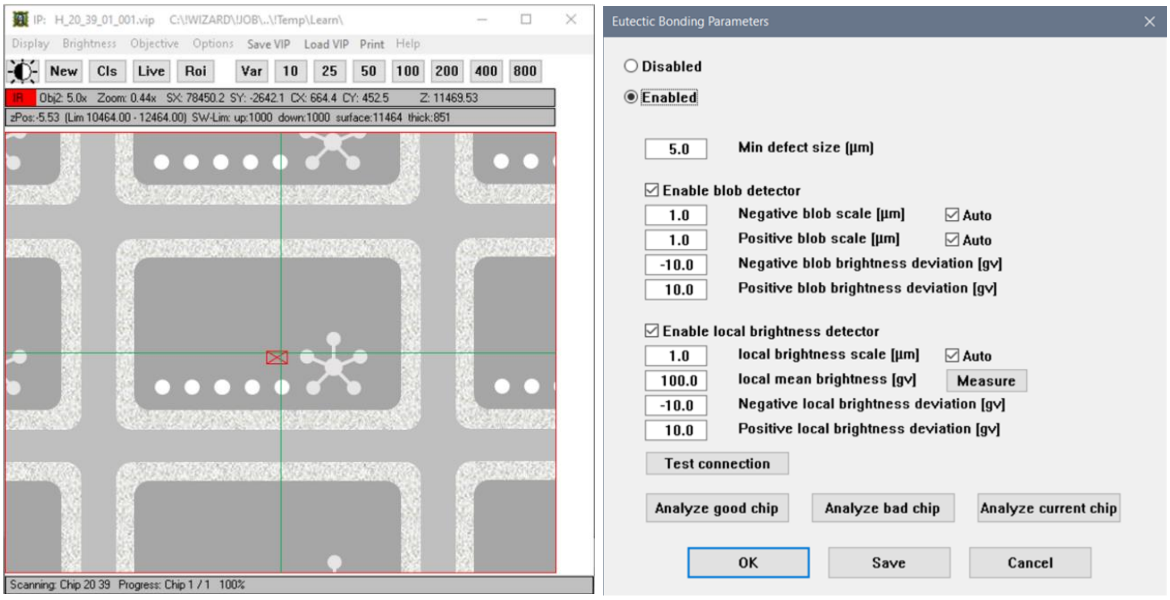
Selection	Description
YES	To drive to another chip to remeasure the mean brightness
NO	To stay at current chip to remeasure the mean brightness
CANCEL	To interrupt the job learning. The Wizard stops.

If you answered **YES** to the previous question to accept the mean brightness, the next **Question** window asks whether go to the learned chip with defect.



Selection	Description
YES	To go to learned chip with defects
NO	To go to other chip with defects using the joystick
CANCEL	To interrupt the job learning. The Wizard stops.

The **Eutectic Bonding Parameters** window is shown. Defects are detected by two independent engines, blob detector and local brightness detector**.** The detectors can be activated by **Enable blob detector** and **Enable Local brightness detector** independently in the **Eutectic Bonding Parameters** window. Note that the **Min defect size** parameter is currently not active.



The blob detector spots regions of different brightness (blobs) with respect to the surrounding background. It is implemented as a bandpass spatial filter. For **blob detector**, the following parameters can be defined.

Selection	Description
Negative blob scale [µm]	Minimum spatial scale of the detected blobs. The sensitivity of the detector is reduced for blobs smaller than this dimension, e.g. small grains. Can be automatically chosen by the algorithm by activating the **Auto** checkbox.
Positive blob scale [µm]	Maximum spatial scale of the detected blobs. The sensitivity of the detector is reduced for blobs bigger than this dimension. Can be automatically chosen by the algorithm by activating the **Auto** checkbox
Negative blob brightness deviation [gv]	Lowest (negative) allowed deviation of the local brightness from the local background. The closer to zero, the more sensitive the detector is for dark blobs.
Positive blob brightness deviation [gv]	Highest allowed deviation of the local brightness from the local background. The closer to zero, the more sensitive the detector is for bright blobs.

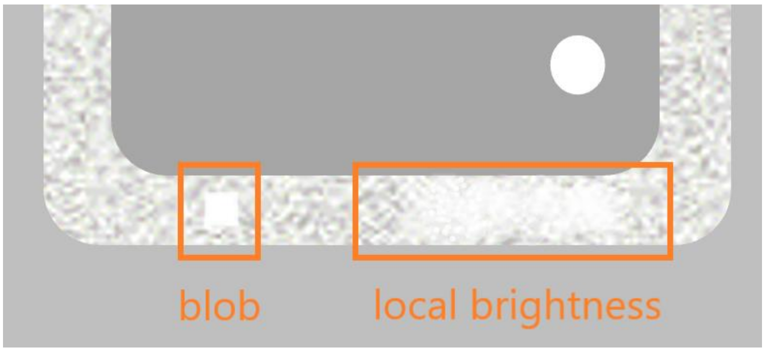
The local brightness detector is sensitive to regions of different brightness with respect to the nominal brightness learned on a "good" chip. Its usage is twofold:

- To spot regions of low or irregular grain density. The local brightness is a proxy of the density of grains.
- To spot regions of different average brightness.

For **Local brightness detector**, the following detector parameters can be defined.

Selection	Description
Local brightness scale [µm]	The detector is most sensitive for defects of this dimension. Can be automatically chosen by the algorithm by activating the **Auto** checkbox.
Local mean brightness [gv]	The nominal brightness to which the actual brightness is compared. By pressing the **Measure** button, the average brightness of the current chip is measured and used.
Negative local brightness deviation [gv]	Allowed deviation of the local brightness of darker areas from the nominal brightness. The closer to zero, the more sensitive the detector is for regions darker than the nominal value.
Positive local brightness deviation [gv]	Allowed deviation of the local brightness of brighter areas from the nominal brightness. The closer to zero, the more sensitive the detector is for regions brighter than the nominal value.

This is an example of defects detected by the blob and the local brightness detectors:



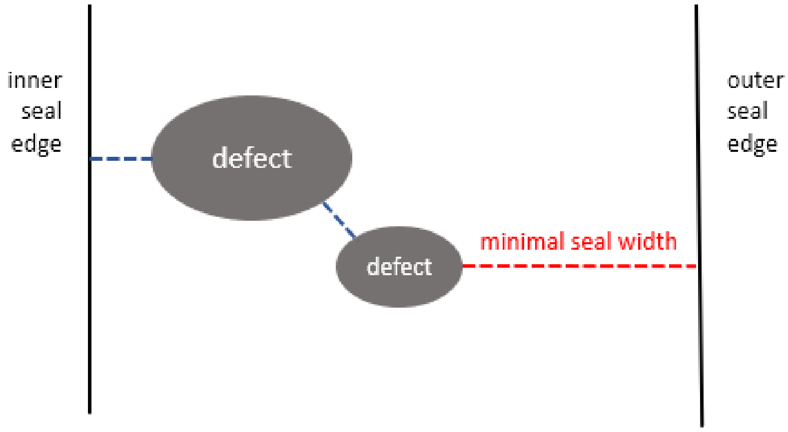
Adjust the value in the entry field and click **Analyze bad chip** to test the defect detection in the **IP** window. Click **Analyze good chip** to test if the setting is not too sensitive and finds defects in a good chip. For this test the previously saved images of a good and a bad chip are used. **Analyze current chip** evaluates a new image from camera.

Click **OK** to accept the defined parameters and to continue.

After defect detection, the following parameters are extracted:

- Average brightness of the whole seal (gv: gray values from 0 to 255)
- Average brightness of each seal side (top, bottom, left, right) (gv)
- Minimal width of each seal side (top, bottom, left, right) (µm)

  The minimal width is defined by considering the shortest path across the seal. In simple cases, this path will be a straight line from the inner to the outer seal edge, and the minimal width is the length of this line. If instead defects are present, these are not considered to contribute to the shortest path, so the latter will be a set of hops between defects and seal edges. In this case, the minimal seal width is defined to be the longest of these hops.

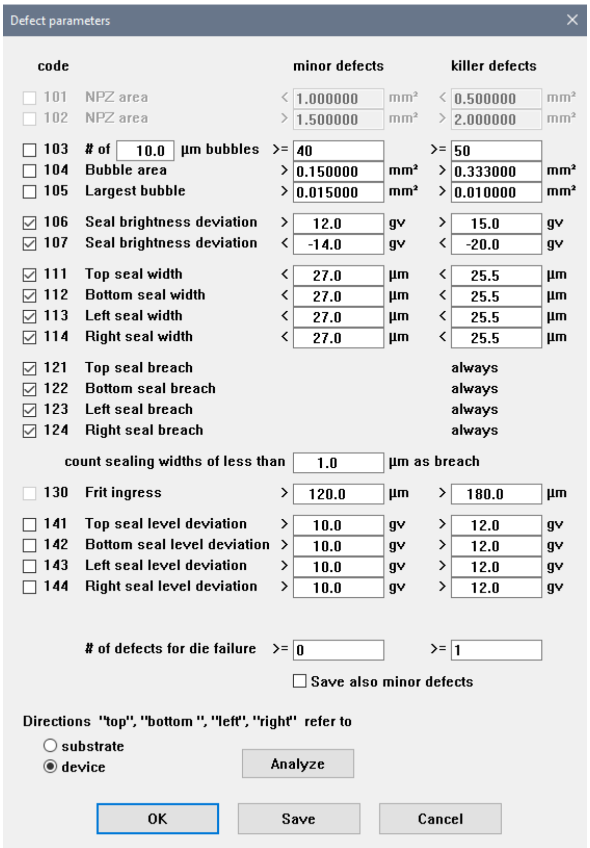


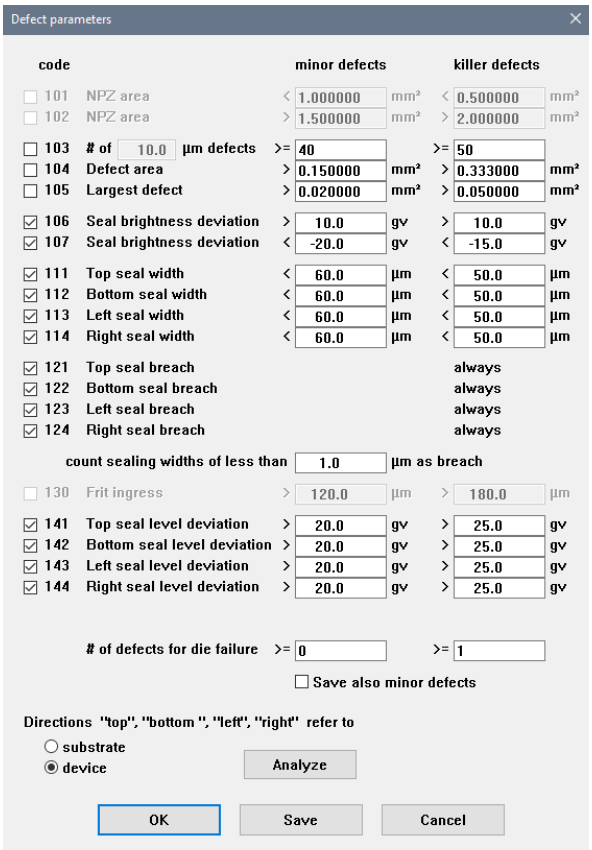
- number of defects
- area covered by defects (µm²)
- area of the largest defect (µm²)

3.5.3.9 Defect Criteria

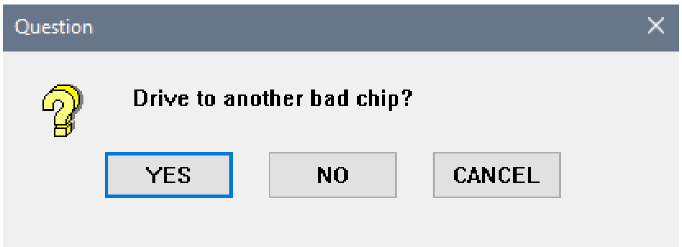
The **Defect Parameters** window is shown and the defect criteria can be set here. During the criteria setting, click the **Analyze** button to test the defined defect criteria. The defect is checked with the criteria parameters row by row.

The left window is for **Frit Sealing** and right window is for **Advanced Eutectic Bonding**. NPZ area is only active if NPZ filling is enabled.





After clicking **OK** to accept the defect criteria, the **Question** window asks whether to drive to another bad chip to test processing parameters again (see 3.5.3.9) for advanced eutectic bonding.



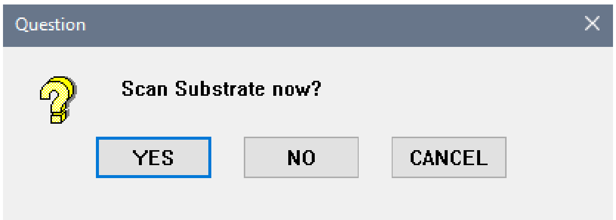
Selection	Description
YES	To drive to another bad chip in **Free mode**.
NO	To continue with the scan focus option
CANCEL	To interrupt the job learning. The Wizard stops.

By clicking **NO** in the previous question or for frit sealing, the **Question** window asks whether to keep the recent scan focus option. The procedure of the focus and scan speed settings are the same as described in device inspection (see 3.5.1.3 and 3.5.1.9).

All sealing parameters are learned and user can continue with a test scan.

3.5.3.10 Scanning

After all the parameters are defined, the **Question** window asks whether to scan the substrate now. The wafer scan is the same as in device inspection (see 3.1.5.10).



After the scan is completed, offline review can be performed as described for inspection job (see 2.6.1)

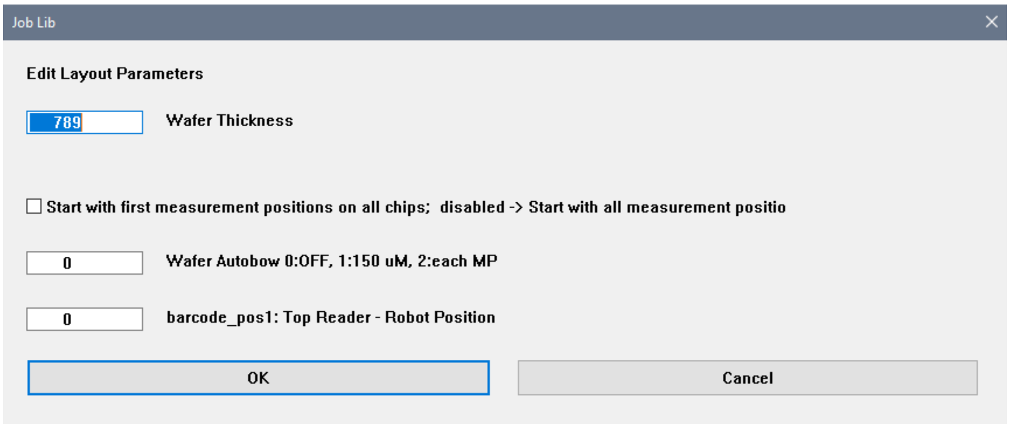
3.6 Quick Editing of Job Parameters

If a job is selected in the **Applic** window and loaded by either **# ASK #**, **# EXECUTE #** or **# LEARN #**, the job parameters can be quickly be checked or changed by pressing the **F9** function key. To work with this job parameters, you should be very familiar with the job and the tool. Prior to change a job parameter, the job needs to be learned by teaching the job. For example, the job parameters are layout, alignment, global measurement and each of the measurement position parameters.

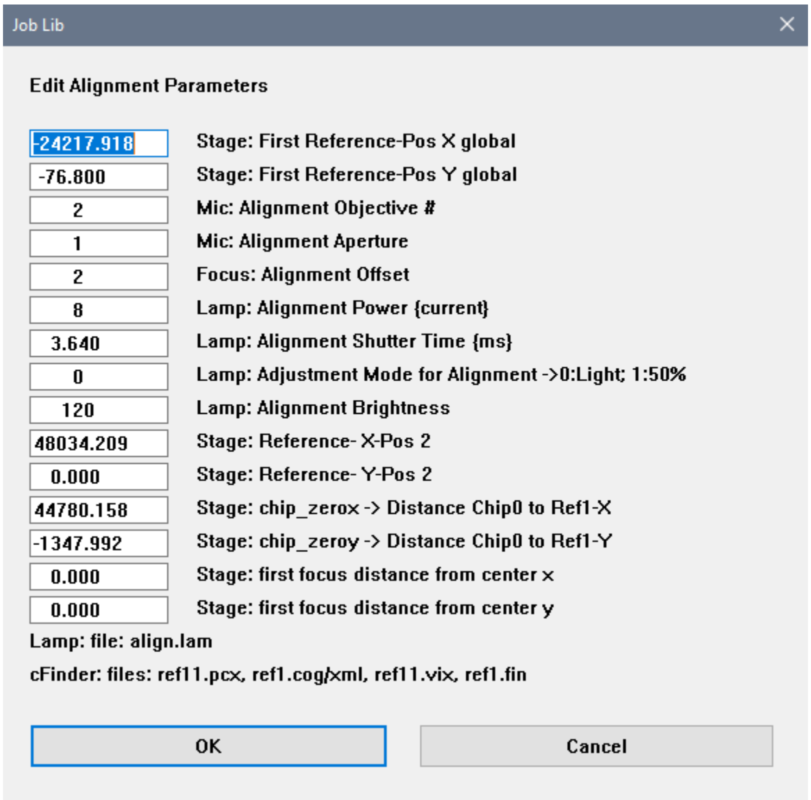
As a guideline, it makes no sense to change the selection of objectives here because other parameters depend on this setting. The necessary changes of the lamp control, finder and measurement ROI can be done more conveniently by teaching the job. A possible application is a subsequent change of shutter time for scanning.

3.6.1 CD/OVL/FT Parameters

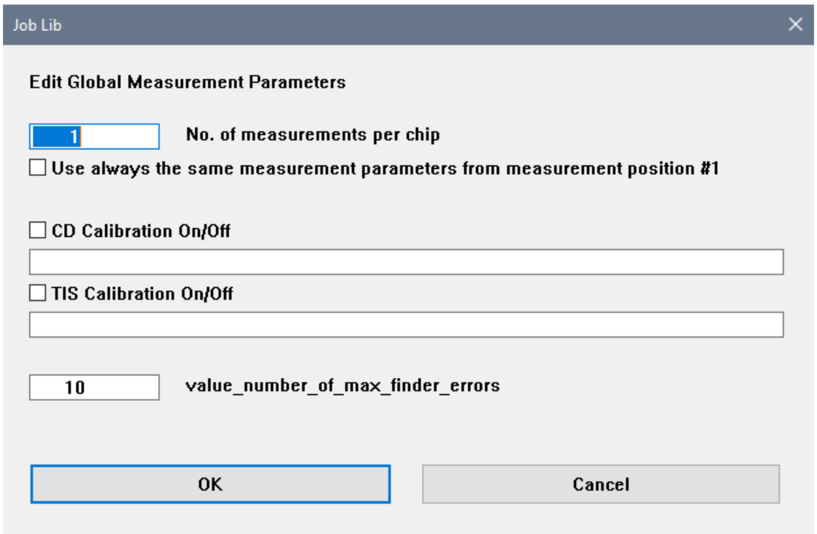
By pressing the **F9** function key, the first **Job Lib** window is shown to editing the **Layout Parameters** of the job (e.g. wafer thickness, processing sequence of the measurement positions and barcode position). Click **OK** to continue to the next **Job Lib** window, one after another. If you click **Cancel**, the changes are not stored. **OK** stores the parameters in this dialog to the INI file(s). User can cancel the next dialog if this was the only change that had to be done. There is no need to click through all dialogs.



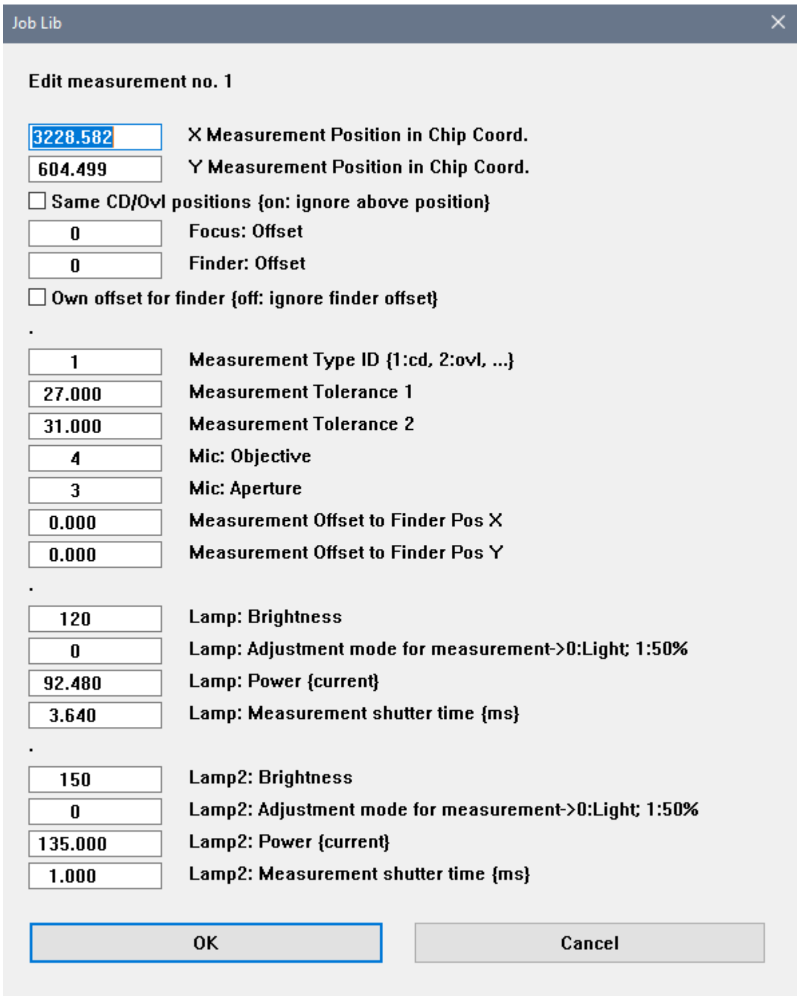
The next **Job Lib** window is to edit the **Alignment Parameters** (e.g. brightness and alignment position).



The next **Job Lib** window is to edit the **Global Measurement Parameters**. The CD Calibration or TIS correction file can be easily assigned to the job.



The next **Job Lib** window is to edit the **Measurement Parameters** (e.g. measurement position if the laser focus is positioned too close an structure edge or nearly out of FOV, brightness and measurement tolerance). The same window is shown for each of the measurement positions of the chip.



After confirming the last measurement position with **OK**, the quick editing is ended.

3.6.2 Defect Inspection Parameter

This is the same as the quick editing for CD/OVL/FT parameters and done by pressing the **F9** function key. The first two **Job Lib** windows are used to edit the **Layout Parameters** and the **Alignment Parameters**. Click **OK** to continue to the next **Job Lib** window, one after another.

Subsequently, different parameter windows are shown according to the inspection, device inspection, blank or sealing inspection. It is not recommended to make the changes here, because some changes may need the chip image to be displayed in the **IP** window.